BT601 - VIROLOGY

Lesson 1-105

Compiled by Fasiha Arooj

Lesson 1: Viruses definition and meaning

Virus name

The name “virus” has been derived from a Latin word meaning “slimy liquid” or “poison.” Viruses are microscopic parasites that lack the capacity to thrive and reproduce outside of a host body. These can multiply only in living cells of animals, plants, or bacteria. Latin word meaning poison or venom given by Beijerinck (1897).

WHAT ARE VIRUSES?

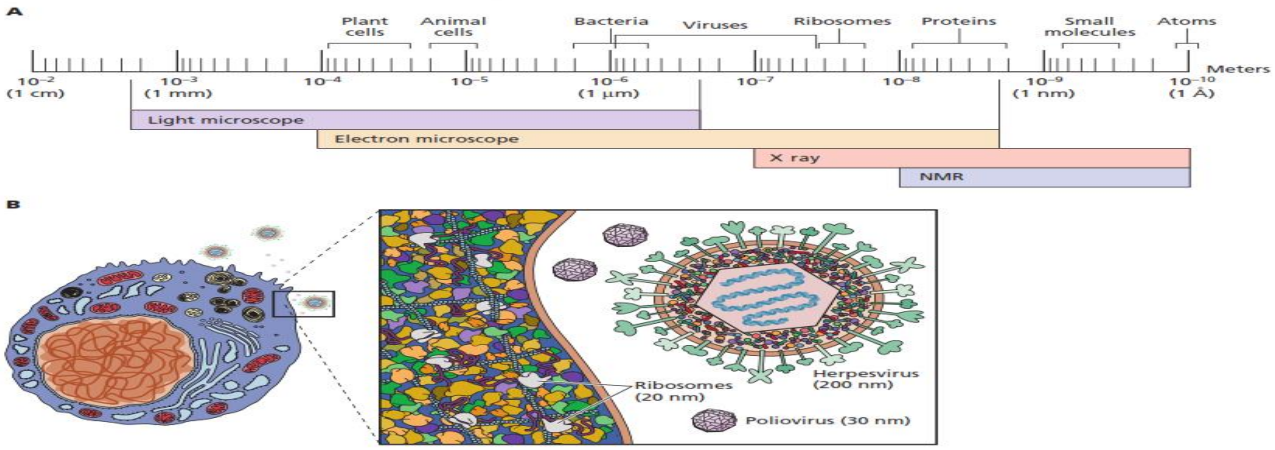
Viruses are submicroscopic, obligate intracellular parasites. Most are too small to be seen by optical microscopes, and they have no choice but to replicate inside host cells. This simple but useful definition goes a long way toward describing viruses and differentiating them from all other types of organism. However, this short definition is not completely adequate.

Viruses are unique in nature. They are the smallest of all self-replicating organisms, historically characterized by their ability to pass through filters that retain even the smallest bacteria. In their most basic form, viruses consist solely of a small segment of nucleic acid encased in a simple protein shell. Viruses have no metabolism of their own but rather are obliged to invade cells and parasitize subcellular machinery, subverting it to their own purposes.

Virus Size

- Submicroscopic

- Smaller than bacteria



- Smaller than the smallest cell

Figure: (A) Sizes of animal and plant cells, bacteria, viruses, proteins, molecules, and atoms are indicated. The resolving powers of various techniques used in virology, including light microscopy, electron microscopy, X-ray crystallography, and nuclear magnetic resonance (NMR) spectroscopy, are indicated. Viruses span a broad range from that equal to some small bacteria to just under ribosome size. **(B)** Illustration of the size differences among two viruses and a typical host cell.

Lesson 2: Viruses and their discovery

Discovery of Virus

Human knowledge of virus diseases goes back a long way, although it is only much more recently that we have become aware of viruses as distinct from other causes of disease. The first written record of a virus infection is a hieroglyph from Memphis, the capital of ancient Egypt, drawn in approximately 3700 BC, which depicts a temple priest showing typical clinical signs of paralytic poliomyelitis.

Edward Jenner in May, 1796, used cowpox-infected to successfully vaccinate 8-year-old James Phipps.

Although initially controversial, vaccination against smallpox was almost universally adopted worldwide during the nineteenth century.

In 1892, **Dimitri Iwanowski**, a Russian botanist, showed that extracts from diseased tobacco plants could transmit disease to other plants after being passed through ceramic filters fine

enough to retain the smallest known bacteria. Unfortunately, he did not realize the full significance of these results.

A few years later (1898), Martinus Beijerinick confirmed and extended Iwanowski's results on tobacco mosaic virus (TMV) and was the first to develop the modern idea of the virus, which he referred to as contagium vivum fluidum ("soluble living germ").

In 1898 Beijernick Studied filtered diseased plant juices and found the caused disease in healthy plants

Wendell Stanley (1935) **discovered that: Viruses are composed of protein and nucleic acid Studied Diseased** tobacco plants TMV. He discovered isolates of virus and won Nobel award.

Friedrich Loeffler and Paul Frosch (1898) showed that a similar agent was responsible for foot-and-mouth disease in cattle, but, despite the realization that these new-found agents caused disease in animals as well as plants, people would not accept the idea that they might have anything to do with human diseases.

This resistance was finally dispelled in 1909 by **Karl Landsteiner and Erwin Popper**, who showed that poliomyelitis was caused by a "filterable agent"—the first human disease to be recognized as being caused by a virus.

Frederick Twort (1915) and Felix d'Herelle (1917) were the first to recognize viruses that infect bacteria, which d'Herelle called bacteriophages ("eaters of bacteria"). Investigation of bacteriophages established the foundations for the field of molecular biology, as well as fundamental

Insights into how viruses interact with their host cells.

Lesson 3: History of Virology

THE HISTORY OF VIROLOGY

The first written record of a virus infection is a hieroglyph from **Memphis,** the capital of ancient Egypt, drawn in approximately 3700 BC, which depicts a temple priest showing typical clinical signs of paralytic poliomyelitis.

Pharaoh Ramses V (فرعون), who died in 1196 BC and whose well-preserved mummified body is now in a Cairo Museum, is believed to have died from smallpox—the comparison between the pustules on the face of this mummy and those of more recent patients is startling.

Smallpox was endemic in China by 1000 BC. In response, the practice of variolation was developed. Recognizing that survivors of smallpox outbreaks were protected from subsequent infection, people inhaled the dried crusts from smallpox lesions like snuff or, in later modifications, inoculated the pus from a lesion into a scratch on the forearm. Variolation was practiced for centuries and was shown to be an effective method of disease prevention, although risky because the outcome of the inoculation was never certain.

Edward Jenner was nearly killed by variolation at the age of seven. Not surprisingly, this experience spurred him on to find a safer alternative treatment. On May 14, 1796, he used cowpox-infected material obtained from the hand of Sarah Nemes, a milkmaid from his home village of Berkeley in Gloucestershire, England, to successfully vaccinate 8-year-old James Phipps. Although initially controversial, vaccination against smallpox was almost universally adopted worldwide during the nineteenth century.

In 1892, **Dimitri Iwanowski**, a Russian botanist, showed that extracts from diseased tobacco plants could transmit disease to other plants after being passed through ceramic filters fine enough to retain the smallest known bacteria. Unfortunately, he did not realize the full significance of these results. A few years later (1898), Martinus Beijerinick confirmed and extended Iwanowski's results on tobacco mosaic virus (TMV) and was the first to develop the modern idea of the virus, which he referred to as contagium vivum fluidum ("soluble living germ").

In 1898 Biernacki Studied filtered diseased plant juices and found the caused disease in healthy plants

Wendell Stanley (1935) **discovered that: Viruses are composed of protein and nucleic acid Studied Diseased** tobacco plants TMV

In the 1930s and subsequent decades, pioneering virologists such as Salvador Luria, Max Delbruck, and others used these viruses as model systems to investigate many aspects of virology, including virus structure, genetics, and replication.

These relatively simple agents have since proved to be very important to our understanding of all types of viruses, including those of humans which can be much more difficult to propagate and study. The further history of virology is the story of the development of experimental tools and systems with which viruses could be examined and which opened up whole new areas of biology, including not only the biology of the viruses themselves but inevitably also the biology of the host cells on which they are dependent.

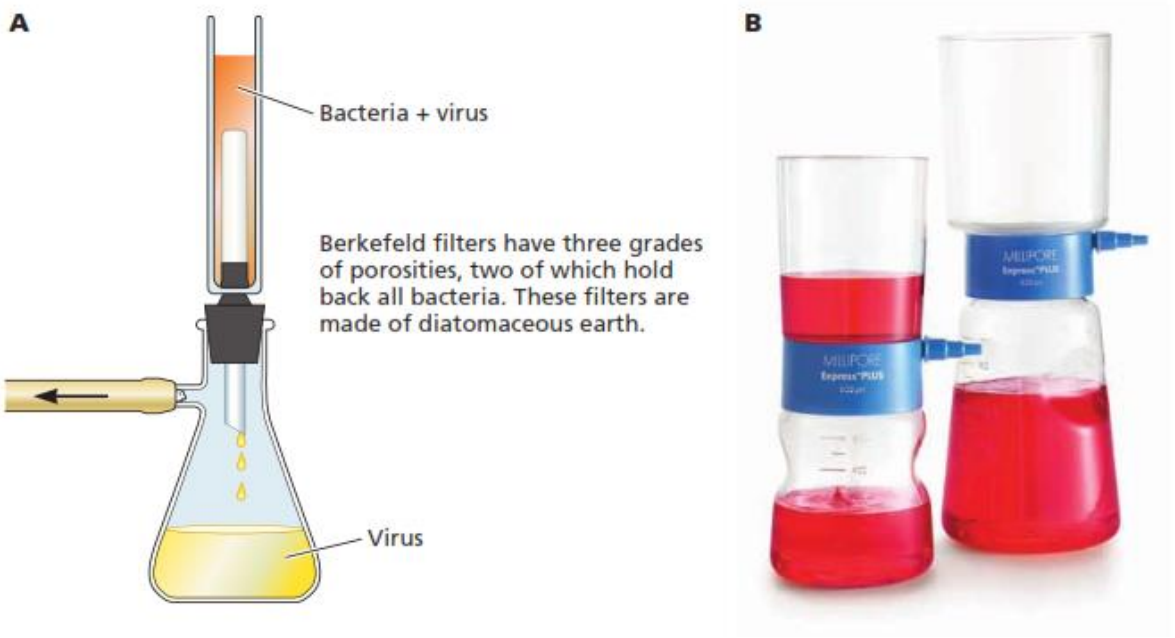


Figure: Filter systems used to characterize/purify virus particles. The earliest, the Berkefeld filter, was invented in Germany in 1891. The smallest pore size retained bacteria and allowed virus particles to pass through. Such filters were probably used by Ivanovsky, Loeffler, and Frosch to isolate the first plant and animal viruses.

Lesson 4: Importance of Virology

- **UNIQUE PROPERTIES**
- Living as well as non-living characteristics
- Unable to cultivate in laboratory easily
- Simplest structure
- Complex interaction with host
- Replication within host
- **To study an important class of pathogens**

Many types of pathogens cause disease in humans. The most familiar are viruses and bacteria. Viruses cause diseases ranging from AIDS and smallpox to the common cold. They are essentially fragments of nucleic acid (DNA or RNA) instructions, wrapped in a protective shell of

proteins and (in some cases) membrane. They use the basic transcription and translation machinery of their host cells for their replication.

Of all the bacteria we encounter in our lives, only a small minority are dedicated pathogens. Much larger and more complex than viruses, bacteria are usually free-living cells, which perform most of their basic metabolic functions themselves, relying on the host primarily for nutrition.

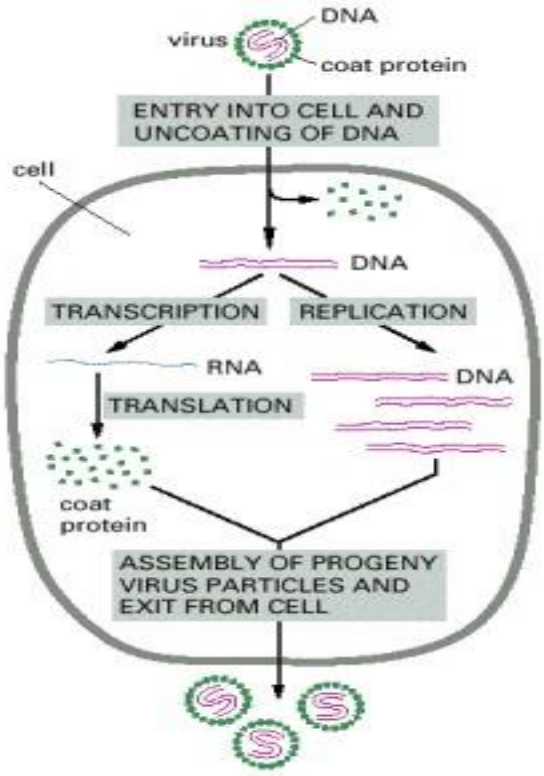
Some other infectious agents are eukaryotic organisms. These range from single-celled fungi and protozoa, through large complex metazoa such as parasitic worms.

Some rare neurodegenerative diseases, including mad cow disease, are caused by an unusual type of infectious particle called a prion, which is made only of protein. Although the prion contains no genome, it can nevertheless replicate and kill the host.

Even within each class of pathogen, there is striking diversity. Viruses vary tremendously in their size, shape, and content, and the same is true for the other pathogens. The ability to cause disease (pathogenesis) is a lifestyle choice, not a legacy shared only among close relatives

Virus life cycle

Bacteria, fungi, and eucaryotic parasites are cells themselves. Even when they are obligate parasites, they use their own machinery for DNA replication, transcription, and translation, and they provide their own sources of metabolic energy. Viruses, by contrast, are the ultimate hitchhikers, carrying little more than information in the form of nucleic acid. The information is largely replicated, packaged, and preserved by the host cells. Viruses have a small genome, made up of a single nucleic acid type—either DNA or RNA. The genome is packaged in a protein coat.



Viruses replicate in various ways. In general, replication involves (1) disassembly of the infectious virus particle, (2) replication of the viral genome, (3) synthesis of the viral proteins by the host cell translation machinery, and (4) reassembly of these components into progeny virus particles.

To study viral cultivation in lab

a. Tissue culture

b. Embryonated egg

c. Live animal

Lesson 6: Living or non-living

Viruses are infectious agents with both living and nonliving characteristics. They can infect animals, plants, and even other microorganisms.

Living Characteristics of Viruses:

a. They reproduce at a fantastic rate, but only in living host cells.

b. They can mutate.

Nonliving Characteristics of Viruses

a. They are acellular, that is, they contain no cytoplasm or cellular organelles.

b. They carry out no metabolism on their own and must replicate using the host cell's metabolic machinery. In other words, viruses don't grow and divide. Instead, new viral components are synthesized and assembled within the infected host cell.

c. The vast majority of viruses possess either DNA or RNA but not both.

CONCLUSION

Viruses are:

"Exceptionally complex aggregation of non-living chemicals"

OR

"Exceptionally simple living organisms"

Recently, viruses have been declared as living entities based on the large number of protein folds encoded by viral genomes that are shared with the genomes of cells. This indicates that viruses likely arose from multiple ancient cells.

Lesson 7: Origin of viruses

The evolutionary history of viruses represents a fascinating topic for virologists and cell biologists. Because of the great diversity among viruses, biologists have struggled with how to classify these entities and how to relate them to the conventional tree of life. They may represent genetic elements that gained the ability to move between cells. They may represent

previously free-living organisms that became parasites. They may be the precursors of life as we know it.

Where Did Viruses Come From?

There is much debate among virologists about this question. Three main hypotheses have been articulated:

1. The virus-first hypothesis states that viruses predate or coevolved with their current cellular hosts.

2. The progressive, or escape, hypothesis states that viruses arose from genetic elements that gained the ability to move between cells

3. The regressive, or reduction, hypothesis asserts that viruses are remnants of cellular organisms

a. The virus-first hypothesis (Co-evolution Theory)

They may have originated and evolved with the other life forms. One postulate showed that viruses existed in a precellular world as self-replicating units. Over time these units, they argue, became more organized and more complex.

b. The Progressive Hypothesis (Escape gene Theory)

According to this hypothesis, viruses originated through a progressive process. Mobile genetic elements, pieces of genetic material capable of moving within a genome, gained the ability to exit one cell and enter another. To conceptualize this transformation, let's examine the replication of retroviruses, the family of viruses to which HIV belongs.

Retroviruses have a single-stranded RNA genome. When the virus enters a host cell, a viral enzyme, reverse transcriptase, converts that single-stranded RNA into double-stranded DNA. This viral DNA then migrates to the nucleus of the host cell. Another viral enzyme, integrase, inserts the newly formed viral DNA into the host cell's genome. Viral genes can then be transcribed and translated. The host cell's RNA polymerase can produce new copies of the virus's single-stranded RNA genome. Progeny viruses assemble and exit the cell to begin the process again.

c. The Regressive Hypothesis (Reduction Theory)

In contrast to the progressive process just described, viruses may have originated via a regressive, or reductive, process. Microbiologists generally agree that certain bacteria that are obligate intracellular parasites, like *Chlamydia* and *Rickettsia* species, evolved from free-living ancestors. Indeed, genomic studies indicate that the mitochondria of eukaryotic cells

and *Rickettsia prowazekii* may share a common, free-living ancestor. It follows, then, that existing viruses may have evolved from more complex, possibly free-living organisms that lost genetic information over time, as they adopted a parasitic approach to replication.

No Single Hypothesis May Be Correct

Where viruses came from is not a simple question to answer. One can argue quite convincingly that certain viruses, such as the retroviruses, arose through a progressive process. Perhaps today's viruses arose multiple times, via multiple mechanisms. Perhaps all viruses arose via a mechanism yet to be uncovered. Today's basic research in fields like microbiology, genomics, and structural biology may provide us with answers to this basic question.

Lesson 8: General properties of Viruses

- **VIRUS**

A virus is a sub-microscopic particle that can infect living cells. Viruses are much smaller than prokaryotes, ranging in size from about 20–300 nanometers (nm), though some can be larger. Prokaryotes are typically 0.5–5.0 micrometers (µm) in length. For example, if a virus was about the size of three soccer balls lying side-by-side, then a prokaryote would be about the size of soccer field.

VIRUS	SIZE
Foot and Mouth Disease virus	27nm
Pox virus	300nm

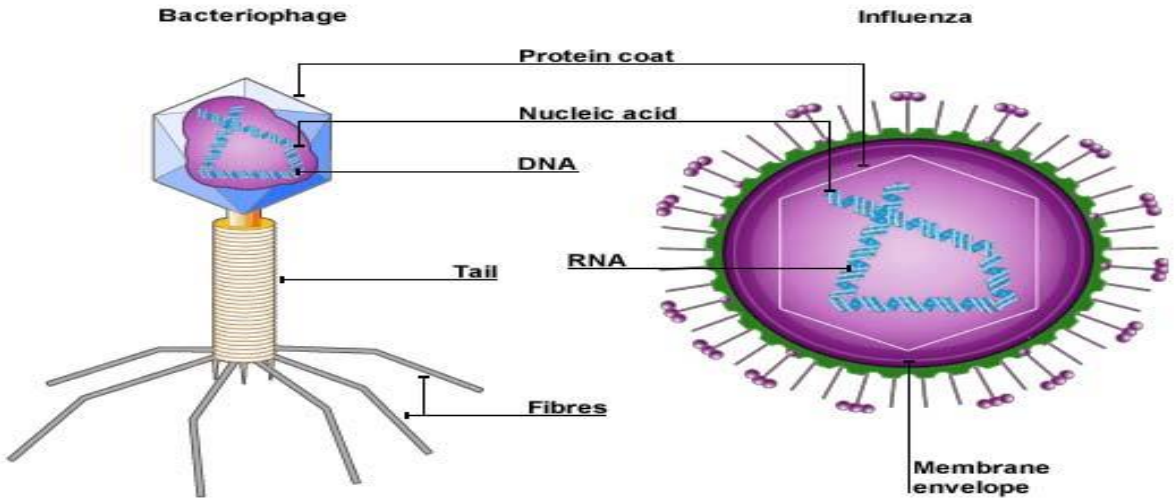
- **Structure**

An individual virus is called a virion. It is a tiny particle much smaller than a prokaryotic cell. Because viruses do not consist of cells, they also lack cell membranes, cytoplasm, ribosomes, and other cell organelles. Without these structures, they are unable to make proteins or even reproduce on their own. Instead, they must depend on a host cell to synthesize their proteins and to make copies of themselves. Although viruses are not classified as living things, they share two important traits with living things. They have genetic material, and they can evolve.

- **Simple Viruses**

- a protective protein coat
- a nucleic acid genome
  - Deoxyribonucleic Acid (DNA)

    or



  - Ribonucleic Acid (RNA)

- **WHY VIRUSES NEED HOST?**
- No enzymes of their own
- No metabolism
- Require host machinery
- **COMPLEX VIRUSES**

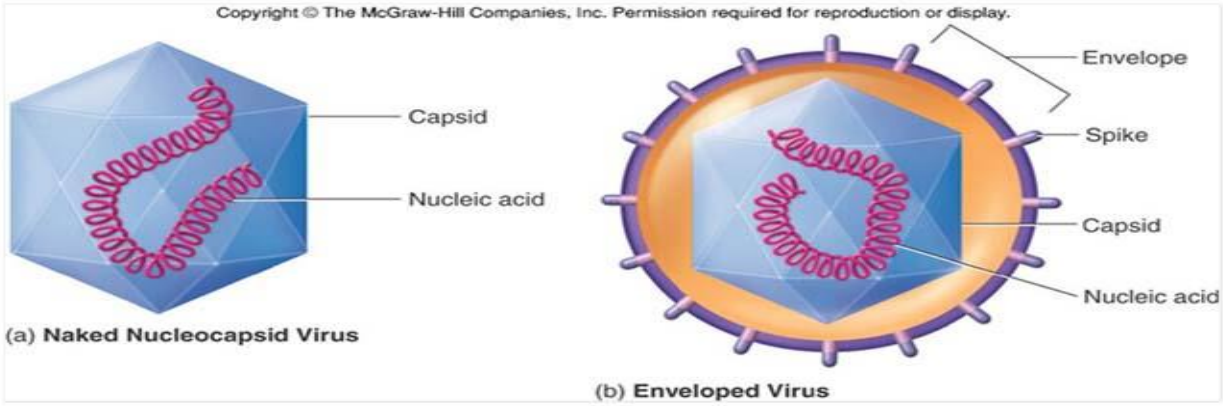
Simple viruses only contain protein coat and nucleic acid genome. Complex viruses also contain some additional contents e.g.

- Lipid
- Polysaccharides
- Trace elements

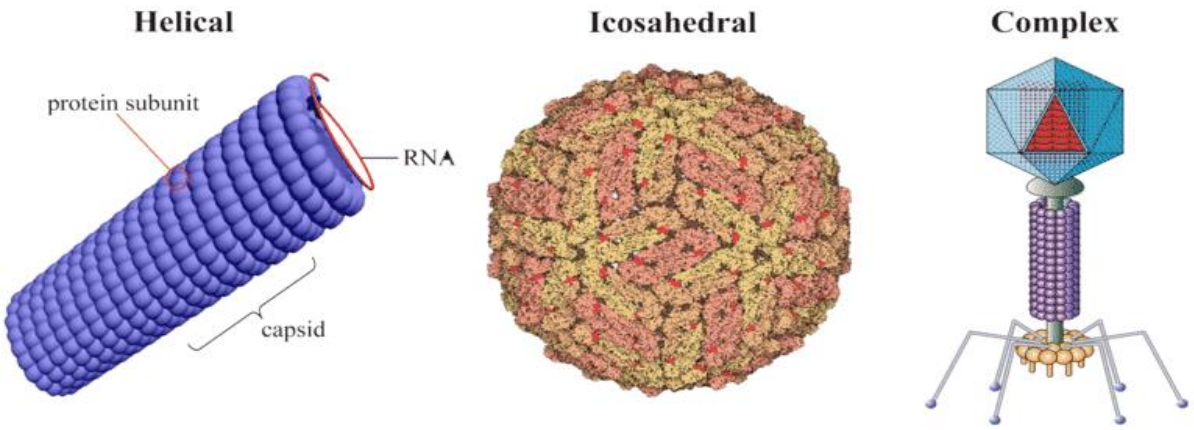
Lesson 9: General properties of Viruses 2

Structure of enveloped and non-enveloped viruses. Non-enveloped viruses are composed of capsid protein and nucleic acid (DNA or RNA), viz. nucleocapsid. whereas enveloped viruses are composed of an envelope and nucleocapsid. Some viruses may have spikes to help attach to the host cell. Most viruses infect only specific host cells

Envelope

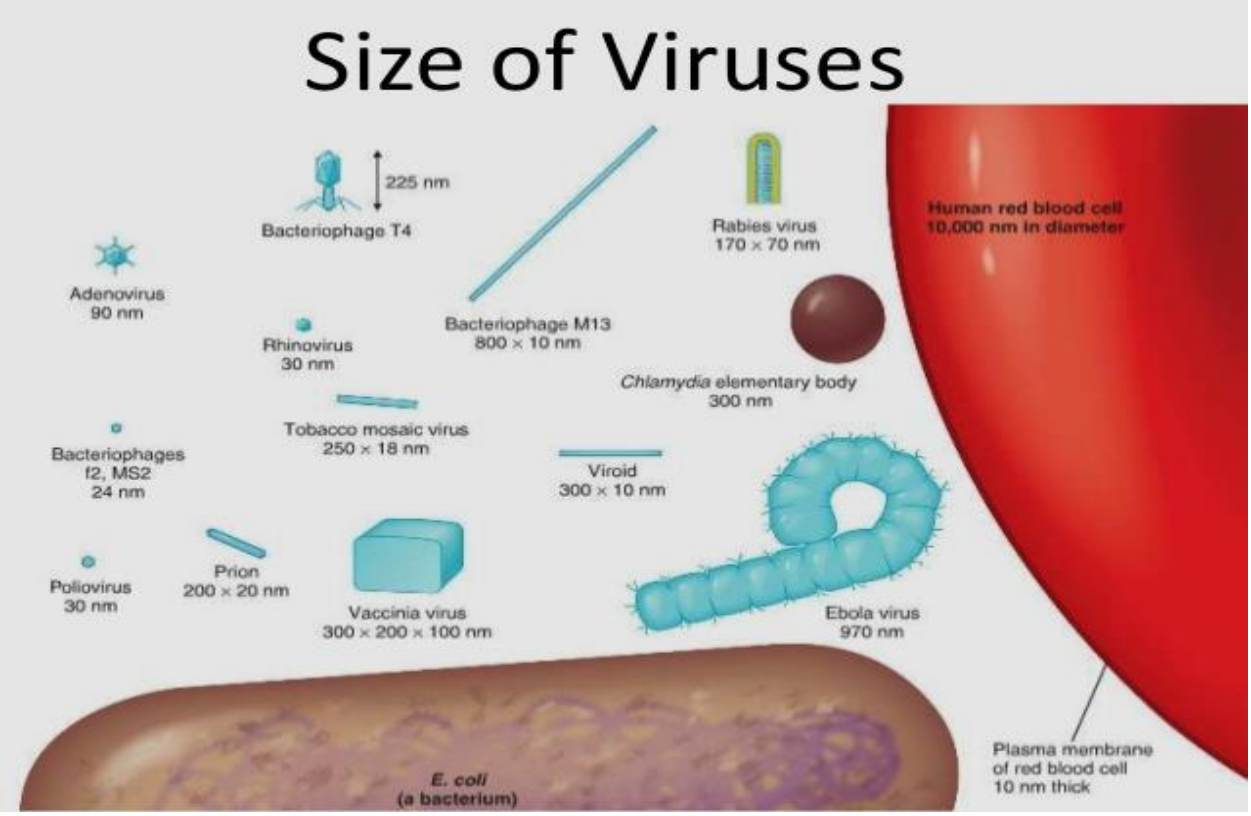


- **MORPHOLOGY**
- Viruses come in a variety of shapes.
- Helical Viruses: Protein subunits interact with each other and with the nucleic acid to form a coiled, ribbon-like structure: *e.g*. tobacco mosaic virus, influenza virus, rabies virus. Ebola virus. structures with helical symmetry.
- Polyhedral shapes: The capsid shell is made of repeating subunits of viral protein (may be one kind of subunit or several, according to the virus). - All faces of the icosahedron are identical. like the influenza virus
- Complex shapes: Most viruses have either icosahedral or helical capsids, but some viruses do not fit into either category. Poxviruses and large bacteriophages are two important examples.



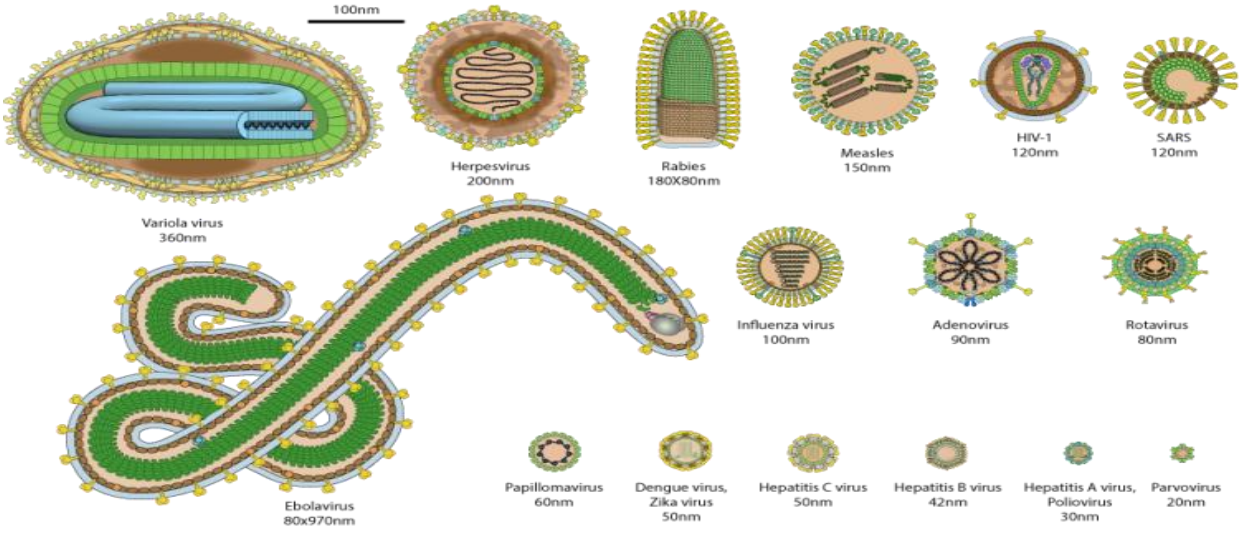
Lesson 10: Viral size

Viruses are usually much smaller than bacteria with the vast majority being submicroscopic. While most viruses range in size from 5 to 300 nanometers (nm) (1 nanometre = 10-9 meters). In recent years a number of giant viruses, including Mimiviruses and Pandoraviruses with a diameter of 0.4 micrometers (µm), have been identified. By contrast, the smallest bacteria are about 400 nanometers in size.



Viruses are much smaller than the cells they infect. Indeed, it was their remarkable smallness that led to their discovery in the first place. Researchers were puzzled by remnant infectious elements that could pass through filters small enough to remove pathogenic bacterial cells. This led to the hypothesis of a new form of biological entity. These entities were subsequently identified as viruses.

- **Most common human viruses with their relative size**



Lesson 11: Functional chemistry of viruses

Chemical Constitution of Virus

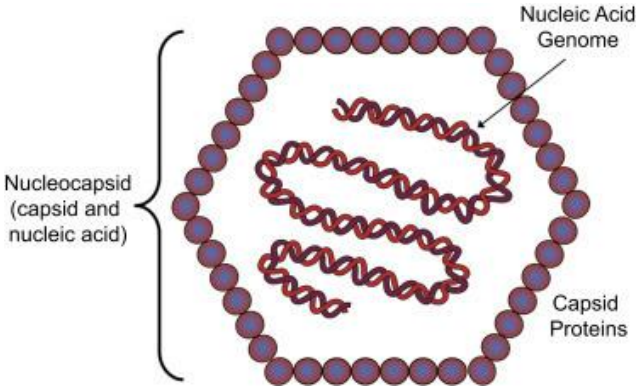
Viruses are mainly composed of **proteins** and **nucleic acids**.

Proteins: The constituent building blocks of proteins are the amino acids, and most of the common amino acids are found in viruses. The analyses of viral proteins have shown that viral proteins are not histones, but are rather ordinary in composition, being even, in some cases, quite acidic in character.

Nucleic acids: Ribonucleic acid (RNA) or deoxyribonucleic acid (DNA) is found in viruses. Only RNA has been found in the viruses affecting higher plants, whereas RNA or DNA, have been reported to be present in highly purified preparations of animal viruses. Nucleic acids, largely because of their primary phosphate groups, are customarily regarded as very reactive compounds.

However, in viruses, they are chemically inert. A few enzymes may also be found in viruses.

Some other **miscellaneous constituents of viruses** include nucleoproteins, lipids, polysaccharide, copper, biotin, and flavin.



The protein subunits in a virus capsid are redundant, that is, there are many copies in each particle. Damage to one or more capsid subunits may make that particular subunit nonfunctional, but rarely does limited damage destroy the infectivity of the entire particle. This helps make the capsid an effective barrier. This combination of strength, flexibility, and small size means that it is physically difficult (although not impossible) to break open virus particles by physical pressure.

SPECIFICITY

The outer surface of the virus is also responsible for recognition of and inter-action with the host cell. Initially, this takes the form of binding of a specific virus-attachment protein to a cellular receptor molecule. However, the capsid also has a role to play in initiating infection by delivering the genome in a form in which it can interact with the host cell. In some cases, this is a simple process that consists only of dumping the genome into the cytoplasm of the cell. Most viruses infect specific host cells i.e. are host specific.

Host specificity is due to:

- specific attachment sites on the host cells called receptors
- Receptor sites for bacteriophage are found in bacterial cell walls or fimbrae or flagella
- Animal cell membranes contain receptors for animal viruses
- availability of cellular factors required for viral multiplication in the host cells

INFECTIVITY

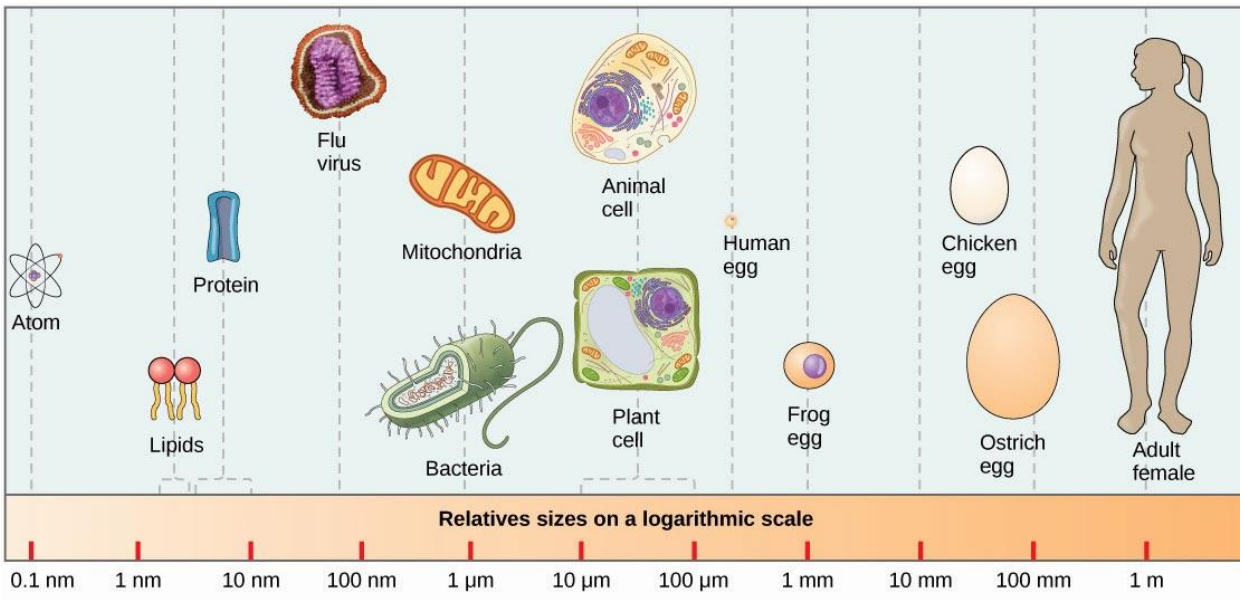
For infectivity in host cells:

1. Viral nucleic acid
2. Viral structural sub components (spikes or glycoproteins)
3. Enzymes induced in host cells (polymerase enzymes)

Lesson 12: Visualization of Viral particles

Viruses are a diverse group of pathogens that have taken widely different life-cycle and genome storage approaches. Perhaps unsurprisingly, assembled viral particles that make up the infectious unit are highly diverse in shape and size. Virion sizes can range from 400 nm in diameter for DNA viruses such as mimivirus and poxviruses to small virions of ~25nm for polio virus. Morphology can also vary widely, from filamentous ebolavirus and marburgvirus virions to pleomorphic viruses such as measles to highly regular “brick” shaped poxvirions.

Characterization of virus particles is typically performed using higher resolution approaches such as electron microscopy or atomic force microscopy. These approaches require purification of virions away from their normal millieu, requiring significant levels of expertise, and can only enumerate small numbers of particles per field of view.



Variation in size also effect on

- Shape greatly varies
- Diversity
- Effects pathogenicity

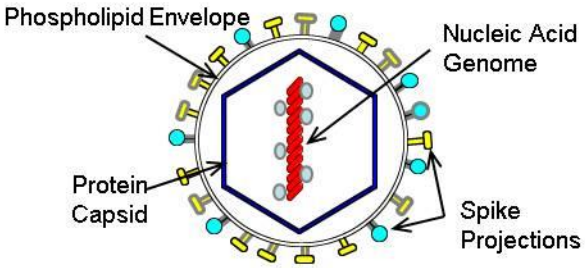
Lesson 13: Biochemistry and cellular background

A virus can be simply defined as an **obligate intracellular parasite** (**Intracellular**= Inside the cell; **obligate**= dependent on host cell; **Parasite**= which harms its host). Each viral particle, or **virion**, consists of a single nucleic acid, RNA or DNA, encoding the viral genome surrounded by a protein coat, and is capable of replication only within the living cells of bacteria, animals or plants. Viruses are classified into different orders and families by consideration of the type of nucleic acid present (RNA or DNA), whether the nucleic acid is single- or double-stranded, and the presence or absence of an envelope.

Structure: A typical virus consists of a protective protein coat, known as a **capsid**. The capsid shape varies from simple helical and icosahedra forms to more complex structures with tails. The capsid provides protection for the viral genome against the environment and functions in receptor recognition, targeting the virus to a susceptible host and cell type.

Some viruses have a **phospholipid envelope**, derived from the infected host's cell membrane, that surrounds the protein capsid. Inserted into the lipid envelope there are usually viral

encoded proteins know as **spike projections** – these are typically glycoproteins and are also involved in receptor recognition and viral tropism.



A typical enveloped virus

Major Criteria for Classifying Human Virus Families

ds, double stranded; ss, single stranded.

Type of genomic nucleic acid	DNA or RNA
Nucleic acid strandedness	ds, ss, partially ds
Sense of ss nucleic acid	+, −, − with ambisense
Capsid morphology	Icosahedral, helical, or complex
Envelope	Present or absent
Genome segmentation	Number of segments
Size of virion and/or genome	For example, large-genome DNA viruses (e.g., poxviruses, herpesviruses) versus small-genome viruses (e.g., picornaviruses, parvoviruses, hepadnaviruses)

Lesson 14: Living character in host cell

As viruses are obligate intracellular pathogens they cannot replicate without the machinery and metabolism of a host cell. Although the replicative life cycle of viruses differs greatly between species and category of virus, there are six basic stages that are essential for viral replication.

1. Attachment: Viral proteins on the capsid or phospholipid envelope interact with specific receptors on the host cellular surface. This specificity determines the host range (**tropism**) of a virus.

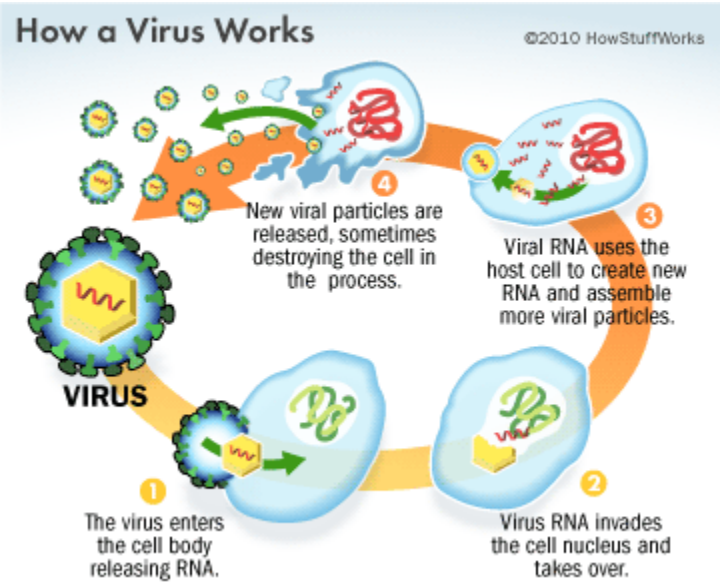
2. Penetration: The process of attachment to a specific receptor can induce conformational changes in viral capsid proteins, or the lipid envelope, that results in the fusion of viral and cellular membranes. Some DNA viruses can also enter the host cell through receptor-mediated endocytosis.

3. Uncoating: The viral capsid is removed and degraded by viral enzymes or host enzymes releasing the viral genomic nucleic acid.

4. Replication: After the viral genome has been uncoated, transcription or translation of the viral genome is initiated. It is this stage of viral replication that differs greatly between DNA and RNA viruses and viruses with opposite nucleic acid polarity. This process culminates in the *de novo* synthesis of viral proteins and genome.

5. Assembly: After *de novo* synthesis of viral genome and proteins, which can be post-transcriptionally modified, viral proteins are packaged with newly replicated viral genome into new virions that are ready for release from the host cell. This process can also be referred to as maturation.

6. Virion release: There are two methods of viral release: **lysis** or **budding**. Lysis results in the death of an infected host cell, these types of viruses are referred to as **cytolytic**. An example is *variola major* also known as smallpox. Enveloped viruses, such as influenza A virus, are typically released from the host cell by budding. It is this process that results in the acquisition (حصول) of the viral phospholipid envelope. These types of viruses do not usually kill the infected cell and are termed **cytopathic viruses**.



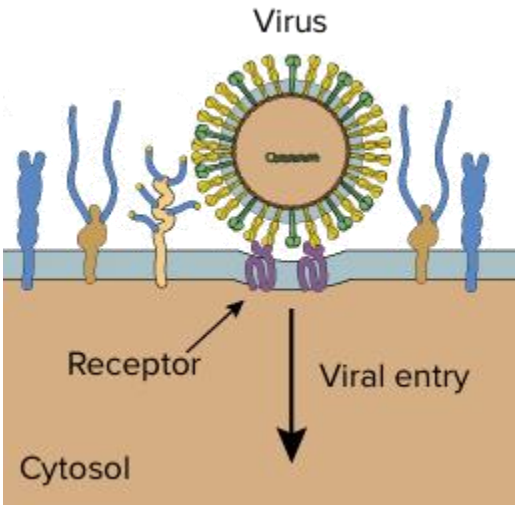
Lesson 15: Host range of viruses

Host: A host is an organism that harbors a parasite or a mutual or commensal symbiont, typically providing nourishment and shelter.

Host range of viruses:

Host range of a virus is one of the biological properties of the virus. A ***virus' host range*** is the range of cell types and host species a ***virus*** is able to infect. And within which that virus can replicate. Hosts play an important role in the biological characterization of a virus. A few of the viruses have the ability to infect only a limited number of hosts while a few other viruses have the ability to infect large number of hosts belonging to several genera and families.

The host range of the virus will depend upon the presence of specific receptors on host cell. **Viral** infection begins when proteins on the surface of a virion bind to specific receptor proteins on the surface of **host** cells. The specificity of this interaction determines the **host range of a virus**. If a host lacks the receptor for a virus, or if the host cell lacks some component necessary for the replication of a virus, the host will inherently be resistant to that virus. For example, mice lack the receptors for polio viruses and thus are resistant to polio virus. Similarly, humans are inherently resistant to plant and many animal viruses.



Host range conditions the transmission dynamics and survival of pathogens and is predicted to be a major factor in their evolution.

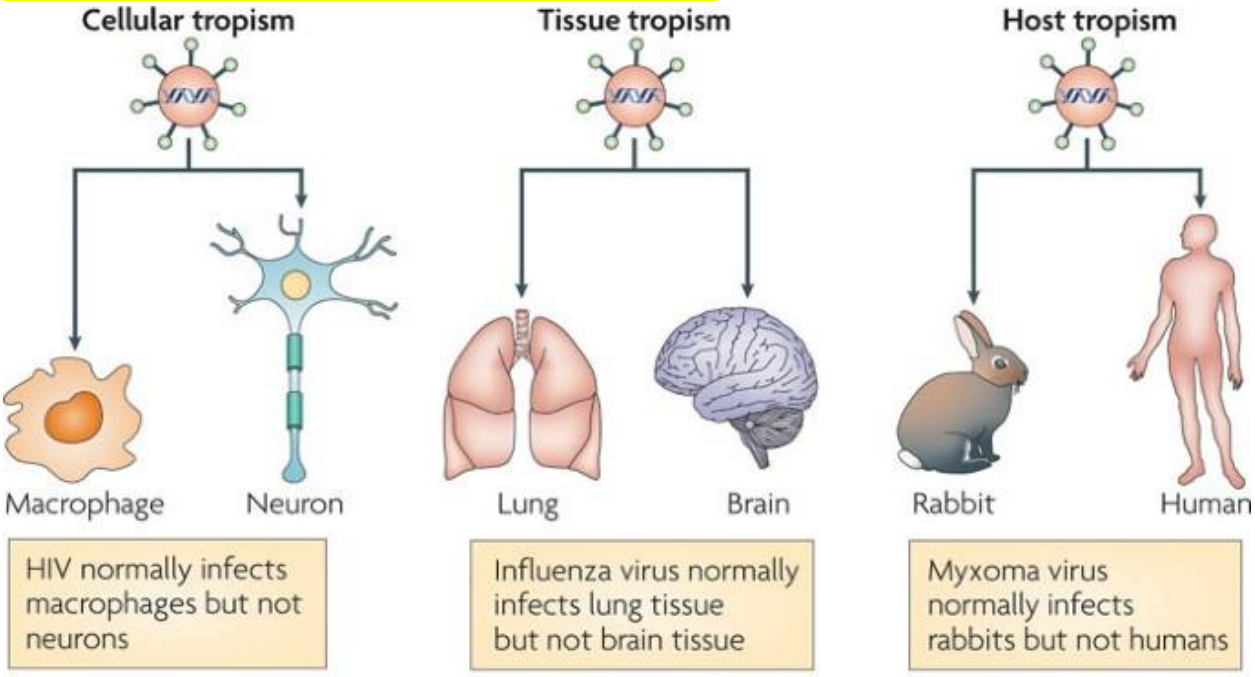
Viral host range

- Plants
- Vertebrates

- Invertebrates
- Protists
- Bacteria
- Fungi

Lesson 16: Viral specificity

In a general sense, viral tropism refers to the ability of a given virus to productively infect a particular cell (**cellular tropism**), tissue (**tissue tropism**) or host species (**host tropism**). For example, if a particular virus can productively infect rabbits but no other vertebrate hosts such as humans, that virus would be said to have a host tropism limited to rabbits. Similarly, if the same virus could productively infect rabbit macrophages but not any other rabbit cells, that virus would be said to have a cellular tropism limited to macrophages.
Viral tropism can be divided into three distinct categories depending on the physiological level at which it is measured. Tropism in which the virus replicates in one cell type but not another is known as **cellular tropism**, tropism in which the virus replicates in a particular tissue or organ but not another is known as **tissue tropism**, and tropism in which the virus replicates in one host species but not another is known as **host tropism**.



Nature Reviews | Immunology

Infection of vertebrate hosts by most viruses begins with the breaching of the natural barriers that are designed to protect the host from challenge, such as the skin, mucus, saliva, stomach

acid and tears. Once the virus finds its first sensitive cellular target, often in the local tissue or through the respiratory or gastrointestinal tracts, the infection itself is usually initiated by binding of the virion to the cell surface, often with the aid of host cell surface receptors that are hijacked from their normal cellular functions. For example, HIV-1 requires the presence of the T cell co-receptor to bind and enter cells.

In virology, Tissue tropism is the cells and tissues of a host that support growth of a particular virus or bacteria. Some viruses have a broad tissue tropism and can infect many types of cells and tissues. Other viruses may infect primarily a single tissue.

VIRUS	HOST
Tobacco Mosaic Virus	Tobacco plant
Rabies	Mammals
Swine flu	Pigs and humans
Foot and Mouth disease	Hooved animals
Bird Flu	Birds and human
Myxoma virus	Rabbits

Zoonotic infections: Occasionally, probably owing to virus mutation or recombination, a virus crosses the species barrier from its current or long-term evolutionary host to a new host species (new host tropism), a phenomenon that is referred to as a zoonotic infection when it occurs in humans.

Lesson 17: Taxonomy I

Taxonomy (nomenclature) is assigning scientific names according to agreed international scientific rules. The official taxonomic groups (from the largest to the smallest are):

- ¦ Kingdom (e.g., animals, plants, bacteria; does not apply to viruses)
- ¦ Phylum (e.g., vertebrates; does not apply to viruses)
- ¦ Class (group of related orders; does not apply to viruses)
- ¦ Order (group of related families)
- ¦ Family (group of related genera)
- ¦ Genus (group of related species)

- ¦ Species, the smallest taxon

International Committee on Taxonomy of Viruses

In 1966, the International Committee on Nomenclature of Viruses was established and produced the first unified scheme for virus classification. In 1973, this committee expanded its objectives and renamed itself the International Committee on Taxonomy of Viruses (ICTV). The ICTV has adopted a formal nomenclature for viruses, specifying sufixes for the various taxa, and rules for written

descriptions of viruses. Names for genera, subfamilies, families, and orders must all be single words, ending with the suffixes -virus, -virinae, -viridae, and -virales, respectively. Species names may contain more than one word and have no specific ending.

Table: Taxa Used to Classify Viruses

Taxon	**Notes**	**Example**
Order	Ends in -virales suffix; only about half of viruses are currently classified in orders.	*Picornavirales*
Family	Ends in -viridae suffix; subfamilies are indicated with -virinae suffix.	*Picornaviridae*
Genus	Ends in -virus suffix.	*Enterovirus*
Species	Generally, the "common name" of the virus. Classifying and cataloging anything below the species classification (such as subtypes, serotypes, strains, isolates, or variants) is the responsibility of the specific field.	*Rhinovirus A* (Serotypes include Human rhinovirus 1, which includes strains human rhinovirus 1A and human rhinovirus 1B)

In ICTV classification, members have several properties in common e.g.

- Host
- Ecological niche
- Pathogenicity
- Mode of transmission
- Habitat

Lesson 18: Taxonomy II

Baltimore classification system

One classification scheme was developed in the 1970s by Nobel laureate David Baltimore. The Baltimore classification system categorizes viruses based on the type of nucleic acid genome and replication strategy of the virus. The system also breaks down single-stranded RNA viruses into those that are positive strand (+) and negative strand (−). Positive-strand (also positive-sense or plus-strand) RNA is able to be immediately translated into proteins; as such, messenger RNA (mRNA) in the cell is positive strand. Negative-strand (also negative-sense or minus-strand) RNA is not translatable into proteins; it first has to be transcribed into positive-strand RNA. Baltimore also took into account viruses that are able to reverse transcribe, or create DNA from an RNA template, which is something that cells are not capable of doing. Together, the seven classes are

- • class I: dsDNA viruses
- • class II: ssDNA viruses
- • class III: dsRNA viruses
- • class IV: positive-sense ssRNA viruses
- • class V: negative-sense ssRNA viruses
- • class VI: RNA viruses that reverse transcribe
- • class VII: DNA viruses that reverse transcribe

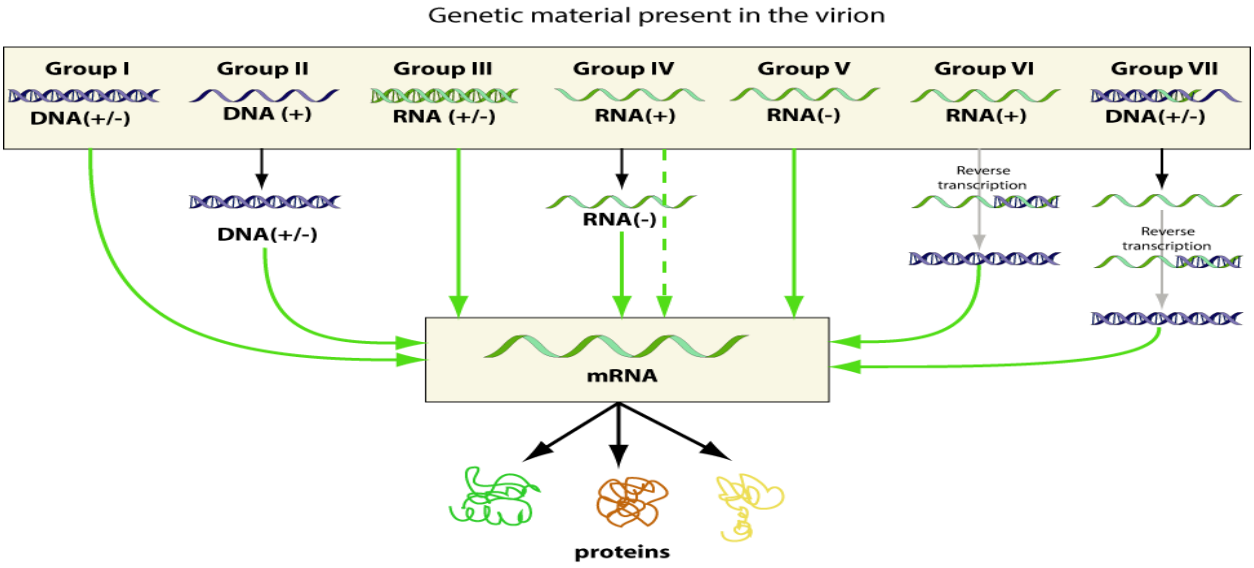
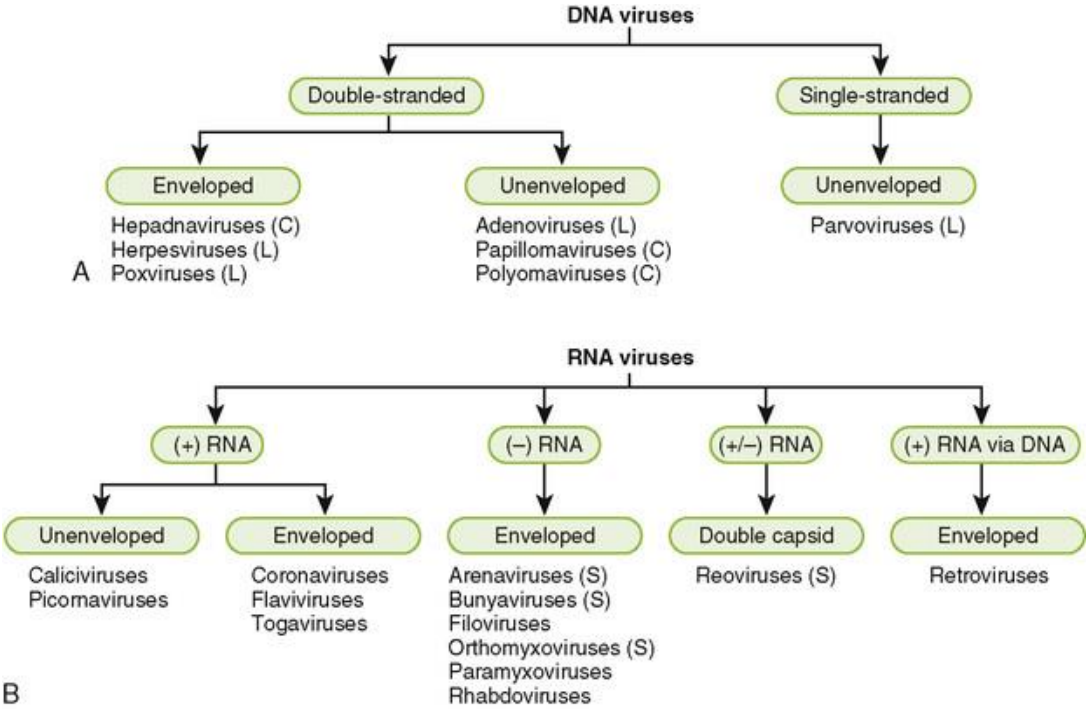


FIGURE: The Baltimore classification, a virus classification scheme based on the form of nucleic acid present in virion particles and the pathway for expression of the genetic material as messenger RNA. The original scheme contained groups I through VI and has been expanded to

accommodate DNA-containing, reverse transcribing viruses. Viruses containing Ambi sense single-stranded RNA genomes are grouped under negative sense single-stranded RNA viruses.



Lesson 19: Plant virus transmission

Plant viruses are obligate parasites, often causing the death of their host, so it is necessary for them to spread from plant to plant and to be introduced into living cells. Plant cells have a robust cell wall and viruses cannot penetrate them unaided.

THE BARRIERS

- External surface
- Structural differentiation of host
- Internal barriers
- Direct transmission only through wounds or poorly protected tissues

Transmission by Vegetative Propagation:

It is one of the chief methods of transmission of virus diseases especially of Potato, Rose, Sugarcane, fruit trees and many ornamentals.

The vegetative parts, the infected plants such as the tubers, bulbs, roots, offshoots, buds and scions which are used for propagation, will contain the virus present in the parent. The new plants raised by the above-mentioned vegetative methods are nearly always infected.



Seed Transmission of Virus

The seeds are important in the spread of a few viruses of legumes, wild cucumber, tomatoes, and curly top virus of beet sugar. In the latter case the seeds carry a high percentage of the virus. The virus, however, does not enter the embryo. It is carried in a portion of the seed of the diseased plants.

Vector Transmission of Virus

Organisms involved in the transport of virus from one plant to another are called ***vectors.*** Transmission by vectors involves some transitory biological interaction between vector and virus, but in many cases a specific vector will interact only with a particular virus.

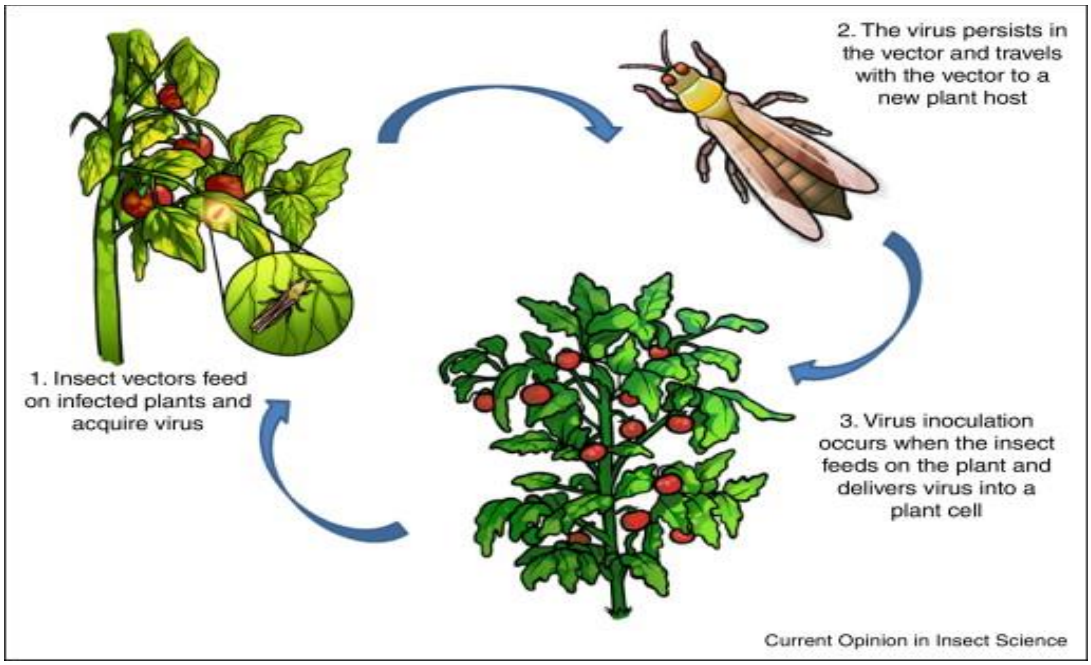
The major vectors of plant viruses are:

- **Aphids**: Aphids are the most common vector of plant viruses.
- **Whiteflies**: transmit viruses from several genera but particularly those in the genus *Begomovirus*.
- **Hoppers**
- **Thrips**
- **Mites**
- **Beetles**



Virus-vector relationships:

- At one extreme, the association occurs within the feeding apparatus of the insect, where the virus can be rapidly adsorbed and then released into a different plant cell. The feeding insect loses the virus rapidly when feeding on a non-infected plant. Such a relationship is termed "non-persistent".
- At the other extreme, the virus is taken up into the vector, circulates within the vector body and is released through the salivary glands. The vector needs to feed on an infected plant for much longer and there is an interval (perhaps several hours) before it can transmit. Once it becomes viruliferous, the vector will remain so for many days and such a relationship is therefore termed "persistent" or "circulative". In some examples of this type (e.g. some hoppers and thrips), the virus multiplies within the vector and this is termed "propagative".



Lesson 20: Animal virus transmission

Viruses are obligate intracellular parasites and cannot grow on inanimate media. They need living cells for replication, which can be provided by inoculation in live animals among other methods used to culture viruses (cell culture or inoculation of embryonated eggs).

Laboratory Animals

In the early 1900s, when viruses were first isolated, freezers and cell cultures were not available and it was necessary to maintain virus stocks by continuous passage from animal to animal. This practice not only was inconvenient but also, led to the selection of viral mutants. Cell culture has largely supplanted the use of animals for propagating viruses, but some viruses cannot yet be grown in this way.

Experimental infection of laboratory animals has always been, and will continue to be, obligatory for studying the processes by which viruses cause disease. Understanding how the immune system or any complex organ reacts to a virus cannot be achieved without research on living animals. The development of viral vaccines, antiviral drugs, and diagnostic tests for veterinary medicine has also benefited from research on diseases in laboratory animals. The laboratory animals used include monkeys, rabbits, guinea pigs, rats, hamsters, and mice.



Viruses can also be inoculated in lab animals by following routes

- Subcutaneous
- intracerebral
- intraperitoneal
- Intradermal
- intraocular

After inoculation, virus multiplies in host and develops disease. The animals are observed for symptoms of disease and death.

Advantages of Animal Inoculation

- Clinical manifestations are determined
- Primary isolation of viruses
- Used for study of immune response

Lesson 21: Microscopy- introduction

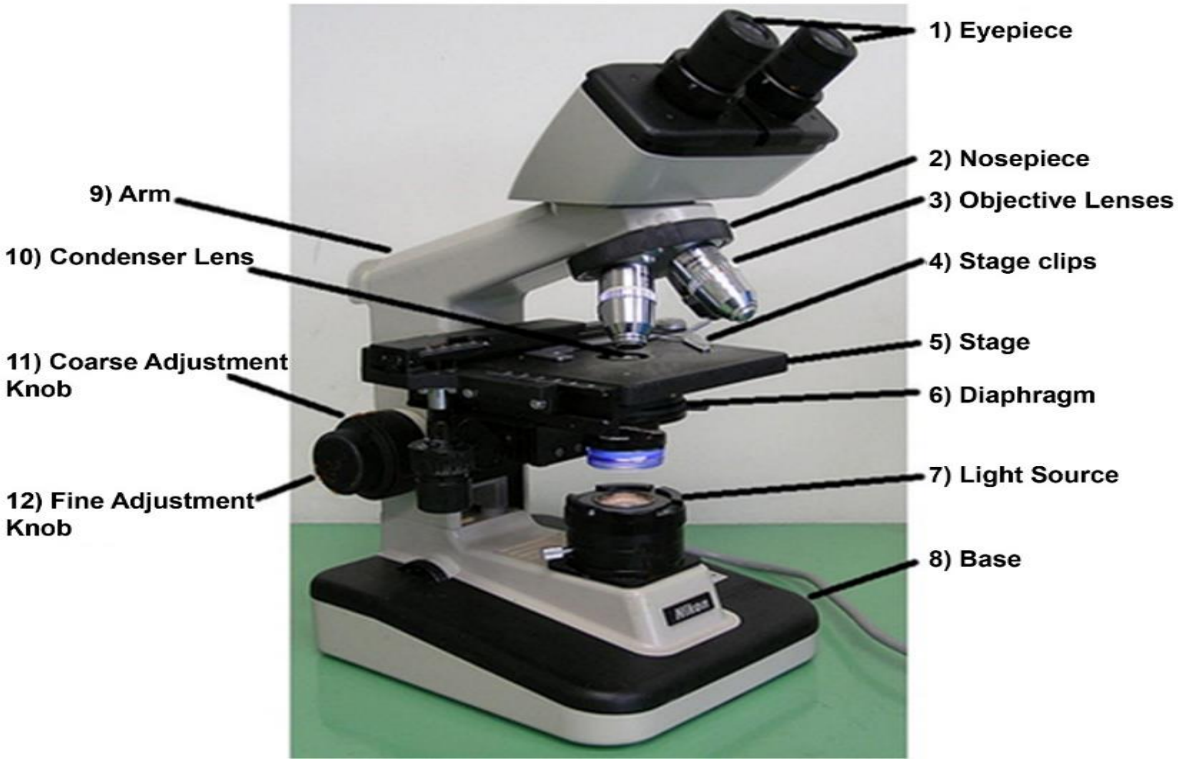
Microscope: An instrument used to see objects that are too small for the naked eye.

Microscopy: The science of investigating small objects using such an instrument is called microscopy

Microscopic: Microscopic means invisible to the eye unless aided by a microscope

- **PARTS OF COMPOUND MICROSCOPE**
- **Ocular lens/ Eyepiece** having10x magnification power, using which you can observe the magnified image of object.
- **Arm** of microscope, which connects eyepiece to base of microscope.
- **Stage** where we place our specimen slide to be observed.
- **Stage clips** to hold the slide in proper place during observation, there are stage clips which fix the slide in its particular position.
- **Objective lenses** of 4X, 10X, 40X,100X magnification powers that are mounted on a rotating turret called as nosepiece. Nosepiece can be rotated to change the lens, which clicks into position when the lens is set. The four lenses are color coded as well to give an idea of their magnification power.
- The lowest power objective lens is coded red and has a magnification of 4x.
- The lens coded yellow has a magnification power of 10x.
- The high-power magnification lenses, coded blue and green respectively, have a power of 40X and 100x. However, 100x is also called as oil immersion lens and its reason will be explained shortly.
- **Condenser:** Whose function is to focus the light rays onto the specimen under observation.
- **Diaphragm** along with its control lever. This is used to control the size of aperture through which light passes. Alternately the intensity of light can also be controlled through another knob known as the illumination control knob.
- **Illuminator/Light source** from where the light emits. Light is then passed to the condenser lens to be focused on the specimen.

- **Coarse adjustment knob** that moves the stage up and down and for focusing the specimen.
- **Fine adjustment knob:** As you know every person have specific vision power so while observing every individual focus the specimen at different position. The Fine adjustment knob here is used to deal with such differences and helps to focus the object without moving the specimen or slide. This in simple terms is used for someone who otherwise wears glasses.
- **Base** provides basal support to the microscope.

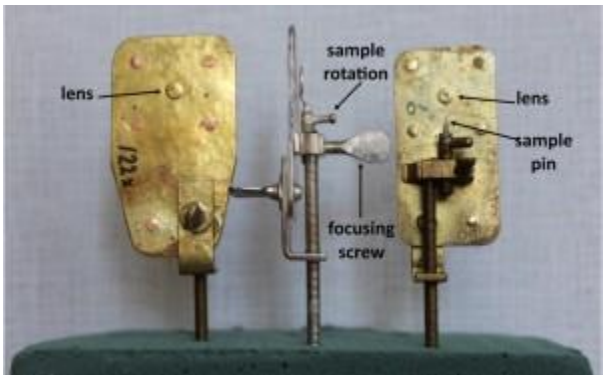


1: Parts of compound microscope

Lesson 22: History of Microscopy

FIRST MICROSCOPE

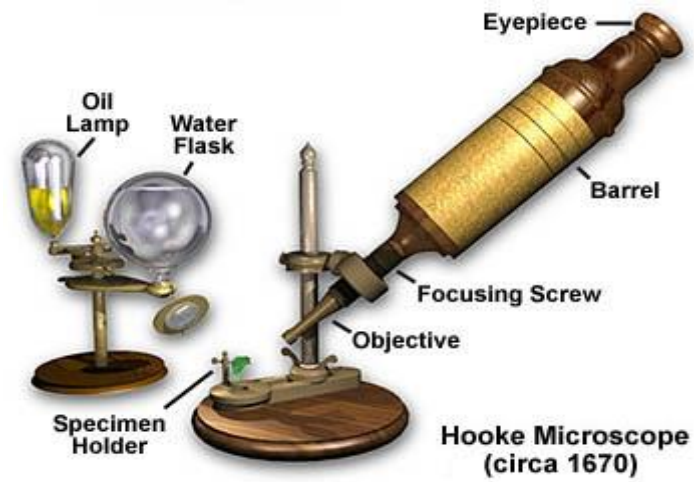
- Antonie van Leeuwenhoek in 17th century builds a simple microscope with one lens to examine blood, yeast and insects. He is the first to describe cells and bacteria. He invents new methods for making lenses that allow for magnifications of up to 270 times.





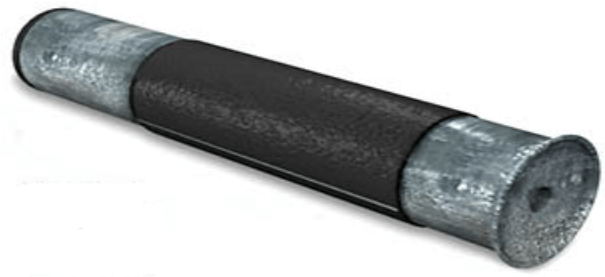
Robert Hooke:

An English scientist named Robert Hooke made key observations. He is reputed to have observed strands of fungi among the specimens of cells he viewed. He was first to publish the drawings of microbes in 1665.



COMPOUND MICROSCOPE

Zaccharias Janssen made the first compound microscope.



BETTER MICROSCOPE

In 1830 a better microscope introduced by Joseph Lister. Joseph Jackson Lister reduces spherical aberration (which produces imperfect images) by using several weak lenses together at certain distances to give good magnification without blurring the image.



Lesson 23: Light Microscopy

Light microscopy refers to the use of any kind of microscope that uses visible light to observe specimens.

TYPES

- Compound light microscopy
- Dark field Microscopy
- Phase-contrast Microscopy
- Differential interference contrast (DIC) Microscopy
- Fluorescence Microscopy
- Confocal Microscopy

COMPOUND LIGHT MICROSCOPY

Brightfield Microscope is also known as the **Compound Light Microscope.** It is an optical microscope that uses light rays to produce a dark image against a bright background. It is the standard microscope that is used in Biology, Cellular Biology, and Microbiological Laboratory studies.

This microscope is used to view fixed and live specimens, that have been stained with basic stains which gives a contrast between the image and the image background. It is specially designed with magnifying glasses known as lenses that modify the specimen to produce an image seen through the eyepiece.



MAGNIFICATION

- Objective lens **X** Eyepiece lens
- It is the ratio of the size of an object seen under microscope to the actual size observed with unaided eye.

RESOLUTION

- refers to the ability of the lenses to distinguish two points that are a specified distance apart
- The resolving power of human eye is 0.25 mm
- The light microscope can separate dots that are 0.25µm apart.
- The electron microscope can separate dots that are 0.5nm apart.

Lesson 24: Dark Field Microscopy

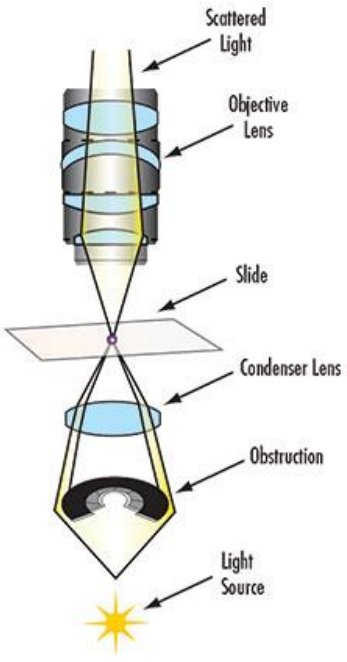
Dark Field Microscopy

Dark Field illumination is a technique used to observe unstained samples causing them to appear brightly lit against a dark, almost purely black, background.

In darkfield microscopy the condenser is designed to form a hollow cone of light, as opposed to brightfield microscopy that illuminates the sample with a full cone of light. In darkfield microscopy, the objective lens sits in the dark hollow of this cone and light travels around the objective lens, but does not enter the cone shaped area. The entire field of view appears dark when there is no sample on the microscope stage. However, when a sample is placed on the stage it appears bright against a dark background. It is similar to back-lighting an object that may be the same color as the background it sits against - in order to make it stand out.

Uses:
Used for live microorganisms that either are invisible in the ordinary light microscope, cannot be stained by standard methods.

Principle
To view a specimen in dark field, an opaque disc is placed underneath the condenser lens, so that only light that is scattered by objects on the slide can reach the eye (as shown in figure below). Instead of coming up through the specimen, the light is reflected by particles on the slide. Everything is visible regardless of color, usually bright white against a dark background.





Lesson 25: DICT Microscopy

Differential interference contrast (DIC) microscope is a dual-beam interferometer made with polarization optics. It is a good alternative to brightfield microscopy for gaining proper images of unstained specimens that often only provide a weak image in brightfield.

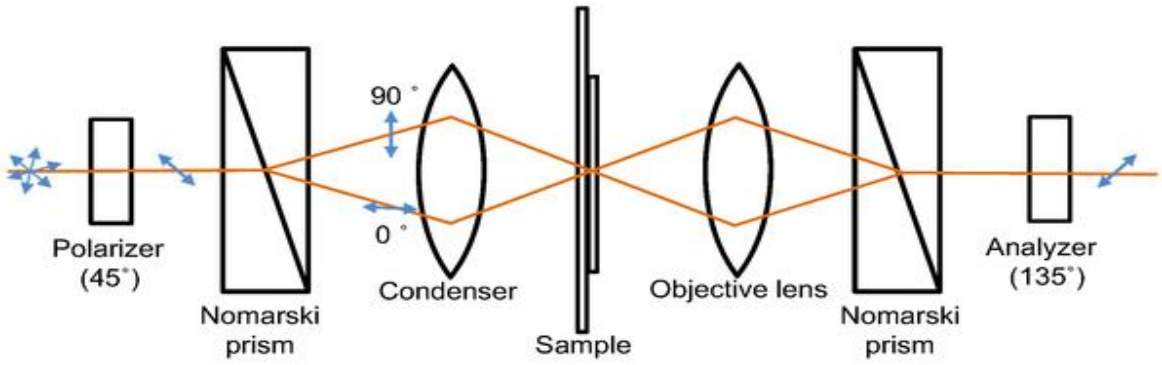
In the mid-1950s, a French optics theoretician named Georges Nomarski modified the Wollaston prism used for detecting optical gradients in specimens and converting them into intensity differences. Today there are several implementations of this design, which are collectively called differential interference contrast (**DIC**). Living or stained specimens, which often yield poor images when viewed in brightfield illumination, are made clearly visible by optical rather than chemical means.

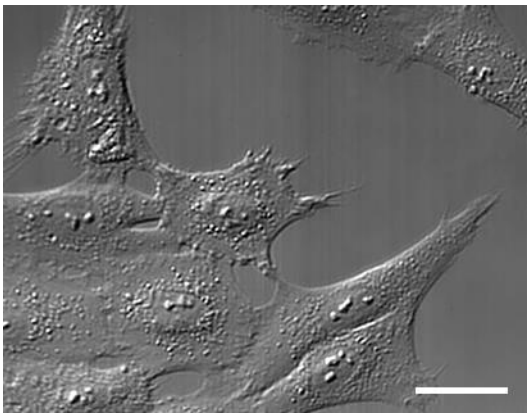
WORKING:

DIC microscope uses two beams of light instead of one. Prisms split each light beam, adding contrasting colors to the specimen. Image is brightly colored and appears nearly three-dimensional.

Principle:

- Unpolarized light from lamp enters into **polarizer** and becomes polarized (vibrating only in one direction perpendicular to the propagation direction of the light beam)
- Then it enters the prism (**Normaski prism**) and gets separated into two beams
- These rays are focused by **condenser** to travel through specimen (**sample**), where their wave paths are altered in accordance with the specimen's varying thickness, slopes, and refractive indices.
- When the parallel beams enter the **objective lens**, they are focused on second prism
- Second prism (**Normaski prism**) recombines the rays forming an image





DIC microscopy of Rat1 cells



DIC microscopy of Microsterias (green algae)

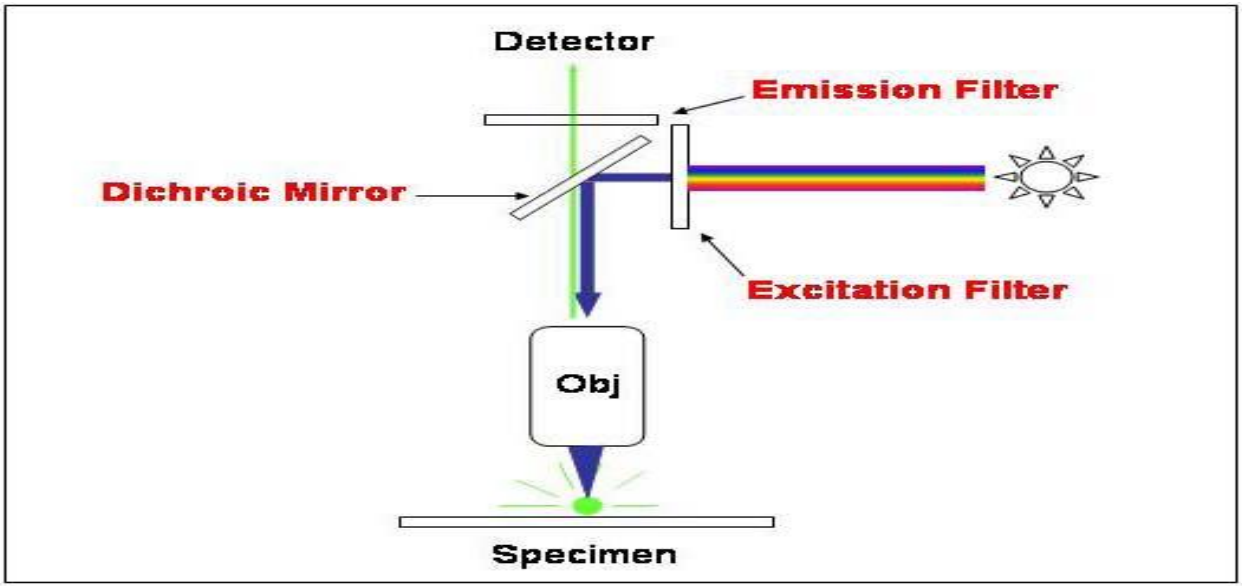
Lesson 26: Fluorescence Microscopy

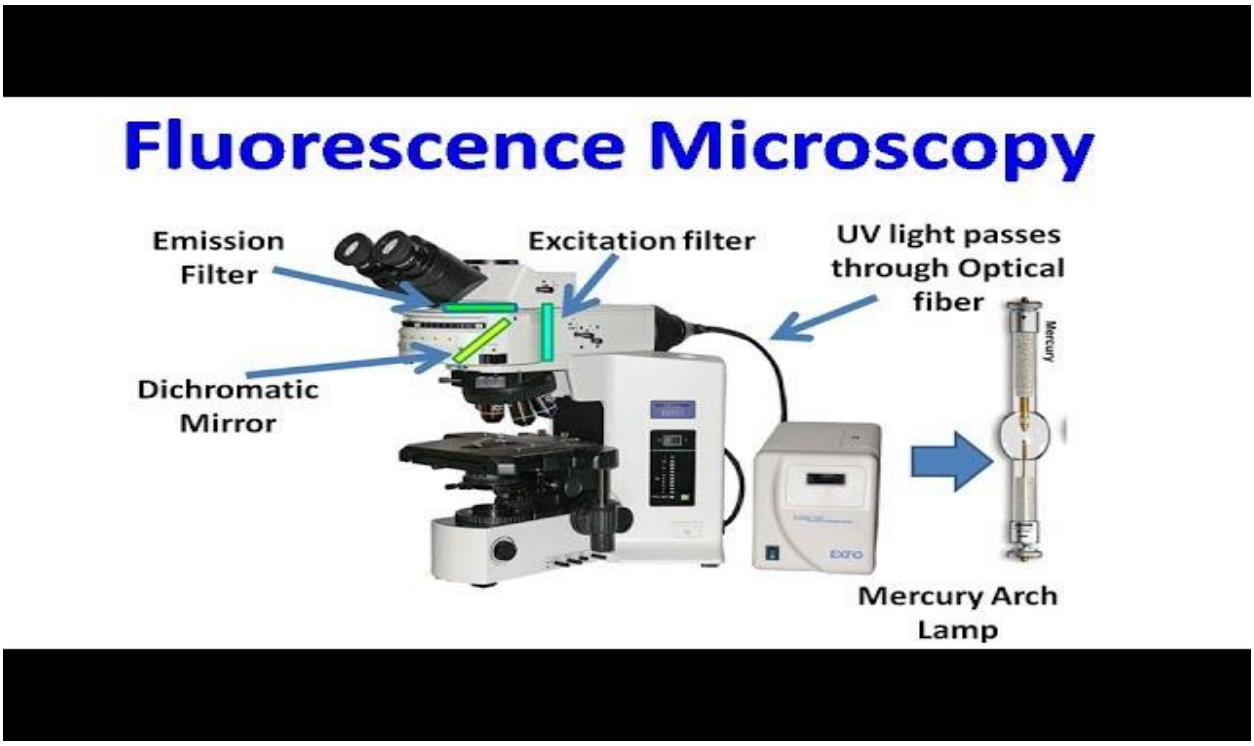
A fluorescence microscope is much the same as a conventional light microscope with added features to enhance its capabilities.

- The conventional microscope uses visible light (400-700 nanometers) to illuminate and produce a magnified image of a sample.
- A microscope that uses fluorescence and phosphorescence instead of/or in addition to reflection and absorption to study properties of a microbe.

PRINCIPLE:

Fluorescence microscopy takes advantage of fluorescence, the ability of substances to absorb short wavelengths of light (ultraviolet) and give off light at a longer wavelength (visible). In most cases the sample of interest is labeled with a fluorescent substance known as a fluorophore and then illuminated through the lens with the higher energy source. The illumination light is absorbed by the fluorophores (now attached to the sample) and causes them to emit a longer lower energy wavelength light. This fluorescent light can be separated from the surrounding radiation with filters designed for that specific wavelength allowing the viewer to see only that which is fluorescing.



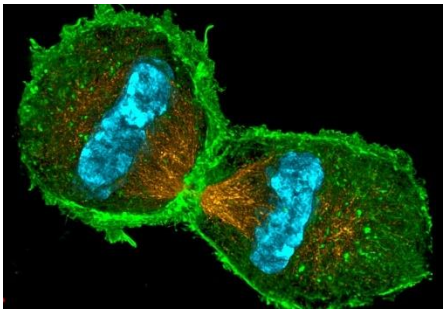


Fluorescent microscopy is often used to image specific features of small specimens such as microbes. It is also used to visually enhance 3-D features at small scales. This can be accomplished by attaching fluorescent tags to anti-bodies that in turn attach to targeted features, or by staining in a less specific manner. When the reflected light and background fluorescence is filtered in this type of microscopy the targeted parts of a given sample can be imaged. This gives an investigator the ability to visualize desired organelles or unique surface features of a sample of interest.

LIGHT SOURCE:

Monochromatic light source usually four types:

- Xenon arc lamps
- Mercury vapor lamps
- Super-continuum source
- High power LED



Lesson 27: Confocal Microscopy

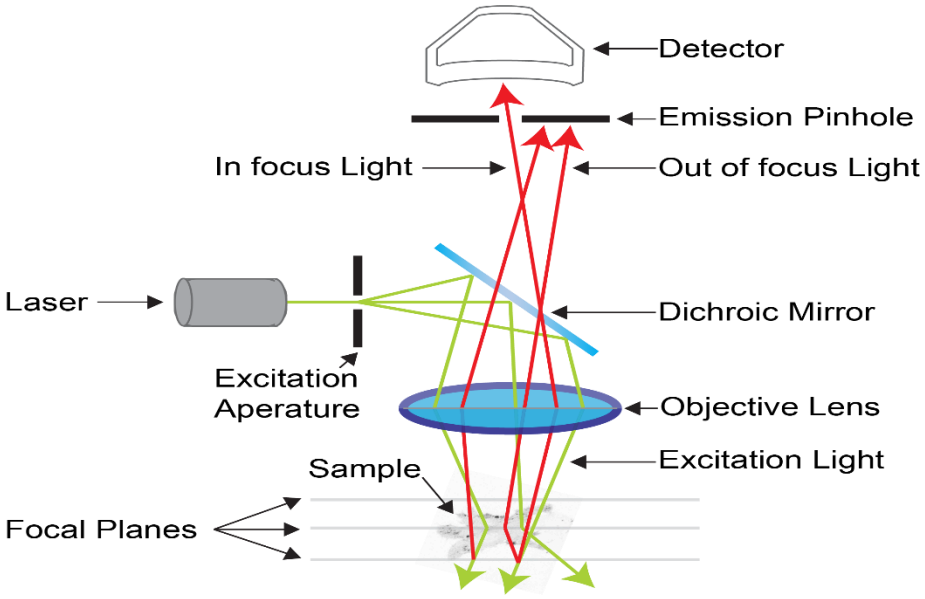
Confocal Microscopy

An optical imaging technique for increasing optical resolution and contrast of a micrograph by using a spatial pinhole to block out-of-focus light in image formation

Principle

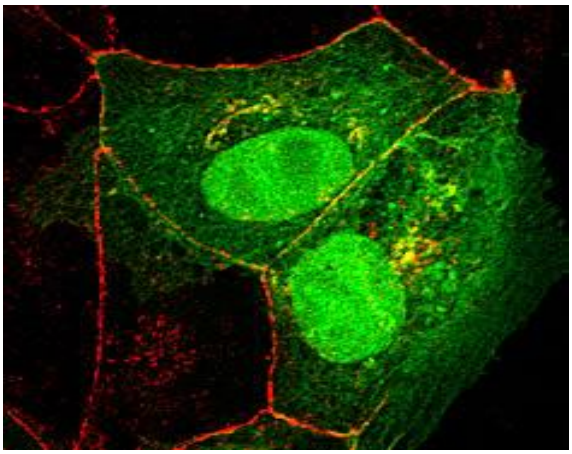
The confocal microscope uses fluorescence optics. Instead of illuminating the whole sample at once, laser light is focused onto a defined spot at a specific depth within the sample. This leads to the emission of fluorescent light at exactly this point. A pinhole inside the optical pathway cuts off signals that are out of focus, thus allowing only the fluorescence signals from the illuminated spot to enter the light detector.

By scanning the specimen in a raster pattern, images of one single optical plane are created. 3D objects can be visualized by scanning several optical planes and stacking them using suitable microscopy deconvolution software (z-stack). It is also possible to analyze multicolor immunofluorescence staining using state-of-the-art confocal microscopes that include several lasers and emission/excitation filters.



Properties

- Each plane corresponds to an image of a fine slice that has been physically cut from a specimen
- Successive planes and regions are illuminated until the entire specimen has been scanned
- Confocal microscopy can image cells in detail only to a depth of less than 100 mm



Types

Laser scanning microscopes - use a sharply focused laser that scans over the sample

Spinning disk microscopes - use a disk with pinholes cut into it that are arranged in the shape of a spiral

Microlens enhanced or dual spinning disk confocal microscope

Programmable array microscopes - use an electronically controlled spatial light modulator that produces a set of moving pinholes.

Lesson 28: Two-photon Microscopy

Principle

Just like widefield or confocal fluorescence microscopy, two-photon microscopy is based on fluorophore excitation, which results in the emission of light. In classic fluorescence microscopy, a fluorophore is excited by absorbing one single photon of a certain wavelength. When using two-photon microscopy, two or three photons of a higher wavelength do the work of one: When they hit the fluorophore at the very same time (typically within several femtoseconds), they are absorbed, resulting in fluorophore excitation and emission of light.

In this process, photons combine their energy, which allows low-energy infrared photons to excite standard fluorophores. The infrared light penetrates tissue more deeply than the standard excitation light used in fluorescence microscopy. Due to its low energy level, infrared light is less damaging, and therefore especially useful when working with living samples.

In order to increase the likelihood that two photons hit the fluorophore simultaneously, lasers with very high intensity are needed. Their infrared light only leads to excitation in the focus of the objective, because only in this area, the critical number of photons per time and space is reached. Therefore, all emitted light comes from one focal point in the specimen, strongly reducing background noise. The image is created just as in confocal microscopy: The laser scans across the sample, recording the image intensity point by point.

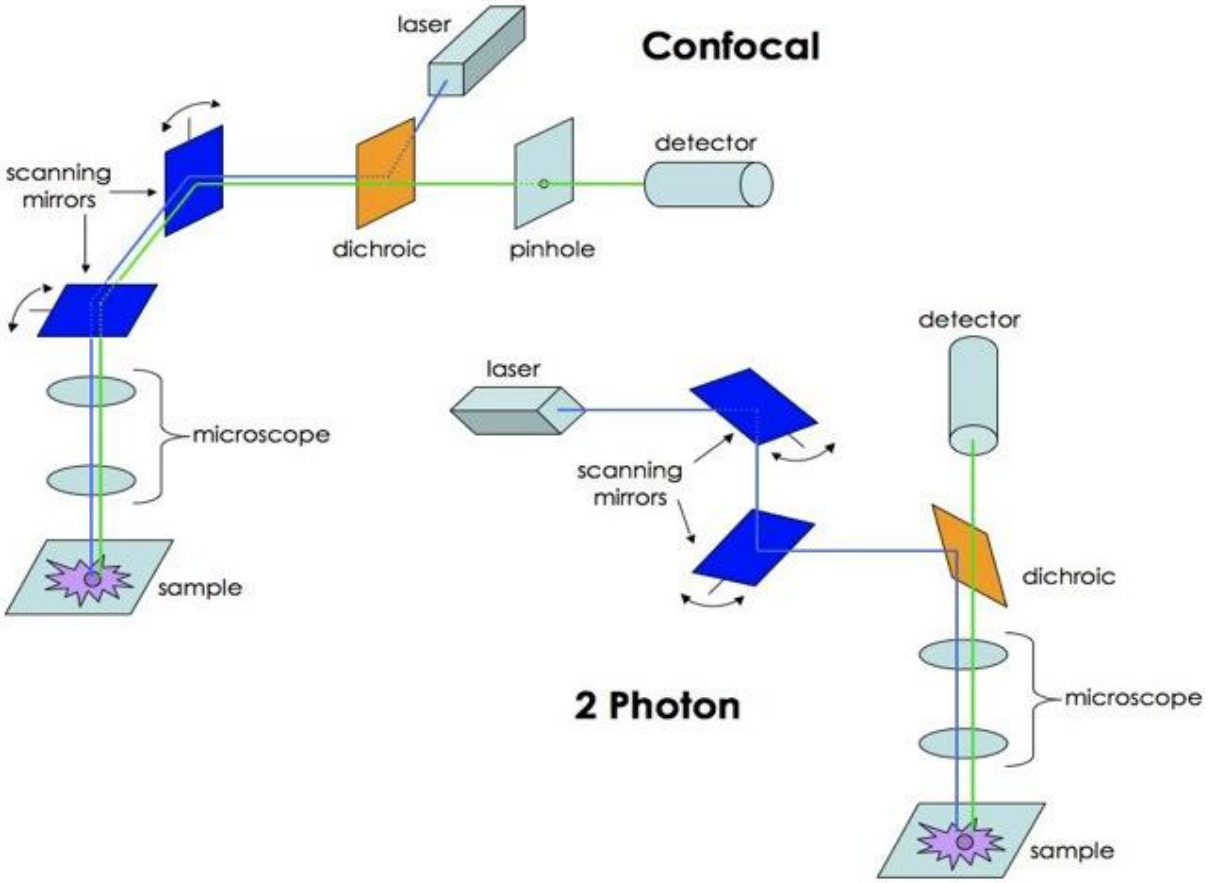
Advantages

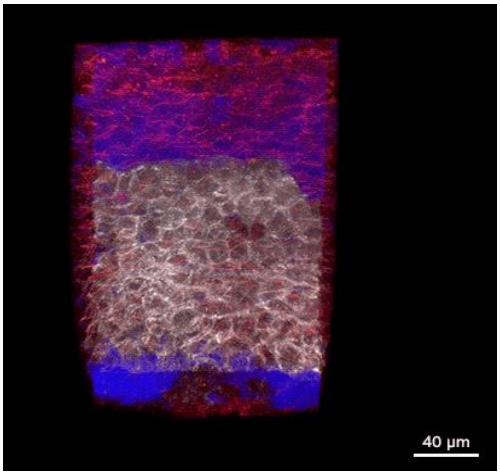
1. It penetrates up to 1 mm into tissue
2. It minimizes phototoxicity because the beam excites just a single focal point at a time.
3. A two-photon microscope has higher sensitivity than a normal confocal microscope, because it uses two photos instead of one

Applications

Two-photon microscopy (also called multiphoton microscopy) can be used for live cell imaging of thin biological specimens, as it has several advantages over confocal microscopy. Molecules can be visualized deeply within the specimen with a maximal penetration depth of about 1 mm. This enables 3D imaging of tissue slices, organoids, whole organs, embryos, or even whole animals. In contrast to confocal microscopy, two-photon microscopy works with higher wavelengths leading to less photobleaching or photodamage, which is especially important when working with living samples.

Working





Rotating 3D rendering of a confocal image stack showing the dorsal germ ring of a zebrafish embryo at the onset of gastrulation (6 hours post fertilization). GFP (white) is expressed in internalizing prechordal plate progenitors and lyn-TagBFP (red) marks the membrane in all cells. The embryo was injected with dextran-rhodamine to label the interstitial fluid (blue).

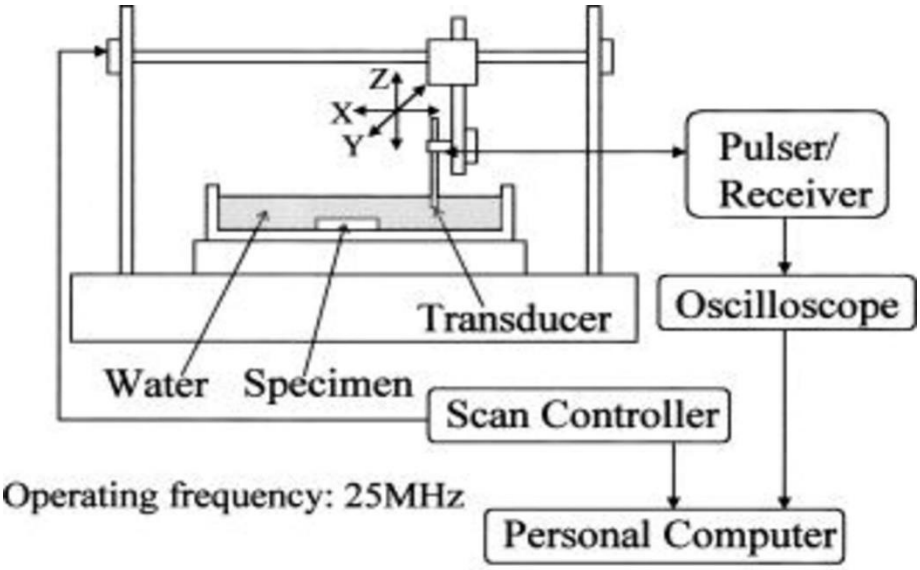
Lesson 29: SAM Microscopy

Scanning acoustic microscopy (SAM), also known as C-SAM or Acoustic Micro Imaging or AMI is a modern, powerful technology for the visualization of the internal structure of materials of a different nature.

- A device that uses focused sound to image an object
- Applicable in biological and medical research
- Developed in 1974

PRINCIPLE:

- Scanning acoustic microscopy (SAM) basically consists of interpreting the action of a sound wave sent through a specimen
- A sound wave of a specific frequency travels through the specimen, and a portion of it is reflected back every time it hits an interface within the material



IMAGE

- Resolution= 1 µm
- Both 2D and 3D image

USES:

- SAM is a non-destructive technique used in failure analysis of complex devices.

- It can obtain high-quality microscopic images of tissues and cells that are comparable with light microscopic images without staining and within only a few minutes.
- SAM is used to study living cells attached to another surface, such as cancer cells, artery plaque, and bacterial biofilms that foul equipment

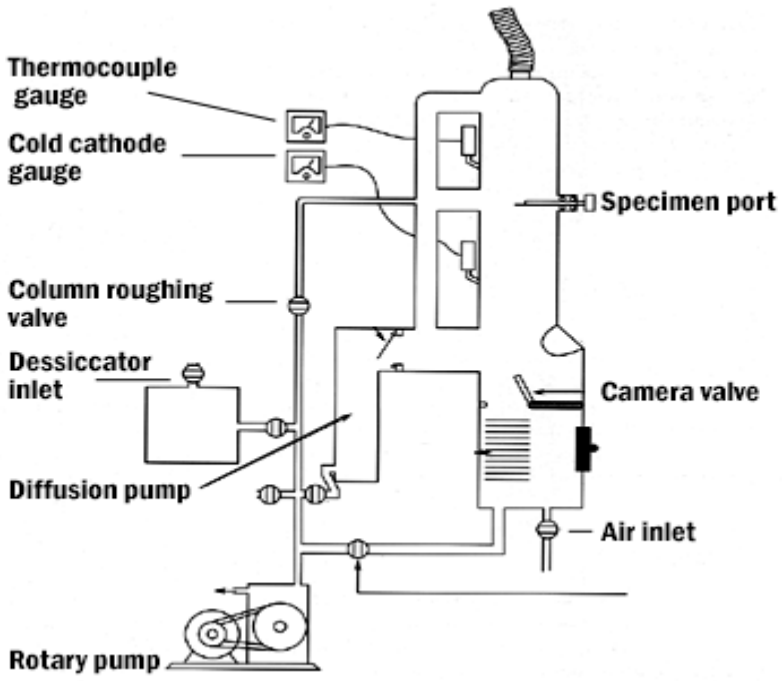
Lesson 30: TEM Microscopy

Transmission Electron Microscopy

The electron microscope is a type of microscope that uses a beam of electrons to create an image of the specimen. It is capable of much higher magnifications and has a greater resolving power than a light microscope, allowing it to see much smaller objects in finer detail. Ultra-thin section of specimen.

PRINCIPLE:

- Focused beam of electrons from an electron gun passes through ultrathin section of the specimen
- Beam gets focused on a small area of the specimen by an electromagnetic condenser lens
- Instead of being placed on a glass slide, the specimen is usually placed on a copper mesh grid



Image

The original form of electron microscopy, Transmission electron microscopy (TEM) involves a high voltage electron beam emitted by a cathode and formed by magnetic lenses. The electron beam that has been partially transmitted through the very thin (and so semitransparent for electrons) specimen carries information about the structure of the specimen. The spatial variation in this information (the "image") is then magnified by a series of magnetic lenses until it is recorded by hitting a fluorescent screen, photographic plate, or light sensitive sensor such as a CCD (charge-coupled device) camera. The image detected by the CCD may be displayed in real time on a monitor or computer.

Transmission electron microscopes produce two-dimensional, black and white images. The transmission electron microscope can resolve objects as close together as 10 pm. The objects are generally magnified 10,000 to 100,000*

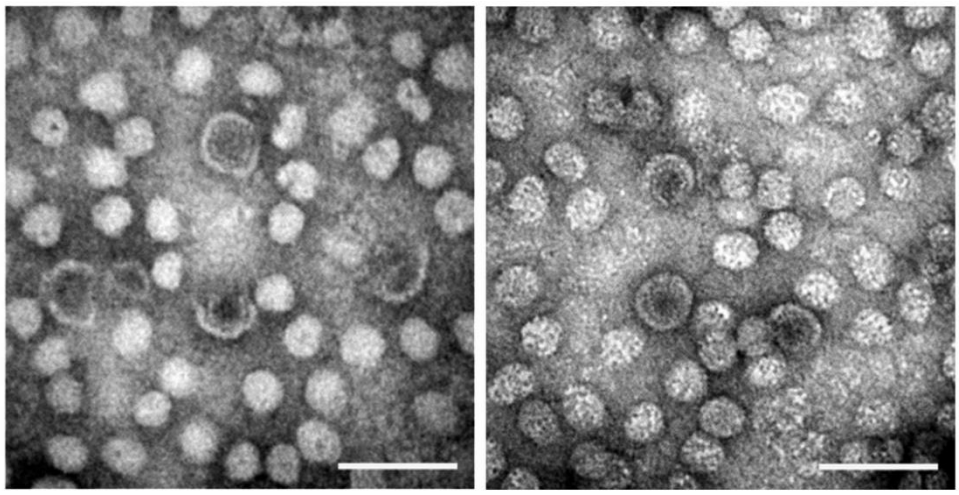


Figure: Transmission electron micrographs of complete hepatitis B virus

DYE:

- Contrast can be greatly enhanced by using a “dye” that absorbs electrons and produces a darker image in the stained region
- Lead, osmium, tungsten, and uranium (Stains)
- These metals can be fixed onto the specimen (positive staining)
- or used to increase the electron opacity of the surrounding field (negative staining)

LIMITATIONS:

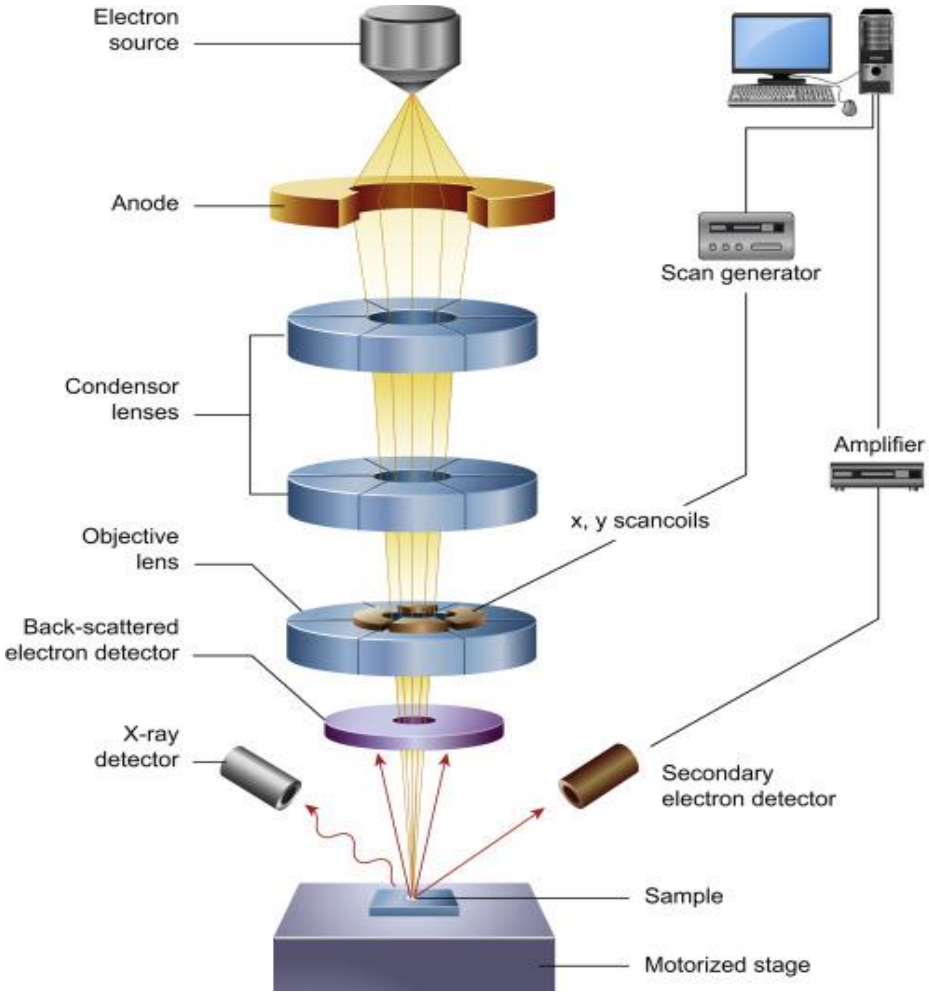
- Only a very thin section of a specimen (about 100 nm) can be studied effectively

- Specimens must be fixed, dehydrated, and viewed under a high vacuum to prevent electron scattering
- These treatments not only kill the specimen, but also cause shrinkage and distortion

Lesson 31: SEM Microscopy

Scanning Electron Microscopy

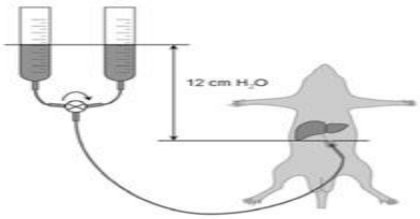
Unlike the TEM, where the electrons in the primary beam are transmitted through the sample, the Scanning Electron Microscope (SEM) produces images by detecting secondary electrons which are emitted from the surface due to excitation by the primary electron beam. In the SEM, the electron beam is scanned across the surface of the sample in a raster pattern (image is composed of a grid of individual pixels, arranged in rows and columns), with detectors building up an image by mapping the detected signals with beam position.



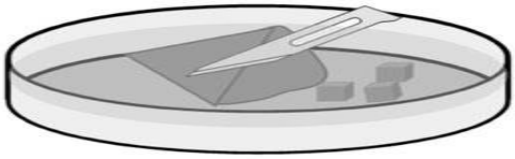
- First of all, an electron source or electron gun located at the top of the SEM column is heated with high-voltage. As a result, it will release electron beams.
- Electron beams are now accelerated down the column and onto a series of electromagnetic lenses. These lenses and tubes are also called solenoids, because they are wrapped in a coil, they create focused electron beams.
- This electron beam is focused onto a specimen.
- A computer is attached to the SEM, which controls the magnification power of the microscope and as well as determines the surface area to be scanned.
- When the incident electrons come in contact with the sample, energetic electrons are released from the surface of the sample. The scatter patterns made by the interaction yields information on the size, shape, texture, and composition of the sample.
- Different types of electrons are emitted from the sample after the interaction between electron beams and sample.

Sample Preparation

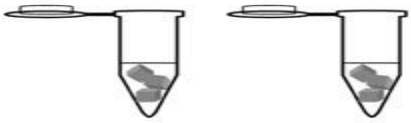
Materials to be viewed in an electron microscope generally require processing to produce a suitable sample. This is mainly because the whole of the inside of an electron microscope is under high vacuum in order to enable the electron beam to travel in straight lines. The technique required varies depending on the specimen, the analysis required and the type of microscope



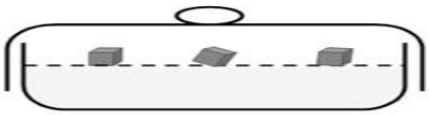
Step I (2-10 min)



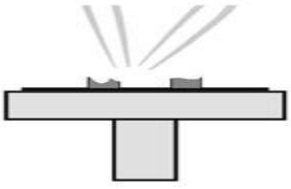
Step II (10 min)



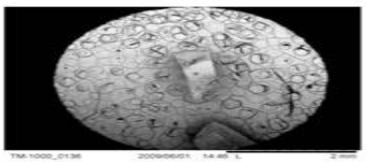
Step III (5×2 min & 10 min)



Step IV (10 min)



Step V (10 min)



Step VI

Figure: Scheme depicting the different steps involved in the preparation of liver tissue for ultrastructural scanning electron microscopy investigation. Step I, perfusion-fixation of the liver with 1.5% glutaraldehyde; Step II, tissue slicing in blocks of 1 mm3; Step III, submerging tissue blocks in an organic reagent with a fast evaporation rate; Step IV, air-drying in a desiccator; Step V, mounting of the samples on stubs and sputter-coating; and Step VI, imaging within a table-top scanning electron microscope

Image

- Provides striking three-dimensional views of specimens
- It can resolve objects as close together as 10 nm, and objects are generally magnified 1000 to 10,000

Uses

- Useful in studying the surface structures of intact cells and viruses
- Also used for fractography (art of qualitatively interpreting the mechanisms of fracture that occur in a specimen)

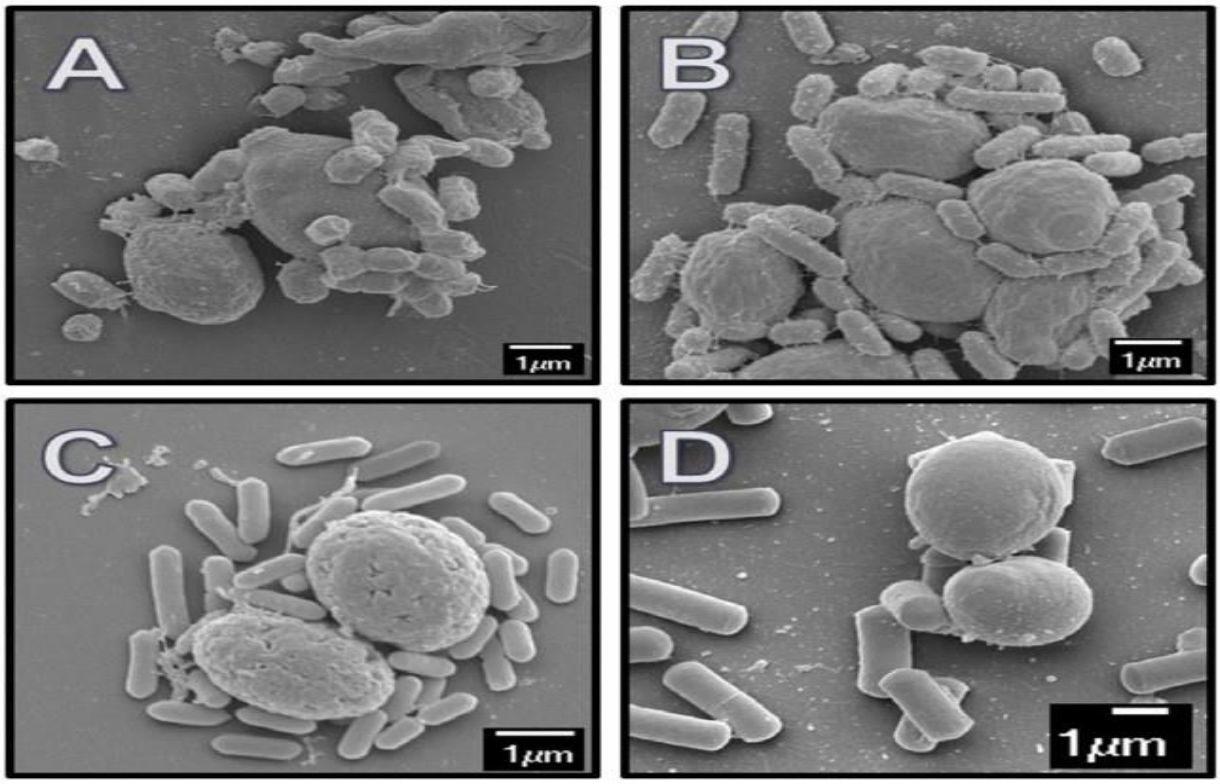


Figure: SEM images of pathogenic bacteria adhered to yeast probiotics and Para probiotics.

Lesson 32: SPM Microscopy

An SPM (Scanning Probe Microscope) is an instrument used for studying surfaces at the nanoscale level. SPMs form images of surfaces using a physical probe that touches the surface of a sample to scan the surface and collect data, typically obtained as a two-dimensional grid of data points and displayed as a computer image. The first SPM was developed in 1982.

Working Principle

An SPM has a probe tip mounted on the end of a cantilever. The tip can be as sharp as a single atom. It can be moved precisely and accurately back and forth across the surface, even atom by atom. When the tip is near the sample surface, the cantilever is deflected by a force. SPMs can measure deflections caused by many kinds of forces, including mechanical contact, electrostatic forces, magnetic forces, chemical bonding, van der Waals forces, and capillary forces. The distance of the deflection is measured by a laser that is reflected off the top of the cantilever and into an array of photodiodes (similar to the devices used in digital cameras). SPMs can detect differences in height that are a fraction of a nanometer, about the diameter of a single atom.

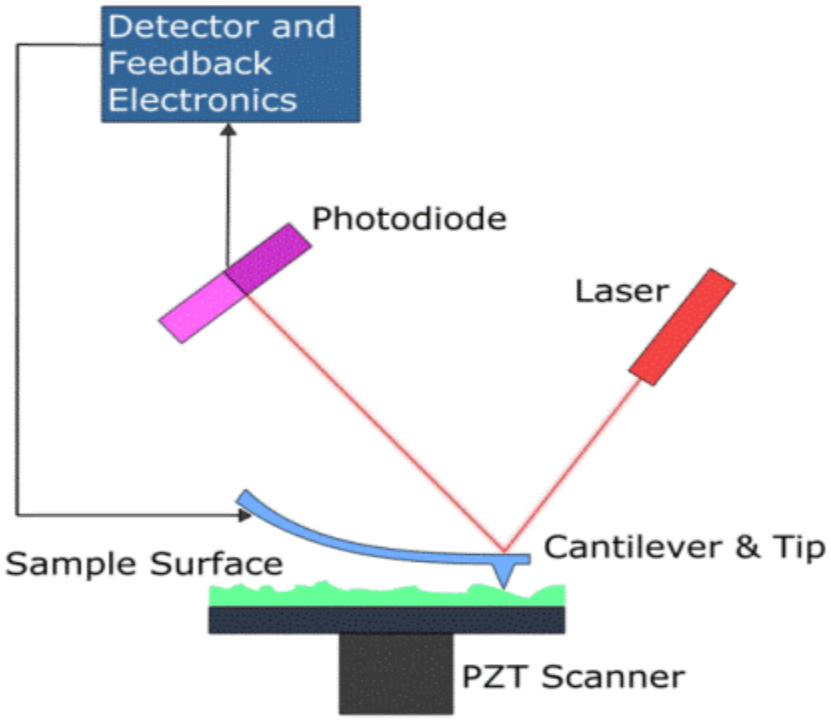


Image formation

The tip is moved across the sample many times. This is why these are called “scanning” microscopes. A computer combines the data to create an image. Result is presented in form of heat waves.

Types

There are several different types of scanning probe microscopes including:

1. Atomic force microscope (AFM): measures the electrostatic force between the tip and the specimen.
2. Scanning tunneling microscope (STM): measures the electrical current between the tip and the specimen.

LIMITATIONS

- Detailed shape of probe is difficult to determine sometimes
- Slower in image acquiring
- Maximum image size is smaller
- Cannot examine buried solid-solid or liquid-liquid interfaces

Lesson 33: Structure and Formation

A virion is a complete, fully developed, infectious viral particle composed of nucleic acid and surrounded by a protein coat that protects it from the environment and is a vehicle of transmission from one host cell to another. Viruses are classified by differences in the structures of these coats.

Nucleic Acid

In contrast to prokaryotic and eukaryotic cells, in which DNA is always the primary genetic material, a virus can have either DNA or RNA—but never both. The nucleic acid of a virus can be single-stranded or double-stranded. Thus, there are viruses with the familiar double-stranded DNA, with single-stranded DNA, with double-stranded RNA, and with single-stranded RNA. Depending on the virus, the nucleic acid can be linear or circular.

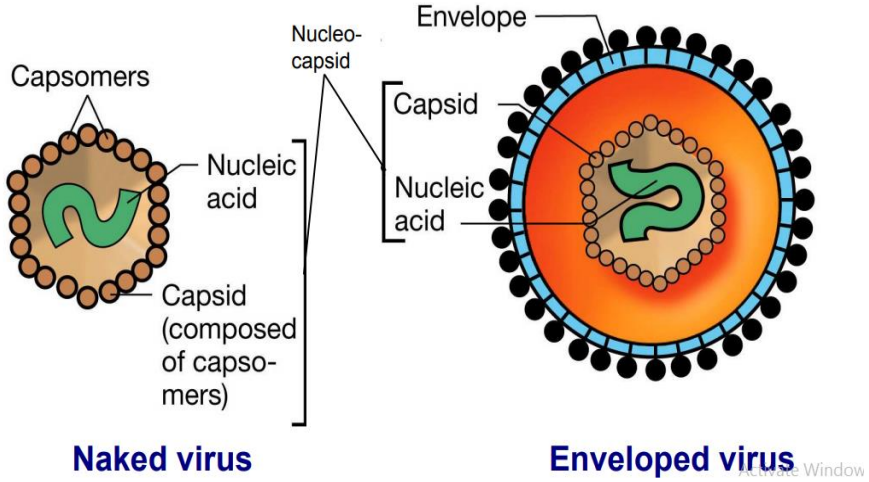
Capsid and Envelope

The nucleic acid of a virus is protected by a protein coat called the capsid. The structure of the capsid is ultimately determined by the viral nucleic acid and accounts for most of the mass of a virus, especially of small ones. Each capsid is composed of protein subunits called capsomeres.

In some viruses, the capsid is covered by an **envelope**, which usually consists of some combination of lipids, proteins, and carbohydrates. Some animal viruses are released from the host cell by an extrusion process that coats the virus with a layer of the host cell's plasma membrane; that layer becomes the viral envelope. In many cases, the envelope contains proteins determined by the viral nucleic acid and materials derived from normal host cell components. Viruses whose capsids are not covered by an envelope are known as nonenveloped viruses.

Functions of Capsid and Envelope

The capsid and envelope play many roles in viral infection, including virus attachment to cells, entry into cells, release of the capsid contents into the cells, and packaging of newly formed viral particles. The capsid and envelope are also responsible for transfer of the viral genetic material from one cell to another. These structures also determine the stability characteristics of the virus particle, such as resistance to chemical or physical inactivation.



Lesson 34: Viral Capsid

Capsid

Regardless of their structural complexity, all virions contain at least one protein coat, the **capsid or** nucleocapsid, which encases and protects the nucleic acid genome.

Functions of Capsid

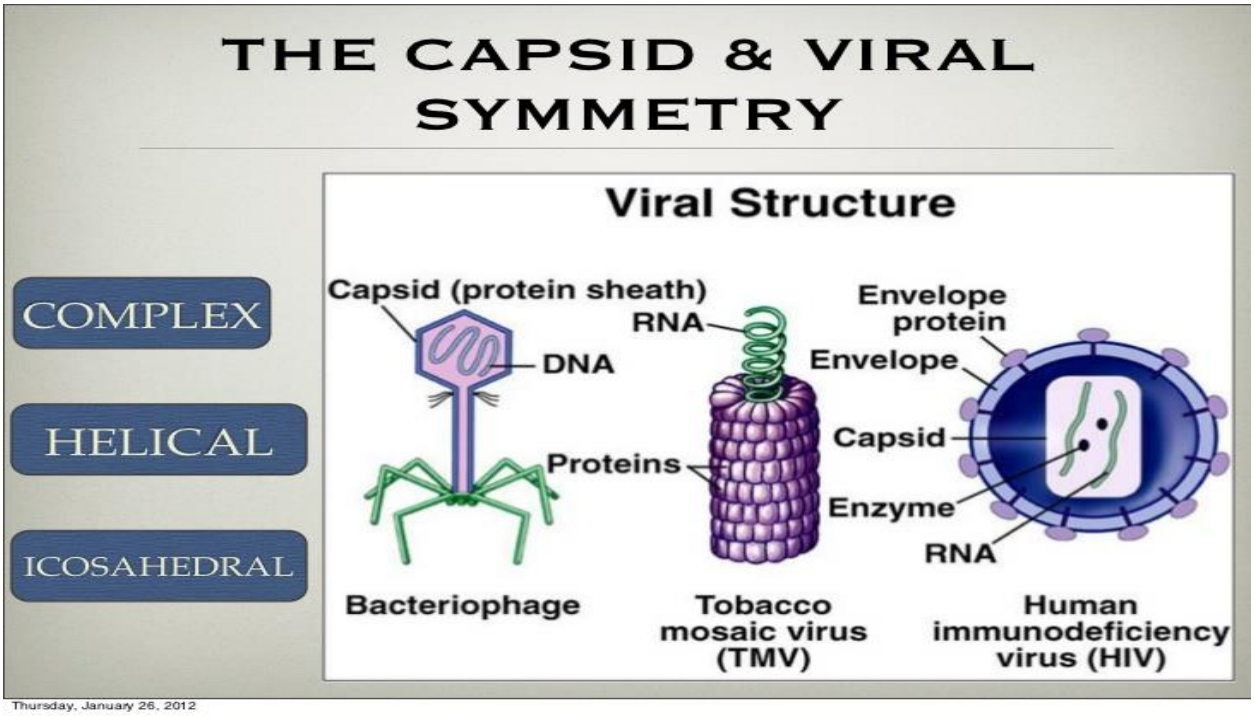
- Protection of the genome
- Prevent digestion by cellular enzymes
- Specific recognition and packaging of the nucleic acid genome
- Delivery of the genome
- Binding to external receptors of the host cell
- Interaction with internal components of the infected cell to direct transport of the genome to the appropriate site

Types of Viral capsid and Symmetry

Capsids are built from identical copies of a small number of viral proteins with structural properties that permit regular and repetitive interactions among them. The repetition of such interactions among a limited number of proteins results in a regular structure, with symmetry that is determined by the spatial patterns of the interactions. In fact, the protein coats of many viruses do display **helical or icosahedral** symmetry.

Helical Viruses

Helical viruses resemble long rods that may be rigid or flexible. e.g. Tobacco mosaic virus



Polyhedral Viruses

Many animal, plant, and bacterial viruses are polyhedral, or many-sided, viruses. The capsid of most polyhedral viruses is in the shape of an icosahedron, a regular polyhedron with 20 triangular faces and 12 corners. The capsomeres of each face form an equilateral triangle. An example of a polyhedral virus is HIV virus

Complex Viruses

Some viruses, particularly bacterial viruses, have complicated structures and are called complex viruses. One example of a complex virus is a bacteriophage. Some bacteriophages have capsids to which additional structures are attached. In this figure, notice that the capsid (head) is polyhedral and that the tail sheath is helical. The head contains the nucleic acid.

Lesson 35: Capsid symmetry and architecture

- Symmetry refers to the way in which capsomere units are arranged in viral capsid.
- Two kinds of symmetry are recognized in the viruses which corresponds to two primary shapes i.e. Rod and spherical shape of virus.
- Rod shaped virus has helical symmetry and spherical shaped virus have icosahedral symmetry.

i. Helical (spiral) symmetry:

- The capsomere and nucleic acid are wined together to form helical or spiral tubelike structure.
- Most of the helical viruses are enveloped and all are RNA viruses.
- The typical virus with helical symmetry is tobacco mosaic virus (TMV), which is an RNA virus with thousands of identical capsomeres arranged in a helix.

ii. Icosahedral (cubical) symmetry:

- An icosahedral is a polygon with 12 vertices (corner), 20 facet (sides) and 30 edges.
- Each facet is an equilateral triangle.
- Icosahedral capsid is the most stable and found in human pathogenic virus e.g. Adenovirus, Picornavirus, Papovavirus, herpes virus etc.

- Icosahedral capsid is of two types;
  - **Pentagon;** Pentagonal capsomere at the vertices
  - **Hexagon;** Hexagonal capsomere at the vertices

iii. Complex symmetry:

- Some viruses are more complex, being composed of several separate capsomere with separate shape and symmetry.
- They do not have either icosahedral or helical symmetry due to complexity of their capsid structure. E.g. Pox virus, Bacteriophage.

Lesson 36: Helical symmetry

- **Helical capsids** are made up of a single type of protein subunit stacked around a central axis to form a helical structure. The helix may have a hollow center, which makes it look like a hollow tube. This arrangement results in rod-shaped or filamentous virions. These virions can be anything from short and very rigid, too long and very flexible.
- Coat protein molecules engage in identical, equivalent interactions with one another and with the viral genome to allow construction of a large, stable structure from a single protein subunit
- The capsomere and nucleic acid are wined together to form helical or spiral tube-like structure.
- Most of the helical viruses are enveloped and all are RNA viruses.
- The typical virus with helical symmetry is tobacco mosaic virus (TMV), which is an RNA virus with 2130 identical capsomeres arranged in a helix.
- In the replication of viruses with helical symmetry, identical protein subunits (protomers) self-assemble into a helical array surrounding the nucleic acid, which follows a similar spiral path. Such nucleocapsids form rigid, highly elongated rods or flexible filaments; in either case, details of the capsid structure are often discernible by electron microscopy. In addition to classification as flexible or rigid and as naked or enveloped, helical nucleocapsids are characterized by length, width, pitch of the helix, and number of protomers per helical turn.

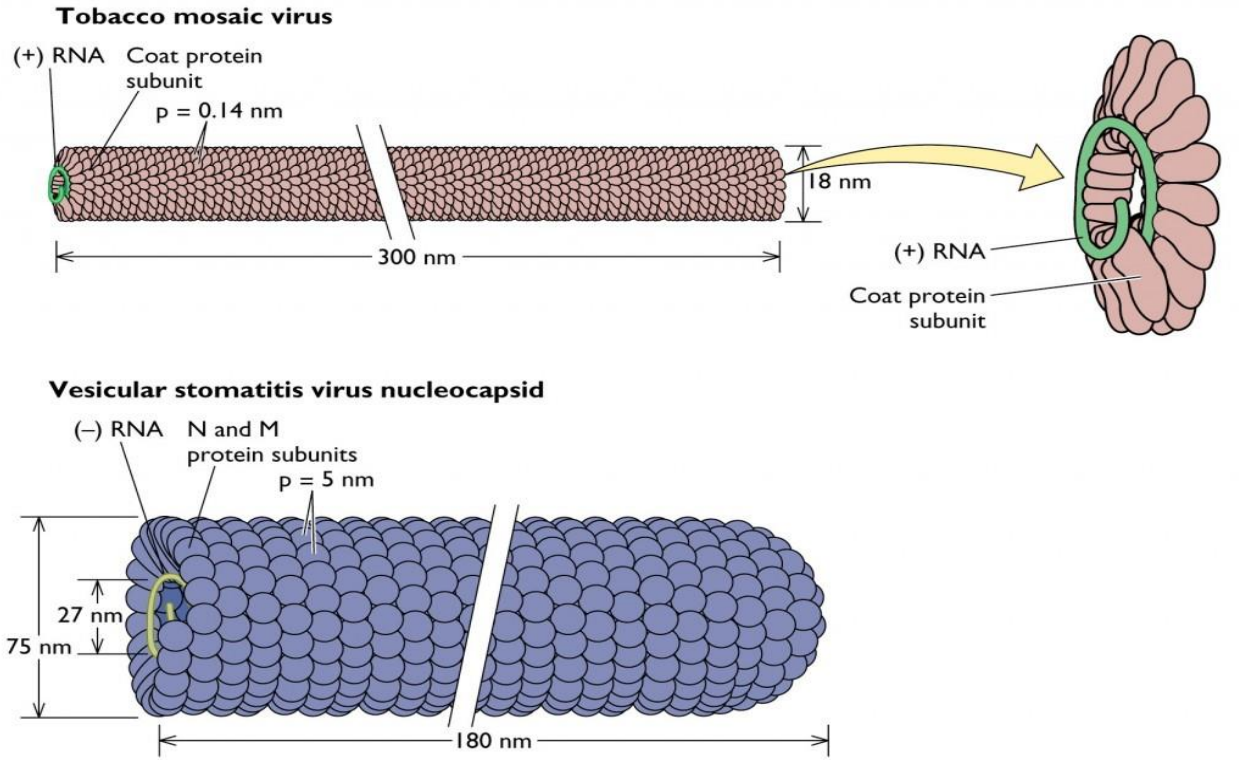


Figure 1: Helical structure of TMV and Vesicular stomatitis virus

Lesson 37: Icosahedral symmetry

Of the two major capsid structures, the icosahedron is by far more prevalent than the helical architecture. In comparison to a helical virus where the capsid proteins wind around the nucleic acid, the genomes of icosahedral viruses are packaged completely within an icosahedral capsid that acts as a protein shell.

An icosahedral capsid is a three-dimensional, 20-sided (facets) structure with 12 vertices and 30 edges. These capsids somewhat resemble a soccer ball.

The subunits can be divided into 12 capsomers that contain five subunits (pentamers) and 20 capsomers that contain six subunits (hexamers).

The proteins that compose the structural unit may form three dimensional structures known as capsomeres that are visible in an electron micrograph. In icosahedral viruses, capsomeres generally take the form of pentons (containing five units) or hexons (containing six units) that form a visible pattern on the surface of the icosahedron.

Adenovirus and picorna virus are two examples of icosahedron that considered among most stable viruses.

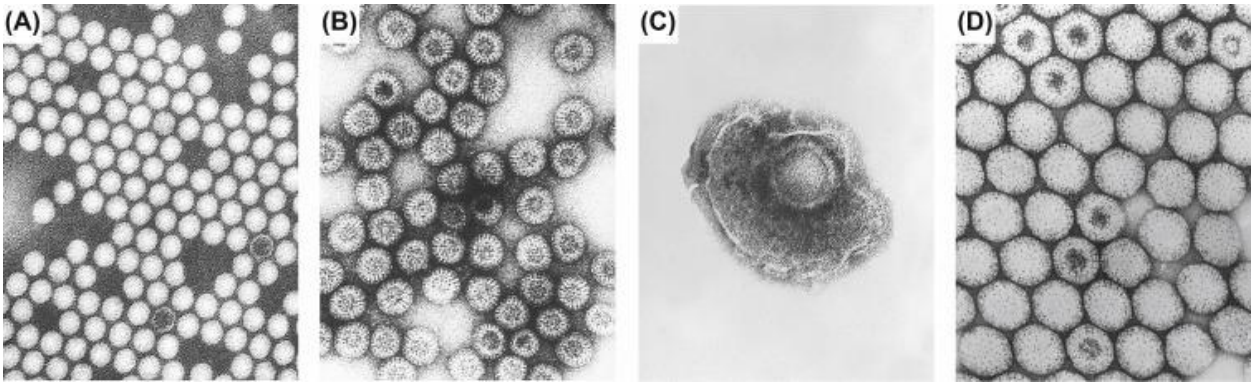


Figure: Electron micrographs of icosahedral viruses. Poliovirus (A), rotavirus (B), varicella–zoster virus (C), the virus that causes chickenpox and shingles, and reovirus (D).

Lesson 38: Complex symmetry

The majority of viruses can be categorized as having helical or icosahedral structure. A few viruses, however, have a complex architecture that does not strictly conform to a simple helical or icosahedral shape. Poxviruses, geminiviruses, and many bacteriophages are examples of viruses with complex structure.

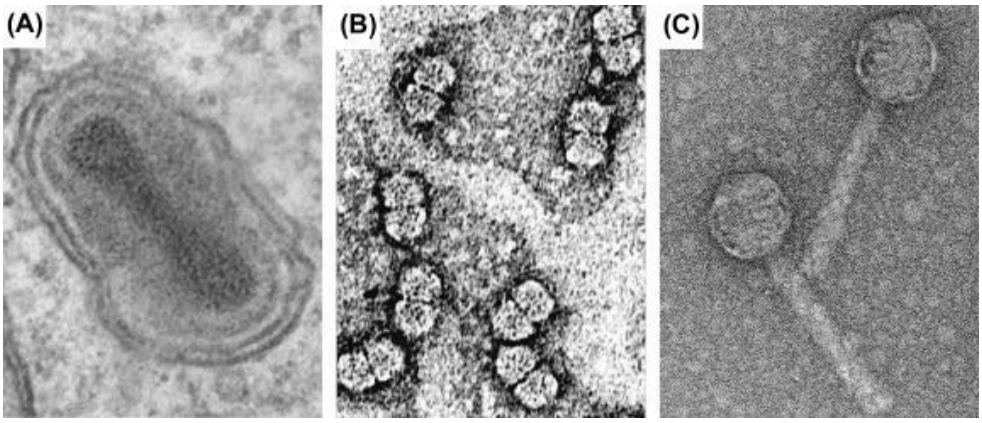


Figure: Electron micrograph of viruses with complex architecture. Vaccinia virus (A), a virus belonging to the poxvirus family, has a complex capsid architecture with a dumbbell-shaped core. Geminiviruses (B) have a double-icosahedron capsid. Bacteriophages, such as P2 (C), often have complex capsid structure.

Poxviruses, including the viruses that cause smallpox or cowpox, are large oval or brick-shaped particles 200–400 nm long. Inside the complex virion, a dumbbell-shaped core encloses the viral DNA and is surrounded by two "lateral bodies," the function of which is currently unknown. Under the electron microscope, sections of poxvirus particle reveal an outer surface of the virion composed of lipid and protein. This layer surrounds the core, which is biconcave (dumbbell-shaped), and two "lateral bodies" whose function is unknown. The core is composed of a tightly compressed nucleoprotein, and the double-stranded DNA genome is wound around it.

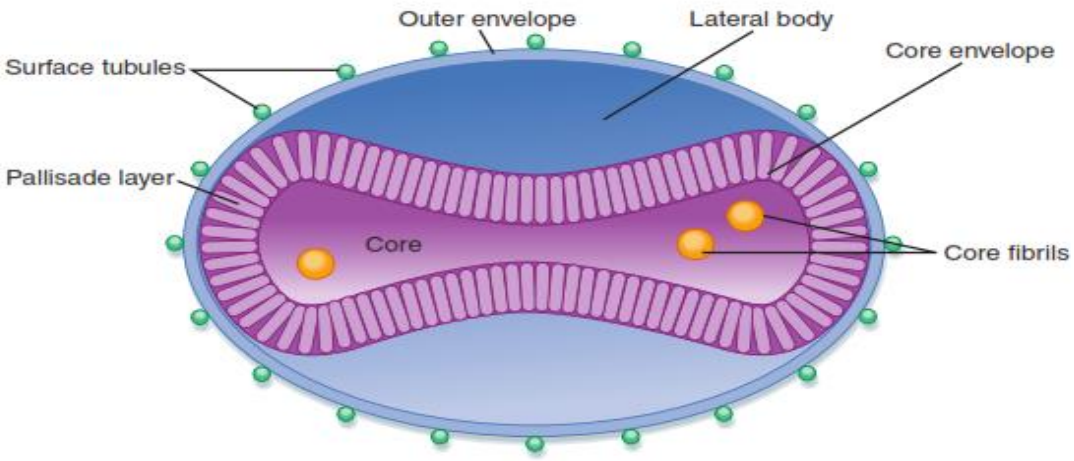


Figure: Poxvirus particles are the most complex virions known and contain more than 100 virus-encoded proteins, arranged in a variety of internal and external structures.

Bacteriophages, also known as bacterial viruses or prokaryotic viruses, are viruses that infect and replicate within bacteria. Many bacteriophages also have complex structure, which has an icosahedral head, containing the nucleic acid, attached to a cylindrical tail sheath that facilitates binding of the bacteriophage to the bacterial cell.

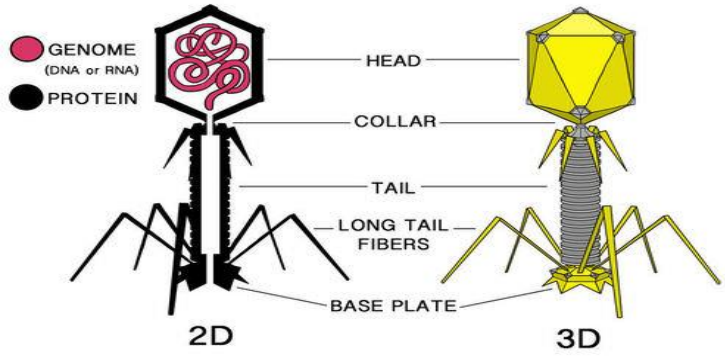
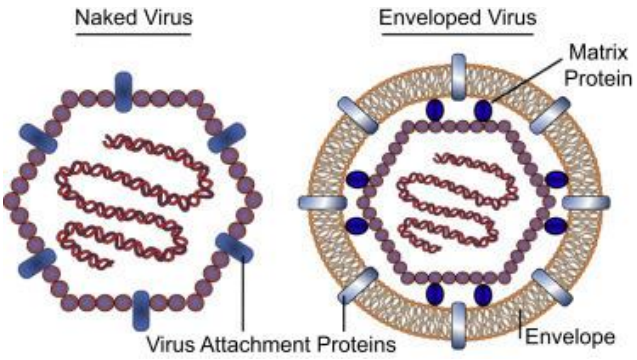


Figure: T4 Bacteriophage

Lesson 39: Enveloped Viruses

Most viruses also have an envelope surrounding the capsid. The envelope is a lipid membrane that is derived from one of the cell's membranes, most often the plasma membrane, although the envelope can also come from the cell's endoplasmic reticulum, Golgi complex, or even the nuclear membrane depending upon the virus. These viruses often have proteins, called matrix proteins, that function to connect the envelope to the capsid inside. A virus that lacks an envelope is known as a nonenveloped or naked virus.

Each virus also possesses a virus attachment protein embedded in its outer-most layer. This will be found in the capsid, in the case of a naked virus, or the envelope, in the case of an enveloped virus. The virus attachment protein is the viral protein that facilitates the docking of the virus to the plasma membrane of the host cell, the first step in gaining entry into a cell.

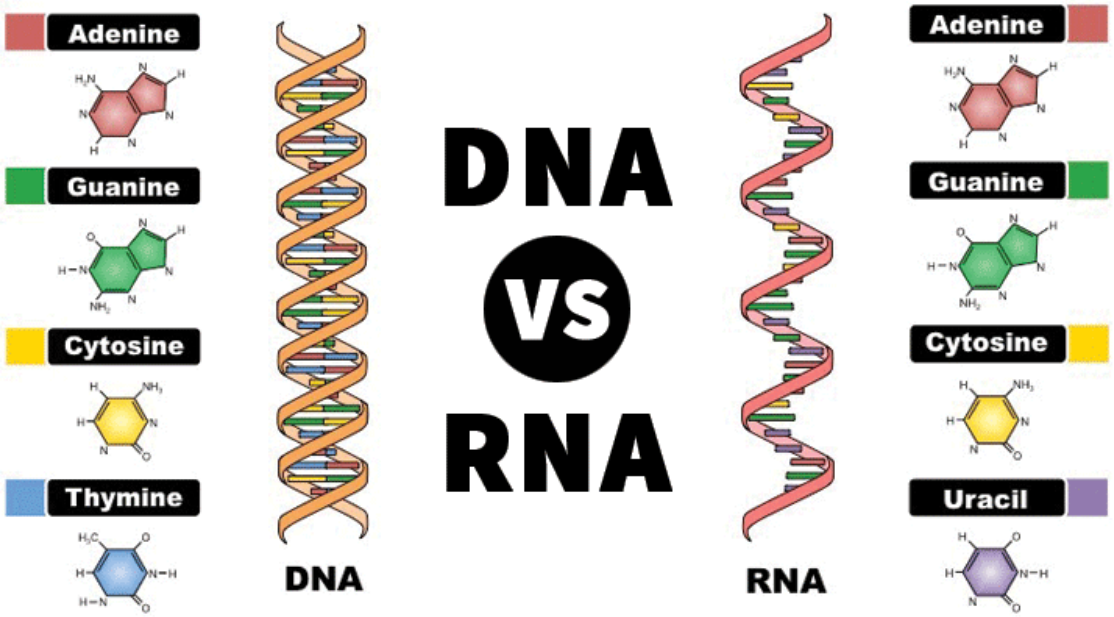


EXAMPLES

- Many animals' viruses
- HIV
- Influenza virus
- Herpesvirus
- Adenovirus
- Hepatitis B virus
- Parvovirus

Lesson 40: Viral Genome

As is true in all forms of life, the nucleic acid of each virus encodes the genetic information for the synthesis of all proteins. In almost all free-living organisms, the genetic information is in the form of double-stranded DNA arranged as a spiral lattice joined at the bases along the length of the molecule (a double helix). In viruses, however, genetic information can come in a variety of forms, including single-stranded or double-stranded DNA or RNA. The nucleic acids of virions are arranged into genomes.



Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are perhaps the most important molecules in cell biology, responsible for the storage and reading of genetic information that underpins all life. They are both linear polymers, consisting of sugars, phosphates and bases, but there are some key differences which separate the two as described in table.

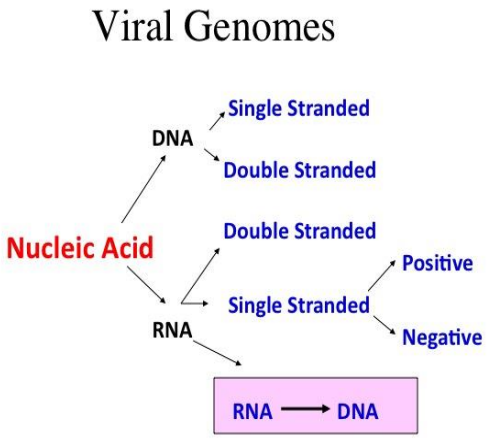
Genome Size

Size of virus genome varies. Size of the DNA viruses is larger than that of RNA viruses. DNA viruses are called large viruses and RNA viruses are small. Smallest genome is of *Circovirus* (2 kilobases) and largest *Pandoravirus* (2 Megabases).

Table: Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA)

Character	DNA	RNA
Full form	Deoxyribonucleic Acid	Ribonucleic Acid
Nucleotides	Each DNA nucleotide contains one of four nitrogenous bases, abbreviated A (adenine), G (guanine), T (thymine), or C (cytosine).	Contains the nitrogenous base uracil in place of thymine.
Sugar	DNA contains deoxyribose sugar.	Contains a different sugar (ribose rather than deoxyribose) in its nucleotides.
Base Pairs	Adenine and Thymine pair (A-T) Cytosine and Guanine pair (C-G)	Adenine and Uracil pair (A-U) Cytosine and Guanine pair (C-G)

With respect to the genome, viruses are broadly divided into DNA viruses and RNA viruses. Both DNA and RNA viruses can either single stranded or double stranded, with a circular, linear or segmented arrangement. DNA and RNA viruses are distinguished by their features, such as monopartite or multipartite. In monopartite, their genome is having a single nucleic acid molecule. All double stranded DNA genomes contain only a single nucleic acid molecule and few of the viruses with single stranded genomes are reported to have multiple segments. In contrast, RNA viral genomes are generally multipartite, with more frequency for single stranded RNA viruses. Additionally, single stranded virus genomes may be either positive sense (+) where the RNA present in the genome will of the same polarity as mRNA and will encode the genes or negative sense (−), where the entire ssRNA genome must be copied and the copied strand is transcribed. Some single stranded viruses are ambisense (a mixture of + and − sense).



Lesson 41: Receptors

Introduction:

Virus receptors are key components of the early events involved in cell infection. To infect a target cell, the virus must attach to a viral receptor or receptor complex. Virus receptors may be highly specialized proteins with limited tissue distribution, such as complement receptors, growth factor receptors, or neurotransmitter receptors, or more ubiquitous components of cellular membranes, such as integrins and other intercellular adhesion molecules, glycosaminoglycans, or sialic acid–containing oligosaccharides.

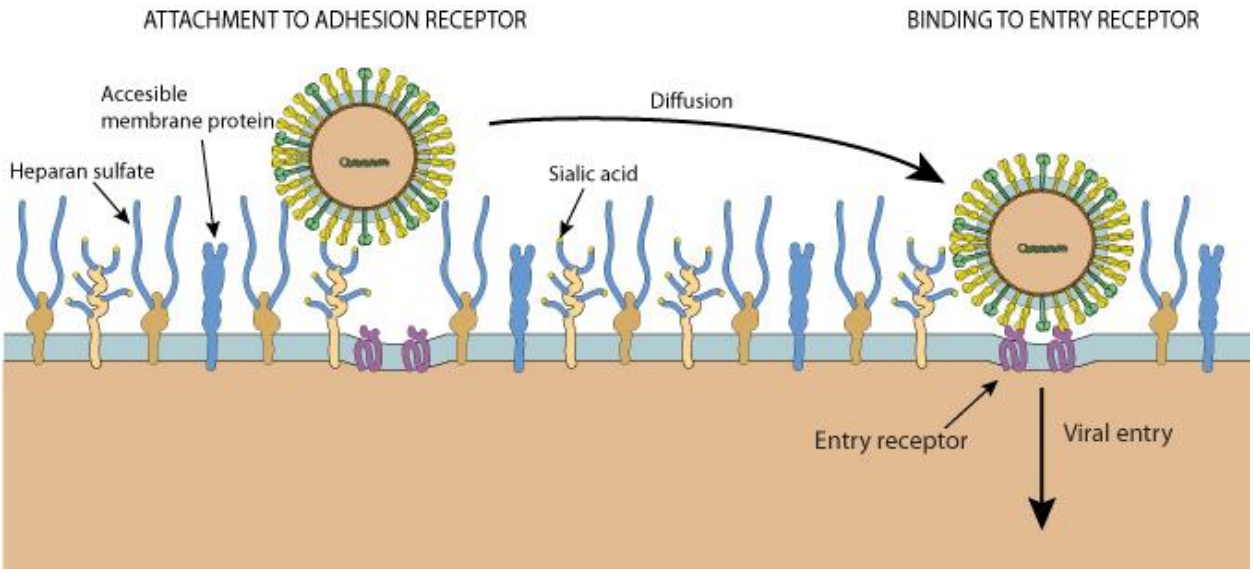
Cells undergoing differentiation may also alter the expression of viral receptors, and receptors may be modulated at different stages of cell growth.

Importance:

The interaction of a virus with its cellular receptor initiates a chain of dynamic events that will enable entry of the virus into the cell. Virus–receptor interactions play a key regulatory role in viral host range, tissue tropism, and viral pathogenesis. For example, receptors for poliovirus are known to confer susceptibility on human and some primate cells, while cells of other species lack receptors and are resistant to infection

Receptor recognition:

Viruses utilize elegant strategies to orchestrate attachment to one or multiple receptors to cross the plasma membrane and access the necessary host cell machinery. The viral attachment protein can be viewed as the “key” that unlocks the cells by interacting with the “lock”—the receptor—on the host-cell surface, initiating critical downstream steps in the viral life cycle. Viral receptors function not only as attachment moieties but also as entry factors, coordinators of viral trafficking, and activators of signaling events, and in many cases, viruses utilize multiple receptors to carry out different functions within the virus life cycle. Thus, the virus must first find the lock, and then use a specific viral “key” or “keys” to unlock the cell. In some cases, viruses must open numerous locks, much like a doorknob plus a deadbolt lock, in order to invade the cell.



Both non-enveloped and enveloped viruses share the same main steps and routes of virus entry (which begin with attachment to cell) surface receptors and end with the delivery of the viral genome to the cell cytoplasm. After binding to receptors (which can be proteins, carbohydrates or lipids) viruses use two main routes to enter the cell

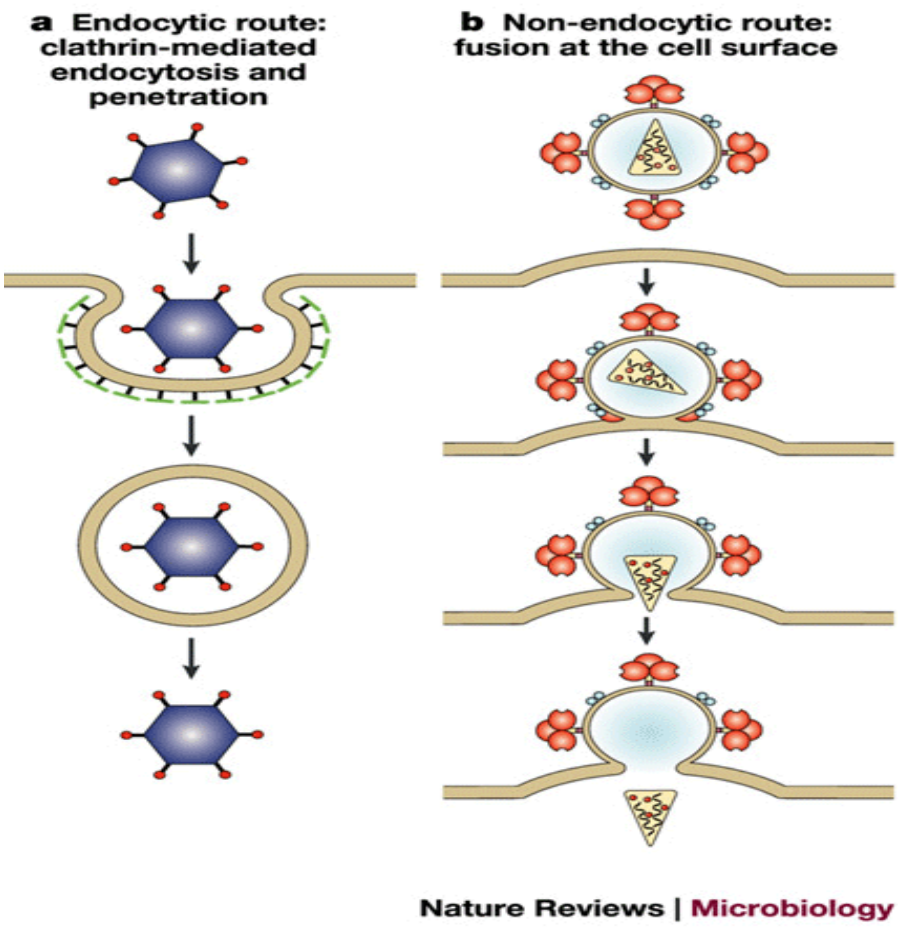
1)endocytic

2) non-endocytic routes.

The **endocytic route** is usually by transport in clathrin-coated vesicles or pits, but non-clathrin-coated pits are also used.

The **non-endocytic route/ viral penetration** involves directly crossing the plasma membrane at neutral pH.

Membrane fusion the process is regulated and is mediated by membrane proteins once the membranes are in close proximity to each other. For both enveloped and non-enveloped viruses, entry into cells involves important conformational changes of the viral ENTRY PROTEINS or the host-cell receptors, which are induced by low endosomal pH.



Lesson 42: Interaction of Virus and Host cells

Host-virus interactions

Resistance to and recovery from viral infections depend on the interactions that occur between the virus and the host. The defenses mounted by the host may act directly on the virus or indirectly on virus replication by altering or killing the infected cell. Non-specific host defenses function early in an encounter with a virus to prevent or limit infection, while the specific host defenses function after infection in recovery to provide immunity for subsequent challenges.

Nonspecific Defenses

Most viral infections are limited by defenses that are antigen nonspecific and/or specific. Nonspecific defenses act sooner than specific defenses. Some are always in place (anatomic barriers, nonspecific inhibitors, and phagocytic cells); others are evoked by the infection (fever, inflammation, and interferon).

Anatomic Barriers: Anatomic barriers are located at body surfaces (skin and mucosa) or within the body (endothelial cells and basement membranes). They are partly effective in preventing virus spread but may be breached by large numbers of virus, by trauma, by increased permeability, by replication of virus in endothelial cells, or by transportation of virus in leukocytes.

Nonspecific Inhibitors: Body fluids and tissues normally contain soluble viral inhibitors. Most prevent viral attachment, some directly inactivate viruses, and others act intracellularly. These inhibitors may be overwhelmed by sufficient virus.

Phagocytosis: Viruses may be phagocytosed to different degrees by polymorphonuclear leukocytes and macrophages. The effect of phagocytosis may be virus inactivation, persistence, or multiplication; consequently, the result may be clearance of virus, transportation to distant sites, or enhanced infection.

Fever: Replication of most viruses is reduced by even a modest rise in temperature. During viral infection, fever can be initiated by several endogenous pyrogens, such as interferon.

Inflammation: Inflammation inhibits viral replication through (1) elevated local temperature, (2) reduced oxygen tension, (3) metabolic alterations, and (4) acid production. The effects of these mechanisms are often additive.

Pathogenesis

Viral pathogenesis describes the processes by which viral infections cause diseases and involves virus–host interactions at the cellular and systemic level that determine whether a virus will cause a disease, what form that disease takes, and how severe the disease will be. Though the pathogenesis of each virus is unique, there are several common points in the virus life cycle that are shared between all pathogenic viruses

The type of virus infection and the virus-induced effects on cells are dependent on the virus, the cell type and species, and often the physiologic state of the cell. Virus-host cell interactions may produce 1) cytocidal (cytolytic) infections; 2) persistent infections, in which the virus or its genome resides in some or all of the cells without killing most of them or 3) transformation, in which the virus does not kill the cell, but produces genetic, biochemical, physiologic, and morphologic changes that may result in the acquisition of malignant properties.

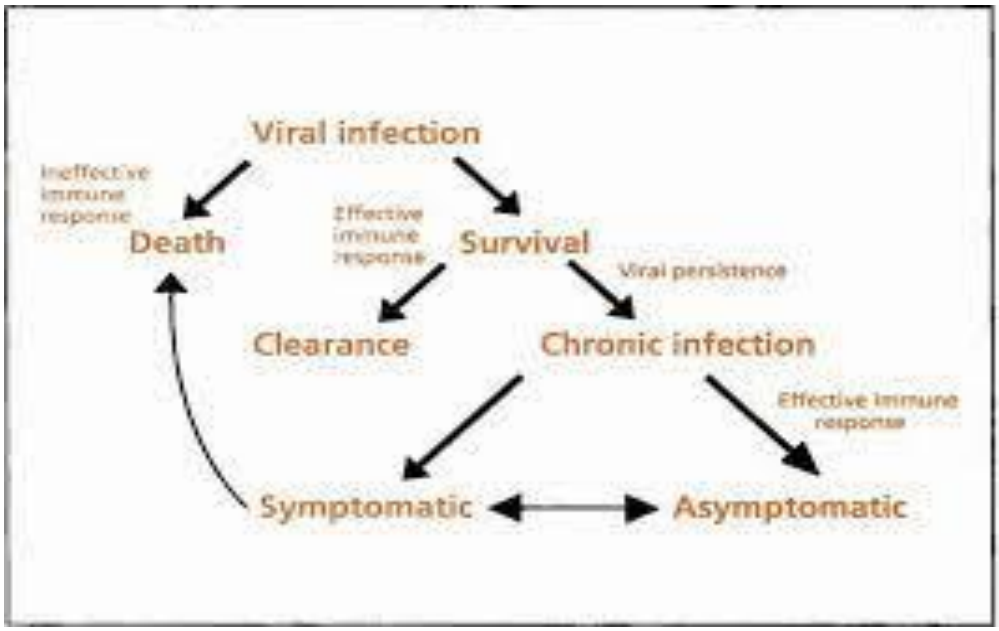


Fig: Host–virus interactions. Adapted from Immunology IV: Clinical Applications in Health and Disease by Joseph A. Bellanti.

1) Cytocidal Infections

Cytocidal (cytolytic) infections, in which production of new infectious virus kills the cell. Infection by cytocidal viruses is usually associated with changes in cell morphology, in cell physiology and sequential biosynthetic events. Many of these changes are necessary for efficient virus replication.

2) Persistent Infections

Some viruses evolved the ability to remain in specific cells for long periods of time. These infections include: latent, chronic, and slow virus infections. The type of persistent infection usually influences the extent of cellular changes.

Latent Infection: Latent infections are characterized by restricted expression of the episomal or integrated virus genome. The viral genomic product(s) are associated with few, if any, changes in the latently infected cell.

Chronic Infection: The cellular effects of chronic infection are usually the same as those of acute cytocidal infections, except that production of progeny may be slower, intermittent or limited to a few cells. The long-term cellular changes may result in severe disease, immune suppression or may trigger immune responses to damaged, or undamaged cells or tissues.

Slow Infection: This type of virus-cell interaction is characterized by a prolonged incubation period, without significant morphological and physiological changes of infected cells. A slow

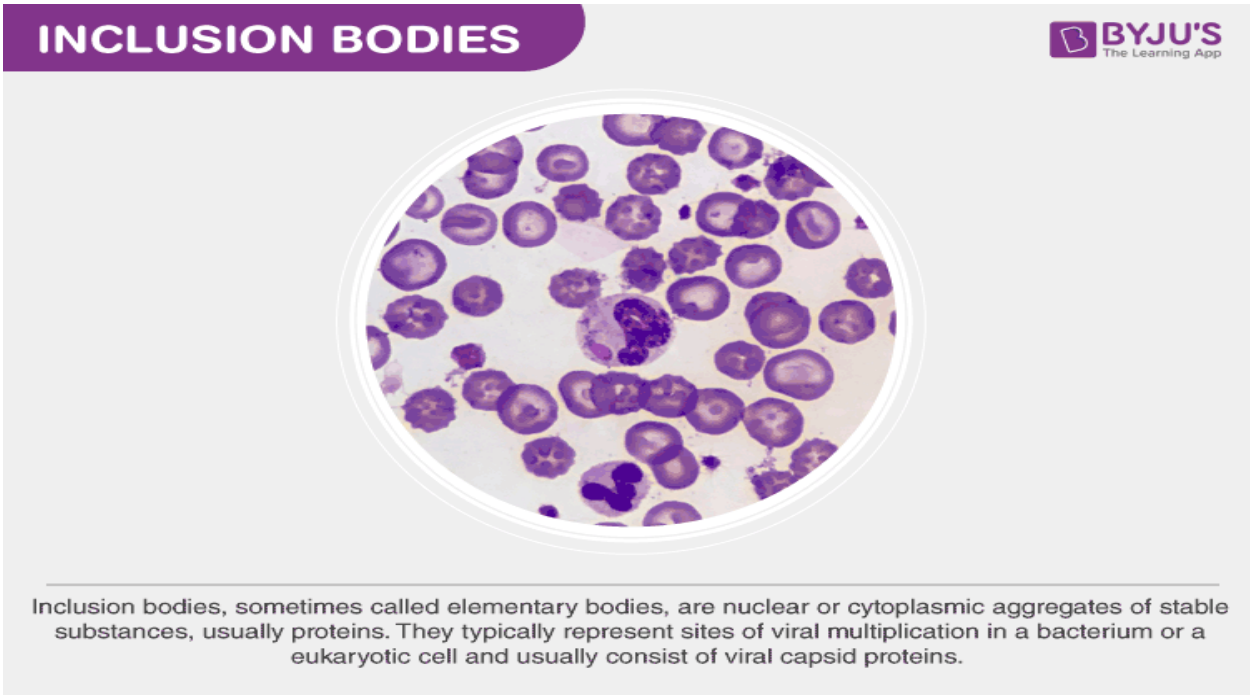
progression of cellular injury may take years and is followed by extensive cellular injury and disease.

3) Transforming Infections

DNA or RNA tumor viruses may mediate multiple changes that convert a normal cell into a malignant one. RNA tumor viruses usually transform cells to a malignant phenotype by integrating their own genetic material into the cellular genome and may also produce infectious progeny. DNA tumor virus infections are often cytocidal; thus, transformation is associated with abortive or restrictive infections in which few viral genes are expressed.

Stages of Transformation: Transformation involves at least two processes: first, the cell gains the capacity for unlimited cell division (immortalization), and second, the immortalized cells acquire additional heritable genetic changes by which the cell is able to produce a tumor in an appropriate host.

The **inclusion bodies** are tiny particles found freely suspended and floating within the cytoplasmic matrix. Therefore, also referred to as cytoplasmic inclusions. These cell inclusions are formed with decreasing pH and from the pool of soluble fusion proteins within the cell. They are the elementary bodies, formed during infectious diseases or within the virus-infected cells such as rabies, herpes, measles, etc.



Entrance point of viruses

- Respiratory tract
- Digestive tract
- Skin
- Conjuctivae
- Sexual contact
- Vectors

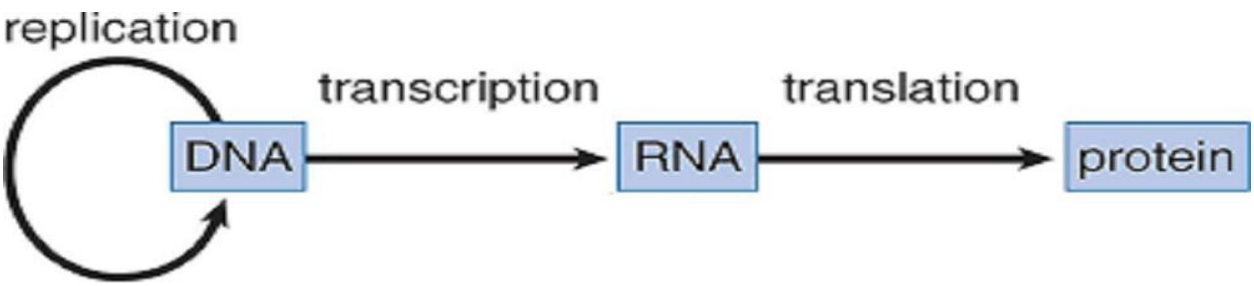
Lesson 43: Interaction of Proteins and Genome

History

- In Late 19th century, after invention of microscope, it was possible to have microscopic observations. Since then, researchers have used variety of procedure to understand the relation that play a crucial role in biological processes

Regulation of biological function

Viruses are clever entities and they are minimalists. Most viruses carry very little within their capsid (and envelope if they have one) and the majority of their insides is genetic material made of either strand(s) of deoxyribonucleic acid (DNA) or ribonucleic acid (RNA). The viral genome (genetic material) has the instructions, genes, to take over the host cell and to build new copies of itself. This process of host cell takeover is known as "molecular hijacking." When a virus replicates it has to build entire viral particles and it has to not only make the proteins that comprise the viral capsid (and enclose it with an envelope), but it must also replicate its genetic material.



The Central Dogma (figure) describes the transfer of genetic information (blueprints) into function in a cell. DNA, which is the makeup of the genome of most organisms, can be replicated, but also transcribed into RNA molecules, which are then translated into proteins. Proteins go on to perform functions, such as the capsid of a virus is made of proteins that provide a structure to encapsulate the viral genome.

The host cell has the following:

- DNA polymerase – the enzyme needed to replicate DNA (DNA --> DNA) ·
- RNA polymerase - the enzyme required to transcribe DNA into RNA (DNA --> RNA) ·
- Ribosomes – the entities responsible for translating RNA into proteins (RNA --> Proteins).
- Phospholipids that make up membranes

Viruses are fully reliant on the translation machinery of their host cells to produce the polypeptides that are essential for viral replication. Consequently, viruses recruit host ribosomes to translate viral mRNAs, typically using virally encoded functions to seize control of cellular translation factors and the host signaling pathways that regulate their activity. This not only ensures that viral proteins will be produced, but also stifles innate host defenses that are aimed at inhibiting the capacity of infected cells for protein synthesis. Despite the diverse functions that viruses encode for their propagation, they remain exquisitely dependent on the translational machinery of the host cell.

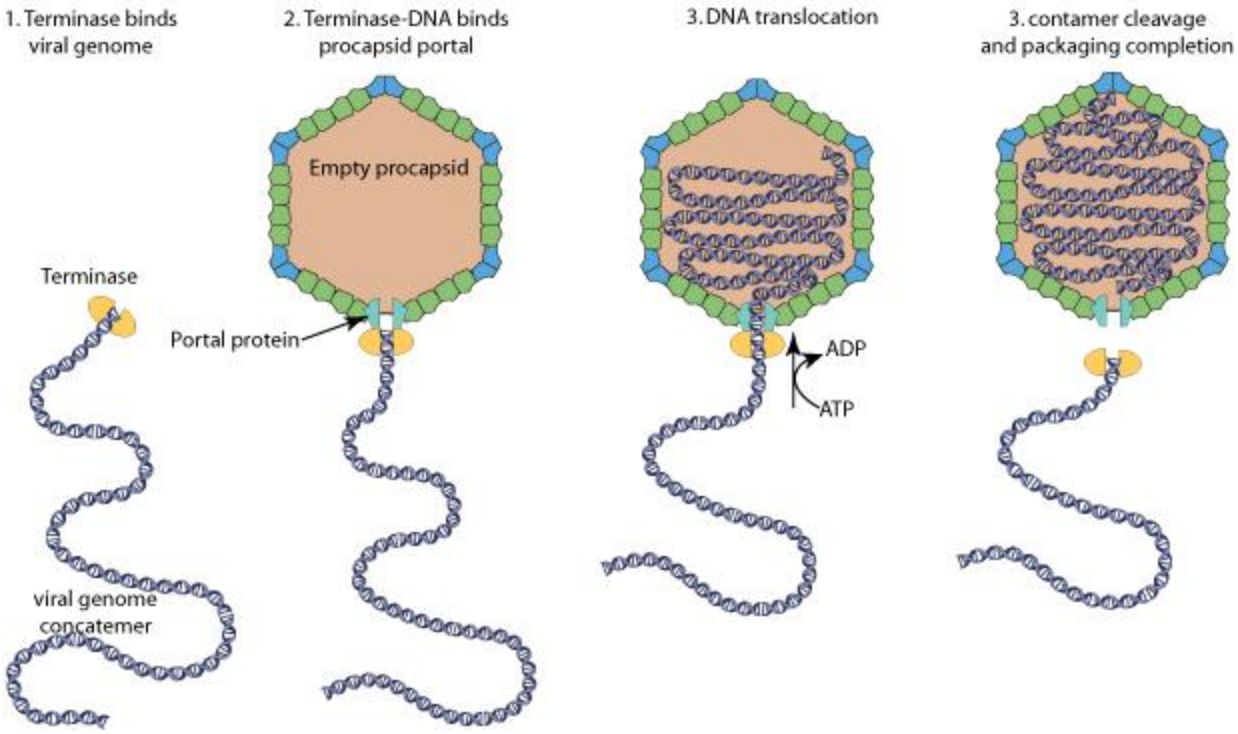
Structure of nucleic acid binding proteins

Proteins, which are specialized in binding to DNA in sequence specific manner, come in different sizes, structures and shapes. All types of proteins contain a region to interact with DNA in sequence specific manner (DNA binding motif), which is most important for identifying the site.

Lesson 43: Genome Packaging

Large dsDNA bacteriophages, herpesviruses, adenoviruses and micro viruses encode a powerful DNA-translocating machinery that encapsulates a viral genome into a pre-assembled capsid or procapsid. The packaging machine is often composed of a portal structure, which provides a gate for DNA entry, and an ATP-driven motor. This **motor** is composed of the large subunit whose ATPase activity fuels DNA translocation, and most frequently, a small subunit that binds to the viral packaging site. DNA cleavage can be coupled to genome packaging.

Schematic diagram of genome packaging in dsDNA viruses



Double-stranded DNA viruses have evolved yet other strategies for packaging their own genome. Most bacteriophages and herpesviruses replicate their genomes as head-to-tail **concatemers** (a long, continuous DNA molecule consisting of multiple copies of the same DNA sequence linked end-to-end). A "**terminase**" complex of two proteins, which is also a part of the packaging motor, recognizes a specific sequence or structure on the concatemeric DNA and makes the initial cut to generate the free end at which packaging is initiated. After one or slightly more than one genome length of DNA has been packaged into the head, the same nuclease makes another cut to terminate packaging. Hence, historically, these proteins that are required for the generation of genome termini were named "terminases".

Certain dsDNA viruses such as bacteriophage f29 and adenoviruses employ protein-primed DNA replication and do not produce concatemers. These viruses recognize their own DNAs through the terminal primer proteins that are covalently linked to the genome ends.

Lesson 45: Introduction to viral Genomics

Virus genome structures and nucleotide sequences have been intensively studied in recent decades because the power of recombinant DNA technology has focused much attention in this area.

Whatever the composition of a virus genome, they must follow one rule. Because viruses are obligate intracellular parasites only able to replicate inside the appropriate host cells, the genome must contain information encoded in a way that can be recognized and decoded by the particular type of host cell. The genetic code used by the virus must match or at least be recognized by the host organism. Similarly, the control signals that direct the expression of virus genes must be appropriate to the host. Many of the DNA viruses of eukaryotes closely resemble their host cells in terms of the biology of their genomes.

Some DNA virus genomes are complexed with cellular histones to form a chromatin-like structure inside the virus particle. Once inside the nucleus of the host cell, these genomes behave like miniature satellite chromosomes, controlled by cellular enzymes and the cell cycle: Split genes containing noncoding introns, protein-coding exons, and spliced mRNAs were first discovered in adenoviruses by Roberts and Sharp in 1977.

Main genes

All viruses have these:

Replicase – an enzyme that replicates genome

Capsid – a protein that protects the genome

Other protein genes

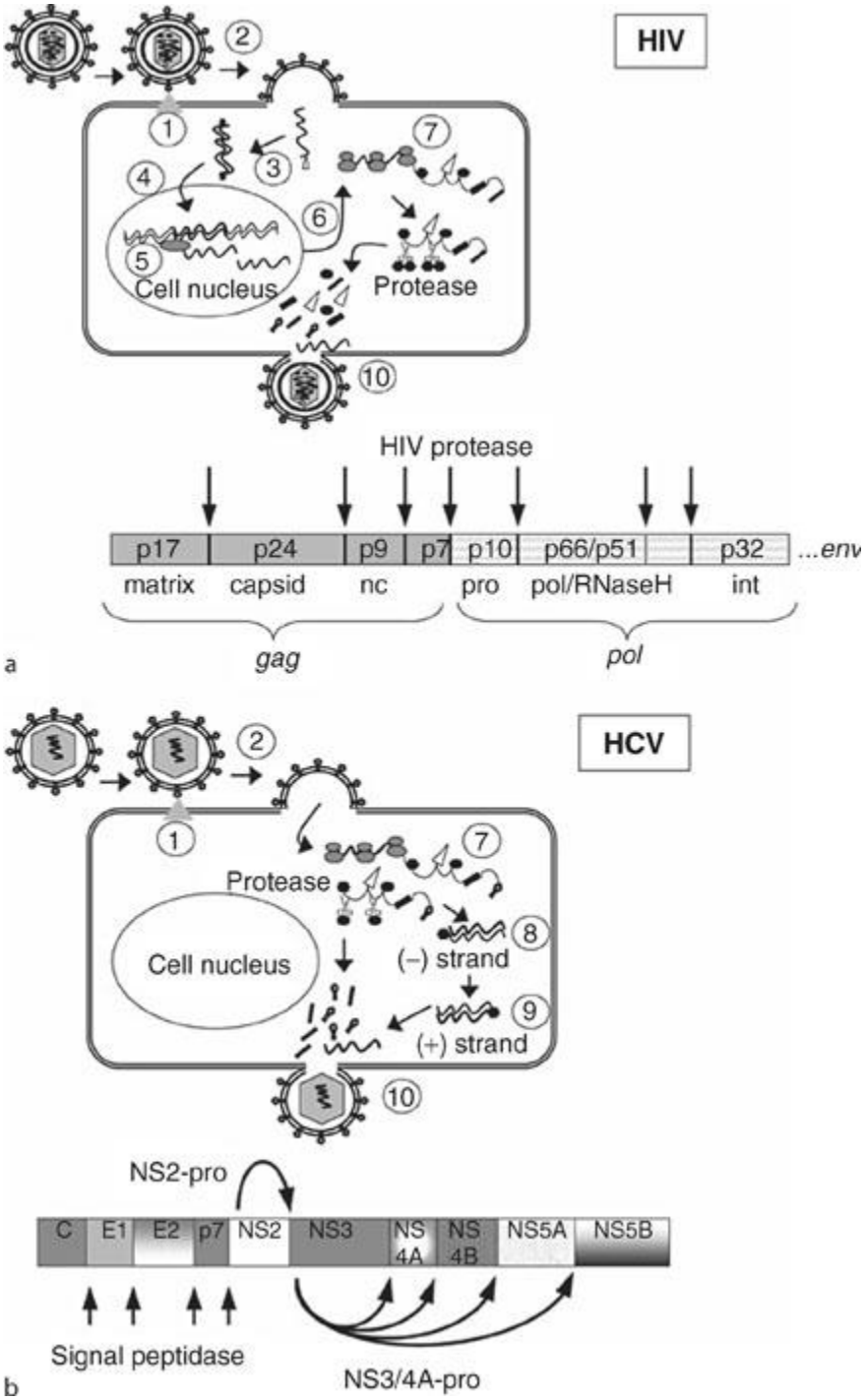
In addition to the proteins specified above, viruses can also carry other types of proteins that have specific function. These include:

- Protease
- Glycoprotein
- Host-shut-off proteins
- Anti-host defense proteins

Viral proteases

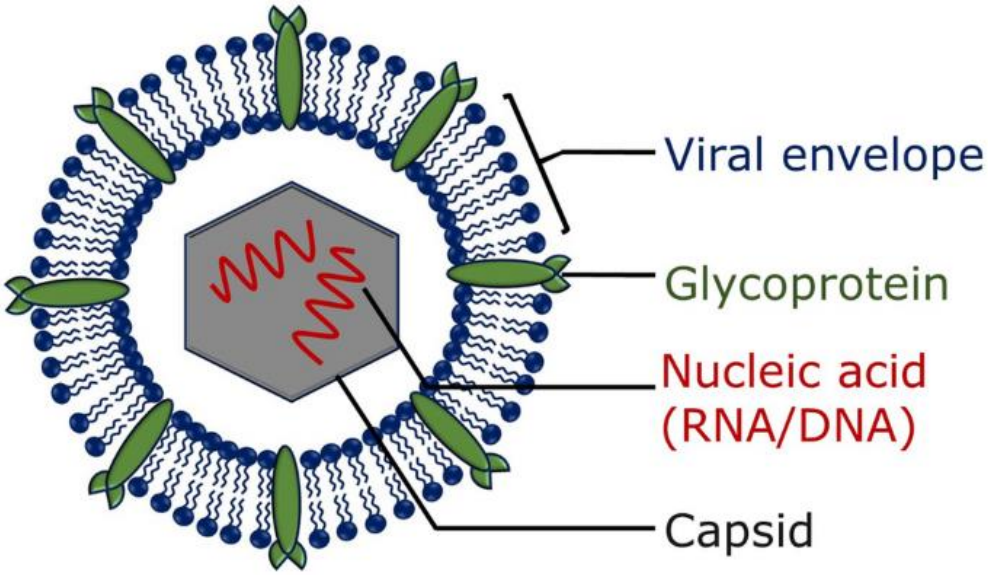
Viral proteases are enzymes encoded by the genetic material (DNA or RNA) of viral pathogens. The role of these enzymes is to catalyze the cleavage of specific peptide bonds in viral

polyprotein precursors or in cellular proteins. In most cases these proteolytic events are essential for the completion of the viral infectious cycle. Selective recognition of these sequence patterns by a complementary substrate binding site of the enzyme ensures a high degree of specific recognition and cleavage



Glycoproteins

- Only in enveloped viruses
- Allows viral entry into host cell
- Targets specific type of host cells
- Aids in viral assembly



Host-shut-off proteins

- As name indicates
- Shut off host activities
- Only viral genes are processed

Anti-host defense proteins

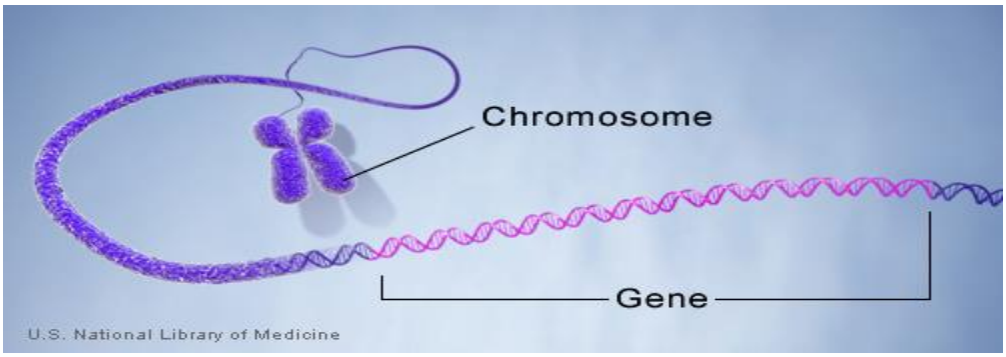
- Number and type of these genes vary
- Prevent host defense mechanism from stopping viral replication

Lesson 46: Terms

Terminologies used in viral genetics

GENE

A unit of DNA that carries information for the bio-synthesis of a specific product



GENOME

The nucleic acid comprising the entire genetic information of an organism.

MUTATION

A permanent inheritable change in a single gene that results in the existence of two or more alleles occurring at the same locus

MUTANT

An organism or cell carrying a mutation. An alternative phenotype to the wild-type; the phenotype produced by a non-wildtype allele

PSEUDOTYPES

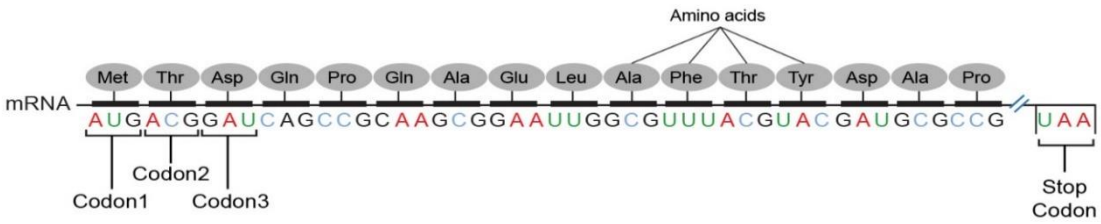
These are the viruses that have acquired the genotype of one parent and phenotype of another.

RECOMBINANT

It is a new combination of genes not found together on the chromosome in either parent which results from crossing over. It may also be produced by genetic engineering.

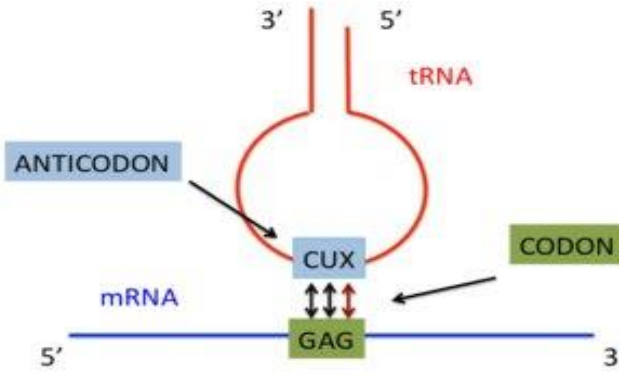
CODON

A sequence of 3 bases that specifies a particular amino acid; a building block of the cell's genetic material, DNA and RNA.



ANTI-CODON

It is a sequence of three bases in tRNA that is complementary to a codon in mRNA. It enables tRNA to sequence amino acids in the order specified by mRNA



RIBOSOMES

These are the complexes of rRNA and protein in cytoplasm that serve as platforms for translation for mRNA into protein.

PLASMID

A circular segment of DNA that encodes a separate set of genes than those present in chromosomes. Independently replicate and encode a product Plasmids are most often found in bacteria, but they are also useful to scientists as vectors.

TRANSCRIPTION

It is the process in the nucleus of cell for synthesizing single-stranded RNA by RNA polymerase using DNA as a template.

TRANSLATION

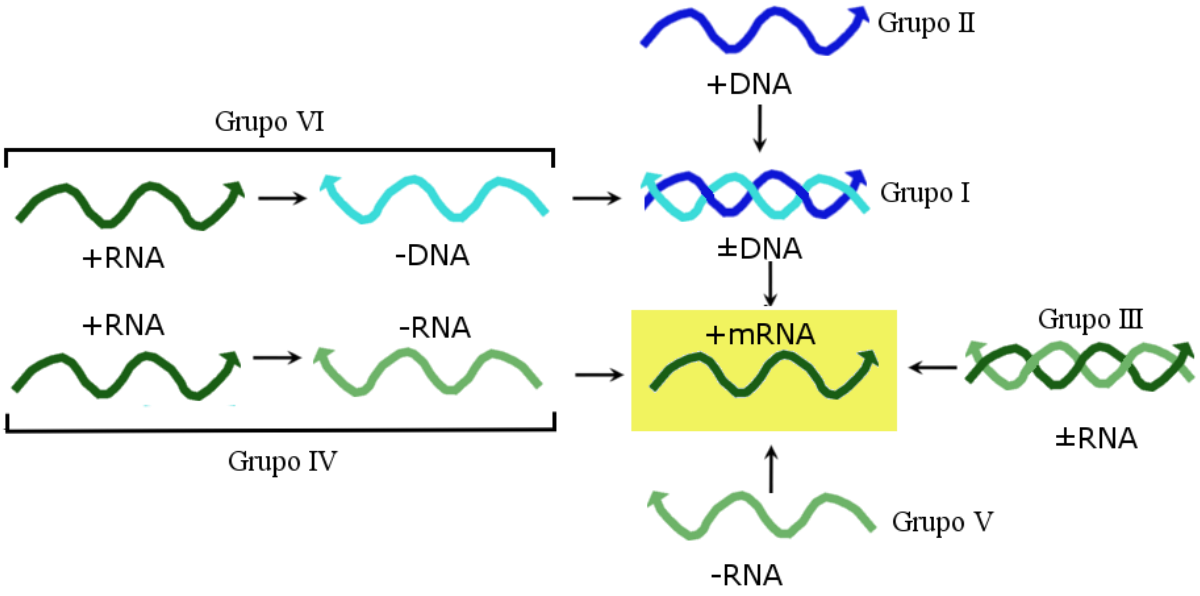
It is the process of translating the codon sequence in mRNA into polypeptides with the help of tRNA and ribosomes

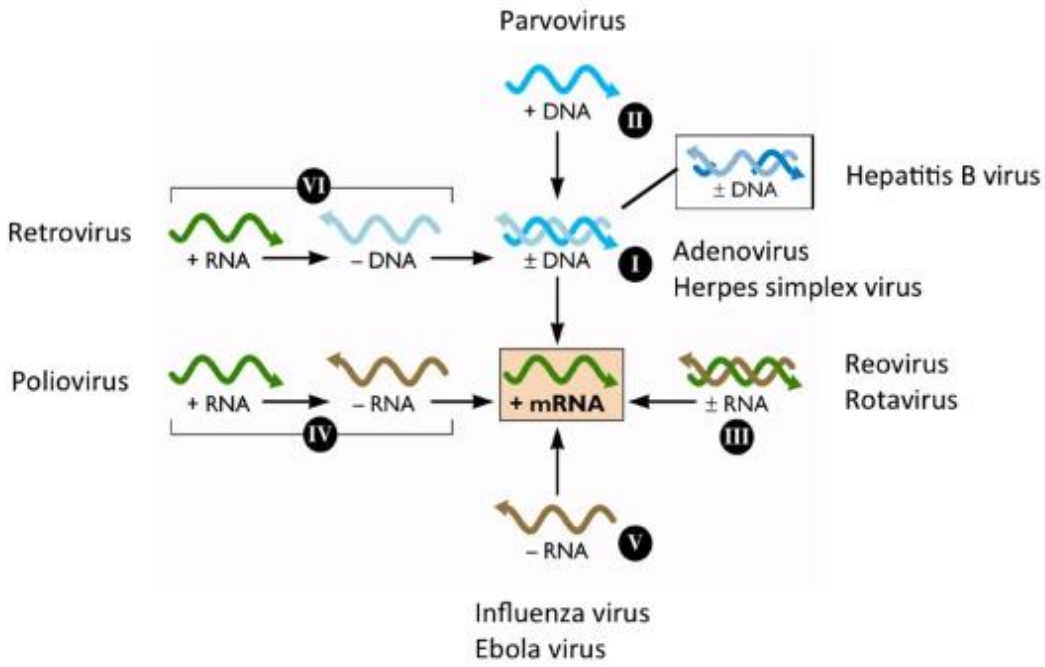
Lesson 47: Molecular Genetics

The nature of the viral genome (RNA or DNA; segmented or non-segmented) plays an important role in the genetics of the virus

Viral genetics, the study of the genetic mechanisms that operate during the life cycle of viruses, utilizes biophysical, biological, and genetic analyses to study the viral genome and its variation. The virus genome consists of only one type of nucleic acid, which could be a single or double stranded DNA or RNA. Single stranded RNA viruses could contain positive-sense (+RNA), which serves directly as mRNA or negative-sense RNA (-RNA) that must use an RNA polymerase to synthesize a complementary positive strand to serve as mRNA.

Class I viruses contain a single molecule of double stranded DNA and are exemplified by adenovirus, simian virus 40 (SV40), herpes viruses, and human papillomaviruses. **Class II viruses** are also called parvoviruses and are made of single stranded DNA that is copied in to double stranded DNA before transcription in the host cell. **Class III viruses** are double stranded RNA viruses that have segmented genomes which means that they contain 10-12 separate double stranded RNA molecules. The negative strands serve as template for mRNA synthesis. **Class IV** viruses, typified by poliovirus, have single plus strand genomic RNA that serves as the mRNA. **Class V viruses** contain a single negative strand RNA which serves as the template for the production of mRNA by specific virus enzymes. **Class VI** viruses are also known as retroviruses and contain double stranded RNA genome. These viruses have an enzyme called reverse transcriptase that can both copy minus strand DNA from genomic RNA catalyze the synthesis of a complementary plus DNA strand. The resulting double stranded DNA is integrated in the host chromosome and is transcribed by the host's own machinery. The resulting transcripts are either used to synthesize proteins or produce new viral particles. These new viruses are released by budding, usually without killing the host cell. Both HIV and HTLV viruses belong to this class of viruses.



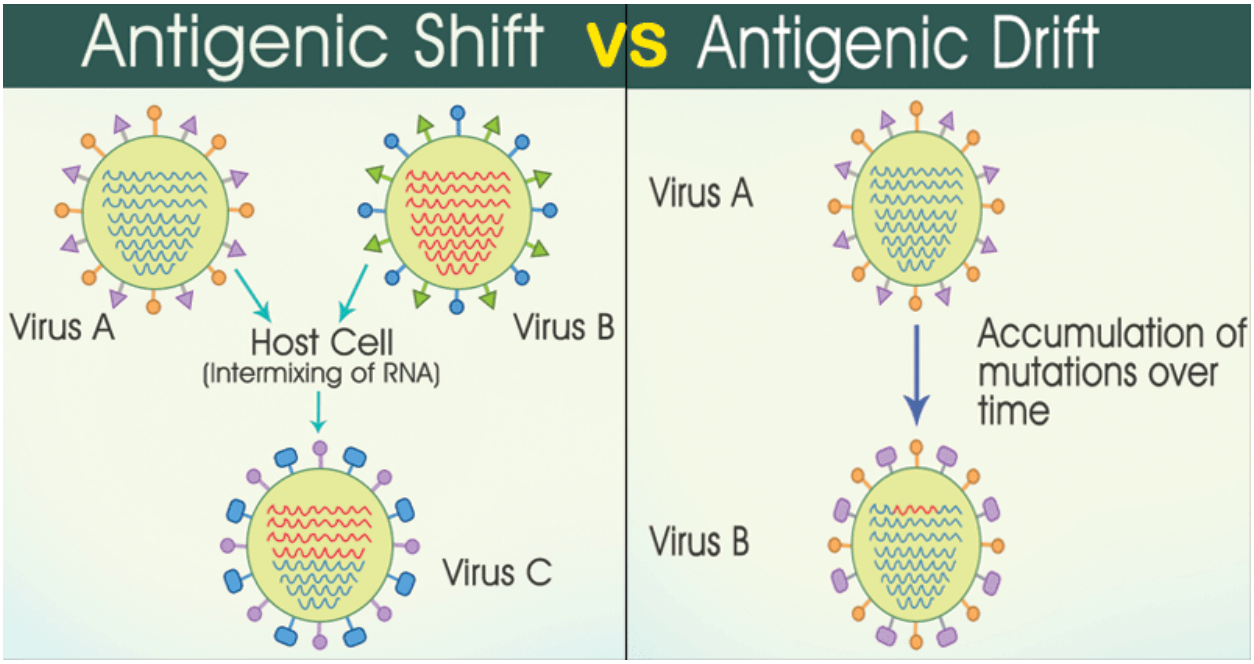


Virus genetics is studied by either investigating genome mutations or exchange of genetic material during the life cycle of the virus. The frequency and types of genetic variations in the virus are influenced by the nature of the viral genome and its structure.

Mutations in the virus genome could either occur spontaneously or be induced by physical and chemical means. Spontaneous mutations that arise naturally as a result of viral replication are either due to a defect in the genome replication machinery or to the incorporation of an

analogous base instead of the normal one. Induced virus mutants are obtained by either using chemical mutants like nitrous oxide that acts directly on bases and modify them or by incorporating already modified bases in the virus genome by adding these bases as substrates during virus replication. Physical agents such as ultra-violet light and x rays can also be used in inducing mutations

Viral genetics



S. N	Characteristics	Antigenic Shift	Antigenic Drift
1.	Definition	Antigenic shift refers to the gene recombination occurring when influenza viruses re-assort.	Mutations causing minute changes in antigens on the surface of the virus is termed antigenic drift.
2.	Result in	Forms a new sub-type (Subtype A + Subtype B –> New Subtype).	Forms a new strain of the virus.
3.	Genome changes	Large change in nucleotides of RNA.	Small mutation of RNA.
4.	Virus involved	One or more viruses are involved.	Only one virus is involved.

Lesson 48: Techniques in Virology

Genetic Techniques in Virology

Types

- In-vitro (Physical analysis)
- In-vivo (viral genotype and phenotype)

The terms “in vitro” and “in vivo” are common in the virology literature. In vitro means “in glass” and refers to experiments carried out in an artificial environment, such as a glass test tube.

In-vitro (Physical analysis)

The use of cell culture to detect viruses is being increasingly supplanted by molecular methods such as the polymerase chain reaction, DNA microarrays, and high-throughput sequencing, especially for discovery of new viruses associated with human diseases. These methods can be used to identify viruses that cannot be propagated in cell culture. Following are different techniques to use for physical analysis.

- Isolation of nucleic acids from virus preparations
- Direct analysis of DNA using restriction endonucleases enzyme
- Use of vectors in DNA sequencing
- Electron microscopy
- RNA sequence analysis using Reverse transcriptase
- Molecular cloning
- Gel electrophoresis
- Hybridization of complementary nucleotide sequences

Polymerase chain reaction. In this technique, specific oligonucleotides are used to amplify viral DNA sequences from infected cells or clinical specimens. Amplification is done in cycles, using a thermostable DNA polymerase. Each cycle consists of primer annealing, extension, and thermal denaturation carried out by automated cycler machines. The result is exponential amplification of the target sequence that is located between the two DNA primers. Clinical laboratories use

PCR assays to detect evidence for infection by a single type of virus (singleplex PCR), while screening for the presence of up to 30 different viruses can be done using multiplex PCR.

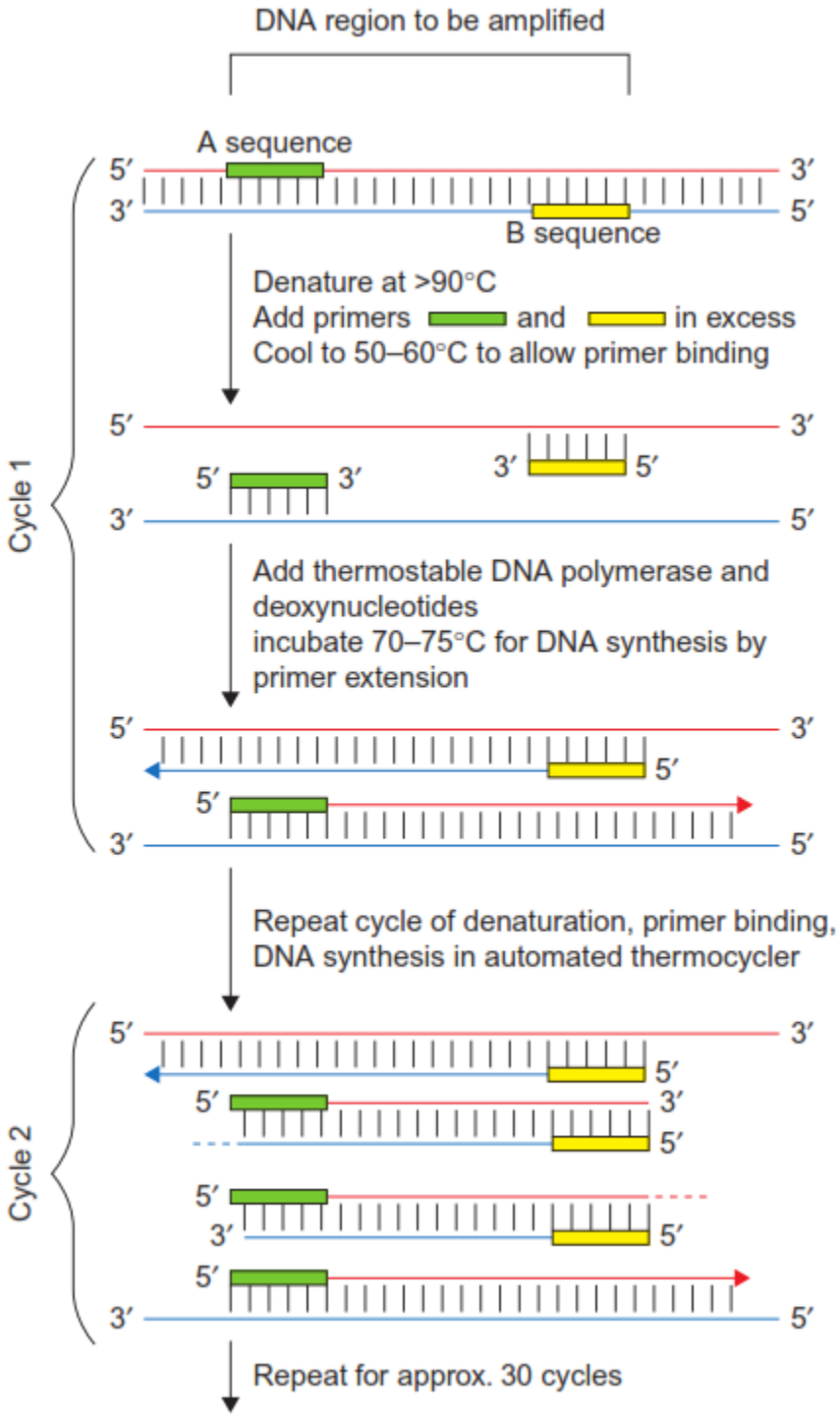


Figure: Amplification of part of a DNA sequence by the polymerase chain reaction. Oligonucleotide primers must first be made according to the sequences of either end of the portion of DNA to be amplified. After the DNA has been denatured by heating, the primers can

hybridize to the complementary sequences on the opposite strand. In the presence of heat resistant. DNA polymerase and deoxynucleotide triphosphates, two new copies of the desired region are produced. The cycles of melting, annealing, and extension are repeated rapidly; each time, the amount of target DNA sequence doubles. After the first few cycles, virtually all the templates consist of just the short region chosen for amplification.

The development of **reverse transcriptase polymerase chain reaction (RT-PCR)** methods to detect RNA sequences was a major advance in cell biology and viral diagnostics. For RT-PCR, the RNA is first transcribed into cDNA using a DNA polymerase capable of using RNA as a template, such as retrovirus reverse transcriptase. Newer reverse transcriptase enzymes have been developed that permit synthesis of the cDNA strand at higher temperatures

In contrast to conventional PCR, **real-time PCR** can be used to quantitate the amount of DNA or RNA in a sample. In this procedure, also called quantitative PCR, the amplified DNA is detected as the reaction progresses, for example, by the use of fluorescent dyes that intercalate nonspecifically into DNA. The number of cycles needed to detect fluorescence above background can then be compared between standard and experimental samples.

High-throughput sequencing. Sequencing of thousands to millions of DNA molecules at the same time is a feature of this method, also known as next-generation sequencing to distinguish it from older methods. The newer approaches have not only made sequencing of DNA cheaper and faster but also helped create innovative experimental approaches to study genome organization, function, and evolution. The use of high-throughput sequencing has led to the discovery of new viruses and has given birth to the field of metagenomics, the analysis of sequences from clinical or environmental samples.

Electron microscopy. With few exceptions, virus particles are too small to be observed directly by light microscopy. However, they can be seen readily in the electron microscope. If a sample contains only one type of virus, the particle count can be determined.

Nucleic Acid Probes

Nucleic acid probes are segments of DNA or RNA that have been labeled with enzymes, antigenic substrates, chemiluminescent moieties, or radioisotopes. They can bind with high specificity to complementary sequences of nucleic acid. Probes can be directed to either DNA or RNA targets and can be from 20 to thousands of bases long. The presence and the quantity of hybrids after hybridization is determined by the detection of the label.

Despite having been around for many years, hybridization assays are still not in common use in the clinical virology laboratory. The main reason is that its sensitivity is not usually higher than far simpler conventional techniques.

PHENOTYPIC ANALYSIS

- Standard technique in virology
- Cell-free extracts to translate mRNAs
- Transfection tests
- In-vitro protein formation
- In-vivo protein replication

Lesson 48: Techniques in Virology

Virus latency (or viral latency) is the ability of a pathogenic virus to lie dormant within a cell, denoted as the lysogenic part of the viral life cycle. A latent viral infection is a type of persistent viral infection which is distinguished from a chronic viral infection. A **latent infection** is a phase in certain virus's life cycles in which after initial infection, virus production ceases. However, the virus genome is not fully eradicated. The result of this is that the virus can reactivate and begin producing large amounts of viral progeny without the host being infected by new outside virus, denoted as the lytic part of the viral life cycle stays within the host indefinitely.

Virus latency is not to be confused with clinical latency during the incubation period when a virus is not dormant.

Mechanisms

Episomal latency

Episomal latency refers to the use of genetic episomes during latency. In this type, viral genes are floating in the cytoplasm or nucleus as distinct objects, both as linear or lariat structures. Episomal latency is more vulnerable to marauding ribozymes or host foreign gene degradation than provirus latency. Examples include the family Herpesviridae, exemplified by the Herpes simplex virus which undergoes episomal latency in neuron cells and leaves linear genetic material floating in the cytoplasm.

Proviral latency

Proviral latency begin with the virus genome integrates into the host genome, effectively become a provirus. This requires that the viral gene get into the nucleus and insert itself into

the host genome, the family of which exemplifies this behavior being the Retroviruses. For example, the Retrovirus, HIV.

Advantages include automatic host cell division results in replication of the virus's gene, and the fact that it is near impossible to remove an integrated provirus from an infected cell without killing the cell.

Disadvantages include the need to enter the nucleus (and the need for packaging proteins that will allow for that) and increased difficulty in maintaining the latency.

Maintaining latency

Both proviral and episomal latency may require maintenance for continued infection and fidelity of viral genes. Latency is generally maintained by viral proteins in order to keep the viral genes from being digested by marauding cellular ribozymes or being found out by the immune system, as well as keeping the gene in good condition. Certain proteins may also inhibit apoptosis or induce cell growth and division to allow more copies of the infected cell to be produced.

Latency can be maintained without the need of proteins and an interesting example of this are the endogenous retroviruses. Generally, these types of viruses have been highly evolved with the hosts immune system to be able to survive without the need of any immune system modulating maintenance proteins

Lesson 50: Examples of latent viruses

In latent infections, overt disease is not produced, but the virus is not eradicated. This equilibrium between host and parasite is achieved in various ways by different parasites and hosts.

Chronic Persistent Infections

Enveloped viruses such as paramyxoviruses, some herpesviruses e.g. EBV, retroviruses and arenaviruses appear particularly suited to initiate persistent infections. Infection appears to persist because the virus does not disrupt the essential housekeeping functions of the cells. (DNA, RNA and protein synthesis). Some persistently infected cells, such as in measles [subacute sclerosing panencephalitis (SSPE)] may be assisted by the capacity of humoral antibodies (Abs) to cap viral antigens (Ags) on the cell surface. This promotes the shredding of viral Ags from the cell surface, leaving the cell surface free of viral glycoproteins and thus the infected cell is protected from cytotoxic T cells (CTLs) and K cells of immune system.

2. Latent Occult Viral Infections

Some DNA and RNA viruses may become undetectable following a primary infection only to reappear and produce acute disease. This latency can be accomplished in different ways.

a. HSV - primary infection usually occurs between 6 to 18 months of age following which the virus persists and cannot be found except during recurrent acute episodes. The form in which the latent occult virus persists is uncertain. Virus cannot be isolated from tissue homogenates, but by cocultivation cells of sensory ganglia with susceptible cells. Virus has been detected in the trigeminal, thoracic, lumbar and sacral dorsal root ganglia. These data suggests that the DNA exist in a linear, unintegrated form, perhaps as episomes.

b. Adenovirus - adenovirus infections in humans are usually self-limiting but the virus frequently establishes a latent, persistent infection of the tonsils and the adenoids. Though these tissues fail to yield infectious virus when homogenized and tested in sensitive cell cultures, cultured fragments of about 85% of these "normal" tonsils and adenoids, after a variable time, show characteristic adenovirus-induced CPE and yield infectious virus. Failure to recover infectious virus initially may be due to the paucity of virions, to their association with either Ab or receptor material, or to the absence of mature virions. The latent infection is probably not the result of lysogeny, since DNA in peripheral lymphocytes appears to be in a linear episomal form.

c. Subacute sclerosing panencephalitis (*SSPE*) - latency occurs as a result of incomplete viral production. Immature viral measles virus nucleocapsids are produced.

Latent viral infections affect the incidence and pathogenesis of acute viral disease in several ways. A reactivated virus may spread and initiate an epidemic among susceptible contacts. Viral latency can also be seen in the development of several chronic diseases dependent on the immunological response eg. SSPE and measles. Some latent states induce tumorigenesis.

Lesson 51: Mutants

Mutant

A mutant is an organism or a new genetic character arising or resulting from an instance of mutation, which is generally an alteration of the DNA/RNA sequence of the genome or chromosome of an organism.

DIFFERENCE IN RELEVANT TERMS

"Mutant," "strain," "type," "variant," and even "isolate" are all terms used rather loosely by virologists to differentiate particular viruses from each other and from the original "parental," "wild-type," or "street" isolates of that virus. More accurately, these terms are generally applied as follows:

■ **Strain:** Different lines or isolates of the same virus (e.g., from different geographical locations or patients).

■ **Type:** Different serotypes of the same virus (e.g., various antibody neutralization phenotypes).

■ **Variant:** A virus whose phenotype differs from the original wild-type strain but the genetic basis for the difference is not known, for example, a new clinical isolate from a patient.

- Mutant: virus showing genetic alteration

ORIGIN OF MUTATION

- **Spontaneous mutation:**
    - Very high to low rate of mutation
    - Due to difference in type of genome
    - Error rate higher in RNA dependent RNA polymerase
- **Induced Mutation**
    - In-vivo
    - or in-vitro mutagens

MAIN REASON

- Replication Error
- Chemicals
- Radiations

TYPES OF MUTANT VIRUSES

- Biochemical markers: drug-resistance mutations
- Deletions: similar to non-sense mutations
- Host range: refers to whole host animal

- Non-sense mutation: alteration of coding sequence
- Plaque morphology
- Temperature sensitive
- Cold sensitive
- Reverent

Lesson 52: Spontaneous Mutations

Spontaneous Mutation

Both DNA and RNA viruses undergo spontaneous mutation; however, the spontaneous mutation rate in RNA viruses is usually much higher than in DNA viruses. In general, the mutation rate at a specific site in different DNA viruses ranges from 10^{-8} to 10^{-11} per replication, whereas in RNA viruses it is at least hundred-fold higher, between 10^{-3} and 10 per replication. The difference in mutation rate observed between RNA and DNA viruses is thought to result primarily from differences in the replication enzymes. Specifically, the DNA dependent DNA polymerases used by DNA viruses Contain a proofreading function, Whereas the reverse Transcriptases used by retroviruses and RNA-dependent RNA polymerases used by RNA viruses lack a proofreading function. The difference in spontaneous mutation rate has profound consequences for both the biology of the viruses and for laboratory genetic analysis of viruses. Specifically, RNA viruses exist in nature as ***quasispecies*** that is, populations of virus variants in relative equilibrium with the environment but capable of swift adaptation owing to a high spontaneous mutation rate.

Conversely, DNA viruses are genetically more stable but less adaptable. In the laboratory, the high mutation rate in RNA viruses presents difficulties in routine genetic analysis because mutants easily revert to wild-type virus that can outgrow the mutant virus.

Reasons

Spontaneous mutations arise from a variety of sources, including errors in DNA replication, spontaneous lesions, and transposable genetic elements.

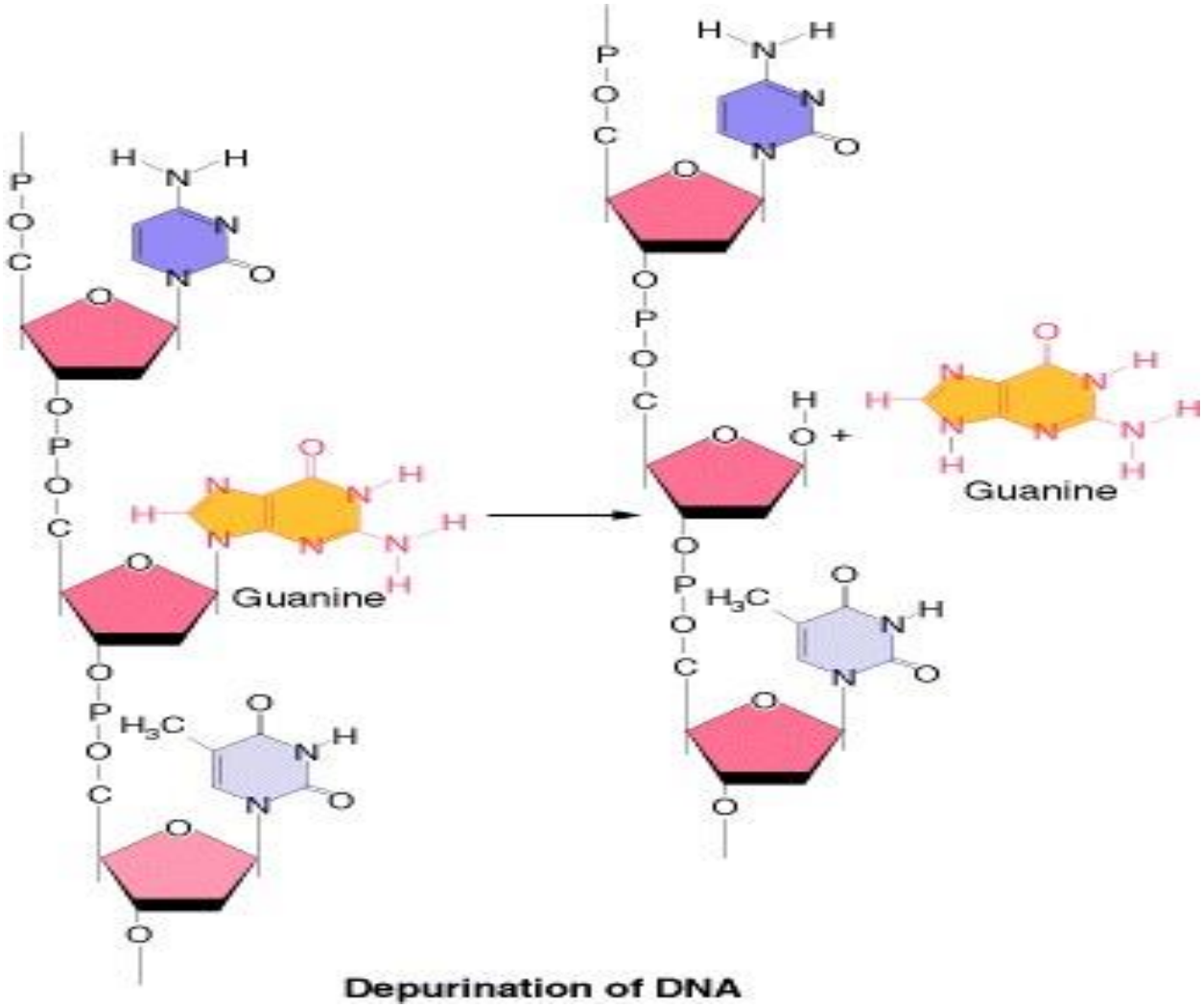
An error in DNA replication can occur when an illegitimate nucleotide pair (say, A–C) forms in DNA synthesis, leading to a base substitution.

Tautomers

Each of the bases in DNA can appear in one of several forms, called **tautomers**, which are isomers that differ in the positions of their atoms and in the bonds between the atoms. The forms are in equilibrium. The keto form of each base is normally present in DNA, whereas the imino and enol forms of the bases are rare. The ability of the wrong tautomer of one of the standard bases to mis pair and cause a mutation in the course of DNA replication

In addition to replication errors, spontaneous lesions, naturally occurring damage to the DNA, can generate mutations. Two of the most frequent spontaneous lesions result from depurination and deamination.

Depurination, the more common of the two, consists of the interruption of the glycosidic bond between the base and deoxyribose and the subsequent loss of a guanine or an adenine residue from the DNA. A mammalian cell spontaneously loses about 10,000 purines from its DNA in a 20-hour cell-generation period at 37°C.



Depurination of DNA

Figure: The loss of a purine residue (guanine) from a single strand of DNA. The sugar-phosphate backbone is left intact

The **deamination** of cytosine yields uracil. Unrepaired uracil residues will pair with adenine in replication, resulting in the conversion of a G–C pair into an A–T pair (a GC → AT transition).

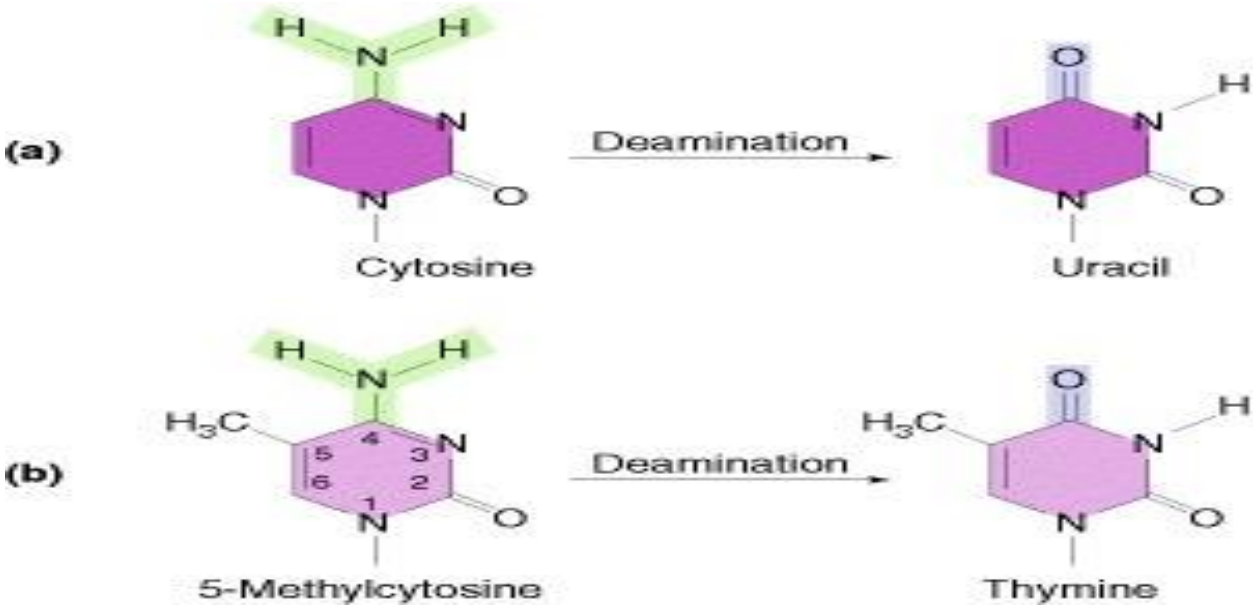


Figure: Deamination of (a) cytosine

Slipped strand mis-pairing: Denaturation of new strand

Replication slipping: error during replication process

Lesson 53: Induced Mutations

INDUCED MUTATION Under most circumstances, the incidence of spontaneous mutations is low enough so that induction of mutation is a practical prerequisite for isolation of virus mutants. It is usually desirable to induce limited, normally single-base changes, and for this purpose, chemical mutagens are most appropriate.

Commonly used chemical mutagens are of two types: *in vitro* mutagens and *in vivo* mutagens.

***In vitro* mutagens** work by chemically altering nucleic acid and can be applied by treating virions in the absence of replication. Examples of *in vitro* mutagens include hydroxylamine, nitrous acid, and alkylating agents, which through chemical modification of specific bases cause mispairing leading to missense mutations.

***In vivo* chemical mutagens** comprise compounds such as nucleoside analogs that must be incorporated during viral replication and thus must be applied to an infected cell. One of the most effective mutagens is the alkylating agent nitrosoguanidine, which although is capable of alkylating nucleic acid *in vitro* is most effective when used *in vivo,* where it works by alkylating guanine residues at the replication fork, ultimately causing mispairing.

Chemical reason

- Hydroxylamine
- Base analogs
- Alkylating agents
- DNA intercalating substances
- DNA cross linkers
- Oxidative damage
- Nitrous acid

Radiation reason

- Ultraviolet rays
- Ionizing radiation

Lesson 54: Types of Mutants

Types of Mutant Viruses

- Biochemical Markers
- Deletions
- Host Range
- Non-Sense
- Plaque morphology
- Temperature sensitive
- Cold Sensitive

- Reverent

Biochemical markers: These include drug resistance mutations, mutations that result in altered virulence, polymorphisms resulting in altered electrophoretic mobility of proteins or nucleic acids, and altered sensitivity to inactivating agents.

■ **Deletions:** Similar in some ways to nonsense mutants but may include one or more virus genes and involve noncoding control regions of the genome (promoters, etc.). Spontaneous deletion mutants often accumulate in virus populations as defective-interfering (DI) particles. These noninfectious but not necessarily genetically inert genomes are thought to be important in establishing the course and pathogenesis of certain virus infections. Genetic deletions can only revert to wild-type by recombination, which usually occurs at comparatively low frequencies. Deletion mutants are very useful for assigning structure function relationships to virus genomes, as they are easily mapped by physical analysis.

■ **Host range:** This term can refer either to whole animal hosts or to permissive cell types in vitro. Conditional mutants of this class have been isolated using **amber-suppressor cells** (mostly for phages but also for animal viruses using in vitro systems).

■ **Nonsense:** These result from alteration of coding sequence of a protein to one of three translation stop codons (UAG, amber; UAA, ocher; UGA, opal). Translation is terminated, resulting in the production of an amino-terminal fragment of the protein. The phenotype of these mutations can be suppressed by propagation of virus in a cell (bacterial or, more recently, animal) with **altered suppressor tRNAs.** Nonsense mutations are rarely “leaky” (i.e., the normal function of the protein is completely obliterated) and can only revert to wild-type at the original site, so they usually have a low reversion frequency.

■ **Plaque morphology:** Mutants may be either large-plaque mutants which replicate more rapidly than the wild-type, or small-plaque mutants, which are the opposite. Plaque size is often related to a temperature-sensitive (t.s.) phenotype. These mutants are often useful as unselected markers in multifactorial crosses.

■ **Temperature-sensitive (t.s.):** This type of mutation is very useful as it allows the isolation of conditional-lethal mutations, a powerful means of examining virus genes that are essential for replication and whose function cannot otherwise be interrupted. Temperature-sensitive mutations usually result from mis-sense mutations in proteins (i.e., amino acid substitutions), resulting in proteins of full size with subtly altered conformation that can function at (permissive) low temperatures but not at (nonpermissive) higher ones. Generally, the mutant proteins are immunologically unaltered, which is frequently useful. These mutations are usually “leaky”—that is, some of the normal activity is retained even at nonpermissive temperatures. Protein function is often impaired, even at permissive temperatures, therefore a high frequency

of reversion is often a problem with this type of mutation because the wild-type virus replicates faster than the mutant.

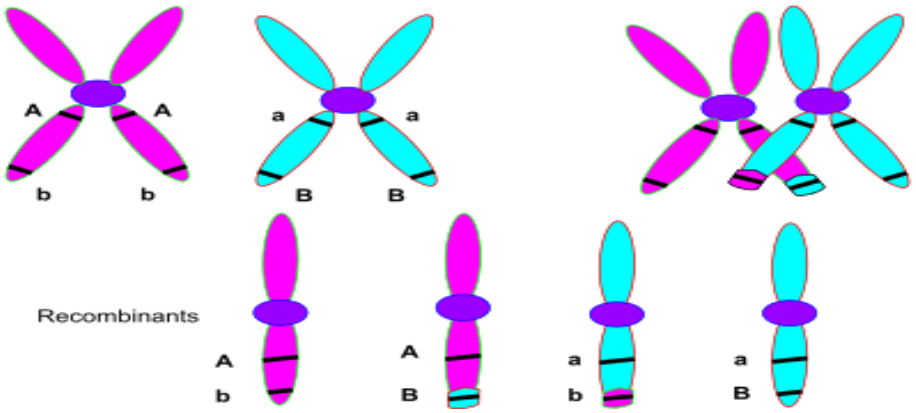
■ **Cold-sensitive (c.s.):** These mutants are the opposite of temperature sensitive mutants and are very useful in bacteriophages and plant viruses whose host cells can be propagated at low temperatures but are less useful for animal viruses because their host cells generally will not grow at significantly lower temperatures than normal.

■ **Revertant:** Reverse mutation is a valid type of mutation in its own right. Most of the above classes can undergo reverse mutation, which may be either simple "back mutations" (i.e., correction of the original mutation) or second-site "compensatory mutations," which may be physically distant from the original mutation and not even necessarily in the same gene as the original mutation.

Lesson 55: Recombination

Recombination is the physical interaction of virus genomes during superinfection that results in gene combinations not present in either parent. Viral recombination occurs when viruses of two different parent strains coinfect the same host cell and interact during replication to generate virus progeny that have some genes from both parents. Recombination generally occurs between members of the same virus type (e.g., between two influenza viruses or between two herpes simplex viruses).

■ **Intramolecular recombination via strand breakage and relegation:** This process occurs in all DNA viruses and in RNA viruses that replicate via a DNA intermediate. It is believed to be caused by cellular enzymes, as no virus mutants with specific recombination defects have been isolated.



EXAMPLES

- Picornavirus
- Togavirus
- Corona virus
- Influenza A virus

RNA VIRUSES

- Randomly with segmented genome
- e.g. orthomyxovirus, reovirus, bunyavirus
- Genome segments show recombination

ADVANTAGE

- Genetic variability in nature
- Rapid adaptation of viruses
- Virulence changes
- Epidemiology of viruses (e.g. Influenza A)

Lesson 56: Viral Suppression

Suppression

It is the inhibition of a mutant phenotype by a second suppressor mutation, which may be either in the virus genome or in that of the host cell. This mechanism of suppression is not the same as the suppression of chain-terminating amber mutations by host-encoded suppressor tRNAs, which could be called **"informational suppression."**

Pseudo revertant:

Genetic suppression results in an apparently wild-type phenotype from a virus which is still genetically mutant—a pseudo revertant. This phenomenon has been best studied in prokaryotic systems, but examples have been discovered in animal viruses—for example, reoviruses, vaccinia, and influenza—where suppression has been observed in an attenuated vaccine, leading to an apparently virulent virus. Suppression may also be important biologically in that it

allows viruses to overcome the deleterious effects of mutations and therefore be positively selected.

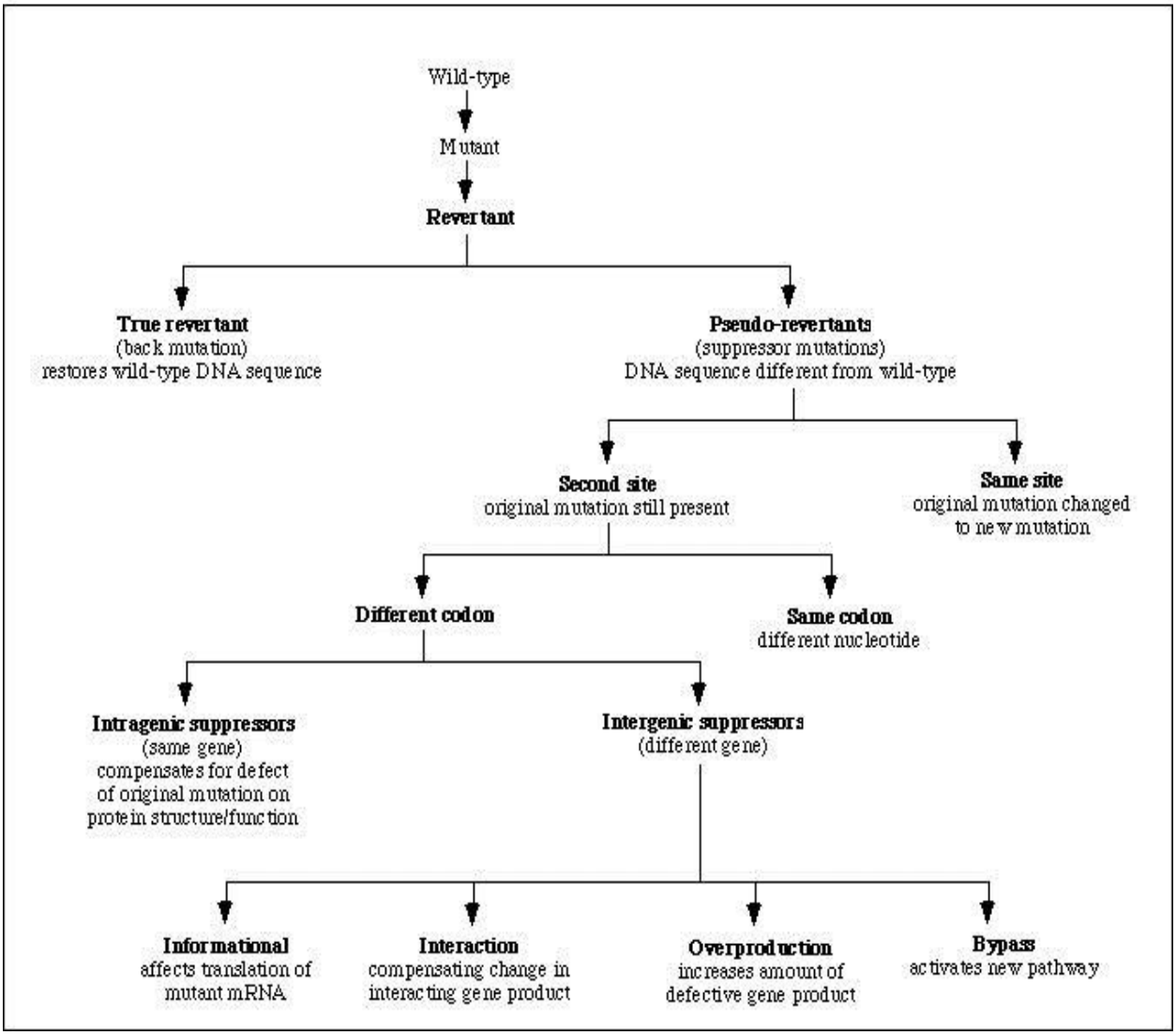
Mutant viruses can appear to revert to their original phenotype by three pathways:

1. Back mutation of the original mutation to give a wild-type genotype/ phenotype (**true reversion**).

2. A second, compensatory mutation that may occur in the same gene as the original mutation, thus correcting it—for example, a second frameshift mutation restoring the original reading frame (**intragenic suppression**).

3. A suppressor mutation in a different virus gene or a host gene (**extragenic suppression**).

SUPPRESSOR ANALYSIS



Lesson 57: Genetic interactions

Genetic interactions between viruses often occur naturally, as host organisms are frequently infected with more than one virus. These situations are generally too complicated to be analyzed successfully. Experimentally, genetic interactions can be analyzed by mixed infection (superinfection) of cells in culture. Two types of information can be obtained from such experiments:

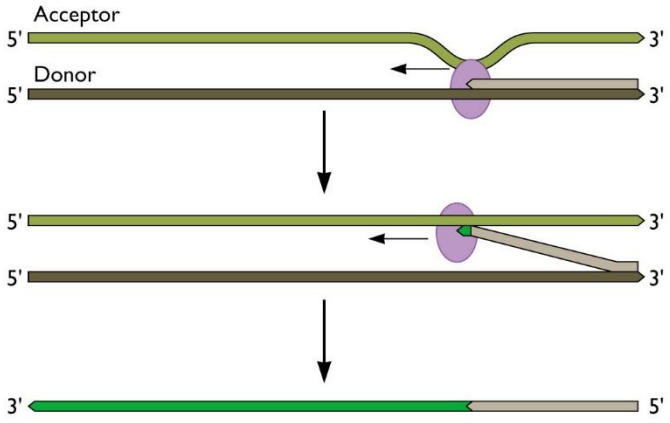
■ The assignment of mutants to functional groups known as **complementation** groups.
■ The ordering of mutants into a linear genetic map by analysis of **recombination** frequencies.

Complementation:
Complementation results from the interaction of virus gene products during superinfection which results in production of one or both of the parental viruses being increased while both viruses remain unchanged genetically. In this situation, one of the viruses in a mixed infection provides a functional gene product for another virus which is defective for that function. If both mutants are defective in the same function, enhancement of replication does not occur and the two mutants are said to be in the same complementation group. The importance of this test is that it allows functional analysis of unknown mutations if the biochemical basis of any one of the mutations in a particular complementation group is known.
Allelic (intragenic) complementation where different mutants have defects in same protein

Non-allelic (intergenic) results from mutants with defects in different genes

Recombination:

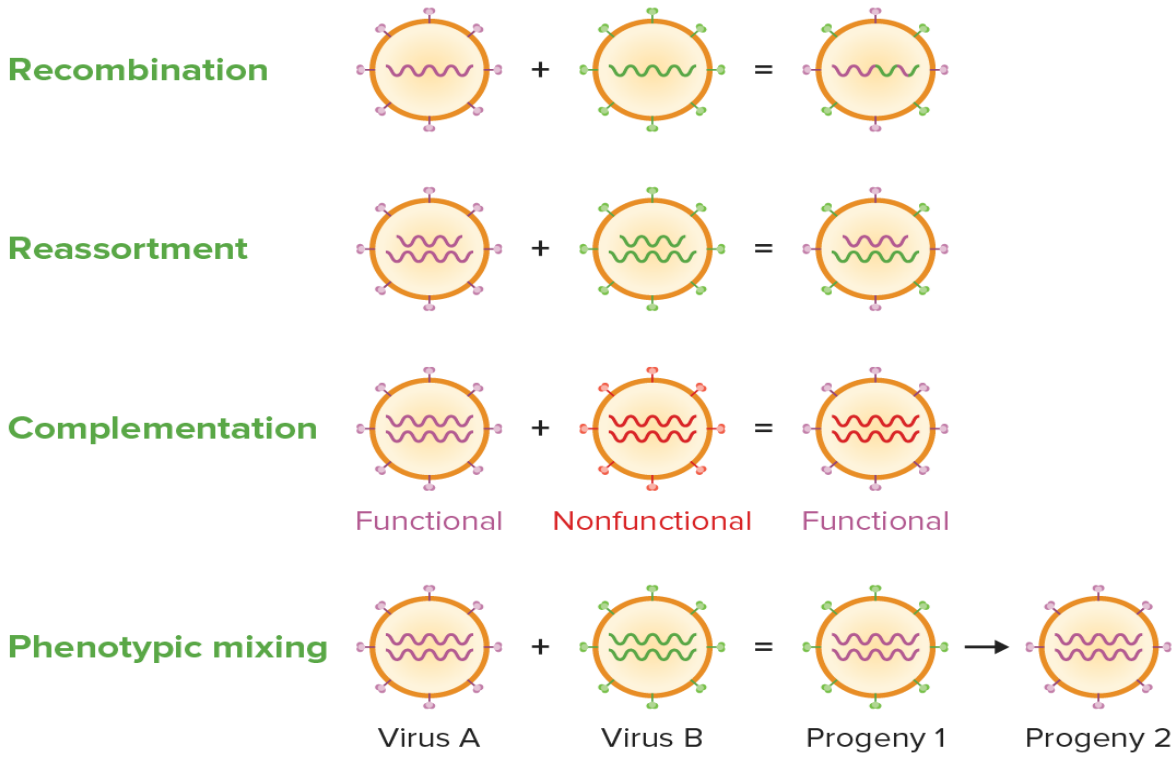
Recombination is the physical interaction of virus genomes during superinfection that results in gene combinations not present in either parent. There are three mechanisms by which this can occur, depending on the organization of the virus genome:



■ **Intramolecular recombination via strand breakage and relegation:** This process occurs in all DNA viruses and in RNA viruses that replicate via a DNA intermediate. It is believed to be caused by cellular enzymes, as no virus mutants with specific recombination defects have been isolated.

■ **Intramolecular recombination by "copy-choice":** This process occurs in RNA viruses, probably by a mechanism in which the virus polymerase switches template strands during genome synthesis. There are cellular enzymes that could be involved (e.g., splicing enzymes), but this is unlikely and the process is thought to occur essentially as a random event. Defective-interfering (DI) particles in RNA virus infections are frequently generated in this way.

■ **Reassortment:** In viruses with segmented genomes, the genome segments can be randomly shuffled during superinfection. Progeny viruses receive (at least) one of each of the genome segments, but probably not from a single parent.



Viruses have evolved diverse strategies for increasing genetic diversity including recombination (the exchange of genes between 2 chromosomes), reassortment (exchange of segments of chromosomes), complementation (the exchange of genetic material from a functional virus with a nonfunctional virus to convert the latter into functional) and phenotypic mixing (mixing of 2 viral genomes a simultaneously infected host cell giving rise to progeny with blended genomes).

Lesson 58: Non genetic interactions

A number of nongenetic interactions between viruses occur which can affect the outcome and interpretation of the results of genetic crosses.

Eukaryotic cells

Eukaryotic cells have a **diploid** genome with two copies of each chromosome, each bearing its own allele of the same gene. The two chromosomes may differ in allelic markers at many loci. Among viruses, only **retroviruses** are truly diploid, with two complete copies of the entire genome, but some DNA viruses, such as **herpesviruses**, have **repeated sequences** and are therefore partially heterozygous.

Heterozygosis

In a few (mostly enveloped) viruses, **aberrant packaging** of multiple genomes may occasionally result in multiploid particles that are heterozygous (e.g., up to 10% of Newcastle disease virus particles). This process is known as heterozygosis and can contribute to the genetic complexity of virus populations.

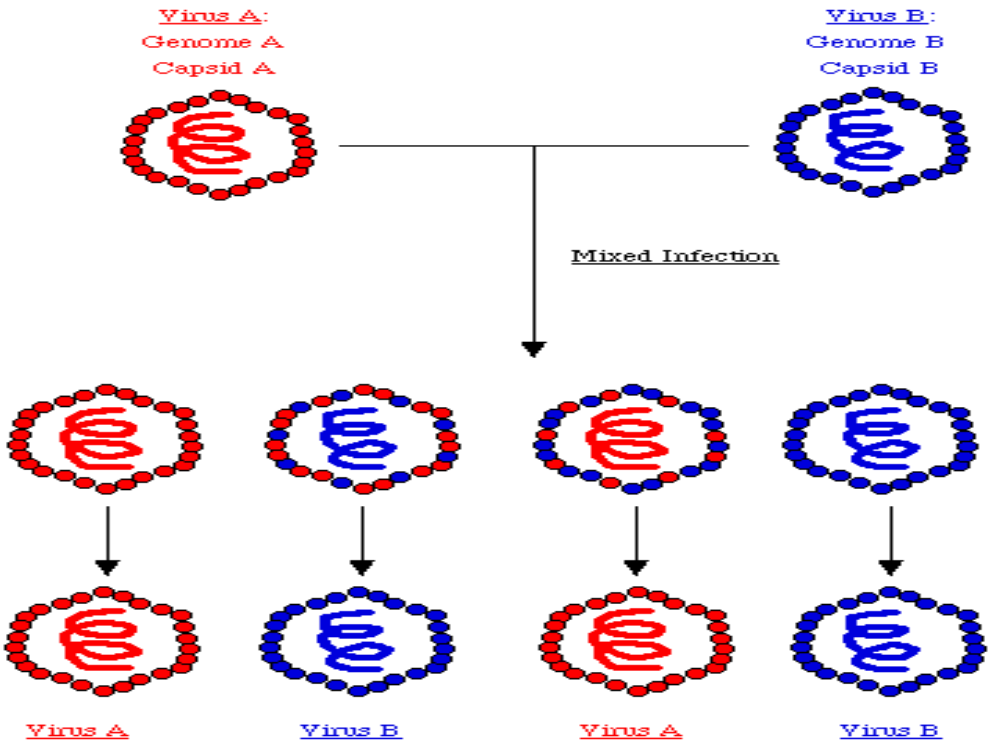
Superinfection

Another nongenetic interaction between viruses that is commonly seen is interference. This process results from the resistance to superinfection by a virus observed in cells already infected by another virus. **Homologous interference** (i.e., against the same virus) often results from the presence of DI particles which compete for essential cell components and block replication. However, interference can also result from other types of mutation (e.g., dominant temperature-sensitive mutations) or by sequestration of virus receptors due to the production of virus-attachment proteins by viruses already present within the cell (e.g., in the case of avian retroviruses).

Phenotypic mixing

Phenotypic mixing can vary from extreme cases, where the genome of one virus is completely enclosed within the capsid or envelope of another (**pseudotyping)**, to more subtle cases where the capsid/envelope of the progeny contains a mixture of proteins from both viruses. This mixing gives the progeny virus the phenotypic properties (e.g., cell tropism) dependent on the proteins incorporated into the particle, without any genetic change. Subsequent generations of viruses inherit and display the original parental phenotypes. This process can occur easily in viruses with naked capsids (nonenveloped) which are closely related (e.g., different strains of enteroviruses) or in enveloped viruses, which need not be related to one another. In this latter

case, the phenomenon is due to the nonspecific incorporation of different virus glycoproteins into the envelope, resulting in a mixed phenotype. Rescue of replication-defective transforming retroviruses by a helper virus is a form of pseudotyping. Phenotypic mixing has proved to be a very useful tool **to examine biological properties** of viruses. Vesicular stomatitis virus (VSV) readily forms pseudotypes containing retrovirus envelope glycoproteins, giving a plaque-forming virus with the properties of VSV but with the cell tropism of the retrovirus. This trick has been used to study the cell tropism of HIV and other retroviruses.



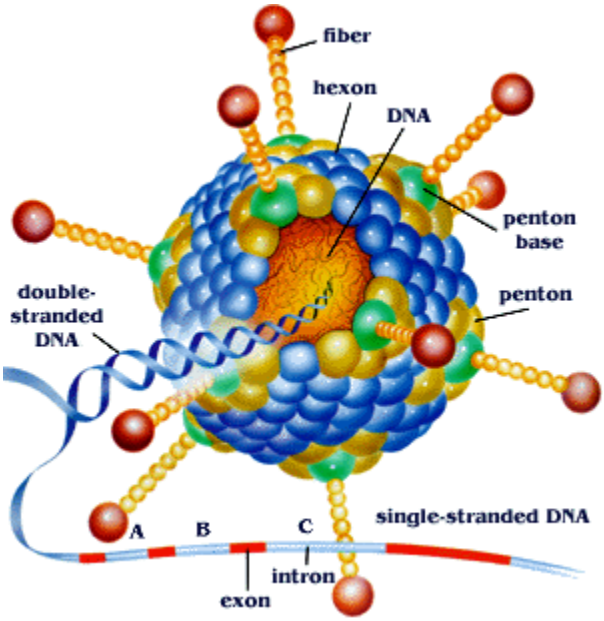
Lesson 59: Large DNA Genome

A number of virus groups have double-stranded DNA genomes of considerable size and complexity. In many respects, these viruses are genetically similar to the host cells that they infect.

Adenoviruses

The genomes of adenoviruses consist of linear, double-stranded DNA of 30-38 kbp, the precise size of which varies between different adenoviruses. These genomes contain 30-40 genes. The

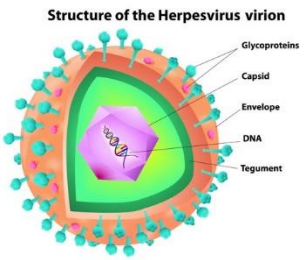
terminal sequence of each DNA strand is an inverted repeat of 100-140 bp, and the denatured single strands can form "**panhandle**" structures. These structures are important in DNA replication, as is a 55-kDa protein known as the terminal protein that is covalently attached to the 5` ends of each strand. During genome replication, this protein acts as a primer, initiating the synthesis of new DNA strands.



Although adenovirus genomes are considerably smaller than those of herpesviruses, the expression of the genetic information is rather more complex. Clusters of genes are expressed from a limited number of shared promoters. Multiply spliced mRNAs and alternative splicing patterns are used to express a variety of polypeptides from each promoter.

Herpesviridae

The **Herpesviridae** is a large family containing more than 100 different members, at least one for most animal species that have been examined to date. There are eight human herpesviruses, all of which share a common overall genome structure but which differ in the fine details of genome organization and at the level of nucleotide sequence. The family is divided into three subfamilies, based on their nucleotide sequence and biological properties.



SUBFAMILIES

Alphaherpesvirinae

- Latent infections in sensory ganglion, genome size 120-180kbp
- Simplexvirus, Varicellovirus, Human herpes virus-I and II,
- Human herpes virus-III

Betaherpesvirinae

- Restricted host range, genome size 140-253 kbp
- Cytomegalovirus, Muromegalovirus, Roseolovirus, Human herpes virus-V, Mouse cytomegalovirus-I, Human herpes virus-VI and VII

Gammaherpesvrinae

- Infection of lymphoblastic cells, genome size 105-175 kbp
- Lymphocryptovirus, Rhadinovirus, Human herpesvirus-IV, Human herpes virus-VIII

Subfamily	Taxonomic name	Common name
Alpha-herpesvirinae	HHV-1	Herpes simplex virus 1 (HSV-1)
	HHV-2	Herpes simplex virus 2 (HSV-2)
	HHV-3	Varicella-zoster virus (VZV)
Beta-herpesvirinae	HHV-5	Human cytomegalovirus (HCMV)
	HHV-6	HHV-6 vairant A or B
	HHV-7	HHV-7
Gamma-herpesvirinae	HHV-4	Epstein-Barr virus (EBV)
	HHV-8	Kaposi's sarcoma-associated herpesvirus (KSHV)

HHV, human herpesvirus.

Herpesviruses have large genomes composed of up to 235 kbp of linear, double-stranded DNA and correspondingly large and complex virus particles containing about 35 virion polypeptides. All encode a variety of enzymes involved in nucleic acid metabolism, DNA synthesis, and protein processing (e.g., protein kinases). The different members of the family are widely separated in terms of genomic sequence and proteins, but all are similar in terms of structure and genome

organization. Some but not all herpesvirus genomes consist of two covalently joined sections, a unique long (UL) and a unique short (US) region, each bounded by inverted repeats. The repeats allow structural rearrangements of the unique region.



The prototype member of the family is herpes simplex virus (HSV), whose genome consists of approximately 152 kbp of double-stranded DNA, the complete nucleotide sequence of which has now been determined. This virus contains about 80 genes, densely packed and with overlapping reading frames. Each gene is expressed from its own promoter. Each gene tends to be expressed from its own promoter.

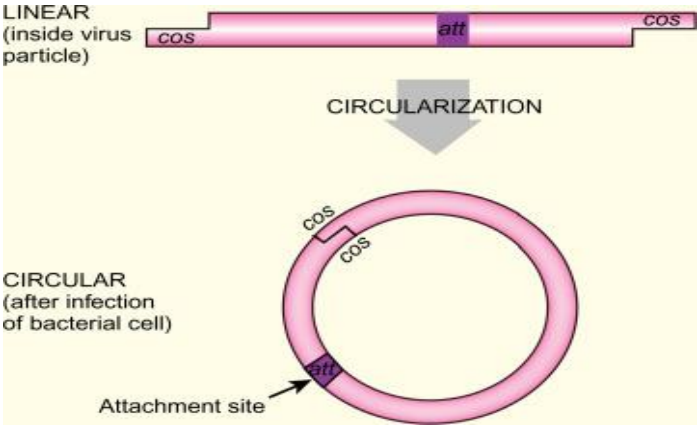
Lesson 60: Small DNA genomes

M-13 PHAGE

The genome of this phage consists of 6.4 kb of single-stranded, positive-sense, circular DNA and encodes 10 genes. Unlike most icosahedral virions, the filamentous M13 capsid can be expanded by the addition of further protein subunits, so the genome length can be increased by the addition of extra sequences in the nonessential intergenic region without becoming incapable of being packaged into the capsid.

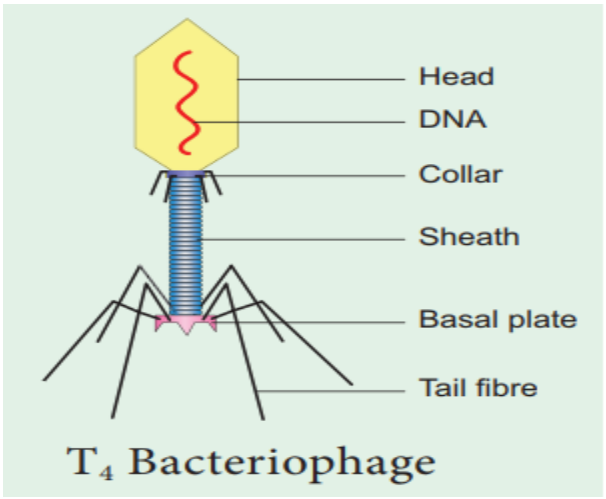
phage λ

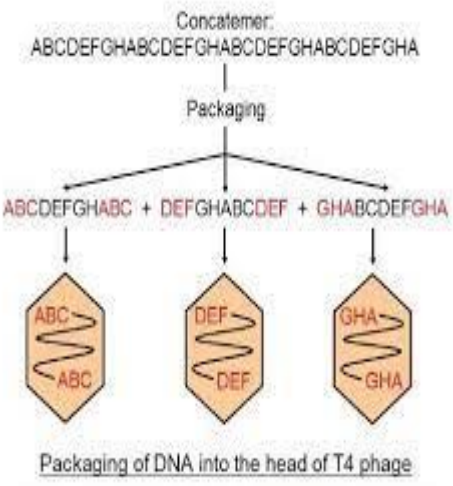
In this bacteriophage, the genome packaging limits are more rigid. phage λ, only DNA of between approximately 95 and 110% (approximately 4654 kbp) of the normal genome size (49 kbp) can be packaged into the virus particle. In the case of phage λ, the substrate packaged into the phage heads during assembly consists of long repetitive strings (concatemers) of phage DNA that are produced during the later stages of genome replication. The DNA is reeled in by the phage head, and when a complete genome length has been incorporated, the DNA is cleaved at a specific sequence by a phage-coded endonuclease. This enzyme leaves a 12-bp 50 overhang on the end of each of the cleaved strands, known as the cos site. Hydrogen bond formation between these “sticky ends” can result in the formation of a circular molecule.



T4 phage

The genome of phage T4 is about 160 kbp double-stranded DNA. Phage T4 illustrates another molecular feature of certain linear virus genome—**terminal redundancy**. Replication of the T4 genome also produces long concatemers of DNA. These are cleaved by a specific endonuclease, but unlike the λ genome the lengths of DNA incorporated into the particle are somewhat longer than a complete genome length. Some genes are repeated at each end of the genome, and the DNA packaged into the phage particles contains repeated information. These examples show that bacteriophage genomes are neither necessarily small nor simple.





Parvovirus

Parvovirus genomes are linear, non-segmented, single-stranded DNA of about 5 kb. Parvovirus genomes are negative-sense, but some parvoviruses package equal amounts of (+) and (-) strands into virions. These are very small genomes, and even the replication-competent

parvoviruses contain only two genes: rep, which encodes proteins involved in transcription, and cap, which encodes the coat proteins. However, the expression of these genes is rather complex, resembling the pattern seen in adenoviruses, with multiple splicing patterns seen for each gene. The ends of the genome have palindromic sequences of about 115 nt, which form "hairpins". These structures are essential for the initiation of genome replication, again emphasizing the importance of the sequences at the ends of the genome, and also determine which strand is packaged into virus particles.

Polyomaviruses

The genomes of polyomaviruses are double-stranded, circular DNA molecules of approximately 5 kbp. The architecture of the polyomavirus genome (i.e., number and arrangement of genes and function of the regulatory signals and systems) has been studied in great detail. Within the particles, the virus DNA assumes a supercoiled form and is associated with four cellular histones. The genomic organization of these viruses has evolved to pack the maximum information (six genes) into minimal space (5 kbp).



Lesson 61: General Classification of Viruses

GENERAL CLASSIFICATION OF VIRUSES

In assigning a virus to a taxonomic group the ICTV considers a range of characteristics. These include host range (eukaryote or prokaryote, animal, plant, etc.), morphological features of the virion (enveloped, shape of capsid or nucleocapsid, etc.), and nature of the genome nucleic acid (DNA or RNA, single stranded or double stranded, positive or negative sense, etc.).

Baltimore classification system

One classification scheme was developed in the 1970s by Nobel laureate David Baltimore. The Baltimore classification system categorizes viruses based on the type of nucleic acid

genome and replication strategy of the virus. The system also breaks down single-stranded RNA viruses into those that are positive strand (+) and negative strand (−). Positive-strand (also positive-sense or plus-strand) RNA is able to be immediately translated into proteins; as such, messenger RNA (mRNA) in the cell is positive strand. Negative-strand (also negative-sense or minus-strand) RNA is not translatable into proteins; it first has to be transcribed into positive-strand RNA. Baltimore also took into account viruses that are able to reverse transcribe, or create DNA from an RNA template, which is something that cells are not capable of doing. Together, the seven classes are

- • class I: dsDNA viruses
- • class II: ssDNA viruses
- • class III: dsRNA viruses
- • class IV: positive-sense ssRNA viruses
- • class V: negative-sense ssRNA viruses
- • class VI: RNA viruses that reverse transcribe
- • class VII: DNA viruses that reverse transcribe

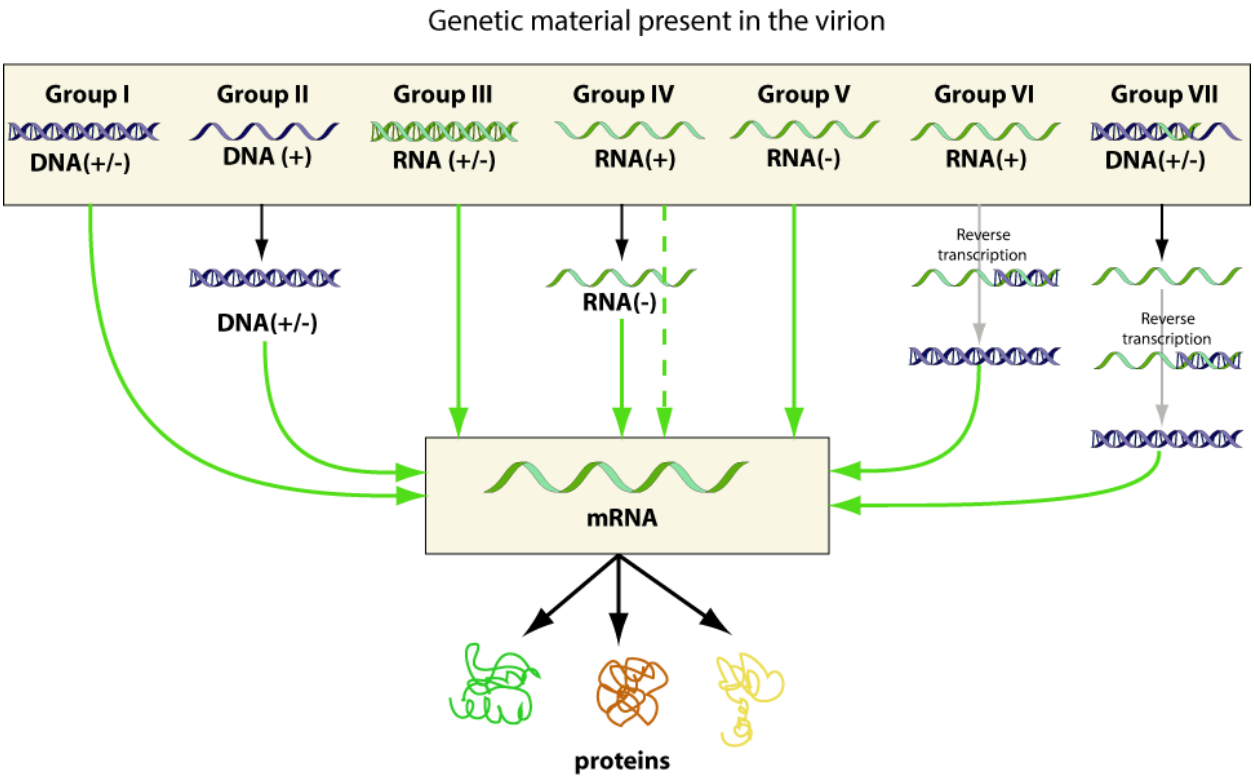
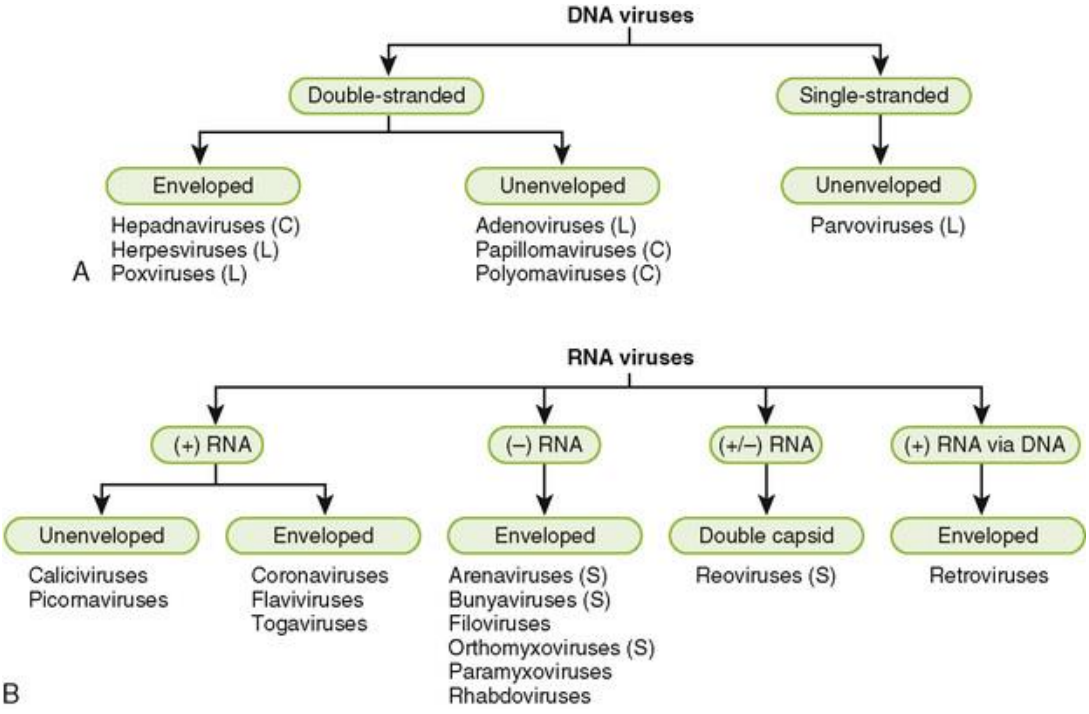


FIGURE: The Baltimore classification, a virus classification scheme based on the form of nucleic acid present in virion particles and the pathway for expression of the genetic material as messenger RNA. The original scheme contained groups I through VI and has been expanded to accommodate DNA-containing, reverse transcribing viruses. Viruses containing ambisense single-stranded RNA genomes are grouped under negative sense single-stranded RNA viruses.



Lesson 62: Positive sense RNA Viruses

CLASSIFICATION OF POSITIVE SENSE RNA VIRUSES

RNA viruses, comprising 70% of all viruses, vary remarkably in genome structure. Because of the error rate of the enzymes involved in RNA replication, these viruses usually show much higher mutation rates than do the DNA viruses. Mutation rates of 10^{-4} lead to the continuous generation of virus variants which show great adaptability to new hosts. The viral RNA may be single-stranded (ss) or double-stranded (ds), and the genome may occupy a single RNA segment or be distributed on two or more separate segments (segmented genomes). In addition, the RNA strand of a single-stranded genome may be either a sense strand (plus strand), which can function as messenger RNA (mRNA), or an antisense strand (minus strand), which is complementary to the sense strand and cannot function as mRNA protein translation. Sense viral RNA alone can replicate if injected into cells, since it can function as mRNA and initiate

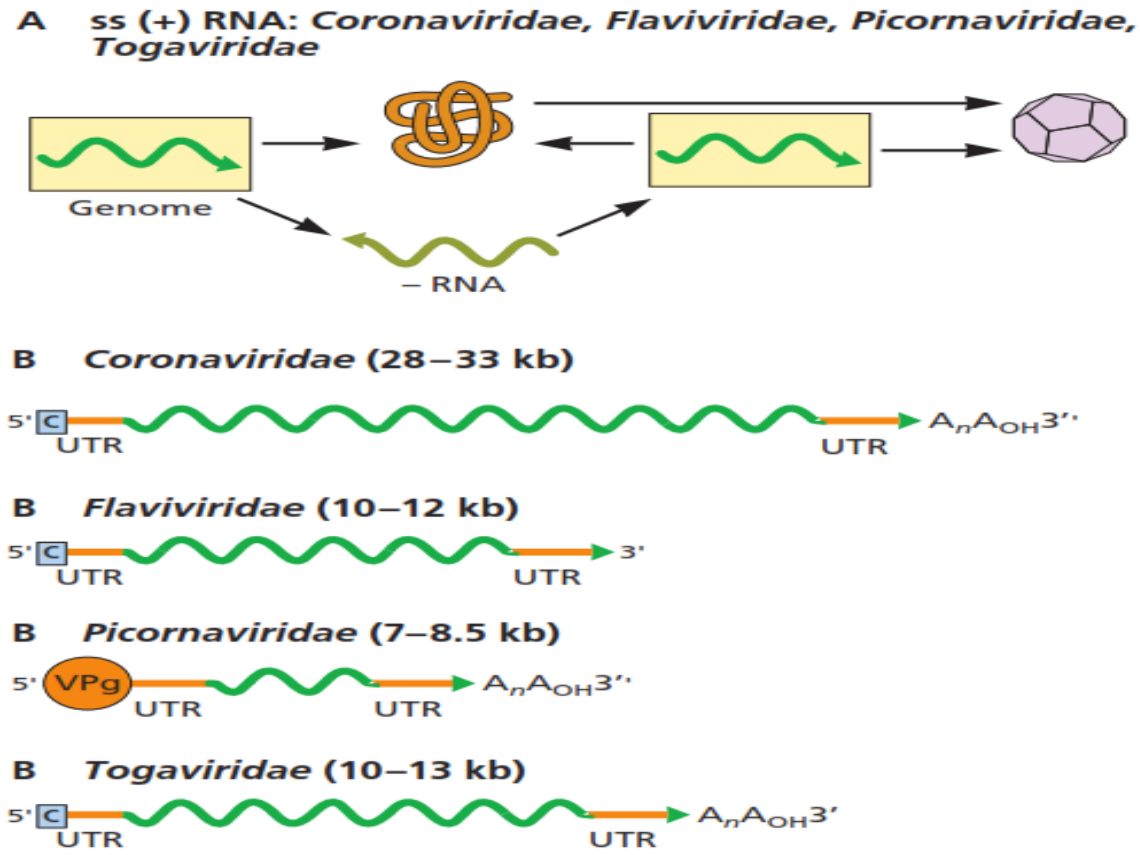
translation of virus-encoded proteins. Antisense RNA, on the other hand, has no translational function and cannot per se produce viral components.

DOUBLE STRANDED RNA VIRUSES

There are eight genuses of viruses with dsRNA genomes. The number of dsRNA segments ranges from 1 (Totiviridae and Endornaviridae , viruses of fungi, protozoa, and plants) to 9 to 12 (Reoviridae, viruses of fungi, invertebrates, plants, protozoa, and vertebrates).

FAMILY	GENUS	TYPE SPECIES
Birnaviridae	Avibirnavirus	Infectious bursal disease virus
	Aquabirnavirus	Infectious pancreatic necrosis virus
Reoviridae	Orthoreovirus	Mammalian orthoreovirus
	Orbivirus	Bluetonue virus-1
	Rotavirus	Rotavirus A
	Coltivirus	Colorado tick fever virus
	Aquareovirus	Aquareovirus A

SINGLE STRANDED RNA VIRUSES



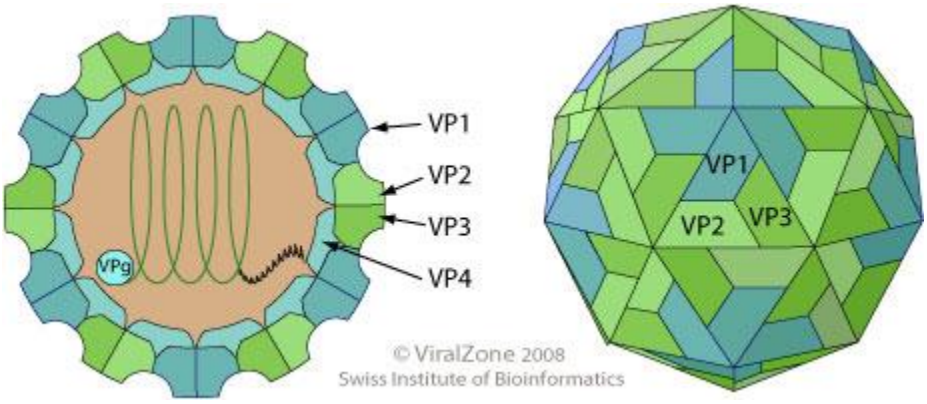
Lesson 63: Picorna Virus properties

PICORNAVIRUS

The **picornaviruses** are small (22 to 30 nm) non-enveloped, single-stranded RNA viruses with icosahedral (cubic) symmetry. The virus capsid is composed of 60 protein subunits, each consisting of four poly-peptides VP1–VP4. Because they contain no essential lipids, they are ether resistant. These are icosahedral viruses with highly diverse positive-sense RNA genomes. They replicate in the cytoplasm. The picornaviruses that affect humans are the enteroviruses, found primarily in the gut; the rhinoviruses, found in the upper respiratory tract and hepatovirus (Hepatitis A virus) in the intestine and liver. Picornaviruses may cause subclinical infections of humans and animals or conditions ranging from mild febrile illness to severe diseases of heart, liver and the central nervous system. Belongs to Group III, Group IV or

Group V of Baltimore Classification system. There are currently 80 species in this family, divided among 35 genera. Five genera infecting humans include:

- Enteroviruses
- Rhinoviruses
- Hepatoviruses
- Parechoviruses
- Kobuviruses



Physicochemical properties

Some species are unstable below pH 7; many are less stable at low ionic strength than at high ionic strength. Virions are insensitive to ether, chloroform, or non-ionic detergents. Viruses are inactivated by light when grown with, or in the presence of photodynamic dyes such as neutral red or proflavin. Thermal stability varies with viruses as does stabilization by divalent cations.

Enteroviruses	Rhinoviruses	Other picornaviruses
• poliomylietis • Echovirus • Bovine enterovirus • Simian Enterovirus • Avian Enterovirus	• Human cold virus • Equine rhinovirus • Bovine rhinovirus • Foot and mouth disease • Vesicular exanthema	• Avian encephalomylietis • Mouse encephalitis • Feline picornavirus

Lesson 64: Picorna Virus genome

Picornavirus Genome

The picornavirus genome consists of one **single-stranded, positive-sense RNA** molecule of between 7.2 kb in human rhinoviruses and 8.5 kb in foot-and-mouth disease viruses, containing a number of features conserved in all picornaviruses:

■ There is a long untranslated region (UTR) at the 5` end which is important in translation, virulence, and possibly encapsidation, as well as a shorter 3` UTR that is necessary for (-) strand synthesis during replication.

■ The 5` UTR contains a “clover-leaf” secondary structure known as the internal ribosomal entry site (IRES).

Both ends of the genome are modified—the 5` end by a covalently attached small, basic protein VPg and the 3` end by polyadenylation. A small protein, VPg (2 to 3 kDa), is linked covalently to the 5’-terminus. A terminal protein provides a hydroxyl group (in a tyrosine or serine residue) to which the first oligonucleotide can be linked, by viral polymerases, via a phosphodiester bond.

Picornaviruses are non-enveloped, with an Icosahedral capsid. The capsid consists of 60 protomers. Each protomer consists of 4 polypeptides. These peptides are called VP (viral protein) 1, 2, 3 and 4. VP2 and VP4 polypeptides originate from one protomer known as VP0 VP0 is cleaved to give the different capsid components.

REPLICATION

The mRNA encodes RNA dependent RNA polymerase. This polymerase makes complementary minus strands of RNA. RNA will act as template to make more plus strands.

An overview of the steps in picornavirus replication attachment,

- entry
- translation
- transcription/genome replication
- Assemble
- exit

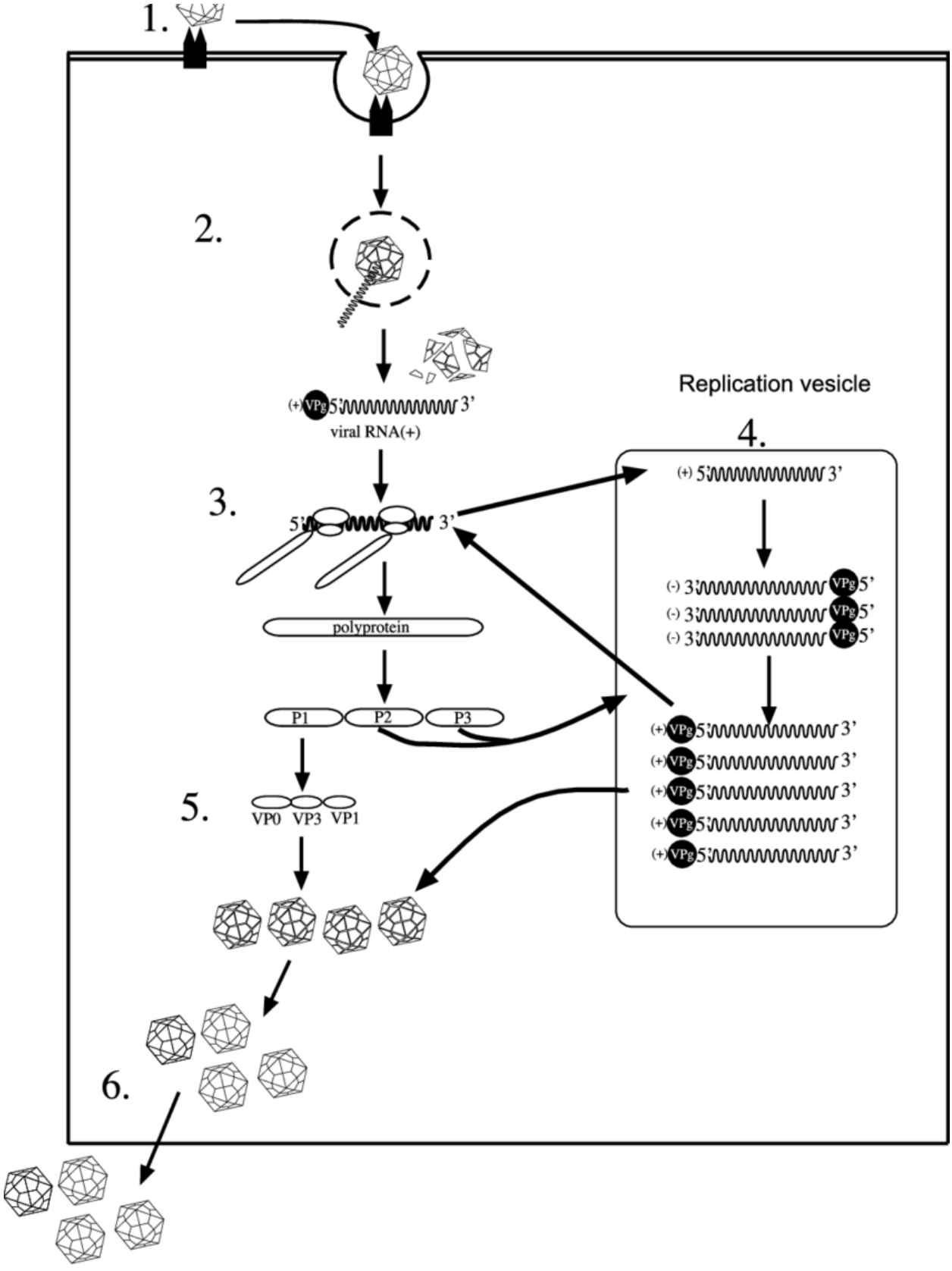

1.
2.
(+) VPg 5' 3'
viral RNA(+)
3.
5' 3'
polyprotein
P1 P2 P3
5.
VP0 VP3 VP1
6.
Replication vesicle
4.
(+) 5' 3'
(-) 3' VPg 5'
(-) 3' VPg 5'
(-) 3' VPg 5'
(+) VPg 5' 3'
(+) VPg 5' 3'
(+) VPg 5' 3'
(+) VPg 5' 3'
(+) VPg 5' 3'

Figure: Life cycle of picornaviruses: 1) Attachment, 2) Entry 3) Translation, 4) Replication, 5) Assembly and 6) Release. Each number is referred to in the text. (Modified from Flint et al., 2000).

Lesson 65: Picorna Virus diseases

Picorna virus important diseases

The Picornaviridae family (picornaviruses) causes a wider range of illnesses than most other, if not all, virus families. Infection with various picornaviruses may be asymptomatic or may cause clinical syndromes such as aseptic meningitis (the most common acute viral disease of the CNS), encephalitis, the common cold, febrile rash illnesses (hand-foot-and-mouth disease), conjunctivitis, herpangina, myositis and myocarditis, and hepatitis.

Polioviruses

Paralytic poliomyelitis can occur without antecedent minor illnesses. A patient may suffer aseptic meningitis with pains in the back and neck muscles for several days without progressing to paralytic poliomyelitis. Poliomyelitis, caused by the enteroviral type species, was one of the first recorded infections. In most cases, no further virus spread occurs, and there is asymptomatic or mild febrile undifferentiated illness, such as fever, malaise, headache, nausea, gastrointestinal disturbances, and sore throat, or combinations of these.

In rare cases in which disease progresses to the neurologic phase, the virus spreads hematogenously to the spinal cord or brain stem or to both. If only scattered nerve cells are destroyed, the patient may develop no visible sign of muscle weakness. More concentrated damage results in flaccid paralysis of the muscles innervated by the affected motor nerves. Muscle involvement peaks a few days after the paralytic phase begins. Paralysis is usually irreversible, and residual paralysis remains for life.

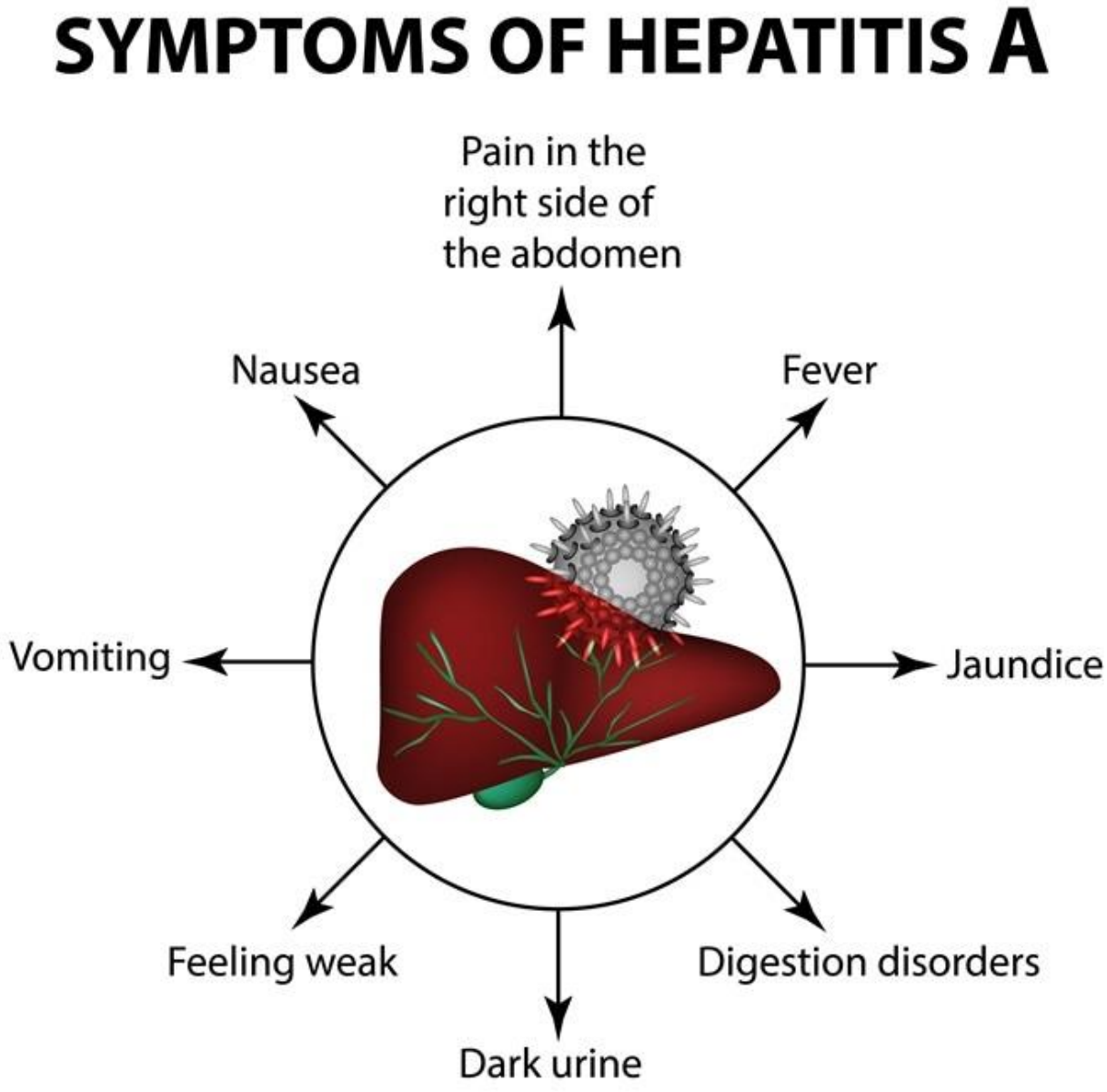
Rhinoviruses

The natural hosts of rhinoviruses are humans and chimpanzees. Rhinoviruses are present in the nose and pharynx, and, unlike enteroviruses, which are acid resistant and have an optimum growth temperature of 36 to 37°C, they are inactivated at pH 3.0 and have an optimum growth temperature of 33°C, the temperature of the nasal epithelium.

Pathologic findings in the common cold consist of inflammatory changes with hyperemia, edema, and inflammation of the columnar epithelial cells lining the nasopharynx. Desquamation of these infected cells coincides with the peak of virus spread.

HEPATITIS A

- Loss of appetite
- Jaundice
- Dark urine
- No long-term consequences



FOOT AND MOUTH DISEASE

- In hooved animals
- Lesions around foot and mouth

- Inability to walk
- Fast spread
- Significant economic damage

Lesson 66: Togavirus properties

Togavirus properties

Introduction

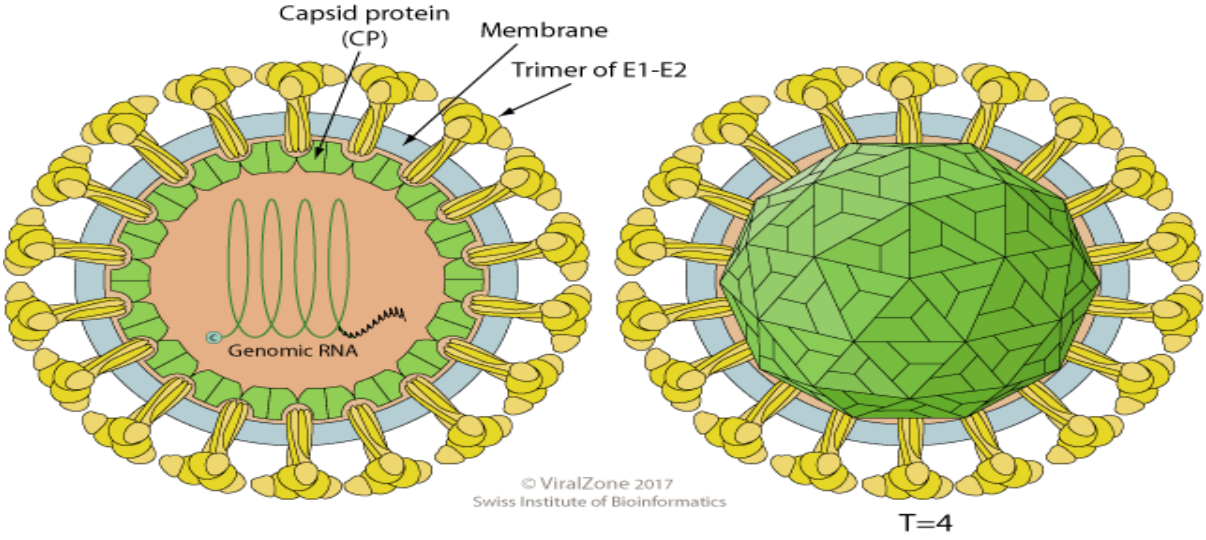
The Togaviridae family is organized into two genera, Alphavirus and Rubivirus. Alphaviruses are mainly arthropod borne viruses and responsible for such diseases in humans as encephalitis, and arthritis. The sole member of the Rubivirus genus is the Rubella virus, which is the causative agent for the common childhood disease, rubella.

Epidemiology/ Geographic distribution

Togaviviridae has a worldwide distribution. The rubella virus, which is the causative agent for the common childhood disease rubella, has a global distribution. Certain alphaviruses may have limited dispersal due to the geographic distribution of the principal arthropod vectors. For example the Western equine encephalitis, Eastern equine encephalitis, and Venezuelan equine encephalitis viruses have principal vectors belonging to Culex species of mosquitoes found in the Americas and therefore have a distribution in this region of the world.

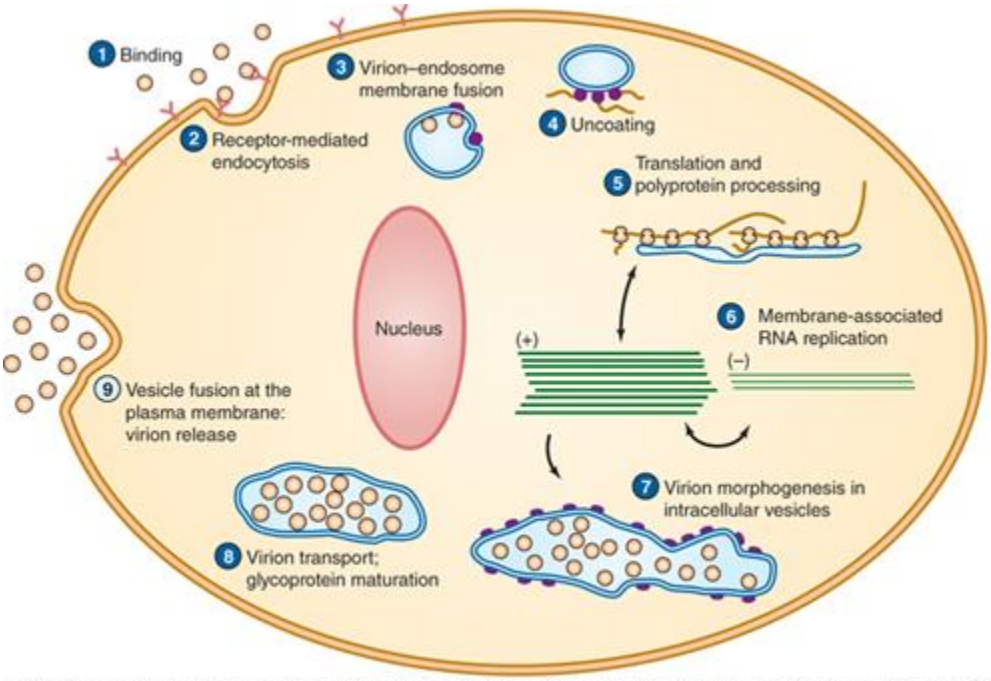
Physical properties of virion

Togavirus particles are comprised of a lipoprotein envelope and a nucleocapsid that possess a positive-sense single-strand RNA genome of 11 kb. These virions are 70 nm in diameter and are spherical to pleomorphic in shape with glycoprotein projections dispersed along the surface.



Replication

1. Attachement of the viral E glycoprotein to host receptors mediates clathrin-mediated endocytosis of virus into the host cell.
2. Fusion of virus membrane with host endosomal membrane. Acidification of endocytic vesicles induces uncoating of the virion resulting in the release of the RNA genome into the cytosol
3. The positive-sense genomic ssRNA is translated into a polyprotein, which is cleaved into non-structural proteins necessary for RNA synthesis (replication and transcription).
4. Replication takes place in cytoplasmic viral factories at the surface of endosomes. A dsRNA genome is synthesized from the genomic ssRNA(+).



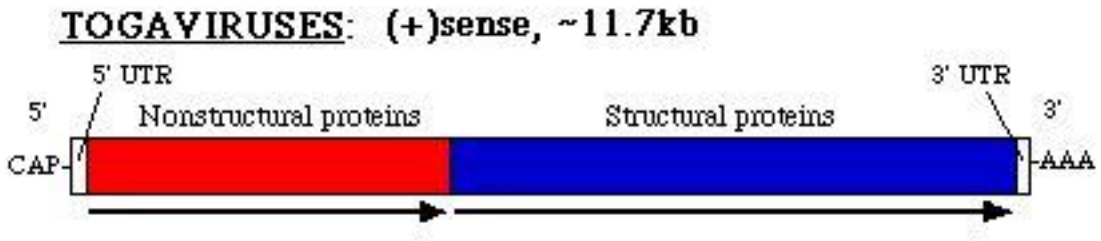
Source: Karen C. Carroll, Stephen A. Morse, Timothy Mietzner, Steve Miller: Jawetz, Melnick, & Adelberg's Medical Microbiology, 27th Edition.
www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

Lesson 67: Togavirus genome

Togavirus genome

Structure:

Togavirus genomes contain a single, linear, monopartite, positive-sense RNA molecule. The complete genome is 9700-11800 nucleotides long, with the rubella virus genome being shorter than that of the alphaviruses. Genomes is capped and polyadenylated, possessing a 5’ cap and a 3' poly (A) tail.



GENE EXPRESSION

The virion RNA is infectious and serves as both genome and viral messenger RNA. The whole genome is translated into a non-structural polyprotein which is processed by host and viral proteases. Structural polyprotein is expressed through a subgenomic mRNA.

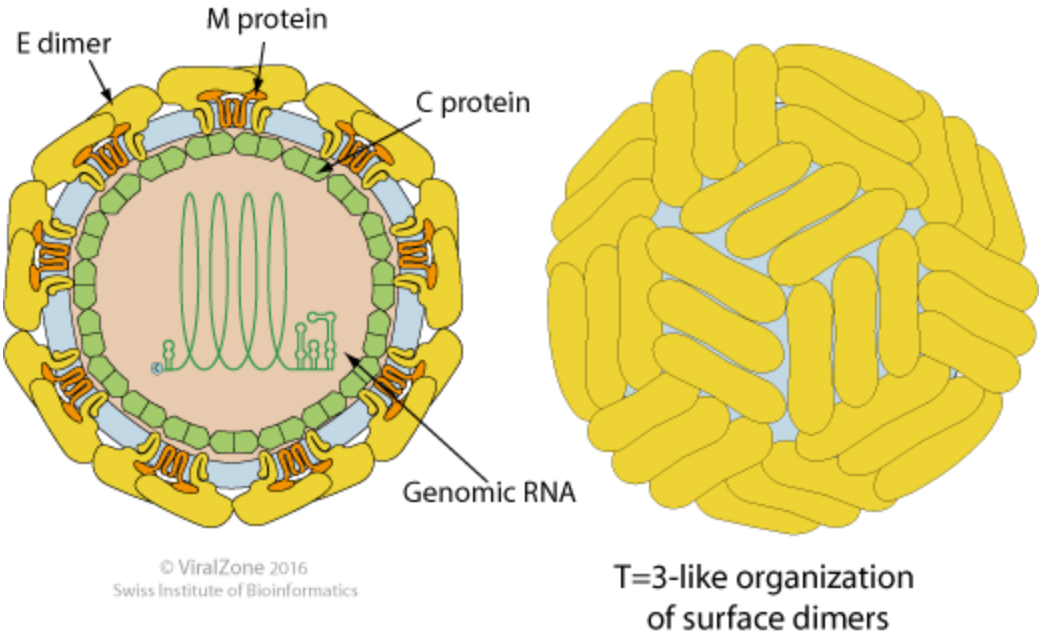
VIRION STRUCTURE

Togavirus virions consist of an envelope and a nucleocapsid. The virions are spheical to pleomorphic in shape, and about 70 nm in diameter. Glycoprotein spikes evenly cover the surface of the virion. The genome is contained in the T=4 icosahedral capsid, which measures 40 nm in diameter.

Lesson 68: Flavivirus properties

Flavivirus properties

Flaviviruses (family Flaviviridae) are small (about 50 nm in diameter), **enveloped, icosahedral** viruses that possess a **positive-strand RNA genome** of 9–12 kb. Most members of this genus, which includes 53 species, are arthropod-borne viruses, with distinct groups infecting mosquitoes or ticks. Mammals and birds are the usual primary hosts. The flavivirus family includes many mosquito-borne viruses, including yellow fever virus, West Nile fever virus, and dengue virus, as well as blood-borne hepatitis C virus (HCV). The genome organization and the mechanism of genome replication of flavivirus are fundamentally related to those of picornavirus. . Other members cause economically important diseases in domestic or wild animals. Additional viruses infect only arthropods or only mammals (*e.g.,* Tamana bat virus).



Virion Morphology

Virions are 40–60 nm in diameter, spherical in shape with a lipid envelope. The capsid is comprised of a single protein and the envelope contains two or three virus-encoded membrane proteins.

Table 1. *Flaviviridae*. Characteristics of the family *Flaviviridae*.

Characteristic	Description
Typical member	yellow fever virus-17D (X03700), species *Yellow fever virus*, genus *Flavivirus*
Virion	Enveloped, 40–60 nm virions with a single core protein (except for genus *Pegivirus*) and 2 or 3 envelope glycoproteins
Genome	9.0–13 kb of positive-sense, non-segmented RNA
Replication	Cytoplasmic, in membrane vesicles derived from the endoplasmic reticulum (ER); assembled virions bud into the lumen of the ER and are secreted through the vesicle transport pathway.
Translation	Directly from genomic RNA containing a type I cap (genus *Flavivirus*) or an internal ribosome entry site (other genera).
Host range	Mammals (all genera); most members of genus *Flavivirus* are arthropod-borne

Characteristic	Description
Typical member	yellow fever virus-17D (X03700), species *Yellow fever virus*, genus *Flavivirus*
Taxonomy	Four genera containing 89 species

DENGUE FEVER VIRUS

- Mosquito borne
    - *Aedes Egyptei*
- Infects bone marrow (break bone fever)
- Low platelet level
- Hemorrhegic fever
- Can lead to renal failure
- Septicemia and death

YELLOW FEVER

- *Aedes* mosquito
- Jaundice
- Backache
- Bloody diarrhea
- Bloody vomiting
- Vaccine available

WEST NILE VIRUS

- Birds are reservoir
- “Culex” Mosquitoes Vector
- Encephalitis
- Meningitis
- Flaccid paralysis
- Seizures and coma

HEPATITIS C

- Transmission: Exposure to infected blood
- Highly antigenic variable
- Jaundice
- 60-80% become chronic cases
- Cirrhosis
- Primary cause of hepatocellular carcinoma
- Interferon treatment

Lesson 69: Flavivirus genome

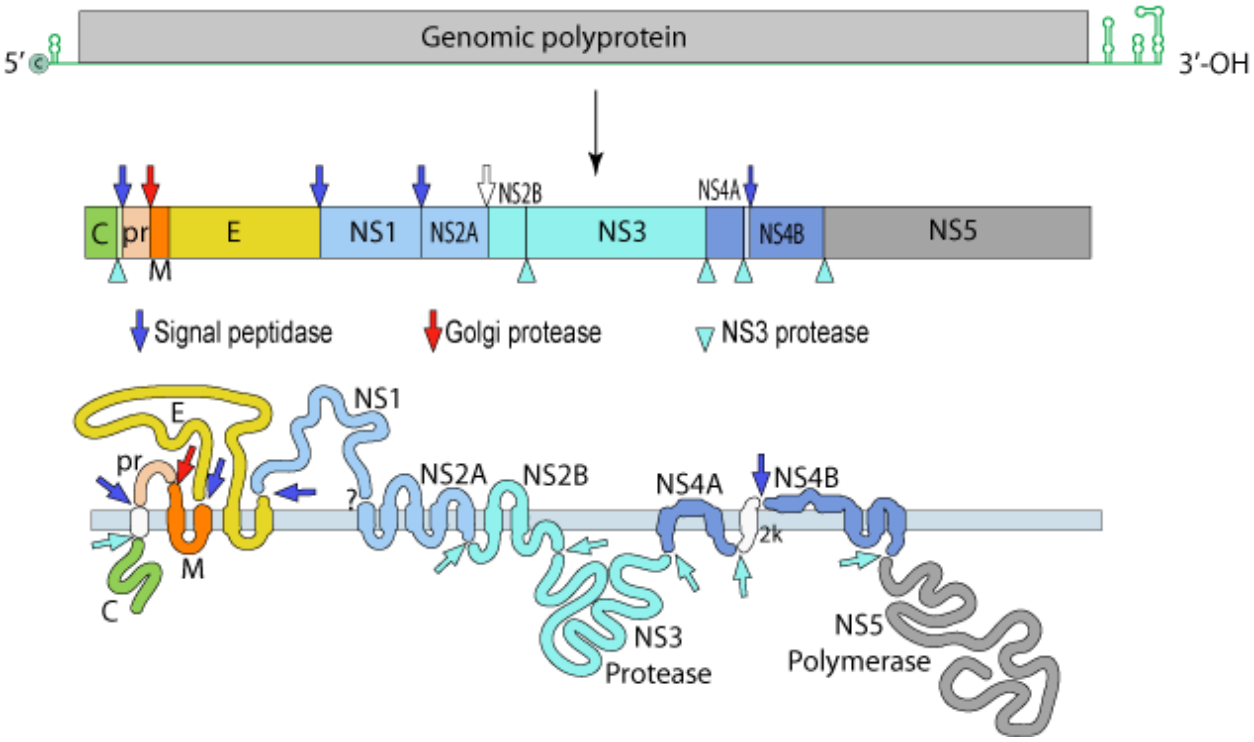
Flavivirus genome

Genomes of *Flaviviridae* members are positive-sense ssRNA. All members of the family lack a 3'-terminal poly(A) tract. Only the genomes of members of the genus *Flavivirus* contain a 5'-terminal type I cap structure, the others possess an internal ribosomal entry site (IRES).

Genome organization and replication

The genomic RNA of all members of the family has a similar organization and is the viral mRNA found in infected cells. It contains a single long open reading frame (ORF) flanked by 5'- and 3'-terminal non-coding regions (NCRs) that form specific secondary structures required for genome replication and translation. Members of the genus *Flavivirus*, but not pestiviruses, hepaciviruses or pegiviruses produce a unique, subgenomic, small (300–500 nt) non-coding RNA that is derived from the 3'-NCR of genomic RNA that is essential for virus replication in cells and modulates pathogenicity in animals. Translation-initiation of genomic RNA is cap-dependent for members of the genus *Flavivirus*, whereas IRES elements are present in viruses of the other genera. Viral proteins are synthesized as part of a polyprotein that is co- and post-translationally cleaved by viral and cellular proteases. The **structural proteins** are contained in the N-proximal portion of this polyprotein and the **nonstructural proteins (NS)** in the remainder. The latter include a serine protease, an RNA helicase and the RdRP. Genome replication occurs in the cytoplasm in association with modified cellular membranes via the synthesis of genome-length negative-strand intermediates. Virion assembly, including acquisition of a glycoprotein-containing lipid envelope, occurs by budding through intracellular membranes. Viral particles are transported in cytoplasmic vesicles through the secretory pathway before they are released by exocytosis, as shown for members of the genus *Flavivirus* and assumed for members of the

other genera. In addition, release of infectious RNA via exosomes has recently been demonstrated.



TWO GROUP HYPOTHESIS

On the basis of morphological information, obtained using electron microscopy , supported the hypothesis of the existence of at least two virus groups.

One group includes non-enveloped virus Family Reoviridae (genera Orbivirus, Coltivirus and Seadornavirus). These are Viruses with an overall diameter of 60–80 nm, Icosahedral symmetry. There are several concentric capsid layers that surround a segmented double-stranded RNA (dsRNA) genome.

Second group includes enveloped viruses (inactivated by ether and deoxycholate), 50–60 nm in diameter. Infectious ssRNA genome of positive polarity, two antigenically distinct subgroups i.e. Group A and Group B.

MEDICAL IMPORTANCE

- Mostly serologically related to each other
- Antibodies of one useful against others
- Antigenic variability of Hepatitis C proteins

- The 17D vaccine is one of the most efficient vaccines ever developed and was derived from a strain of Flavivirus isolated from a man who recovered from infection by the virus.

Lesson 70: Corona virus structure

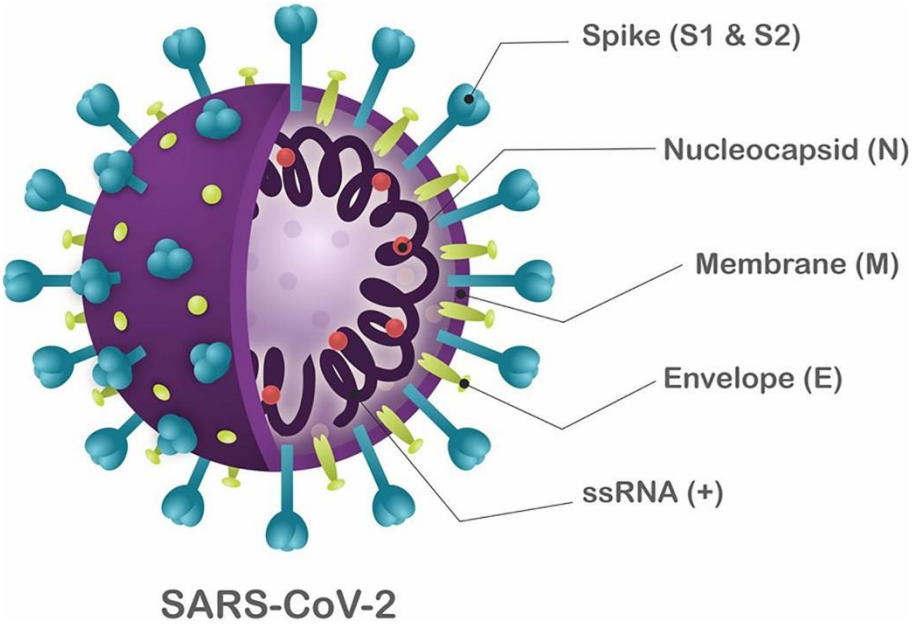
Coronavirus

Introduction

Coronaviruses are found in avian and mammalian species. They resemble each other in morphology and chemical structure: for example, the coronaviruses of humans and cattle are antigenically related. In animals, various coronaviruses invade many different tissues and cause a variety of diseases, but in humans they are only proved to cause mild upper respiratory infections, i.e. common colds. On rare occasions, gastrointestinal coronavirus infection has been associated

Structure

Spherical or pleomorphic enveloped particles containing single-stranded (positive-sense) RNA associated with a nucleoprotein within a capsid comprised of matrix protein. The envelope bears club-shaped glycoprotein projections. The Genomic size of coronaviruses ranges from approximately 26 to 32 Kb, the largest for an RNA virus.



Taxonomy

Order Nidovirales

Family *Coronaviridae*

Subfamily *Coronavirinae*

COMMON COLD

- Second most common cause of common cold
- SARS (severe acute respiratory syndrome)
- GI tract infections in kids

Severe Acute Respiratory Syndrome (SARS)

- Zoonotic origin
- Acute respiratory form
- Outbreaks lead to enormous number of deaths
- Flu and Fever
- Lethargy
- Cough
- Pnemonia

Middle East Respiratory Syndrome (MERS)

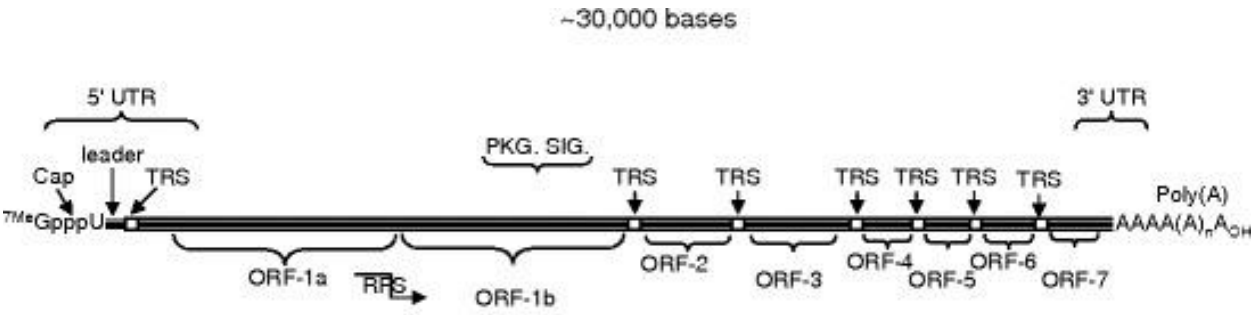
- Fever
- Cough
- Shortness of breath
- Sometimes diarrhea
- Mortality rate: 35%
- Camels are major reservoir hosts

Lesson 71: Corona virus genome

Coronavirus Genome

Members of the family Coronaviridae are large, enveloped, single-stranded RNA viruses. They are the largest known RNA viruses, with genomes ranging from 27-30 kb and a virion of approx. 120 nm in diameter. Virions are roughly spherical and are notable for the large spike (S) glycoprotein that extends from the virus envelope. 27-32kb in size (the largest of all RNA virus genomes)

- Capped
- Polyadenylated
- The leader RNA (65-89 bp) at the 5' end of the genome is also present at the end of each subgenomic RNAs.



TRS - Transcriptional Regulatory Sequence [UCUAAAC FOR MHV]
RFS - Ribosome Shifting Sequence
ORF - Open Reading Frame
5'-UTR - 5' Untranslated Region (~275 bases)
3'-UTR - 3' Untranslated Region (~80 bases)
Leader - Leader sequence (~65 bases)

The family contains two subfamilies, the Coronavirinae and the Torovirinae . The toroviruses are notable for having a helical, doughnut-shaped nucleocapsid.

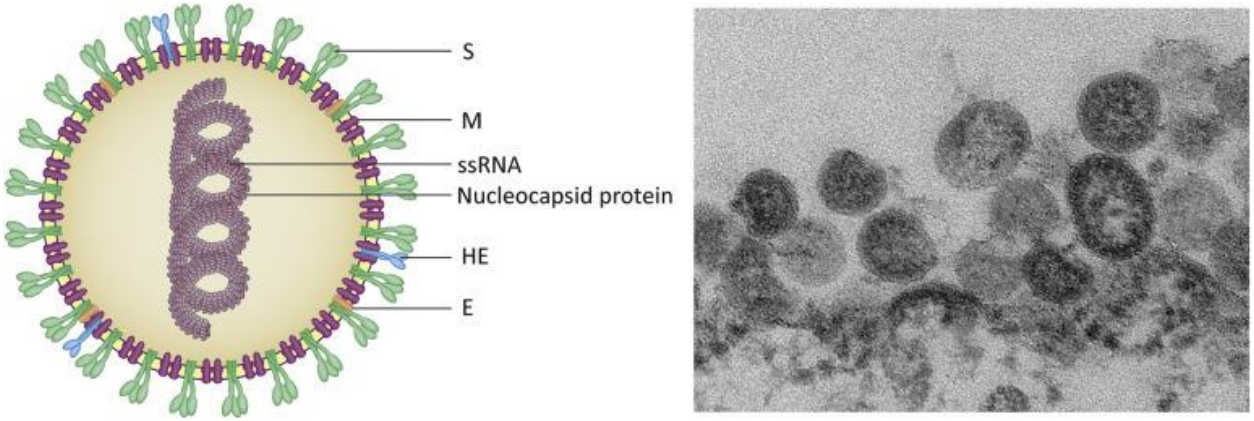


Figure: Virion structure (subfamily *Coronavirinae*) (right). TEM image of MERS-CoV virions in culture (left).

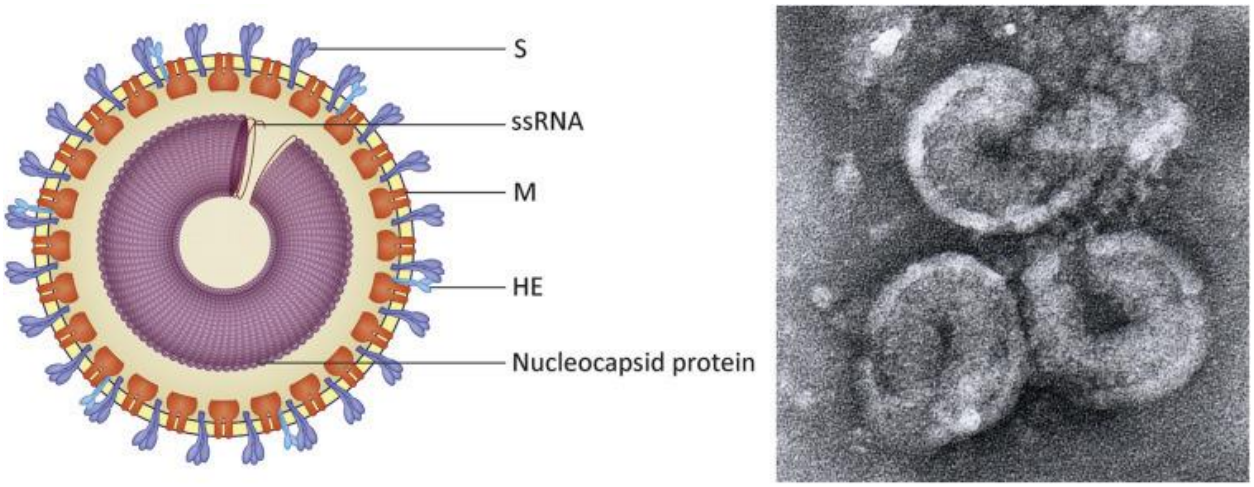


Figure: Torovirus structure (left) and transmission electron micrograph of torovirus particles (right).

REPLICATION

- Attachment of the viral S protein receptors
- Endocytosis of the virus into the host cell
- Fusion of virus membrane with endosomal membrane
- Genome is released into the cytoplasm.
- Synthesis and proteolytic cleavage of the replicase polyprotein.
- A dsRNA genome is synthesized from the genomic ssRNA(+).
- The dsRNA genome is transcribed by providing viral mRNAs/new ssRNA(+) genomes.
- Synthesis of structural proteins encoded by subgenomic mRNAs.
- Assembly and budding at membranes of the endoplasmic reticulum (ER), the intermediate compartments, and/or the Golgi complex.
- Release of new virions by exocytosis.

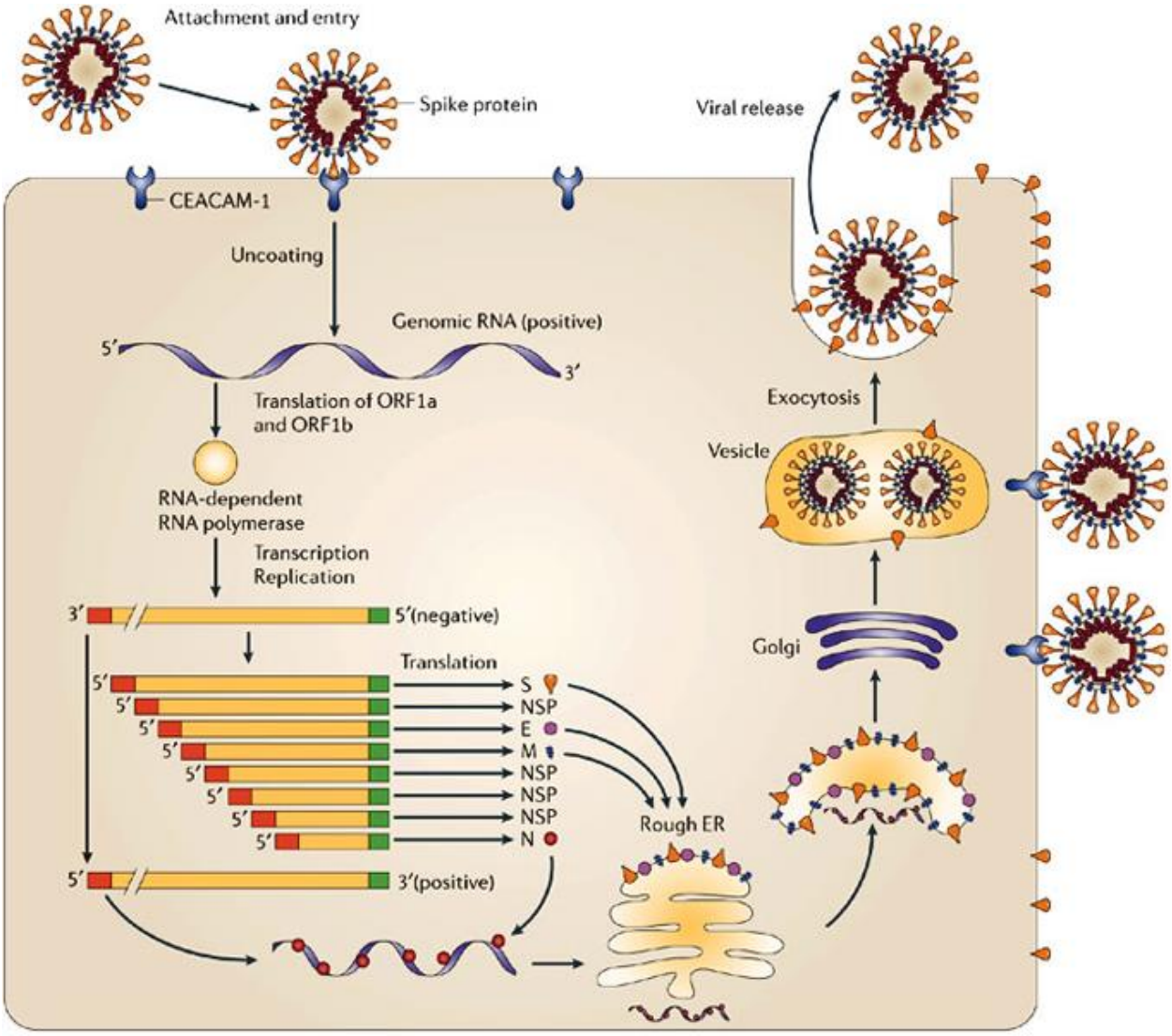


Figure: Summary of mouse hepatitis virus (MHV) replication. MHV binds to the host-cell receptor CEACAM-1 through interaction of the spike (S) glycoprotein. Virus entry into the host cell can occur through fusion with the surface of the host cell, with the subsequent release of the genomic RNA into the cytoplasm. Alternatively, MHV can enter the host cell through the formation of endocytic vesicles, and genomic RNA is released into the cytoplasm following fusion with the vesicle membrane (not shown). Translation of the positive-strand genomic RNA gives rise to a large polyprotein that undergoes proteolytic processing to generate an RNA-dependent RNA polymerase. Through the action of the RNA polymerase, a full-length, antisense negative-strand template is generated. Subgenomic mRNAs are synthesized, presumably from subgenomic negative-strand templates. Translation of subgenomic mRNAs gives rise to structural viral proteins. S glycoprotein is expressed on the surface of the host cell and this might contribute to fusion with neighbouring uninfected cells by binding to CEACAM-1. Virus assembly occurs within vesicles, followed by virus release by fusion of virion-containing vesicles

with the plasma membrane. Released virus can infect other cells and can replicate within the parent cell through binding to CEACAM-1. E, envelope protein; ER, endoplasmic reticulum; M, membrane protein; N, nucleocapsid protein; ORF, open reading frame.

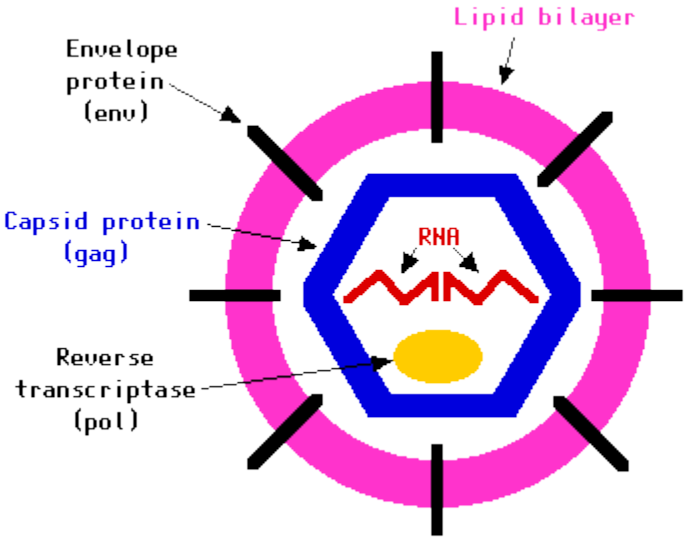
Lesson 72: Retroviruses

Retroviruses

Retroviruses comprise a large and diverse family of enveloped RNA viruses. A typical, "minimal" retrovirus consists of:

- an outer **envelope** which was derived from the plasma membrane of its host
- many copies of an **envelope protein** embedded in the lipid bilayer of its envelope
- a **capsid**; a protein shell containing
- two molecules of **RNA** and
- molecules of the enzyme reverse **transcriptase**

Reverse transcriptase is a DNA polymerase that uses RNA as its template. Thus it is able to make genetic information flow in the reverse (RNA ->DNA) of its normal direction.



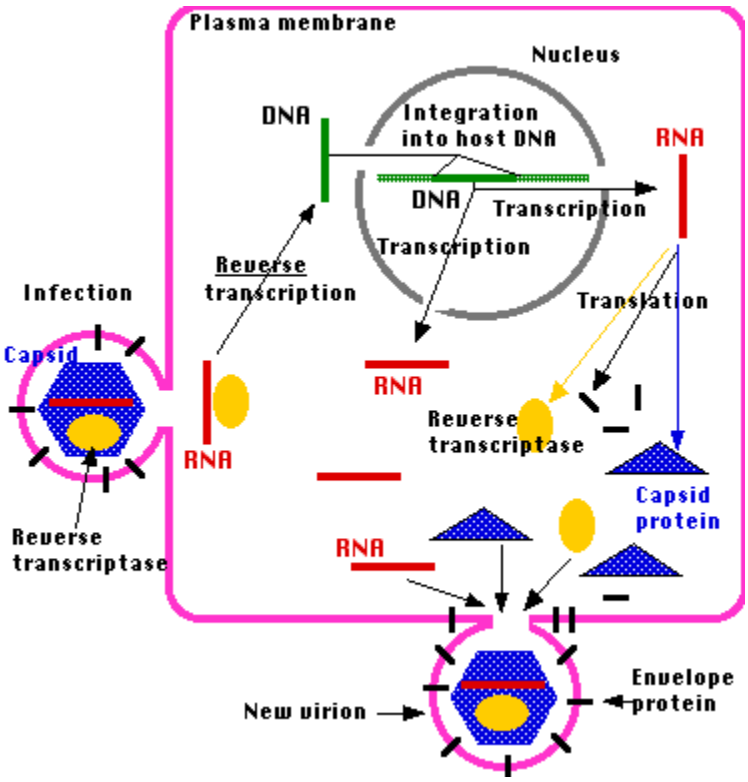
Replication

Infection of a host cell requires that the cell have a surface protein that can serve as a **receptor for the envelope protein** of the retrovirus. When a retrovirus infects a cell its molecules of reverse transcriptase are carried into the cell attached to the viral RNA molecules. The reverse

transcriptase synthesizes DNA copies of the RNA. This dsDNA can then be integrated into the host chromosome - it inserts into the DNA of the host-cell, with the help of the viral enzyme integrase. In this integrated dsDNA state, the virus is called a provirus. These inserts are transcribed by the host's enzymes into fresh RNA molecules which re-enter the cytosol where some are translated by host ribosomes

- *gag* gene is translated into molecules of the capsid protein
- *pol* gene is transcribed into molecules of reverse transcriptase
- *env* gene is translated into molecules of the envelope protein

Other RNA molecules become incorporated into fresh virus particles



The genome of retroviruses consists of RNA not DNA. HIV-1 and HIV-2, the agents that cause AIDS, are retroviruses. In February 1997 it was reported that pig cells contain a retrovirus capable of infecting human cells (at least, in vitro). This is troublesome because of the efforts that are being made to transplant pig tissue into humans (e.g., fetal pig cells into the brains of patients with Parkinson's disease). Transplant recipients must have their immune systems suppressed if the transplant is to avoid rejection. The probability that the original hosts for HIV-1 and HIV-2 where some other primate suggests that retroviruses can move from one species to another.

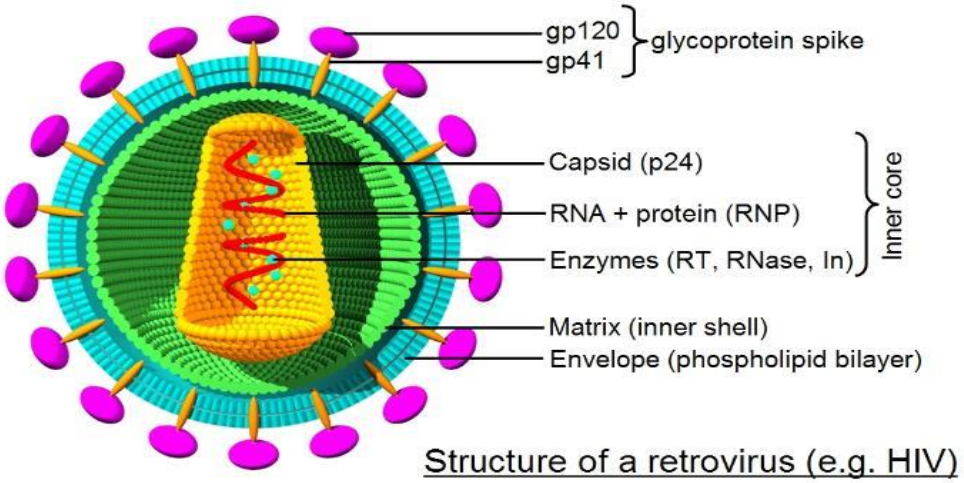
TRANSMISSION

- blood
- Semen
- pre-seminal fluid
- rectal fluids
- vaginal fluids
- breast milk
- **Blood transfusion**
- **Syringes**

Lesson 73: Retroviruses Characteristics

Retroviruses Characters

Retroviruses use RNA as its genetic material. The virions are approx.100 nm in diameter, and their outer lipid envelope incorporates and displays the viral glycoproteins. The shape and location of the internal protein core are characteristic for various genera of the family. The virion RNA is 7–12 kb in size, and it is linear, single-stranded, non-segmented. The hallmark of the family is its replicative strategy which includes as essential steps reverse transcription of the virion RNA into linear double-stranded DNA and the subsequent integration of this DNA into the genome of the cell.



Structure of a retrovirus (e.g. HIV)

A retrovirus of the lentivirus type, e.g. human immunodeficiency virus (HIV). HIV has two (+) ssRNA molecules in its **genome**, represented above as the spirals (red) in the center of the elongated inner core (yellow) which are both identical copies of the genome wound with

packaging proteins to form ribonucleoprotein. There is **enveloped**, that is it is enclosed in a phospholipid bilayer membrane (cyan) derived from the host cell-surface membrane. This envelope contains **glycoprotein** spikes. Just beneath the envelope is a protein shell (inner shell or matrix) which stabilizes the structure of the envelope.

Virion components

1. **Envelope**

Envelope is composed of lipids that is obtained from the host plasma membrane. In this embedded are glycoprotein encoded by the env gene. It provides protection from the extracellular environment via the lipid bilayer, enabling the retrovirus to enter/exit host cells through endosomal membrane trafficking and the ability to directly enter cells by fusing with their membranes.

2. **Nucleic acid**

- dimer RNA
- cap at the 5' end and a poly(A) tail at the 3' end
- terminal noncoding regions for replication
- internal regions that encode virion proteins
- The 5' end includes four regions, which are R, U5, PBS, and L.

3. **Proteins**

- consisting of gag proteins, protease, pol proteins, and env proteins

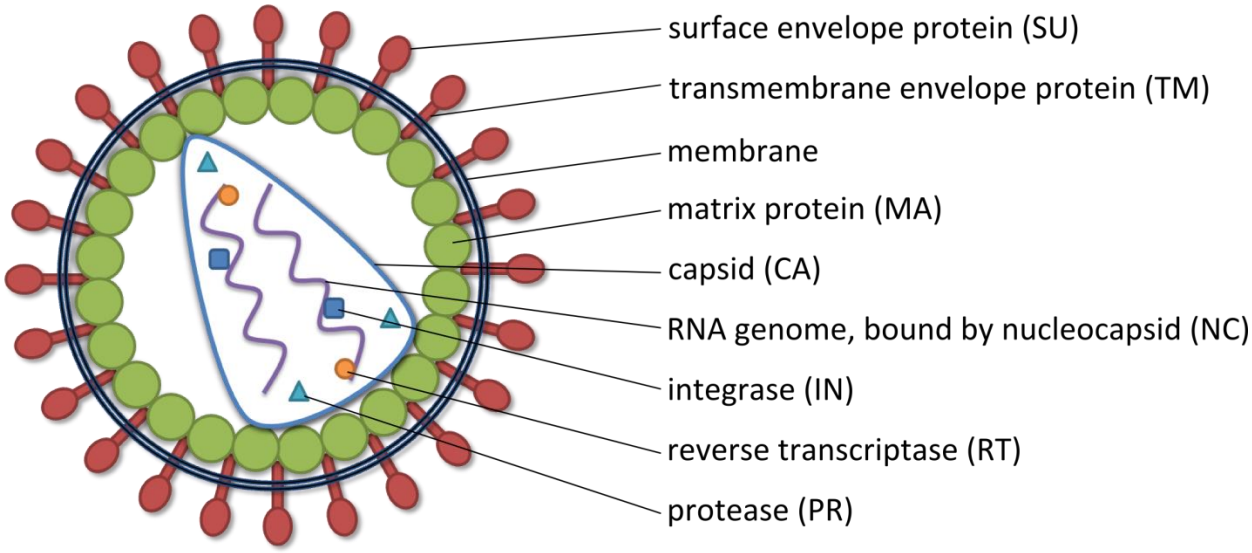
SALIENT CHARACTERISTICS

- Retroviruses are the only animal viruses that integrate into the host cell's genome
- The retroviral genomes can accommodate changes to its configuration.
- The retroviral life cycle begins in the nucleus of an infected cell.
- Retroviruses offer gene therapy researchers aid for delivering genes to target cells at high efficiency that allows for long-term, stable expression of introduced genetic elements

Lesson 74: Retroviruses genome

RETROVIRUS Genome

The viral particles of retroviruses contain two copies of positive-stranded RNA with an associated viral reverse transcriptase (RT) located within an internal core. Also located within this compartment are structural and enzymatic proteins, including the nucleocapsid (NC), capsid (CA), integrase (IN), and protease (PR). The inner core is surrounded by an outer protein layer comprised of the matrix (MA) protein, which is in turn encompassed by the envelope glycoprotein (ENV)-studded, host cell membrane-derived envelope.



The viral RNA contains a cap at its 5`end and a poly A tail at the 3`end. A simple retroviral genome contains three genes. The group-specific antigen glycoprotein (gag) that encodes the structural proteins of the virus: matrix, capsid and nucleocapsid. The pol gene encodes the viral enzymes protease (PR), integrase (IN), and reverse transcriptase (RT). The env gene encodes for the viral Env protein that is composed of two subunits, the surface unit (SU) and the transmembrane (TM) region. Ψ-represents a packaging signal.

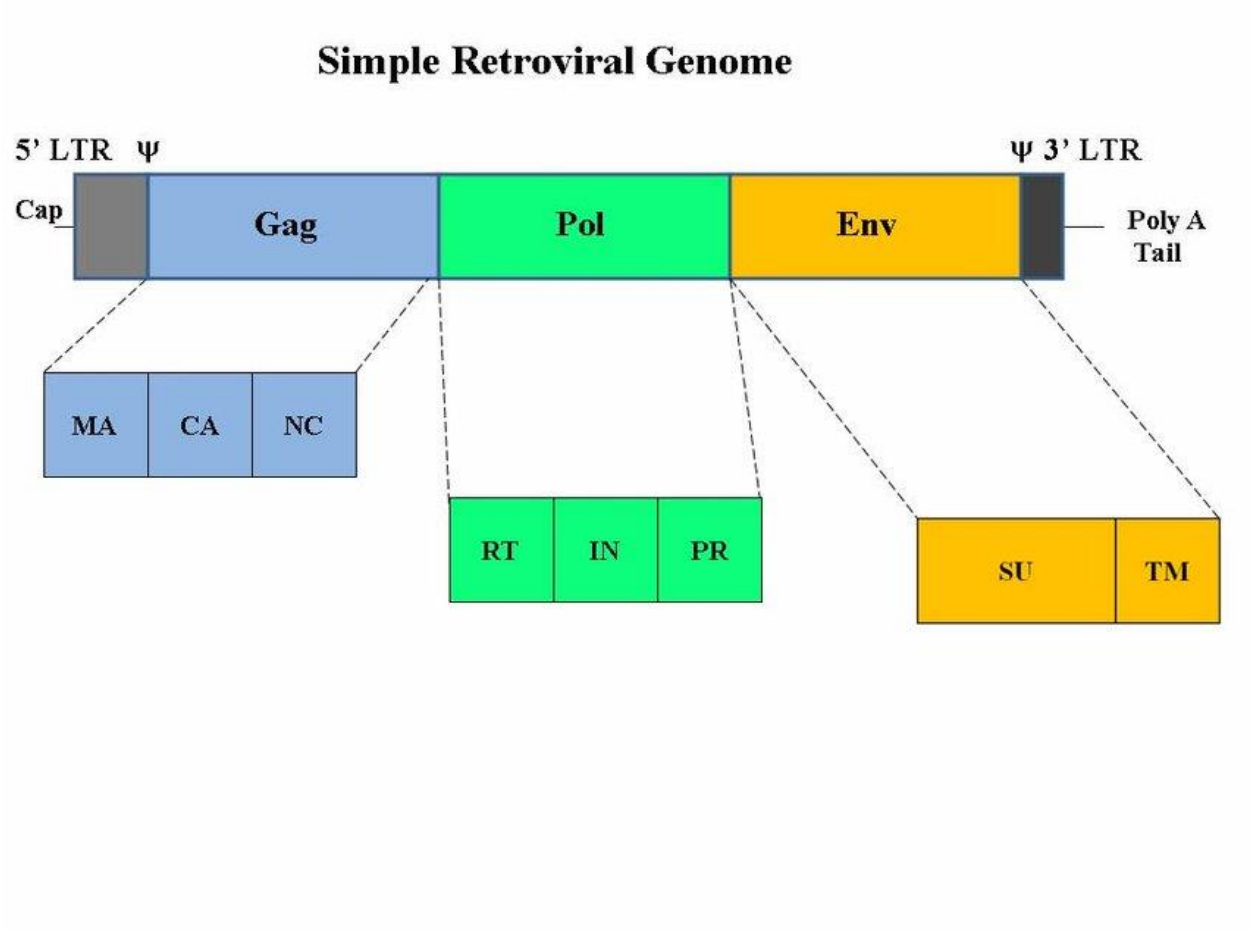


Figure: Structure of the genome of a simple retrovirus.

The retrovirus reverse transcribed and integrated DNA is transcribed into fresh RNA molecules which reenter the cytosol where some are translated by host ribosomes.

- The env RNA is translated into molecules of the envelope protein (gp160). These pass through the endoplasmic reticulum and then the Golgi apparatus where they become glycosylated by enzymes of the host cell.
- Proteases of the host cell then cut gp160 into
    -
        - gp120 which sits on the surface of the virions (and is the target of most of the vaccines currently being tested).
        - gp41, a transmembrane protein associated with gp120.
- the gag and pol genes are translated into a single protein molecule which is cleaved by the viral protease into

- 6 different capsid proteins
- the protease
- reverse transcriptase
- the integrase

Lesson 75: Retroviruses life cycle

Retrovirus Lifecycle

Life cycle of retroviruses

The life cycle of retroviruses is arbitrarily divided into two distinct phases:

1. **The early phase** refers to the steps of infection from cell binding to the integration of the viral cDNA into the cell genome. During the long journey from the cell surface to the nucleus, retroviruses will face multiple obstacles, since in addition to finding a path through the cytoplasm to the nucleus they have to cross two main barriers, the plasma and nuclear membranes, whilst at the same time avoiding or counteracting cellular defenses that can interfere with many of these steps.

- Retroviral particles must bind specifically to their target cells
- cross the plasma membrane
- reverse-transcribe their RNA genome
- Uncoat the cores
- find their way to the nuclear membrane
- penetrate into the nucleus

2. The late phase begins with the expression of viral genes and continues through to the release and maturation of progeny virions. The synthesis of virus proteins only begins efficiently once the provirus is integrated into the host DNA.

- Transcription
- Transduction
- Assembly
- Maturation
- Budding
- Release

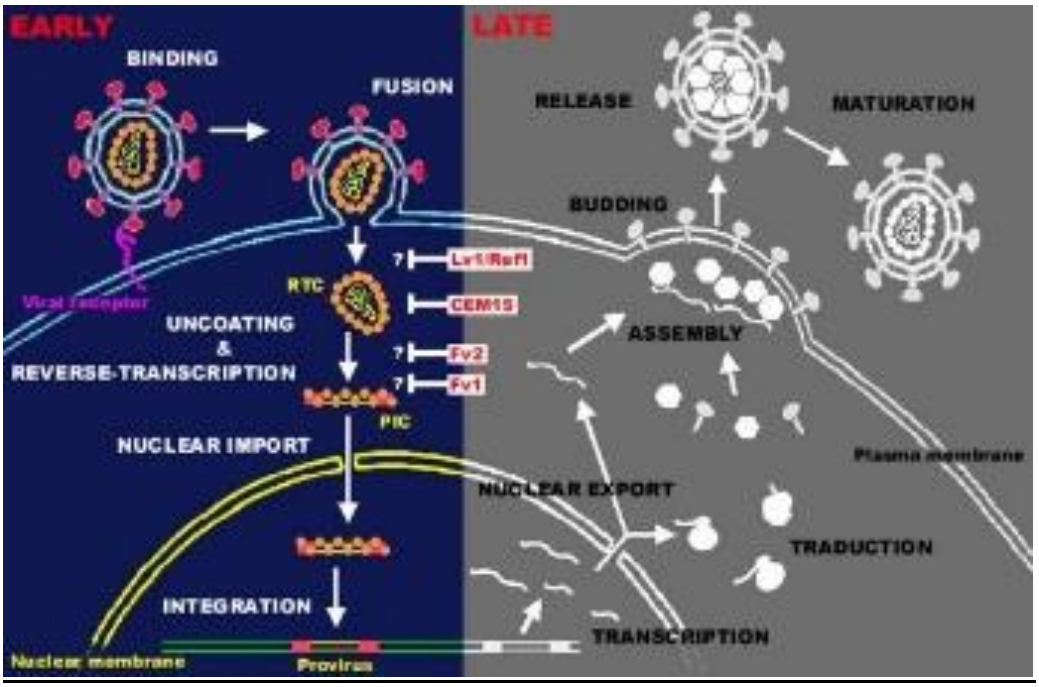


Figure: **The retroviral life cycle**. A schematic view of early and late stages of the retroviral replication cycle is represented. Examples of cellular factors interfering with early steps are indicated: Lv1/Ref1; CEM15, also known as APOBEC3G (apolipoprotein B mRNA-editing enzyme-catalytic polypeptide-like-3G); Fv2; Fv1. The question marks indicate the exact step affected by the restriction factors has not precisely been determined. Lv1 and Ref1 block incoming particles before reverse-transcription whereas Fv1 and Fv2 act at a stage between reverse-transcription and integration. Abbreviations: RTC, reverse transcription complex; PIC, pre-integration complex.

Lesson 76: Positive sense RNA Virus

LUTEOVIRIDAE

- Sobemovirus

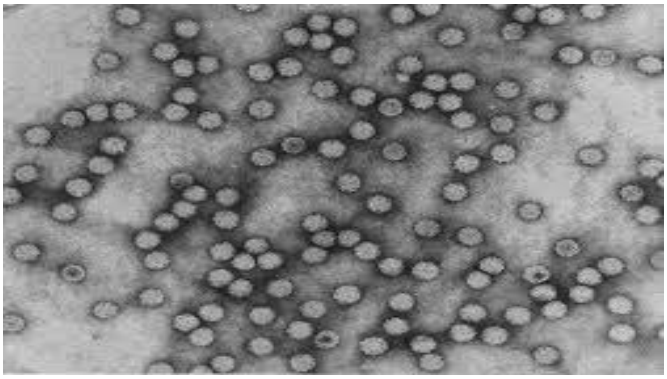


SEQUIVIRIDAE

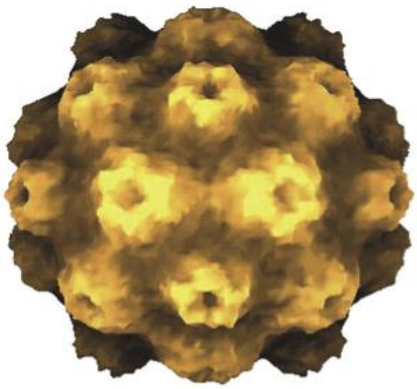
- Sadwavirus
- Chevavirus

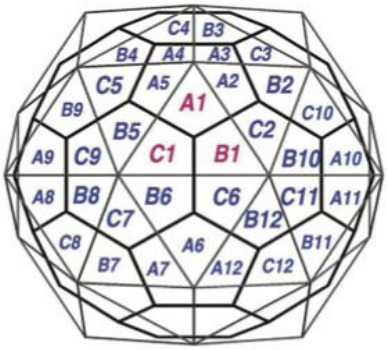


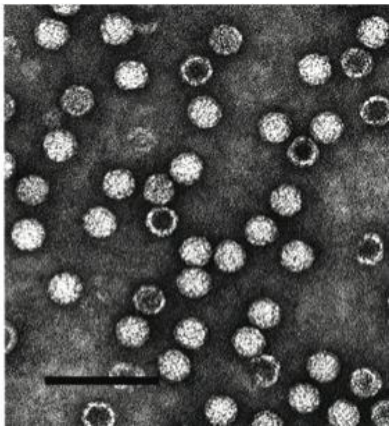
TOMBUSVIRIDAE



TYMOVIRIDAE







COMOVIRIDAE

- Idaeovirus



BROMOVIRIDAE

- Cucumovirus
- Bromovirus
- Ilavirus
- Alfamovirus
- Ourmiavirus
- Tobamovirus
- Hordievirus
- Pecluvirus
- Furovirus
- Pomovirus
- Bemivirus



OTHERS…

- Flexiviridae
- Potyviridae
- Closteroviridae

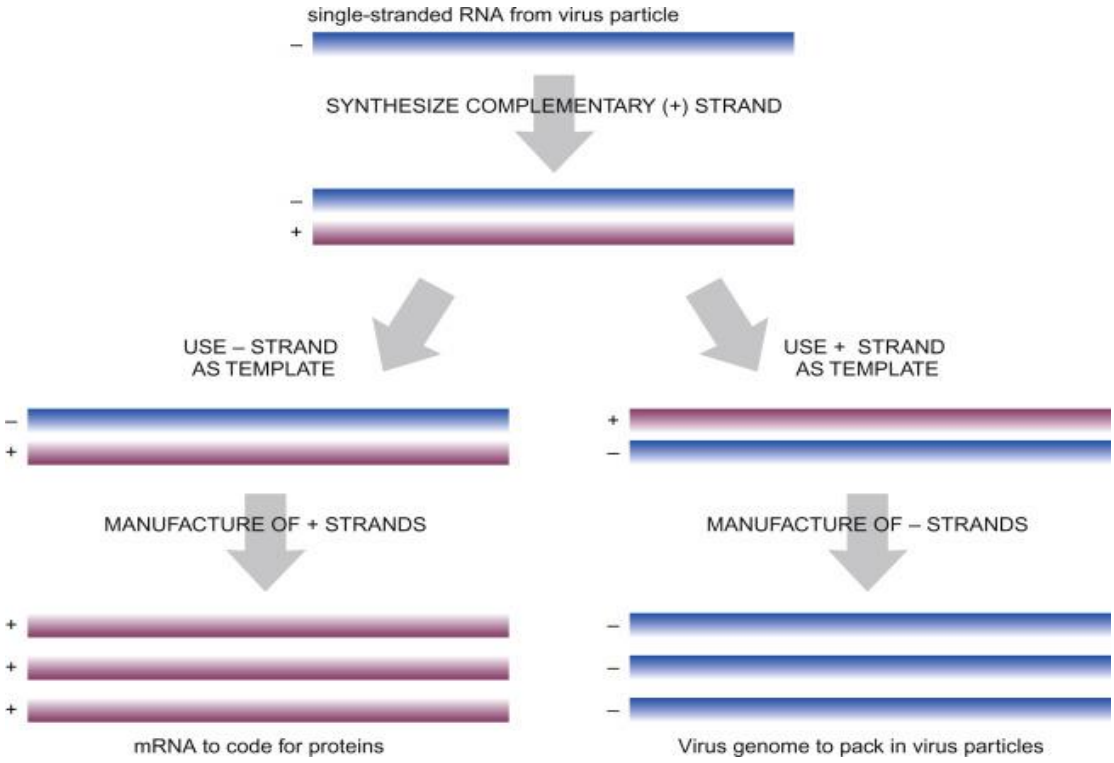
Lesson 77: Negative sense RNA viruses

Negative sense RNA Viruses

Negative-strand RNA virus, also known as an antisense-strand RNA virus, is a virus whose genetic information consists of a single strand of RNA that is the negative or antisense strand which does not encode mRNA (messenger RNA).

The RNA found in a negative-sense virus is not infectious by itself, as it needs to be transcribed into positive-sense RNA. The complementary plus-sense mRNA must be made before proteins can be translated from the viral genome. This RNA negative-strand to positive-strand copying is carried out by an RNA-dependent RNA-polymerase (RNA replicase). Each virion that has one negative-strand copy can be transcribed to several positive-sense RNAs.

The negative-strand RNA viruses of animals are divided into several families and include the agents of well-known diseases such as rabies, mumps, measles, and influenza as well as more exotic emerging pathogens such as Ebola virus. In all of these, the single-stranded RNA in the virus particle is complementary to the mRNA and is therefore the minus strand. These viruses vary in shape and structure but are similar in having an outer envelope derived from the membrane of the host cell where they were assembled.



Family	Subfamily	Genus	Species
Bornaviridae		*Bornavirus*	Borna disease virus
Bunyaviridae		*Bunyavirus* *Hantavirus* *Phlebovirus*	Bunyamweravirus Hantaanvirus Rift valley fever
Orthoviridae		*Influenza virus A,B,C* *Thogotovirus*	Influenza A,B,C Thogotovirus
Paraviridae	*Paramyxovirinae* *Pneumovirinae*	*Respirovirus* *Morbillivirus* *Rubulavirus* *Pneumovirsu* *Metapneumovirus*	Sendai virus Measles Virus Mumps virus Human respiratory synctial virus
Rhabdoviridae		*Lyssavirus* *Vessiculovirus* *Ephemerovirus* *Novirhabdovirus*	Rabies virus Vessicular Stomatitis Virus Bovine Ephemeral fever virus

Lesson 78: Bunyaviridae characters

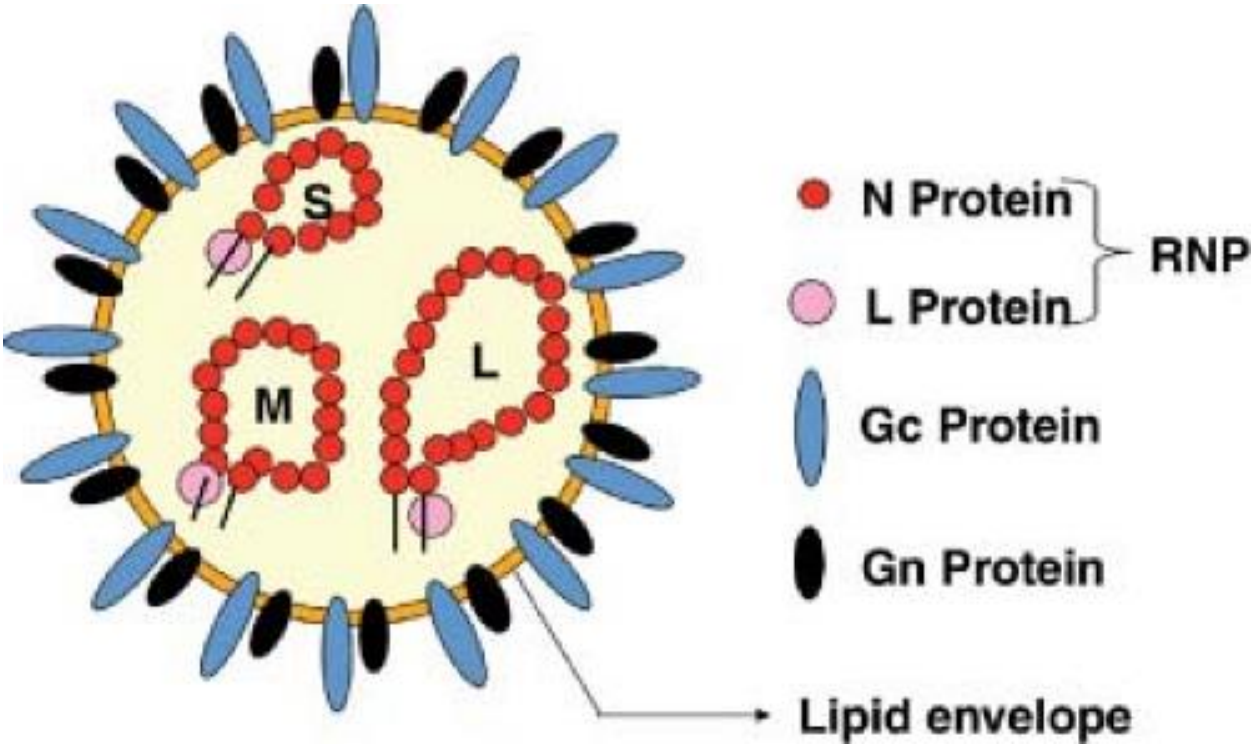
Bunyaviridae Characters

Clinical Manifestations

Bunyaviruses cause fevers sometimes with rash. In addition, Crimean-Congo hemorrhagic fever virus may cause hemorrhage; Rift Valley fever virus may cause hemorrhagic hepatitis, encephalitis, or blindness.

Structure

Bunyaviruses are spherical, enveloped particles 90 to 100 nm in diameter. They contain three segments of antisense (and sometimes ambisense) single-stranded RNA combined with nucleoprotein. Two external glycoproteins form surface projections.



Classification and Antigenic Types

There are five genera 5 genera:

1. Orthobunyavirus
2. Hantavirus
3. Nairovirus

4. Phleovirus
5. Tospovirus

Multiplication

Bunyaviruses replicate in the cytoplasm. Their RNA genome is transcribed to mRNA. The host RNA sequence in some representative viruses primes viral mRNA synthesis. Bunyaviruses mature by budding into vesicles at or near the Golgi apparatus. Reassortment of RNA segments occurs between closely related members.

Epidemiology

The distribution of each disease is determined by the distributions of the vector and vertebrate host. Except for hantaviruses, biologic transmission is by a tick, mosquito, midge, or sand fly vector. Arthropods are infected for life. Transovarial transmission is common in arthropods. Wild or domestic vertebrates usually are needed to maintain the cycle. Humans are usually dead-end hosts for all these viruses except phleboviruses. There is a secondary nosocomial spread of Crimean-Congo hemorrhagic fever. Hantaan virus cycles among rodents, probably by aerosol or fomite transmission from infected rodent urine. Human infection is incidental.

DISEASES

Family: Peribunyaviridae

California encephalitis virus, La Crosse encephalitis virus, Jamestown Canyon virus and Snowshoe hare virus vector: mosquitoes

Family: Nairoviridae

Crimmean Hemmorhegic Congo fever

Vector: ticks

Family:Phenuiviridae

Rift velley fever

Vector: mosquitoes

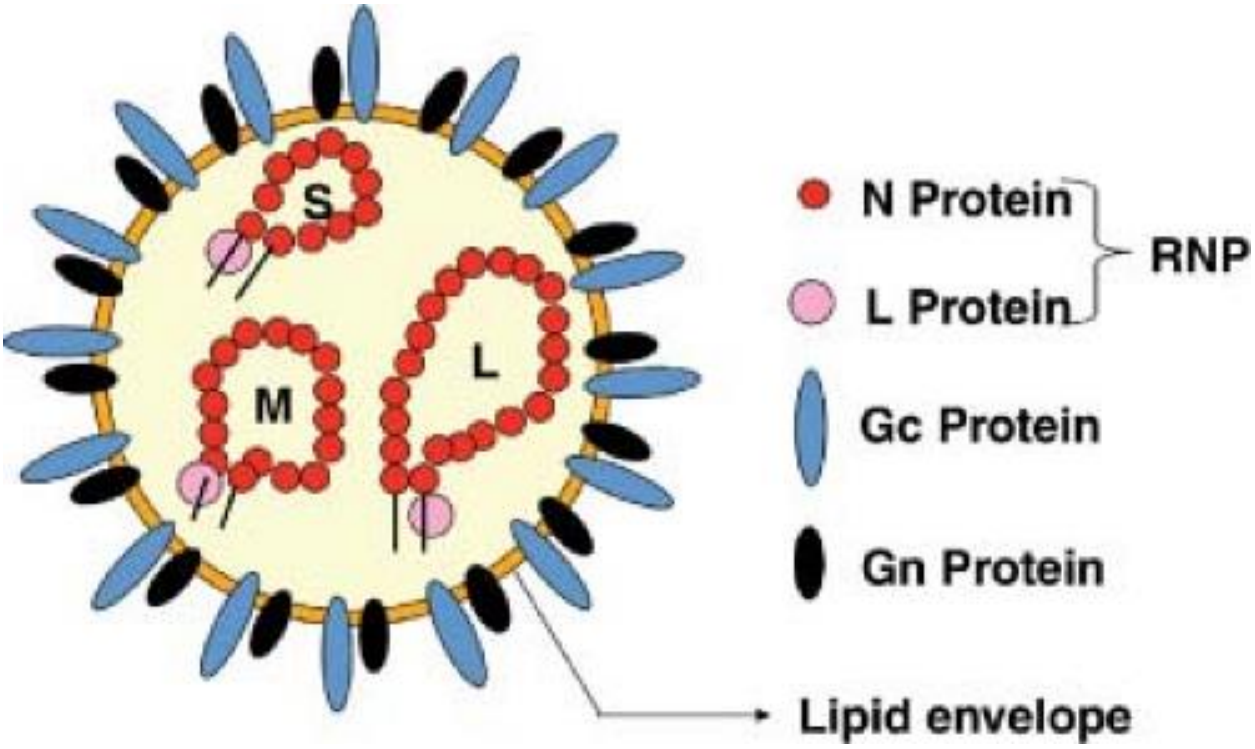
Family:Hantaviridae

- Hantavirus
- Vector: aerosolized excreta from these mammals

Lesson 79: Bunyaviridae genome

Bunyaviridae Genome

Bunyavirus genomes consist of **three RNA segments** called large, medium, and small (L, M, and S, respectively) that encode four structural proteins (RNA polymerase or L protein; two glycoproteins, Gn and Gc; and nucleocapsid protein, N), as well as one or two nonstructural proteins (NSm and NSs). Each genome segment has the same basic structure with the coding region flanked by untranslated regions (UTRs) at the 5′ and 3′ ends. Although bunyaviruses are classed as negative-strand viruses, some viruses have genome segments with an ambisense coding strategy. Phleboviruses, for instance, have S genome segments in which the N open reading frame (ORF) is in the negative sense, whereas the NSs ORF is in the positive sense.



Functions of segments

The L segment encodes the RNA Dependent RNA-polymerase, necessary for viral RNA replication and mRNA synthesis.

The M segment encodes the viral glycoproteins, which project from the viral surface and aid the virus in attaching to and entering the host cell.

The S segment encodes the nucleocapsid protein (N).

Segment	Size
L	6.3-12
M	3.5-6
S	1-2.2

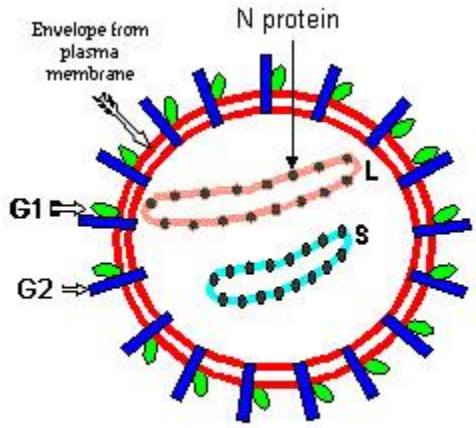
Lesson 80: Arenavirus

Arenavirus

The arenaviruses are a family of viruses that are usually transmitted from rodents to humans and include viruses such as lymphocytic choriomeningitis (LCM), Lassa virus, Junin virus, and Machupo virus.

Structure

The virus is round, oval, or pleomorphic, 60 to 300 nm in size, and enveloped. The genome consists of two distinct single-stranded viral RNA species, called L and S. The arenaviruses have ambisense genomes: the 3' half is antisense, whereas the 5' half is positive-sense. Cellular ribosomes appear to be incorporated into the virion.



Arenavirus particle

Classification and Antigenic Types

Of the fifteen arenaviruses known to infect animals, eight cause disease in humans: Lassa virus, Junin virus, Machupo virus, Guanarito virus, and lymphocytic choriomeningitis virus. All arenaviruses contain a set of internal cross-reacting antigens as well as species-specific envelope antigens.

Multiplication

Arenaviruses are thought to multiply like typical antisense RNA viruses (e.g. bunyaviruses).

Disease

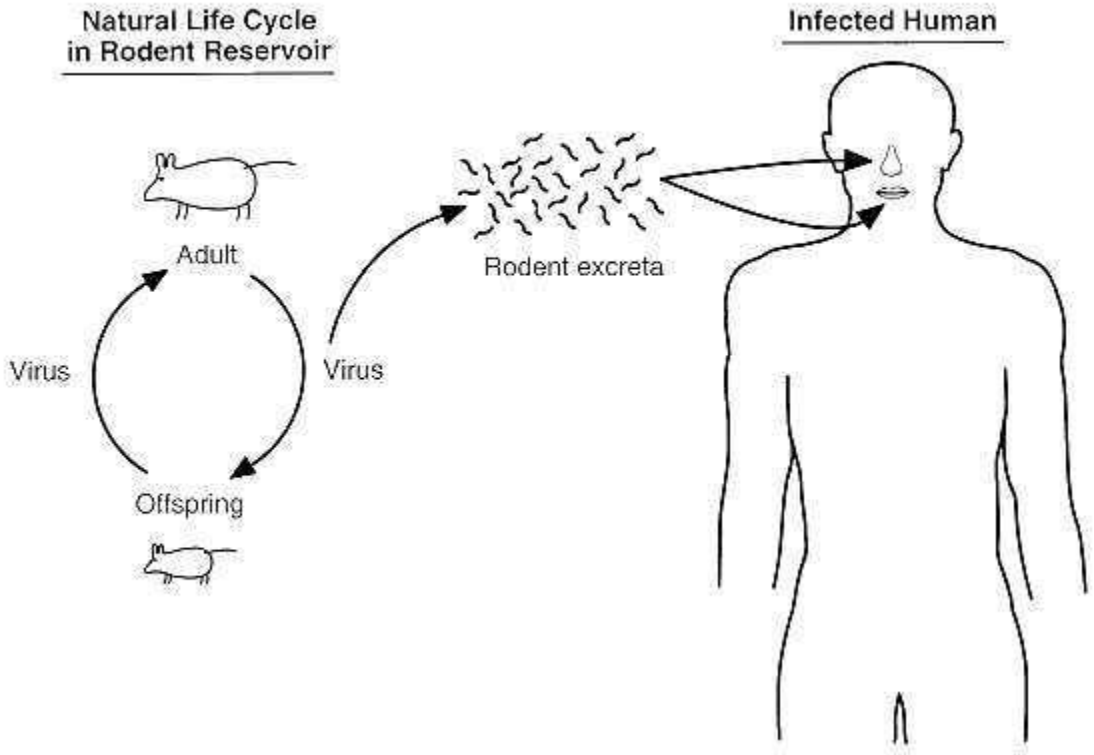
- Lymohocytic Choriomeningitis Virus infection
- Hemmorhegic fever from a number of virus belonging to this family

Pathogenesis

Infection is suspected to be caused by invasion through broken skin or the aerosol/respiratory route. The gross pathology caused by these diseases is unimpressive and is of little help in explicating the pathogenetic mechanism.

Epidemiology

The arenaviruses that affect humans exist in nature as benign infections in restricted rodent hosts; human disease is usually due to contact with rodent excreta.



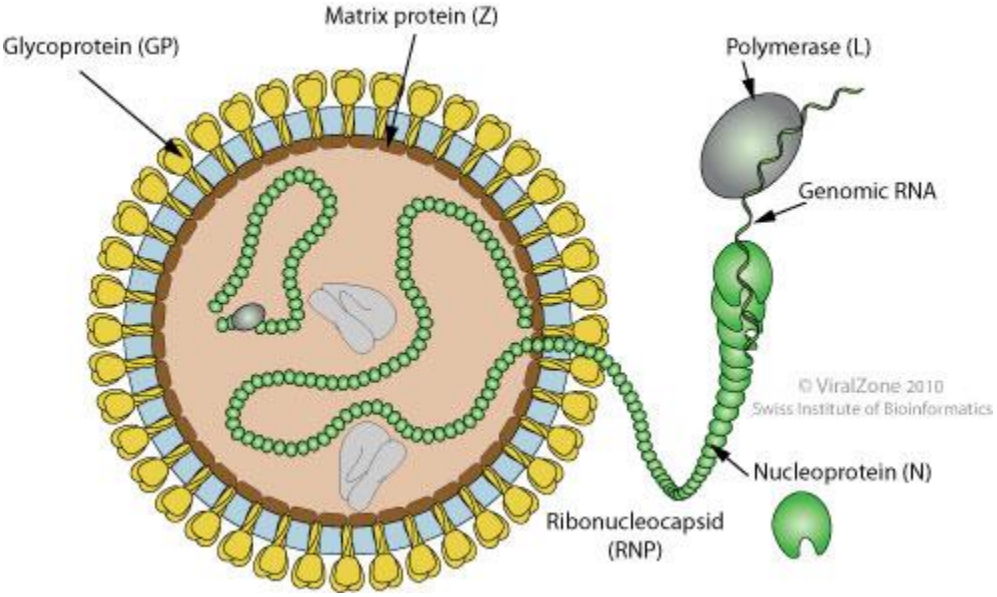
Lesson 81: Arenavirus genome

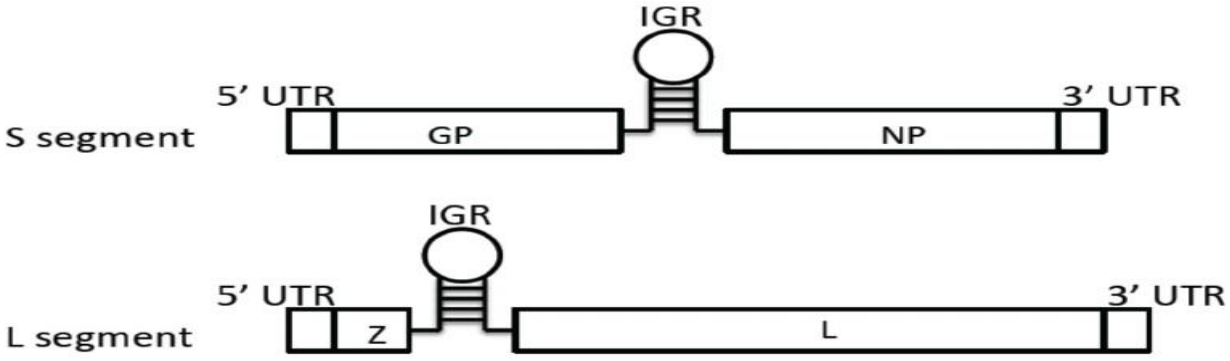
Arenavirus genome

Arenaviruses are enveloped RNA viruses with a single-stranded **ambisense** genome that is composed of **two segments**. The genome that encodes for four proteins.

L segment is about 7.2 kb and S segment 3.5 kb. The genomic L (large) segment encodes the Z matrix protein and the L polymerase protein

S (small) segment encodes the glycoprotein (GP) and nucleoprotein (NP). The genes encoded on each segment are separated by noncoding intergenic regions (IGR). The linear genomes are flanked by the conserved 5’ and 3’ untranslated regions (UTRs).





GENE EXPRESSION

Both segments are ambisens. The viral RNA dependent RNA polymerase (L) binds to a promoter on each encapsidated segment, and transcribes a messager RNA. Transcription is terminated by a strong hairpin sequence at the end of each gene. mRNAs are capped, probably by L protein during synthesis.

- The Z protein forms homo oligomers and a structural component of the virions.
- The formation of these oligomers is an essential step for particle assembly and budding. Binding between Z and the viral envelope glycoprotein complex is required for virion infectivity.
- Z also interacts with the L and NP proteins.
- Polymerase activity appears to be modulated by the association between the L and Z proteins.
- Interaction between the Z and NP proteins is critical for genome packaging.

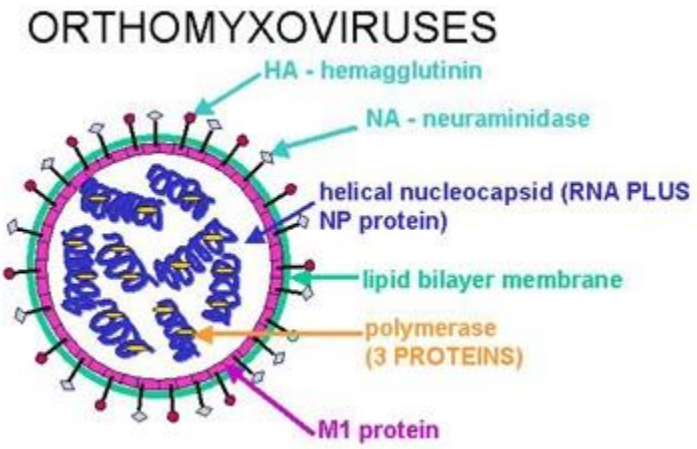
Lesson 82: Orthomyxoviridae

Orthomyxoviridae

Greek word “Orthos” meaning proper and “Myx” meaning mucus

Morphology

- Spherical/ pleomorphic
- Enveloped
- 80-120nm
- Spikes on Influenza A and B
- Labile to heat, pH variations, detergents



Influenza A viruses have a complex structure and possess a lipid membrane derived from the host cell. This envelope harbors the hemagglutinin (HA), the neuraminidase (NA), and the M2 proteins that project from the surface of the virus.

CLASSIFICATION

The family of *Orthomyxoviridae* is defined by viruses that have a negative-sense, single-stranded, and segmented RNA genome. The definition of negative-sense RNA viruses came from work by David Baltimore, who showed that the packaged genome of this class of viruses is complementary to the messenger RNA (mRNA), which is defined as positive.

There are following different genera in the family of *Orthomyxoviridae:*

Influenza viruses A

Influenza viruses B

Influenza viruses C

Influenza viruses D

Thogotovirus

Isavirus

Quaranilvirus

Diseases

- Influenza A infects humans, other mammals, and birds, and causes all flu pandemics
- Influenza B infects humans and seals
- Influenza virus C infects humans, pigs, and dogs

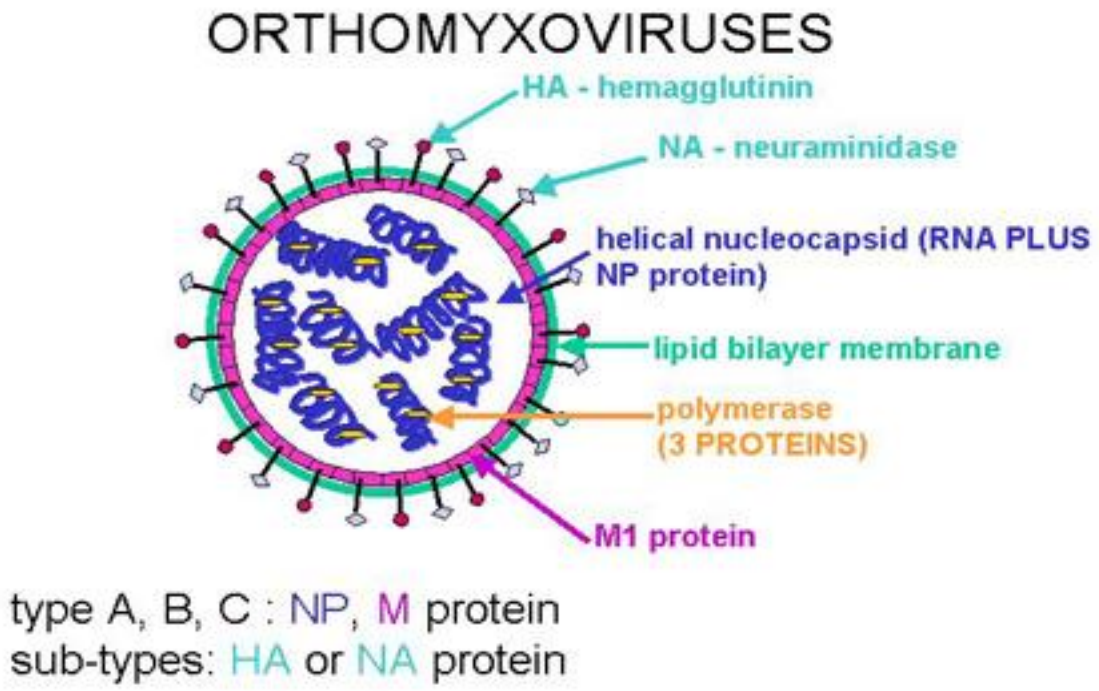
- Influenza virus D infects cattle and pigs

Lesson 83: Orthomyxoviridae genome

Orthomyxoviridae Genome

Influenza viruses are spherical, although filamentous forms may also occur. The structure of a type A or B influenza virus. The antisense RNA genome occurs in eight separate segments containing 10 genes. The segments are complexed with nucleoprotein to form a nucleocapsid with helical symmetry. The nucleocapsid is enclosed in an envelope consisting of a lipid bilayer and two surface glycoproteins, a hemagglutinin and a neuraminidase. Because influenza viruses are enveloped, they are readily inactivated by nonpolar solvents and by surface-active agents. The influenza C virus is less well studied, but is known to contain only seven RNA segments and a single surface glycoprotein.

The segmented genome is encapsidated with each segment in a separate nucleocapsid called as Ribonucleoprotein complexes (RNPs). Terminal sequences of RNA bound to trimeric RdRp. All these RNPs are surrounded by one envelope.



segment	Protein
PB1	Polymerase
PB2	Polymerase
PA	Polymerase
HA	Hemagglutinin
NP	Nucleoprotein
NA	Neuraminidase
M	Membrane protein(s)
NS	non-structural protein(s)

REPLICATION

- Entrance through body fluids
- Virus binds through hemagglutinin-glycoprotein and sialic acid receptors
- Virus enters by endocytosis
- Proteins and polymerase forms complex
- New protein synthesis
- Replication in the nucleus
- Eight segments of genome formed

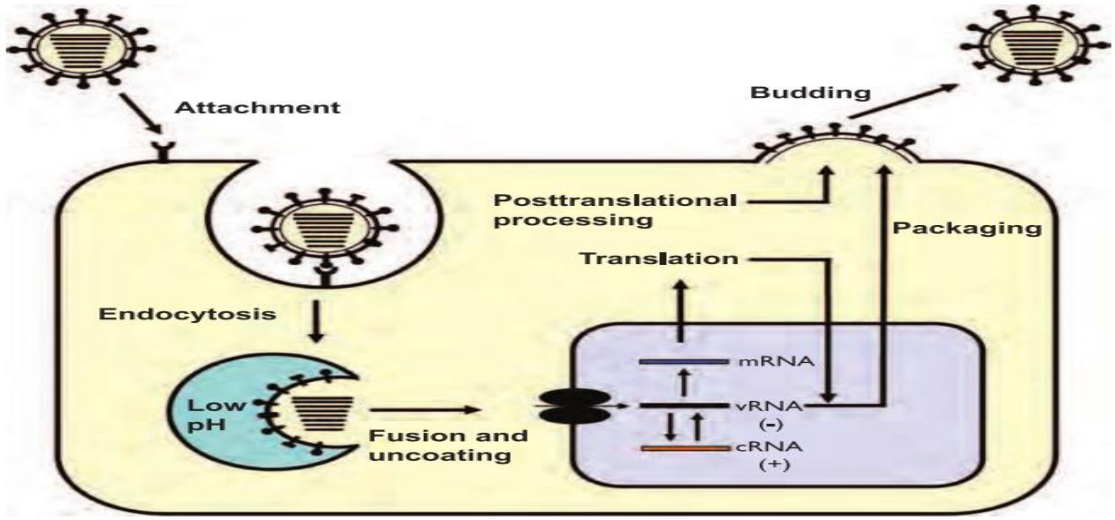


Figure: Illustration of the influenza virus replication cycle. Upon binding at the cell surface, the virus is internalized by receptor-mediated endocytosis. The low pH in the endosome triggers fusion of the viral and endosomal membranes, releasing the viral ribonucleoproteins (vRNPs) into the cytoplasm. vRNPs are imported into the nucleus where they serve as the template for transcription. New proteins are synthesized from viral mRNA. The viral genome (vRNA) is replicated through a positive-sense intermediate (complementary RNA [cRNA]). Newly synthesized viral RNPs are exported from the nucleus to the assembly site at the apical plasma membrane, where virus particles bud and are released.

Lesson 84: Paramyxoviridae

Paramyxoviridae

Nomenclature:

Para means beyond

Myxo means mucus

Paramyxo and orthomyxo were formerly known as myxoviruses

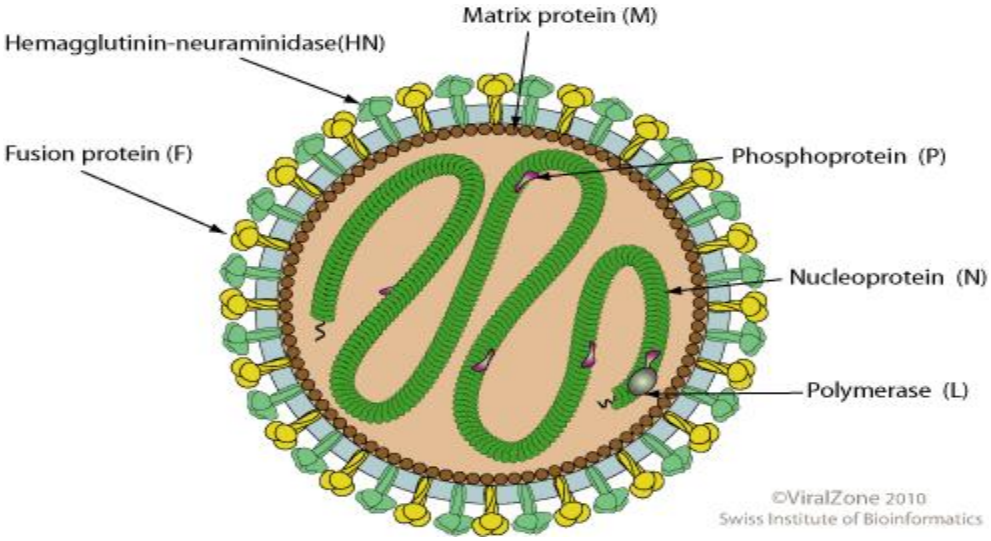
VIRION STRUCTURE

The *Paramyxoviridae* contain a lipid bilayer envelope that is derived from the plasma membrane of the host cell in which the virus is grown. *Paramyxoviridae* are generally spherical, 150 to 300 nm in diameter, but can be pleomorphic in shape, and filamentous forms can be observed. Inserted into the envelope are glycoprotein spikes that extend approximately 8 to 12 nm from the surface of the membrane, and that can be readily visualized by electron microscopy. Inside the viral membrane is the nucleocapsid core (sometimes referred to as the ribonucleoprotein [RNP] core) that contains single-stranded RNA genome.

- Spikes on envelope
- Sensitive to heat, chemicals and radiations
- Host: Vertebrates
- Genera: 7
- Species: 49

PROTEINS

- N – the nucleocapsid protein associates with genomic RNA and protects the RNA from nuclease digestion
- P – the phosphoprotein binds to the N and L proteins and forms part of the RNA polymerase complex
- M – the matrix protein assembles between the envelope and the nucleocapsid core, it organizes and maintains virion structure
- F – the fusion protein projects from the envelope surface as a trimer, and mediates entry
- H/HN/G – the cell attachment proteins span the viral envelope and project from the surface as spikes.
- L – the large protein is the catalytic subunit of RNA dependant RNA polymerase (RDRP)
- Accessory proteins – a mechanism known as RNA editing allows multiple proteins to be produced from the P gene.



Lesson 85: Paramyxoviridae genome

Paramyxoviridae genome

Paramyxoviruses contain non-segmented single stranded RNA genomes of negative polarity and replicate entirely in the cytoplasm. Their genomes are 15 to 16 kB in length, and the genomes contain 6 to 10 tandemly linked genes. A lipid envelope containing two surface glycoproteins (F and a second glycoprotein variously referred to as HN, or H or G) surrounds the virions. Inside the envelope lies a helical nucleocapsid core containing the RNA genome and the nucleocapsid

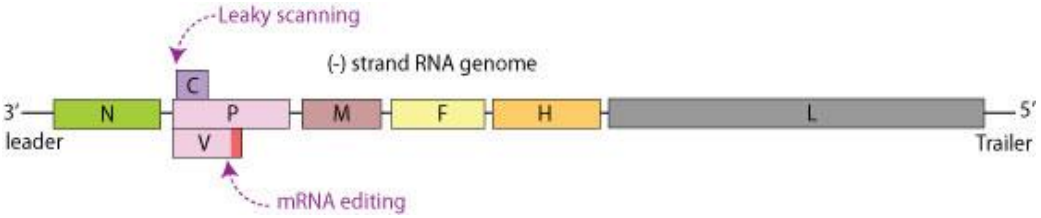
(N), phosphor (P), and large (L) proteins, which initiate intracellular virus replication. Residing between the envelope and the core lies the viral matrix (M) protein that is important in virion architecture, and which is released from the core during virus entry. In addition to the genes encoding structural proteins, paramyxoviruses contain “accessory” genes that are found mostly as additional transcriptional units interspersed with the tandemly linked invariant genes. For the *Paramyxovirinae,* the accessory genes are found mostly as open reading frames (ORFs) that overlap within the *P* gene transcriptional unit.

Extracistronic regions

- A 3’ leader sequence, , which acts as a transcriptional promoter.
- A 5’ trailer sequence, 50–161 nucleotides long

Intergenomic regions

- between each gene
- which are three nucleotides long for morbilliviruses, respiroviruses and henipaviruses
- variable length (1-56 nucleotides) for rubulaviruses



- Characteristic herring bone appearance
- Nucleocapsid core contains:
    - Genomic RNA
    - Nucleocapsid protein
    - Polymerase protein

Lesson 86: Rhabdoviruses

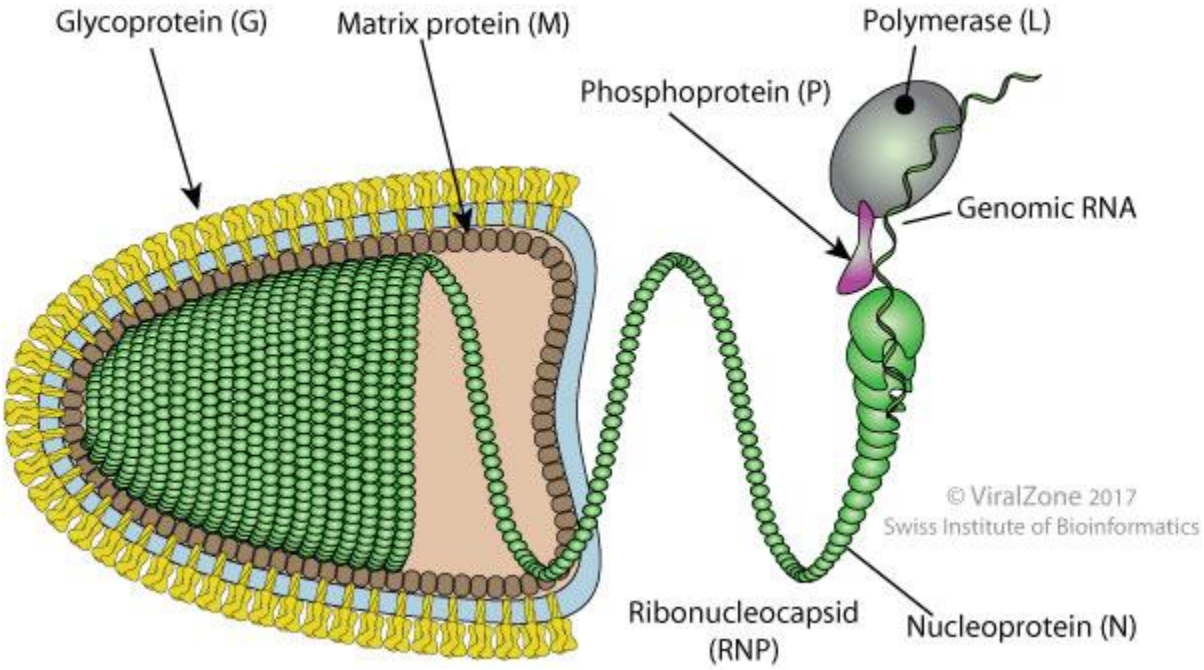
Rhabdoviruses

Nomenclature

Greek word Rhabdos meaning rod, Referring to shape of virus Mononegavirales

Virion Structure

Rhabdovirus is enveloped; bullet shaped which is 180 nm long and 75 nm wide. Certain plant rhabdoviruses are bacilliform in shape and almost twice the length. Genome contain RNA that is single-stranded, negative-sense and unsegmented. The virus genome encodes five proteins. For example, **Rabies virus** is a Rhabdovirus.



Rhabdovirus Proteins

- large protein polymerase (L)
- glycoprotein (G)
- nucleoprotein (N)
- phosphoprotein (P)
- matrix protein (M)

Clinical Manifestations

Rabies virus causes acute infection of the central nervous system. Five general stages are recognized in humans: incubation, prodrome, acute neurologic period, coma, and death. The incubation period is exceptionally variable, ranging from fewer than 10 days to longer than 2 years, but is usually 1–3 months.

Members

The family Rhabdoviridae includes following genera

- Vessicular stomatitis Indiana Virus
- Lyssavirus
- Ephemerovirus
- Cytorhabdovirus
- Nucleorhabdovirus

VESICULOVIRUS

- Infects human, cattle, horses
- Flu-like illness
- Fever
- Siezures
- Diarrhea
- Vomiting
- Coma
- Death

LYSSAVIRUS

- Infects human
- Bat and rodent carrier
- Throughout world
- Causes rabies
- Anxiety, confusion, agitation, paranoia, hallucinations
- Death

EPHEMEROVIRUS

- Infects cattle primarily
- Causes Bovine Ephemeral Fever
- Fever, stiffness, lameness
- Nasal and ocular discharges

- High morbidity and low mortality

CYTORHABDOVIRUS

- Plant viruses
- Bacilliform
- Mature in cell cytoplasm
- lettuce necrotic yellow virus
- barley yellow striate mosaic virus

NUCLEORHABDOVIRUS

- Plant viruses
- Bacilliform
- Linear genomes encoding 5 to 6 proteins
- Nuclear viral replication transmitted by insect bite

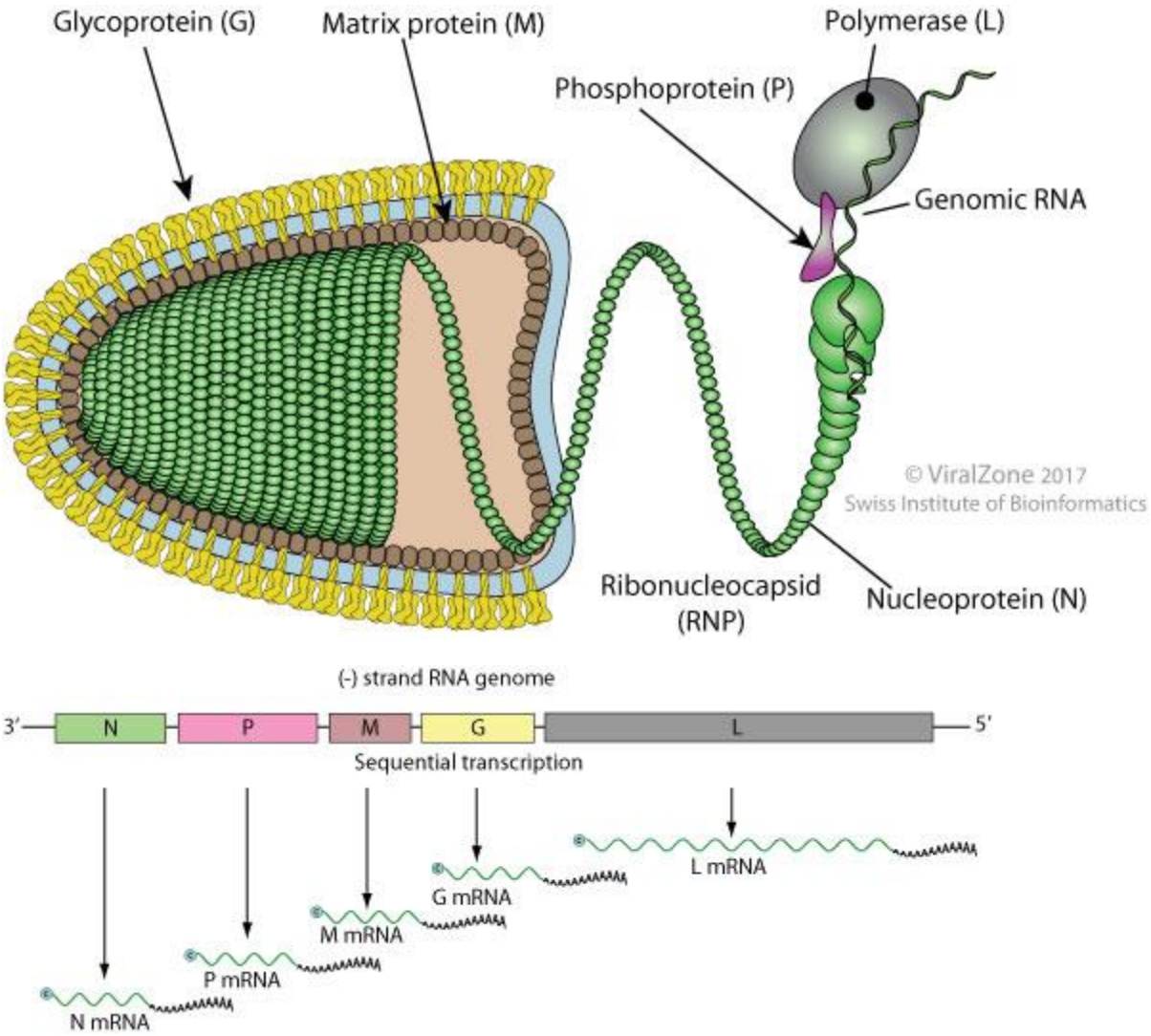
Lesson 87: Rhabdoviruses genome

Rhabdovirus genome

Rhabdovirus contains negative-stranded RNA linear genome which is about 11-15 kb in size. Its genome encodes for 5 to six proteins.

GENE EXPRESSION

The viral RNA dependent RNA polymerase binds the encapsidated genome at the leader region, then sequentially transcribes each genes by recognizing start and stop signals flanking viral genes. mRNAs are capped and polyadenylated by the L protein during synthesis.



The virus genome encodes five proteins associated with either the ribonucleoprotein (RNP) complex or the viral envelope. The L (transcriptase), N (nucleoprotein), and NS (transcriptase-associated) proteins comprise the RNP complex, together with the viral RNA. These aggregate in the cytoplasm of virus-infected neurons and compose Negri bodies, the characteristic histopathologic finding of rabies virus infection. The M (matrix) and G (glycoprotein) proteins are associated with the lipid envelope. The G protein forms the protrusions that cover the outer surface of the virion envelope and is the only rabies virus protein known to induce virus-neutralizing antibody.

REPLICATION

The replication of rabies virus is believed to be similar to that of other negative-stranded RNA viruses. The virus attaches to the host cell membranes via the G protein, penetrates the cytoplasm by fusion or pinocytosis, and is uncoated to RNP. The core initiates primary

transcription messenger RNAs by using the virion-associated RNA-dependent RNA polymerase. Each RNA is then translated into an individual viral protein. After viral proteins have been synthesized, replication of the genomic RNA continues with the synthesis of full length, positive-stranded RNA, which acts as a template for the production of progeny negative-stranded RNA. 1. The ribonucleocapsid binds to the matrix protein and buds at the plasma membrane, releasing new virions.

Lesson 88: Segmented and multipartite genome

Segmented and multipartite genome

The architecture, organization, and packaging of viral genetic information can be divided into three categories: monopartite, segmented, and multipartite viruses.

Segmented virus genomes are those that are divided into two or more physically separate molecules of nucleic acid, all of which are then packaged into a single virus particle. In contrast, although multipartite genomes are also segmented, each genome segment is packaged into a separate virus particle. These discrete particles are structurally similar and may contain the same component proteins, but they often differ in size depending on the length of the genome segment packaged.

Multipartite viruses have genomes segmented in pieces enclosed in different capsids that are independently transmitted. Since all segments have to meet in the host for complementation and completion of the viral cycle, multipartite viruses are bound to fight the loss of genomic information.

ARENAVIRUS – 2 segments

BUNYAVIRUS – 3 segments

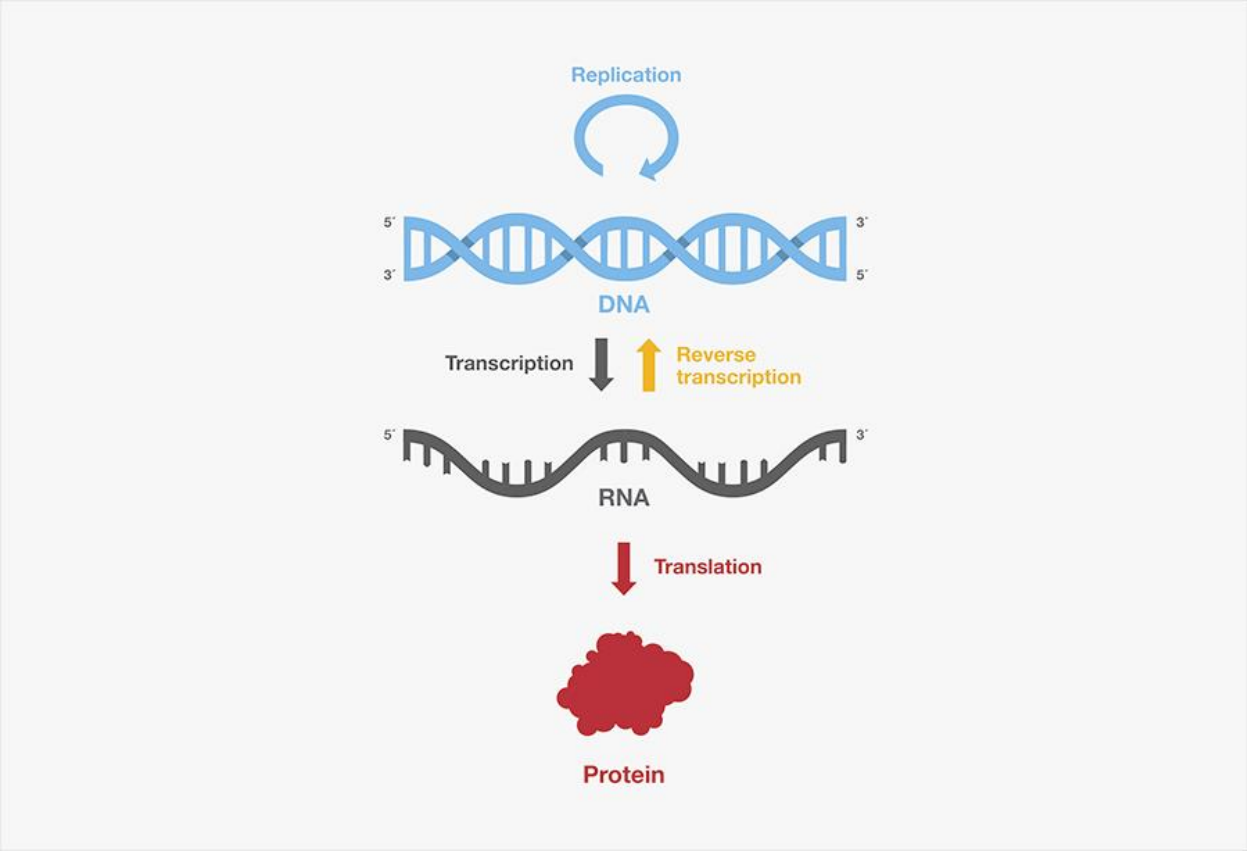
ORTHOMYXOVIRUS – 8 segments

REOVIRUS – 10 segments

Lesson 89: Reverse Transcription

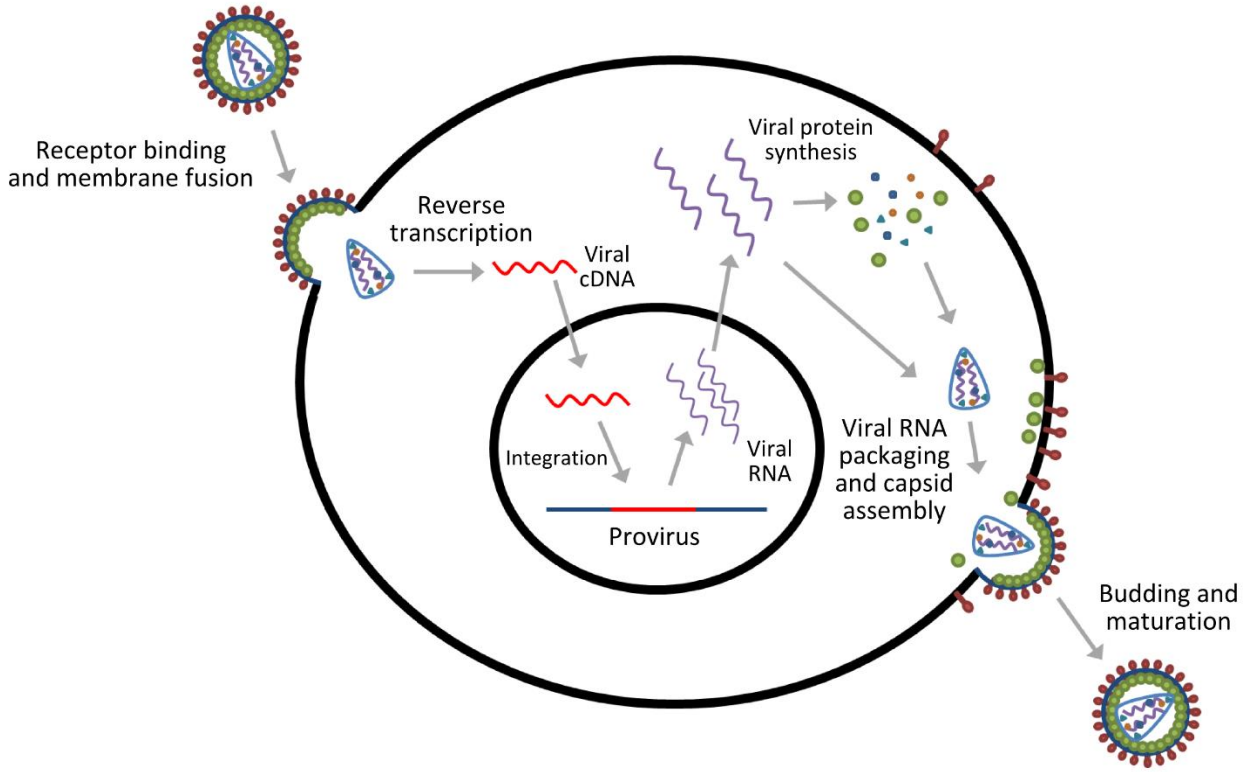
The first successes of molecular biology were the discovery of the double helix structure of DNA and in understanding of the language of the genetic code. The importance of these findings does not lie in their chemistry but in their importance in allowing predictions to be made about the fundamental nature of living organisms. The confidence that flowed from these early triumphs resulted in the development of a grand universal theory, called the “central dogma of molecular biology”—namely, that all cells (and hence viruses) work on a simple organizing

principle: the unidirectional flow of information from DNA, through RNA, into proteins. In the mid-1960s, however, there were rumblings that life might not be so simple. In 1963, Howard Temin showed that the replication of retroviruses, whose particles contain RNA genomes, was inhibited by actinomycin D, an antibiotic that binds only to DNA. The replication of other RNA viruses is not inhibited by this drug. So pleased was the scientific community with an all-embracing dogma that these facts were largely ignored until 1970, when Temin and David Baltimore simultaneously published the observation that retrovirus particles contain an RNA-dependent DNA polymerase: reverse transcriptase. This finding alone was important enough, but like the earlier conclusions of molecular biology, it has subsequently had reverberations for the genomes of all organisms, and not merely a few virus families. It is now known that retrotransposons with striking similarities to retrovirus genomes form a substantial part of the genomes of all higher organisms, including humans. Earlier ideas of genomes as constant, stable structures have been replaced with the realization that they are, in fact, dynamic and rather fluid entities.



Reverse transcriptase, also called RNA-directed DNA polymerase, an enzyme encoded from the genetic material of retroviruses that catalyzes the transcription of retrovirus RNA (ribonucleic acid) into DNA (deoxyribonucleic acid). This catalyzed transcription is the reverse process of normal cellular transcription of DNA into RNA, hence the names reverse transcriptase and

retrovirus. Reverse transcriptase is central to the infectious nature of retroviruses, several of which cause disease in humans, including human immunodeficiency virus (HIV), which causes acquired immunodeficiency syndrome (AIDS), and human T-cell lymphotrophic virus I (HTLV-I), which causes leukemia.



The synthesis of DNA from an RNA template, via reverse transcription, produces complementary DNA (cDNA). Reverse transcriptases (RTs) use an RNA template and a short primer complementary to the 3' end of the RNA to direct the synthesis of the first strand cDNA, which can be used directly as a template for the Polymerase Chain Reaction (PCR). This combination of reverse transcription and PCR (RT-PCR) allows the detection of low abundance RNAs in a sample, and production of the corresponding cDNA, thereby facilitating the cloning of low copy genes. Alternatively, the first-strand cDNA can be made double-stranded using DNA Polymerase I and DNA Ligase. These reaction products can be used for direct cloning without amplification. In this case, RNase H activity, from either the RT or supplied exogenously, is required.

Many RTs are available from commercial suppliers. Avian Myeloblastosis Virus (AMV) Reverse Transcriptase and Moloney Murine Leukemia Virus (M-MuLV, MMLV) Reverse Transcriptase are RTs that are commonly used in molecular biology workflows.

Lesson 90: Transposons

Transposons

Transposable elements (TEs), also known as "jumping genes," are DNA sequences that move from one location on the genome to another. These elements were first identified in 1940 by geneticist Barbara McClintock of Cold Spring Harbor Laboratory in New York. Biologists were initially skeptical of McClintock's discovery. Over the next several decades, however, it became apparent that not only do TEs "jump," but they are also found in almost all organisms (both prokaryotes and eukaryotes) and typically in large numbers. For example, TEs make up approximately 50% of the human genome and up to 90% of the maize genome. They may

- cause mutations
- Increase (or decrease) the amount of DNA in the genome of the cell, and if the cell is the precursor of a gamete, in the genomes of any descendants.

Most transposons eventually become inactive and no longer move.

Types of Transposons

- **Class II transposons/ Simple transposons**: These consist of DNA that moves directly from place to place.
- **Class I transposons**. These are **retrotransposons** that
    - First transcribe the DNA into RNA and then
    - Use reverse transcriptase to make a DNA copy of the RNA to insert in a new location.

Class II Transposons

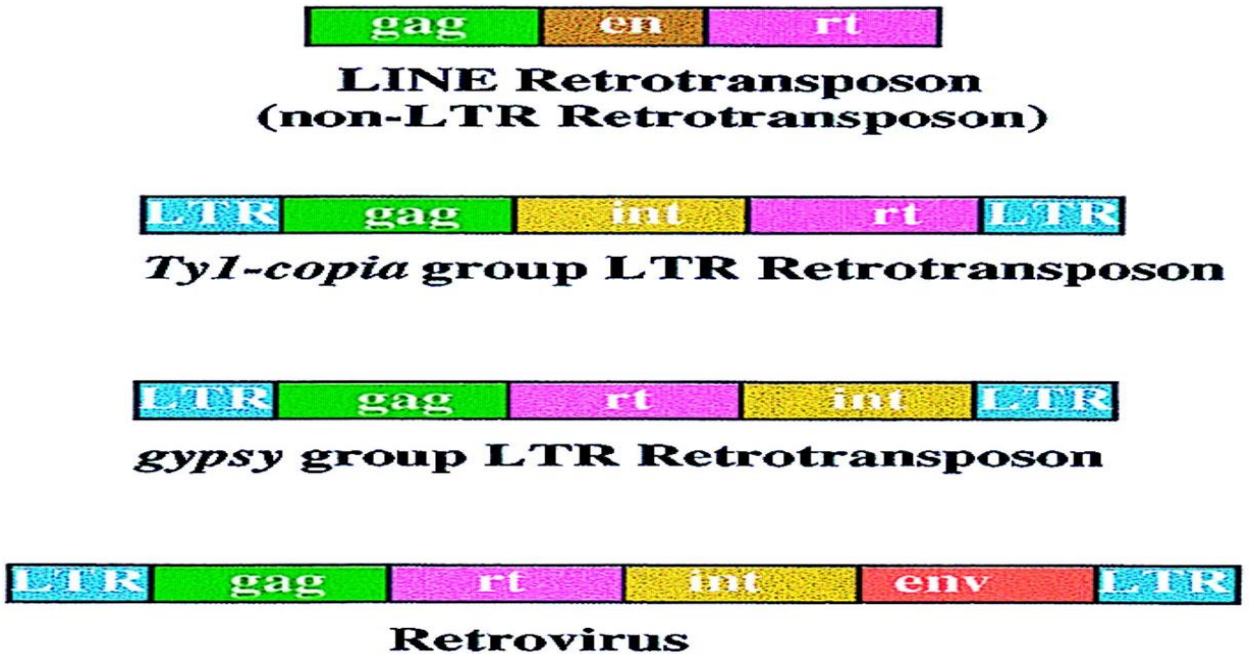
Class II transposons move by a "**cut and paste**" process: the transposon is cut out of its location and inserted into a new location. This process requires an enzyme — a **transposase** — that is encoded within some of these transposons.

Retrotransposons

Retrotransposons represent a highly unique group of transposable elements and form large portions of the genomes of many eukaryotes (organisms with cells containing a clearly defined nucleus). Retrotransposons function by a “copy and paste” mechanism. Thus, they leave behind the original copy and generate a second copy that is inserted elsewhere in the genome. This process results in the insertion of repetitive sequences of DNA throughout the genome and

is the mechanism responsible for the vast spread of transposable elements in many higher organisms.

Retrotransposons may look a lot like retroviruses, but they are not by themselves infectious. The basic difference between a retrotransposon and a retrovirus is the retrotransposon's lack of an envelope glycoprotein gene.



Lesson 91: Isolation and identification of viruses

Isolation, Cultivation and Identification of Viruses

Virus isolation is a very specific method for diagnosis of viral infection, but culture techniques need a week or more for completion and require specialized laboratory equipment and skills.

VIRUS ISOLATION

- Centrifugation
    - Differential centrifugation- high vs low speed to separate cells from viruses
    - Gradient centrifugation - separate by size or density

CULTIVATION

Viruses are obligate intracellular parasites that require living cells in order to replicate. Cultured cells, eggs and laboratory animals may be used for virus isolation. Although embryonated eggs and laboratory animals are very useful for the isolation of certain viruses, cell cultures are the sole system for virus isolation in most laboratories. The development of methods for cultivating animal cells has been essential to the progress of animal virology.

IDENTIFICATION

1. Plaque formation
2. Cytopathic Effects
3. Fluorescent Antibody technique
4. Nucleic Acid hybridization
5. Electron Microscopy

1. **Plaque Formation**

- a visible structure formed within a cell culture
- The bacteriophage viruses replicate and spread, thus generating regions of cell destructions known as plaques

2. **Cytopathic Effects**

    - **Syncytia:** Cell fusion (syncytium or polykaryon formation) involves the fusion of the plasma membranes of four or more cells to produce an enlarged cell with four or more nuclei.
    - **Cell rounding**
    - **Membrane proliferations**
    - **Vacuolization: Vacuoles in cytoplasm**
    - **Inclusion bodies**
    - **Focus (foci)**
    - **Hemadsorption:** adherence of red blood cells to the surface of something (as a virus or cell).

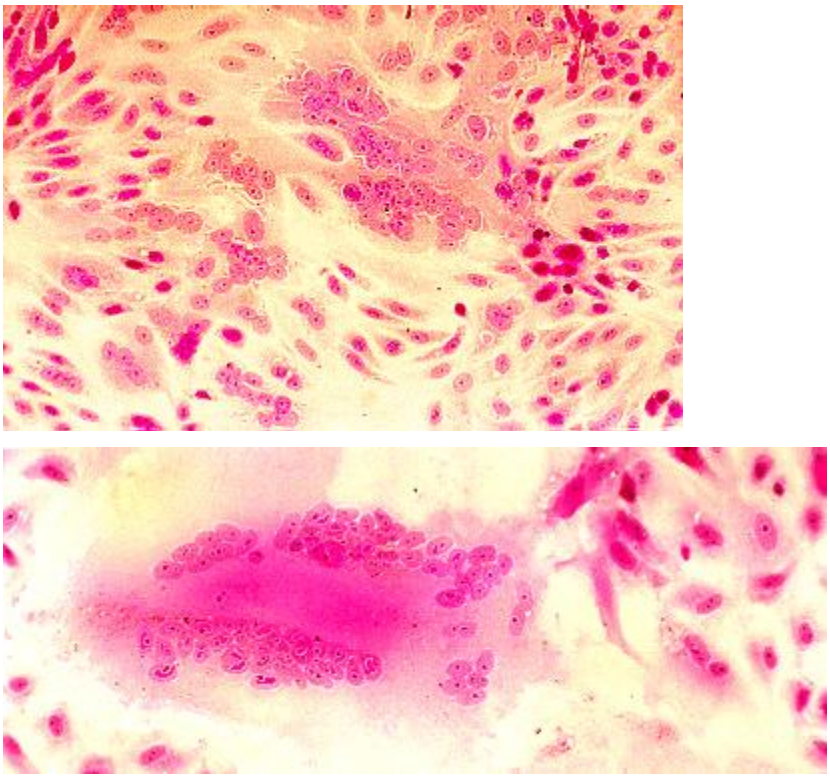


Figure: Cytopathic effects of mumps and measles viruses in cell culture: note the formation of syncytia.

3. **Immunofluorescense**

Immunofluorescence (IF) is widely used for the rapid diagnosis of virus infections by the detection of virus antigen in clinical specimens, as well as the detection of virus-specific IgG or IgA or IgM antibody. The technique makes use of a fluorescein- labelled antibody to stain specimens containing specific virus antigens, so that the stained cells fluoresces under UV illumination.

4. **Nucleic Acid Probes**

Nucleic acid probes are segments of DNA or RNA that have been labeled with enzymes, antigenic substrates, chemiluminescent moeities, or radioisotopes. They can bind with high specificity to complementary sequences of nucleic acid. Probes can be directed to either DNA or RNA targets and can be from 20 to thousands of bases long. The presence and the quantity of hybrids after hybridization is determined by the detection of the label.

5. **Electron Microscopy**

Virus diagnosis by electron microscopy relies on the detection and identification of viruses on the basis of their characteristic morphology. A major advantage of virus diagnosis by EM is the

ability to visualize the virus. By identifying the virus directly, it is possible to perform an examination without a preconceived concept of the etiological agent, in contrast with those assays which require a specific viral probe.

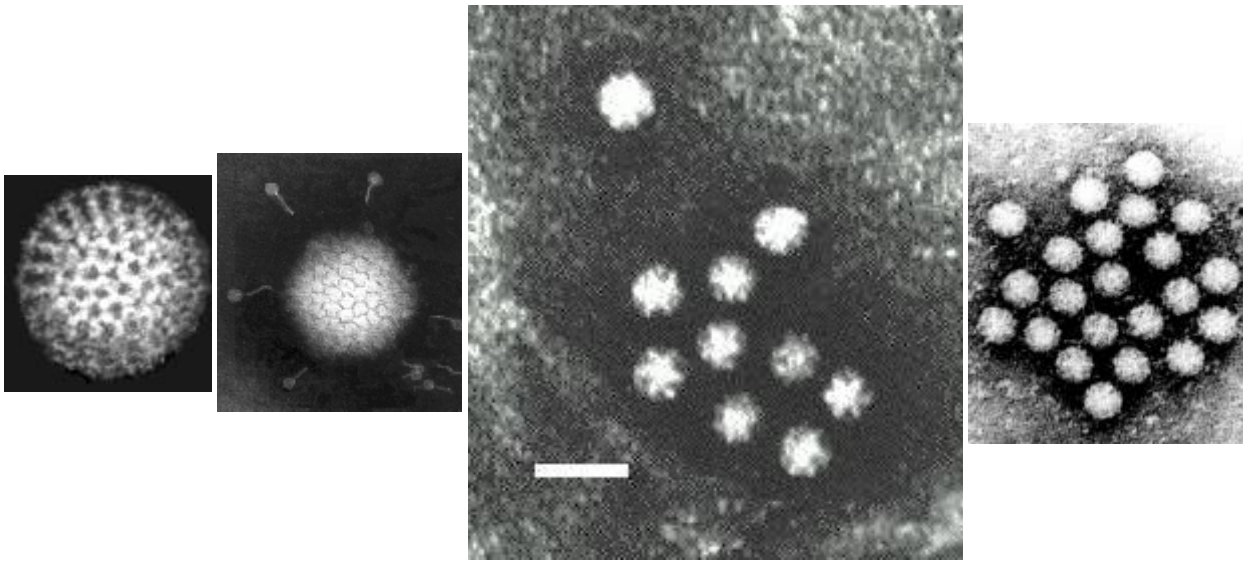


Figure: Electronmicrographs of viruses commonly found in stool specimens from patients suffering from gastroenteritis. From left to right: rotavirus, adenovirus, astroviruses, Norwalk-like viruses.

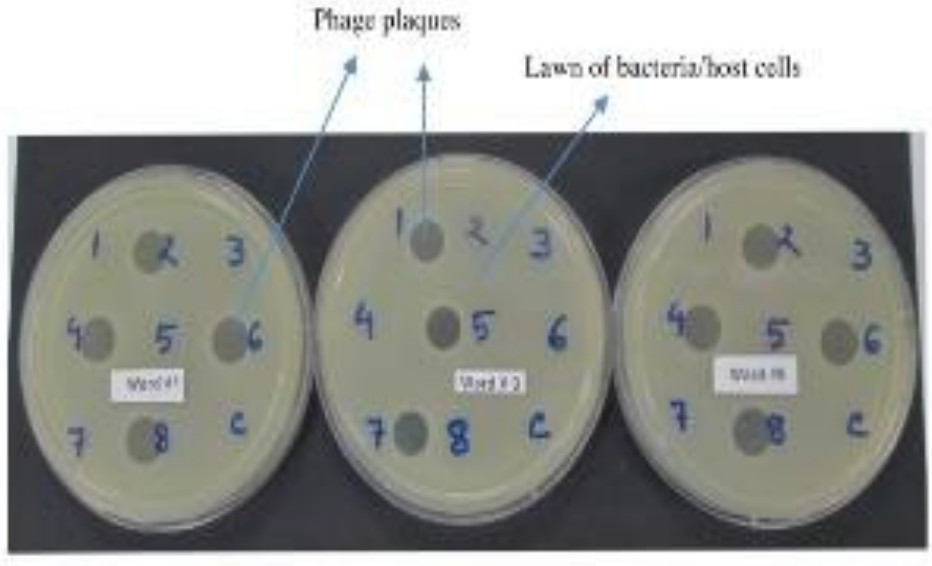
Lesson 92: Cultivation of bacteriophages

Cultivation of Bacteriophages

Bacteriophages, also called phages, are viruses that specifically infect bacteria and we can confirm their presence and quantify them using a tool called the plaque assay. Bacteriophages infect their susceptible hosts by first attaching to the bacterial cell wall and injecting their genetic material. Then, they hijack the cell's biosynthetic machinery to replicate their DNA and produce numerous progeny phage particles, which they then release by lysing and killing the host cell.

Phages can be grown in **liquid** bacterial cultures or in bacterial cultures on **solid** media. The bact eriophages were discovered by accident when a liquid culture of bacteria in a lab was accidental ly invaded by bacteriophages. Liquid cultures become quite cloudy as the numbers of bacteria in crease, but to the researcher’s surprise, a culture that was very cloudy and full of bacterial grow th one dawas perfectly clear the next. If a little of this clear solution was added to another cloud y culture, the same thing happened. Finally it was discovered that a virus was attacking the bact eria and causing them to lyse

The plaque assay is a key laboratory method in virology for the isolation and enumeration of phage particles, as well as for measuring virulence of the bacteriophages, and studying the growth of lytic viruses. A plaque is a small, clear area on an agar plate where the host bacteria have been lysed as a result of the lytic life cycle of the infecting bacteriophages. As the bacteria replicate on the plate they form a "lawn" of confluent growth. Meanwhile, each bacteriophage that adsorbs to a bacterium will reproduce and cause lysis of that bacterium. The released bacteriophages then infect neighboring bacteria, causing their lysis. Eventually a visible self-limiting area of lysis, a plaque, is observed on the plate.



This lytic activity is the basis of a widely used phage enumerating technique known as the plaque assay or double agar layer assay. Here, a bacteriophage mix is first prepared in a molten nutrient broth containing low concentration agar. All bacteria used in the mix should be alive and actively dividing in the log phase of their growth, which will ensure that a large percentage of the bacteria are viable and able to form a dense lawn around the plaques. Next, this molten bacterial-phage agar mix is spread over a more solid, concentrated agar nutrient medium which is already solidified on a Petri dish. On incubation at room temperature, the low concentration agar-phage-bacteria broth also solidifies to form a soft agar overlay.

Here, the bacterial cells can derive additional nutrients from the bottom layer and should rapidly multiply to produce a confluent lawn of bacteria. However, as phage particles are also present in the soft layer, these will infect and replicate their genetic material within the bacteria, culminating in cell lysis, which releases multiple progeny. These phage progeny then infect the neighboring cells, as the semi-solid state of the bacteria-phage layer restricts their movement to more distantly located host cells. This cycle of infection and lysis continues over multiple rounds, killing a large number of bacteria in a localized area. The effect of the

neighboring cells being destroyed, is to produce a single circular clear zone, called a plaque, which can be seen by the naked eye, effectively amplifying the bacteria lytic activity of the phage and enabling their enumeration.

The number of plaques on a Petri dish are referred to as Plaque-Forming Units, or PFUs, and, providing the initial bacteriophage concentration was sufficiently dilute, should directly correspond to the number of infective phage particles in the original sample. This technique can also be used for characterization of plaque morphology, to aid in identification of phage types, or to isolate phage mutants. In this lab, you will learn how to perform the plaque assay for enumerating phages, using the T7 phage of *E. coli* as an example.

Lesson 93: Cultivation of animal viruses

Growing Animal Viruses in Embryonated Eggs

Viruses are obligate intracellular parasites that require living cells in order to replicate. Cultured cells, eggs, and laboratory animals may be used for virus isolation. Although embryonated eggs and laboratory animals are very useful for the isolation of certain viruses, cell cultures are the sole system for virus isolation in most laboratories.

- The process of cultivation of viruses in embryonated eggs depends on the type of egg which is used.
- Viruses are inoculated into chick embryo of 7-12 days old.
- For inoculation, eggs are first prepared for cultivation, the shell surface is first disinfected with iodine and penetrated with a small sterile drill.
- After inoculation, the opening is sealed with gelatin or paraffin and incubated at 36°c for 2-3 days.
- After incubation, the egg is broken and virus is isolated from tissue of egg.
- Viral growth and multiplication in the egg embryo is indicated by the death of the embryo, by embryo cell damage, or by the formation of typical pocks or lesions on the egg membranes
- Viruses can be cultivated in various parts of egg like chorioallantoic membrane, allantoic cavity, amniotic sac and yolk sac. So inoculation can be divided into
- Chorioallantoic membrane inoculation

- Amniotic inoculation
- Allantoic inoculation
- Yolk sac inoculation

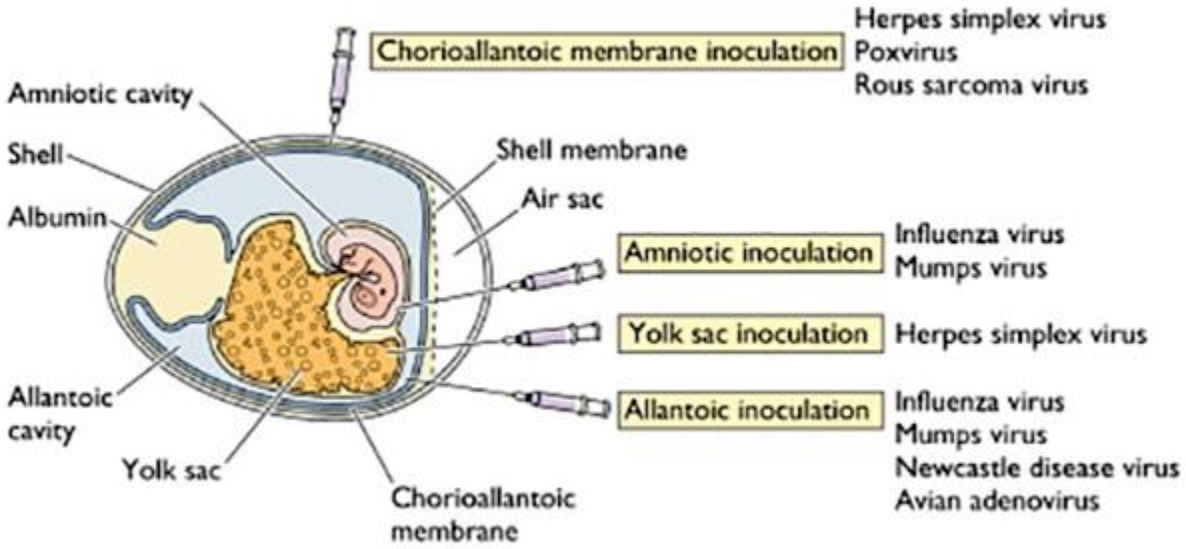


Figure: Inoculation routes into embryonated egg

Advantages

- Convenient
- Inexpensive
- Host for many viruses

Purpose

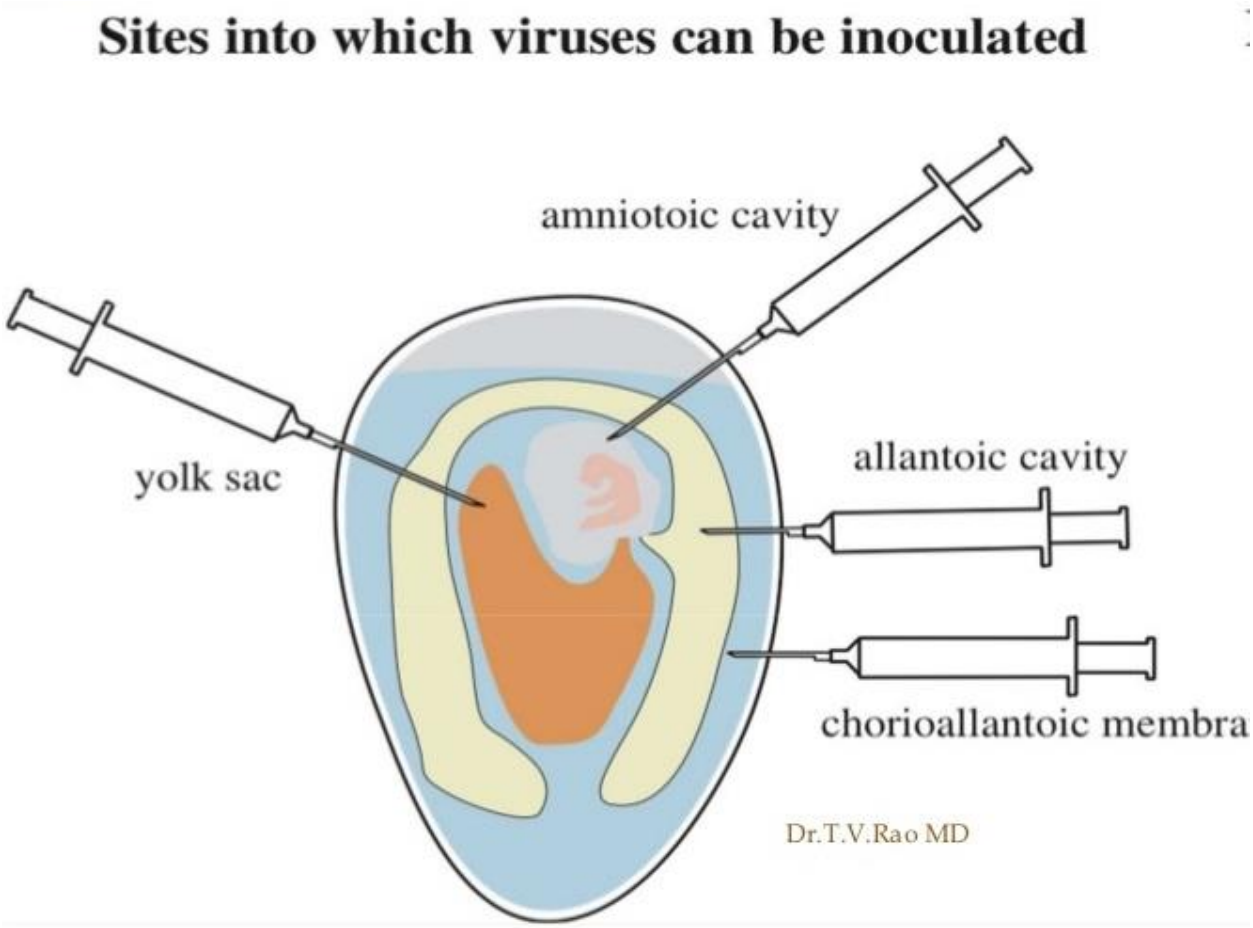
- Isolation and growth of virus
- Growth for vaccination

Lesson 94: CAM inoculation

Chorioallantoic Membrane (CAM) Inoculation

- This inoculation route is mainly for growing poxvirus.
- After incubation, visible lesions called pocks are observed, which is grey white area in transparent CAM.

- Herpes simplex virus is also grown.
- Single virus gives single pocks
- This method is suitable for plaque studies.



The chicken chorioallantoic membrane (CAM) is a simple, highly vascularized extraembryonic membrane, which performs multiple functions during embryonic development. Over the last two decades, interest in the CAM as a robust experimental platform to study blood vessels has been shared by specialists working in bioengineering, development, morphology, biochemistry, transplant biology, cancer research and drug development.

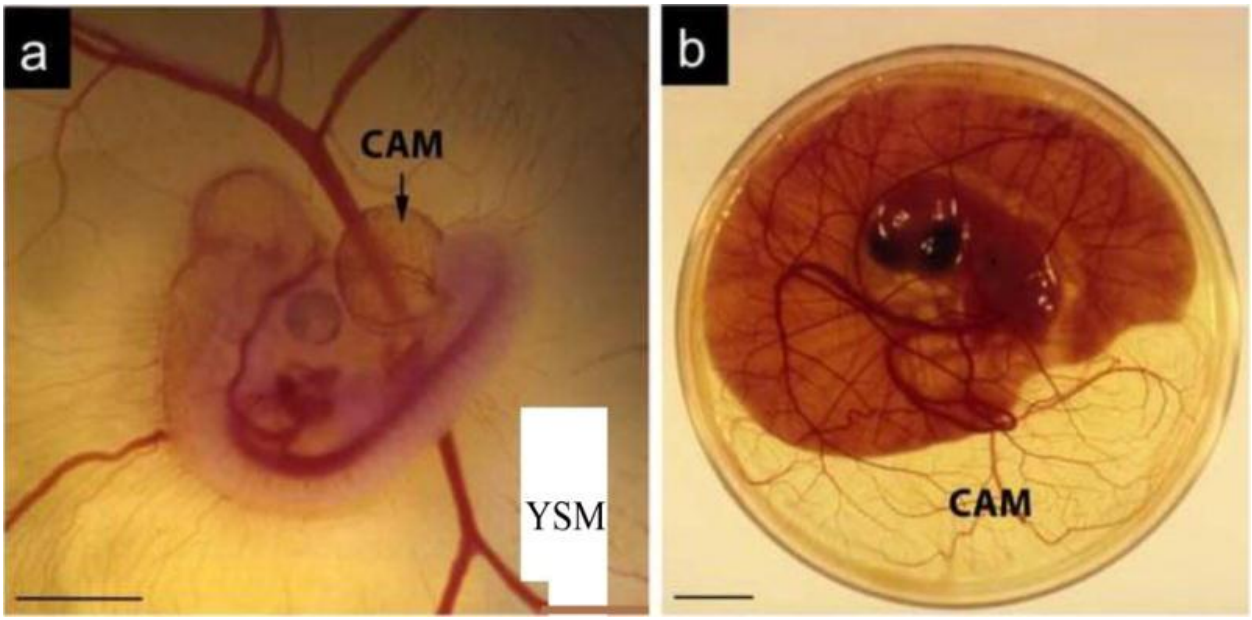


Figure: Chicken embryo and associated extra-embryonic structures. a, Chicken embryo at day 4 post-fertilization the highly vascularized chorioallantoic membrane (CAM) that expands from the hind-gut. The yolk sac membrane (YSM), also highly vascularized, is seen in the background; b Embryo at day 12 post-fertilization into a 10 cm petri dish.

Perform CAM (Chorioallantoic membrane) inoculation using the following procedure:

- Mark an area free from large blood vessels on the side where the embryo is located and the area over the air sac.
- Make one hole on the side and one over the air sac. Puncture the shell membrane taking care not to damage the chorioallantoic membrane.
- Gently apply suction to the hole over the air sac.
- Add 1.0 ml inoculum on the dropped membrane, rotate egg to distribute inoculum over membrane.
- Seal hole with nail polish.
- Candle inoculated eggs daily. Any that die within 24 hours should be discarded.

CAM harvesting:

- Make a tape handle over the area of inoculation (cut a piece of tape, without joining either end, join the middle of the piece together leaving both ends free to adhere to the egg shell with a middle piece that is no longer sticky as the handle sticking up from the egg shell).

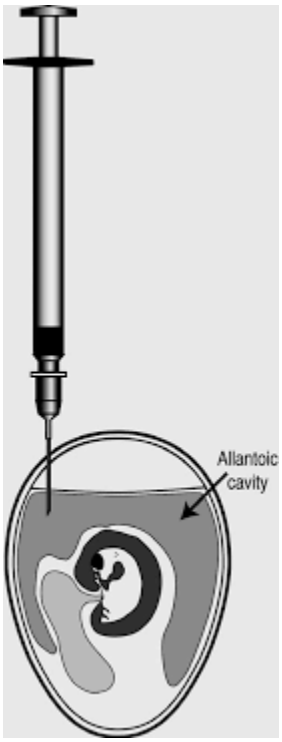
- Cut off the top half of the eggshell, including the infected area, and gently remove the CAM, which is attached to the shell.
- Place the infected CAM in a Petri dish with PBS, spread the membrane flat against the dish and place the dish on a dark surface to facilitate counting of pocks.

Lesson 95: Allantoic Cavity Inoculation

Allantoic Cavity Inoculation

Allantoic Cavity is the most popular and simple method for viral inoculation. Allantoic inoculation is employed for the growth and replication of the influenza virus for vaccine production. This will provide a rich yield of influenza and some paramyxoviruses. Most of avian viruses can be isolated using this method.

- This inoculation is mainly done for production of vaccine of influenza virus, yellow fever, rabies.
- Most of avian viruses can be isolated using this method.



Lesson 96: Intraembryonic, intracranial and intravenous inoculations

Intraembryonic, Intracranial and Intravenous Inoculation

These inoculations of embryos are not widely used.

Embryonic inoculation:

Embryos are inoculated and incubated for a period of one to six days during which they are observed twice a day for the evidence of death of embryo. Some viruses, such as equine encephalomylietis virus kill the embryo within 24 hours while others kill the embryos in 2 to 3 days. Some viruses readily multiply within the embryonic tissue and are seldom lethal.

- Embryo death is accompanied by collapse of the large blood vessels of the chorioallantois so that they are difficult to see upon candling the egg, whereas they are quite distinct while the embryos are alive. Normal movements of the embryo are no longer observed if the embryo is moribund or dead.
- If the embryo is not dead 5 or 6 days of incubation, it is killed or examined for lesions.

Lesson 97: Agglutination and precipitation

Agglutination and precipitation

Viruses maybe sensitized by specific antibody. In the presence of electrolyte the virus particles cling to each other forming aggregates that can be observed through examinations either microscopic or macroscopic.

Flocculuation

Size of virus is intermediate between bacteria and protein molecule. Hence the term, Flocculation is more appropriate.

Applications

Two ways:

1. If virus is known, antibodies can be detected
2. If known serum is available, an unknown virus can be detected

Challenges

Preparation of antigen is difficult

Viruses must be grown in in the animal tissues

Lesson 98: Complement Fixation Test

Complement Fixation

The complement system is a collection of several proteins which interact with each other to initiate a series of reactions to assist in the defense of our immune system.

The complement fixation test is an immunological test used to detect the presence of either specific antibody or specific antigen on the basis of complement fixation. The complement fixation is a system of serum proteins that react with antigen-antibody complexes. If this reaction occurs on a cell surface, it will result in the formation of trans-membrane pores and therefore destruction of the cell.

SHORT PROCEDURE

- Serum is separated from the patient.
- Patients naturally have different levels of complement proteins in their serum. To negate any effects this might have on the test, the complement proteins in the patient's serum must be destroyed and replaced by a known amount of standardized complement proteins.
    - The serum is heated in such a way that all of the complement proteins—but none of the antibodies—within it are destroyed.
    - A known amount of standard complement proteins are added to the serum. (These proteins are frequently obtained from guinea pig serum.)
- The antigen of interest is added to the serum.
- Sheep red blood cells (sRBCs) which have been pre-bound to anti-sRBC antibodies are added to the serum.

RESULT INTERPRETATION

- If the patient's serum contains antibodies against the antigen of interest, they will bind to the antigen in step 3 to form antigen-antibody complexes.
- The complement proteins will react with these complexes and be depleted.
- Thus when the sRBC-antibody complexes are added in step 4, there will be no complement left in the serum.
- However, if no antibodies against the antigen of interest are present, the complement will not be depleted and it will react with the sRBC-antibody complexes added in step 4, lysing the sRBCs and spilling their contents into the solution, thereby turning the solution pink

Lesson 99: Serum Neutralization Test

Serum Neutralization Test

Serum neutralization or plaque reduction assays are serological methods used to confirm the identity of an unknown viral isolate. The procedures involve the use of a known dilution of specific neutralizing antiserum mixed with multiple dilutions of the homologous and suspect virus and subsequent observation of the ability of those viruses to produce Cytopathic effects when inoculated onto a sensitive cell line. Normal serum from the species of animal used to produce the antiserum (usually rabbit or goat) is used as the negative control to account for nonspecific inhibitors of the virus.

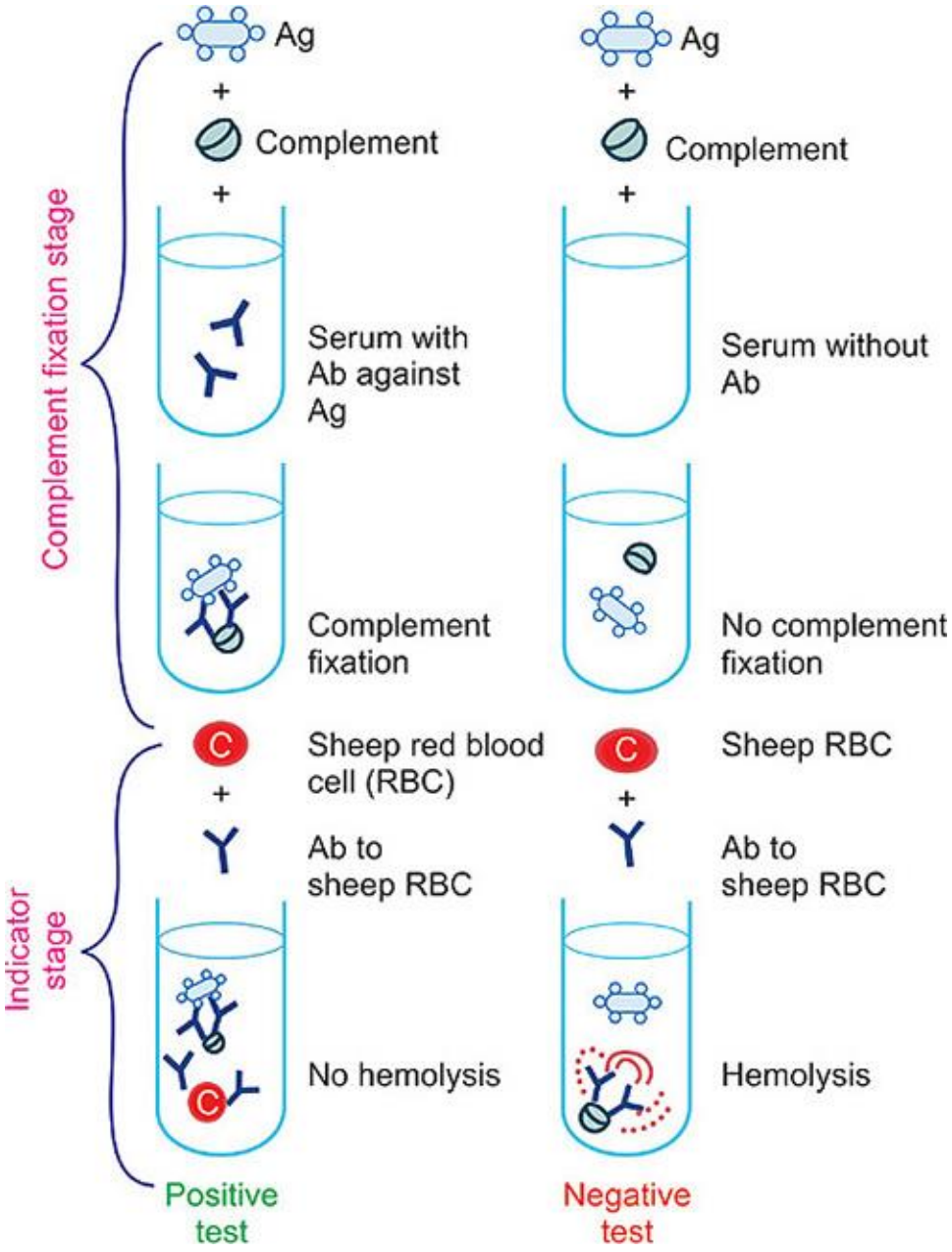
Basic principles

- Wide range of dilutions used
- Equal volumes of viral dilution
- Equal volume of virus dilution and immune serum used
- Inoculation of 5 to 6 animals
- Equal number of controls
- Observation of animals for signs, lesions or deaths

For comparisons

- L.D50 (Lethal dose 50)

- Viral dilutions checked for causing deaths and infectivity
- Titer noted



APPLICATIONS

- Detection of presence of antibodies in serum
- Identification of unknown virus by using known antibody

Lesson 100: Virus Neutralization Test

Virus neutralization test

Neutralization of viruses by their specific antibodies are called virus neutralization tests. Inoculation of viruses in cell cultures, eggs, and animals results in the replication and growth of viruses. When virus-specific neutralizing antibodies are injected into these systems, replication and growth of viruses is inhibited. This forms the basis of virus neutralization test.

If the antibodies specific for the respective virus are inoculated in these mediums, it can halt the growth and replication of viruses. It is the fundamental principle of the virus neutralization test.

The virus contains certain antigenic determinants to which the antibodies initiate the neutralization of the effects of the virus.

Cytopathic Effect (CPE) is the change that occurs in the structure of the cells of the host invaded by the virus. It can result in the death of the cells. e. g. Poliovirus produces cytopathic effects.

Requirements

Serum sample, Known viral suspension, Cell culture suitable for the virus, Diluting solution

Procedure

The general procedure is as follows:

1. The patient's serum is taken and diluted.
2. It is then mixed with the known viral suspension in equal volume and incubated.
3. Then the mixture is inoculated in cell culture.
4. Incubated and observed for cytopathic effects.
5. A control is also prepared to compare the results.

Result Interpretation

If the sample contains neutralizing antibodies, then the viral components will be neutralized by it and cytopathic effects won't be observed after the test. It will be considered a positive test. The occurrence of cytopathic effects will suggest a negative result.

Pros	Cons
• Can be specific	• Very slow
• Sensitive	• Intensive
• Vaccine production/ immunological status	• Required skilled people
	• Depends on cell lines

Lesson 101: Protection Tests

Protection Tests

This test is not strictly serological but use for the identification of those viruses that are used to produce successful vaccines. This procedure also employed for testing commercial serum and vaccine products. Production of either active or passive immunity in an animal followed by a challenge dose of virulent virus. It is also called in-vivo neutralization test.

Vaccine prepared for all types of immunity to different lab animals. When satisfactory immunity is produced, all are challenged with a known virus. Used commercially for Rabies and hog cholera vaccines. Used by injecting in graded doses

Lesson 102: Result interpretation of Protection Tests

HYPOTHETICAL TEST

- Suppose we have three types of vaccines
- We inject them in three different guinea pigs
- Observe for immunity

Immunized guinea pigs	10-3	10-4	10-5	10-6	10-7
Eastern type vaccine	5	5	2	1	0
Western type vaccine	1	0	0	0	0
Venezuelan type vaccine	5	5	3	0	0
Controls	5	5	3	1	0

Vaccine that provides best immunity and no death proves to be the best

Lesson 103: Haemagglutination /Hemagglutination inhibition Test

Haemagglutination test

- Haemagglutination assay is a diagnostic technique used for the detection of viruses, bacteria, and antibodies.
- The antigens present in various viruses or bacteria bind with the sialic acid receptors present on the surface of red blood cells, creating a network of RBCs and viral particles.
- The formation of these lattices depends on the concentration of viruses/ bacteria and RBCs.
- Based on the amount of agglutination formed in the well, the concentration of the antigen can be determined. The antigen concentration can then be used to determine the concentration of the organism present in the sample.

Hemagglutination occurs when measles viruses and red blood cells are mixed. But, if the serum of a person infected with **measles virus** is mixed with RBC and measles virus, there won't be any agglutination of RBC. This phenomenon is known as **hemagglutination inhibition**. This arises because antibodies present in the serum of that infected person reacts with the measles viruses and neutralize them (positive result).

If the patient's serum do not contain antibodies against surface proteins of test virus, there will be presence of hemagglutination as surface molecules are free to hemaaglutinate RBCs (negative result).

RBC Preparation

- Preparation of 1% RBCs
- Equal amount of blood and Normal Saline (NS) centrifuged in Falcon tube
- Remove supernatant and add the same volume of NS again and centrifuge
- Repeat the procedure until clear supernatant
- Remove the supernatant and add ten times of its volume of NS

Haemagglutination Assay:

- 1 A ninety six well plate marked according to sample
- Each row assigned to one sample
- Add 50uL NS to all wells
- Add antigen to first well and make two fold dilution till second last well
- Leave last well as control
- Add 25uL RBCs solution to all wells
- 96 well plate labelled according to samples
- 25uL antiserum added to first well and make two fold dilution till end
- 0.5uL RBCs added to all wells
- Incubation for 30 minutes

Hemagglutination inhibition test

The basis of the HAI assay is that antibodies to that particular virus will prevent attachment of the virus to RBC. Therefore, hemagglutination is inhibited when antibodies are present.

The HAI test may be complicated by the presence of non-specific inhibitors of viral haemagglutination and naturally occurring agglutinins of the erythrocytes. Therefore, the sera should be treated before use or false positive or negative results may arise.

Procedure:

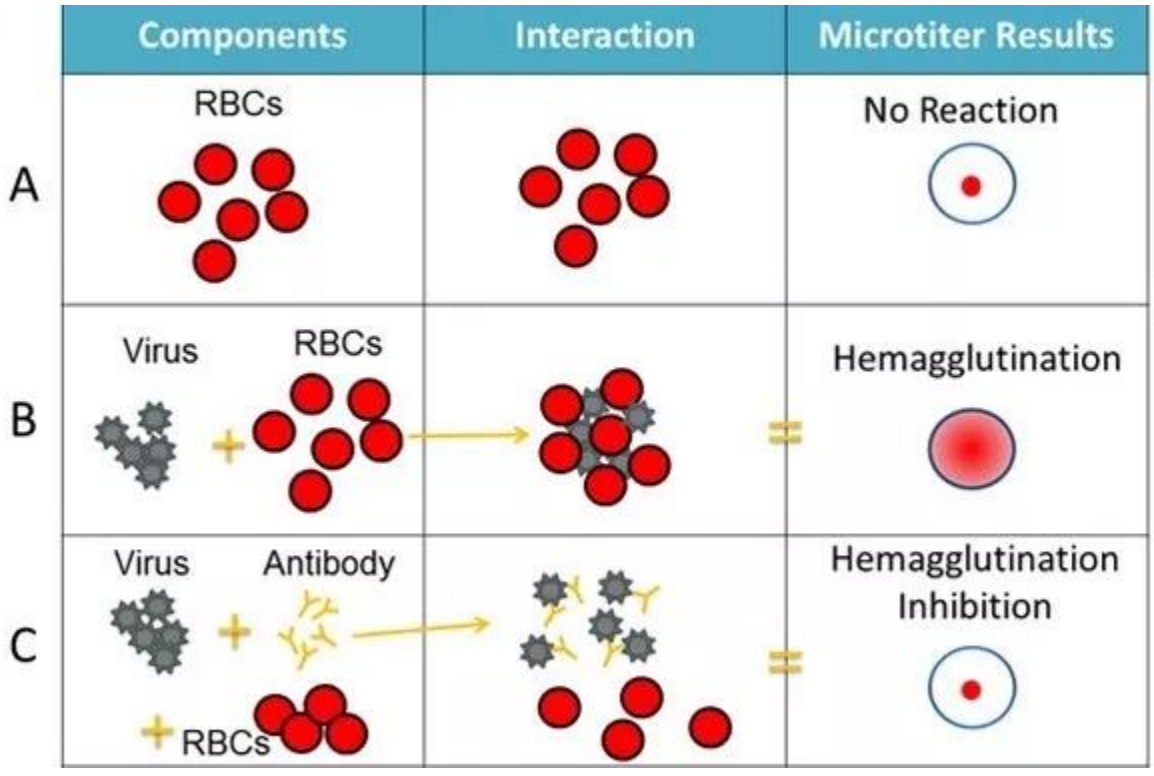
1. Obtain a preparation of virus (e.g. influenza viruses) with known HA titer or determine its HA titer
2. Prepare two-fold dilutions of patient/test serum to be tested e.g. from 1:4 to 1:1024.

3. Add a fixed amount of virus to every well of a 96-well plate, equivalent to 4 HA units (varies according to the virus), except for the serum control wells.
4. Allow the plate to stand at room temperature for 60 minutes (time varies according to specific requirements).
5. Add red blood cells (RBC) and incubate at 4°C for 30 minutes.
6. Read the wells.

Lesson 104 : Result interpretation of HA and HI Tests

Result Interpretations of Hemagglutination (HA) and Hemagglutination Inhibition (HAI) Tests

- Button formation showed no hemagglutination
- Hazy wells showed agglutination



Haemagglutination pattern of serially diluted influenza virus for determination of HA titre

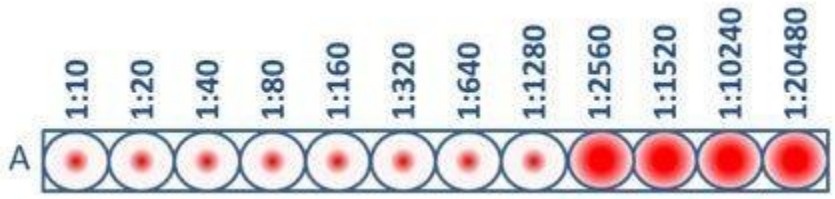
1. Lattice formation upto well 6 shows the presence of one hemagglutination unit (1 HAU) viral load in the well.

2. HA Titre is the reciprocal of the highest dilution upto which haemagglutination was observed.
   Here,
   HA titre = 320
   Absence of hemagglutination from well 7 to well 11 – showing no or insufficient viral load to cause agglutination.
3. Well 12: Test control

Result Interpretations of HAI Test

HAI Titer: *The highest dilution of serum (Ab) that prevents hemagglutination is called the HAI titer of the serum.*

1. If the serum contains no antibodies that react with virus, then hemagglutination will be observed in all wells.
2. Likewise, if antibodies to virus are present, hemagglutination will not be observed until the antibodies are sufficiently diluted.



3. This virus sample has HAI titer of 1280, which means that the greatest dilution of antibody that still blocked hemagglutination from occurring was at 1280 dilution. At this dilution, the antibodies were still capable of recognizing and binding to the antigens on the virus.

Lesson 105 : Cytopathogenecity

Cytopathic effects

Animal and plant viruses are detected by cytopathic effects. Cytopathic effects (CPEs) are distinct observable cell abnormalities due to viral infection. CPEs can include loss of adherence to the surface of the container, changes in cell shape from flat to round, shrinkage of the nucleus, vacuoles in the cytoplasm, fusion of cytoplasmic membranes and the formation of multinucleated syncytia, inclusion bodies in the nucleus or cytoplasm, and complete cell lysis.

Further pathological changes include viral disruption of the host genome and altering normal cells into transformed cells, which are the types of cells associated with carcinomas and sarcomas. The type or severity of the CPE depends on the type of virus involved.

Basic Ingredients

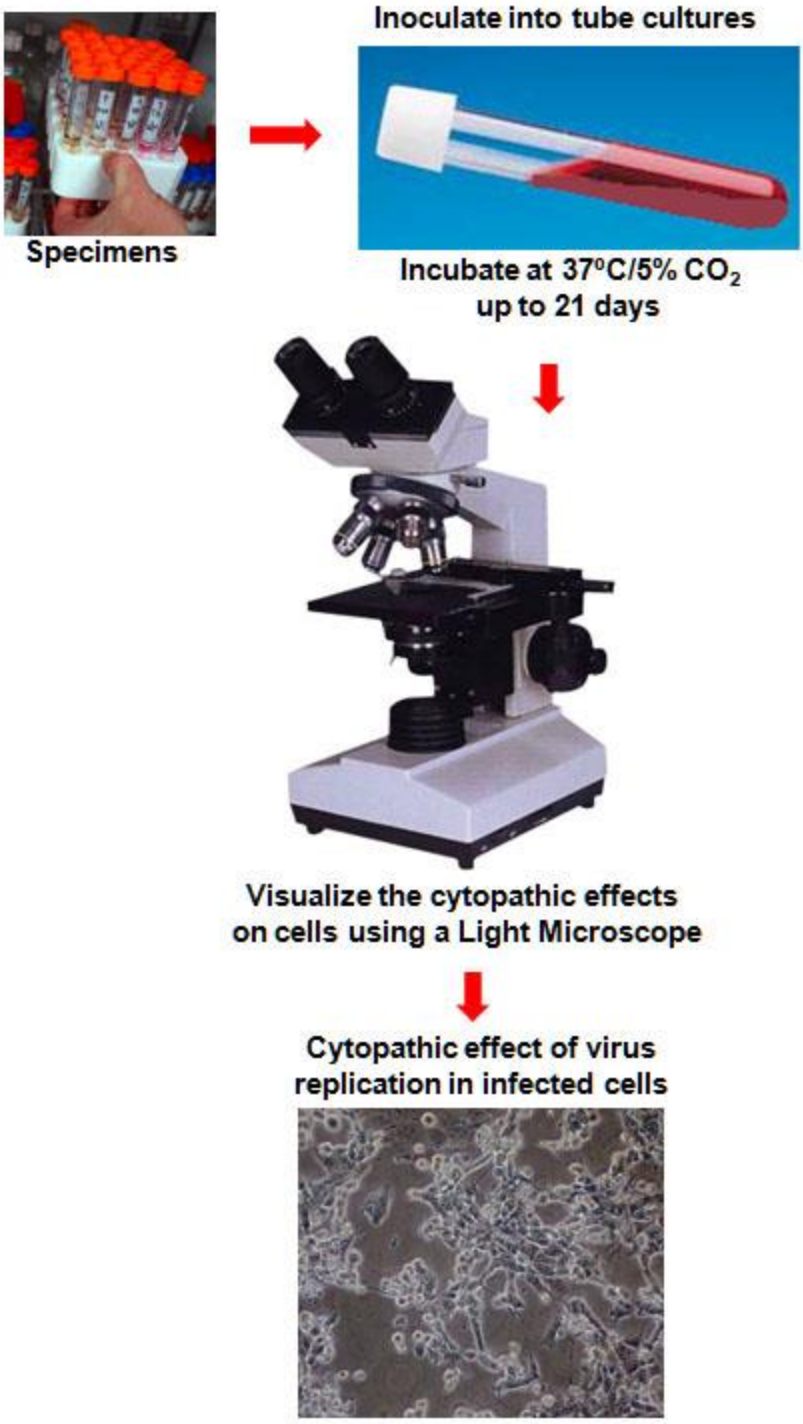
1. Tissue cells
2. Lactalbumin hyrolysate
3. Serum
4. Balanced salt solution

Other Ingredients

1. Ascitic fluid
2. Amniotic fluid
3. Serum ultrafilterate
4. Amino acids
5. Vitamins
6. Glucose
7. Inorganic salts
8. Penicillin and streptomycin

Glassware used must be absolutely clean and free film and toxic substances

Chemicals and distil water must be free from toxic ions


Inoculate into tube cultures
Specimens
Incubate at 37°C/5% CO_2
up to 21 days
Visualize the cytopathic effects
on cells using a Light Microscope
Cytopathic effect of virus
replication in infected cells

Cell death

Well-known virus-induced CPE is **cell death**. Many viruses kill cells either by lysis or by inducing apoptosis. For example, HIV is known for killing $CD4^+$ T lymphocytes, which is the main reason why infected individuals become immunocompromised.

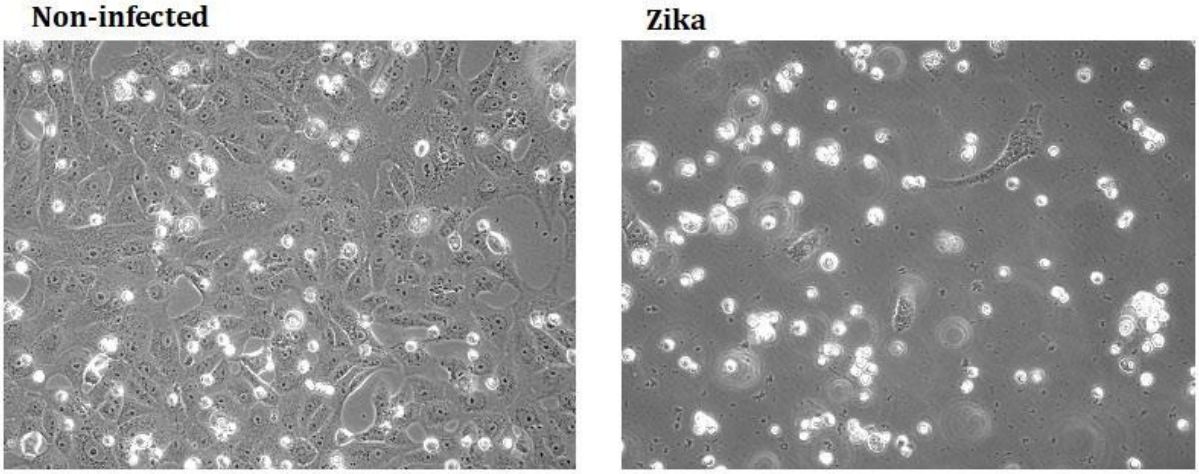


Figure: Vero cells were infected with Zika virus. After 72 h of infection, a clear CPE is evident by the presence of cell debris as a result of cell death.

Syncytium

This is also known as cell fusion. In this Cytopathic effect, the plasma membranes of the host cells fuse and form an enlarged cell. This cell contains at least four nuclei. This can be detected by fixing and staining the host cells. Herpes virus and paramyxovirus can be detected this way.

Inclusion Bodies

Insoluble abnormal structures in the nuclei or cytoplasm of the cells can only be detected through staining. inclusion bodies are areas of altered staining in cells. These inclusions may be large, small, single, multiple, intranuclear, intracytoplasmic, basophilic, or eosinophilic. Chromatin migration causes a thin stained ring around the edge of the nucleus.